Table 502/26-16-00-993-802

SMOKE DETECTOR PAIR	FWD DET SWITCH POSITION	AFT DET SWITCH POSITION	CARGO BAY	LOCATION	EQUIPMENT NUMBER
A1	NORM	A	AFT	FWD LEFT	M2242
B1	NORM	B	AFT	FWD RIGHT	M2245
A2	NORM	A	AFT	MID LEFT	M2243
B2	NORM	B	AFT	MID RIGHT	M2246
A3	NORM	A	AFT	AFT RIGHT (AFT)	M2244
B3	NORM	B	AFT	AFT RIGHT (FWD)	M2247

G. Put the Airplane Back to Its Usual Condition

SUBTASK 26-16-00-860-009

(1) On the P8 cargo fire control panel, set the FWD DET SELECT and AFT DET SELECT switches to NORM position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO BAY SMOKE DETECTOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the cargo bay smoke detector

(2) An installation of the cargo bay smoke detector.

TASK 26-16-01-000-801

2. Cargo Bay Smoke Detector Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the cargo bay smoke detector.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-16-01-100-801	Clean the Cargo Bay Smoke Detectors (P/B 701)

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

D. Prepare for the Removal

SUBTASK 26-16-01-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

E. Cargo Bay Smoke Detector Removal

SUBTASK 26-16-01-860-006

CAUTION DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Before the smoke detector [2] is touched, do the procedure for devices that are sensitive to electrostatic discharge, as follows:

- ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802
- ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-16-01-020-001

(2) Remove the screws [3] that hold the guard assembly [1] to the cargo ceiling.

SUBTASK 26-16-01-020-002

(3) Remove the screws [4] that hold the smoke detector [2] to the detector pan.

SUBTASK 26-16-01-020-003

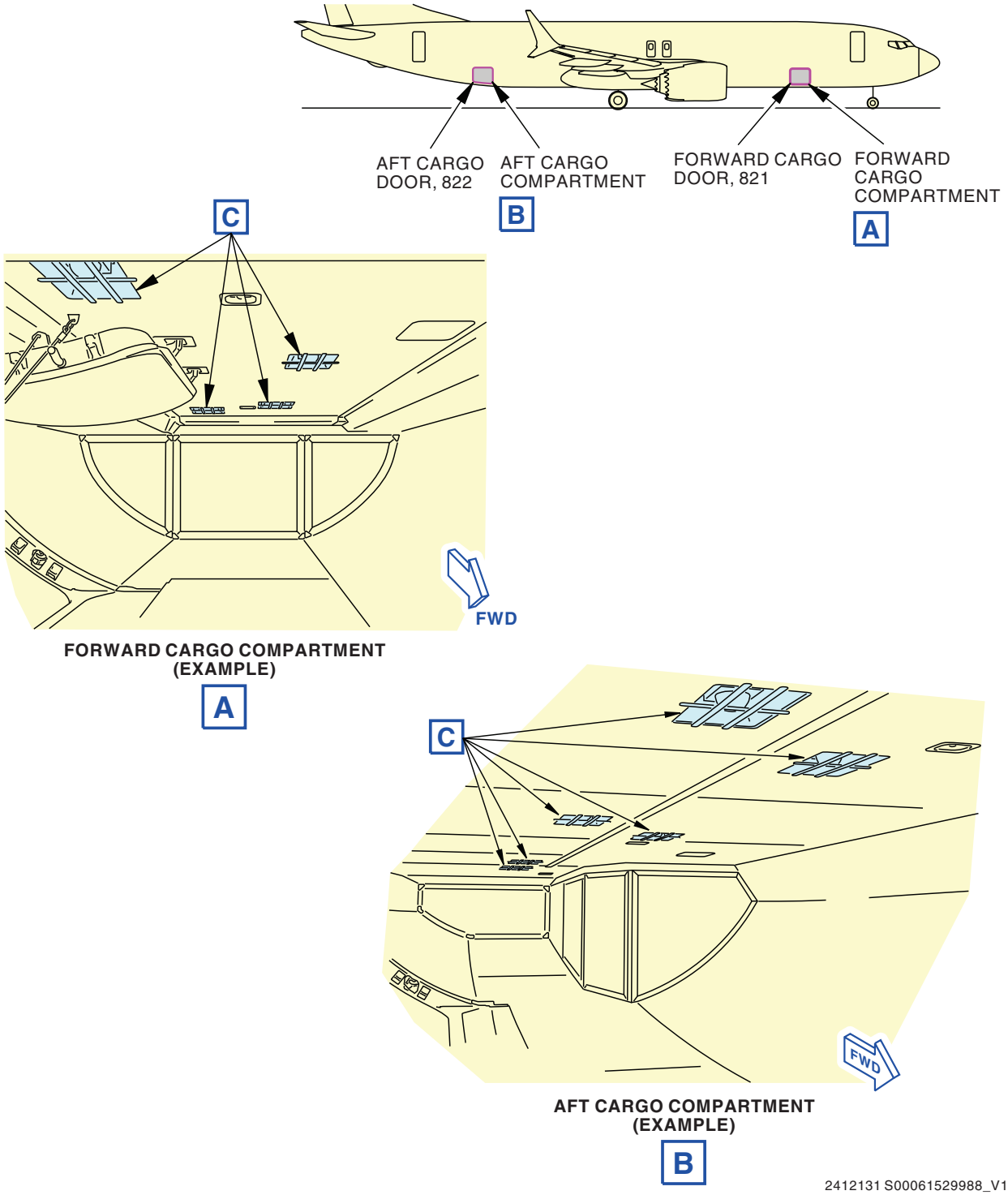
(4) Disconnect the electrical connector from the smoke detector [2].

SUBTASK 26-16-01-160-001

(5) If it is necessary, do this task: Clean the Cargo Bay Smoke Detectors, TASK 26-16-01-100-801.

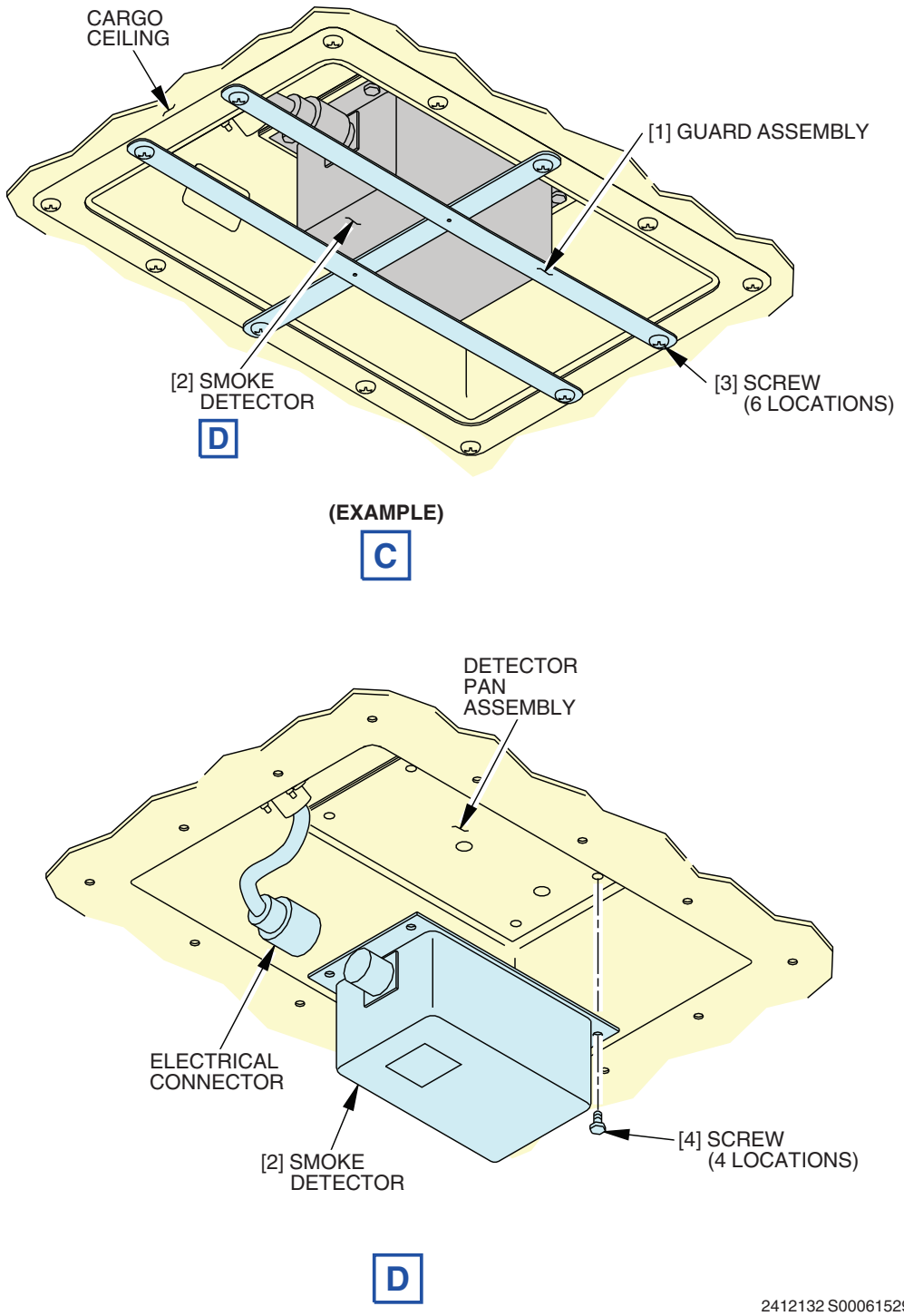
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cargo Bay Smoke Detector Installation
Figure 401/26-16-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cargo Bay Smoke Detector Installation
Figure 401/26-16-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-16-01-400-801

3. Cargo Bay Smoke Detector Installation

(Figure 401)

A. General

(1) This task gives instructions to install the cargo bay smoke detector.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-16-00-710-801	Cargo Bay Smoke Detection - Operational Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Smoke detector	26-16-01-05-210	PTF ALL
		26-16-01-08-220	PTF ALL

D. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

E. Prepare for the Installation

SUBTASK 26-16-01-860-002

(1) Make sure that these circuit breakers are open:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

F. Cargo Bay Smoke Detector Installation

SUBTASK 26-16-01-860-007

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Before the smoke detector [2] is touched, do the procedure for devices that are sensitive to electrostatic discharge, as follows:

- ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802
- ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-16-01-420-001

(2) Connect the electrical connector to the smoke detector [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-16-01-420-002

(3) Put the smoke detector [2] in position on the detector pan.

(a) Install the screws [4].

SUBTASK 26-16-01-420-003

(4) Put the guard assembly [1] in position on the cargo ceiling.

(a) Install the screws [3].

G. Cargo Bay Smoke Detector Installation Test

SUBTASK 26-16-01-860-003

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-16-01-710-001

(2) Do this task: Cargo Bay Smoke Detection - Operational Test, TASK 26-16-00-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO BAY SMOKE DETECTOR - CLEANING/PAINTING

1. General

A. This procedure has a task that cleans the cargo bay smoke detectors.

B. Each cargo bay has smoke detectors installed in the ceiling. The cleaning procedure is the same for each smoke detector.

TASK 26-16-01-100-801

2. Clean the Cargo Bay Smoke Detectors

A. References

Reference	Title
26-16-00-710-801	Cargo Bay Smoke Detection - Operational Test (P/B 501)

B. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle
STD-1086	Gloves - Rubber
STD-3939	Air Source - Regulated, Dry Filtered, 0 to 10 psig

C. Consumable Materials

Reference	Description	Specification
B00065	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002 (Supersedes O-A-396)
B50073	Alcohol - Isopropyl	ASTM D 770
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

E. Clean the Smoke Detector

SUBTASK 26-16-01-860-004

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-16-01-160-002

WARNING

WHEN YOU CLEAN THE SMOKE DETECTOR AND SENSOR, MAKE SURE THAT THE AREA HAS SUFFICIENT AIRFLOW. KEEP THE ALCOHOL AWAY FROM HEAT, SPARKS OR FLAMES. DO NOT BREATHE THE FUMES FROM THE ALCOHOL. PUT ON RUBBER GLOVES. ALCOHOL IS A FLAMMABLE AND POISONOUS SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

WARNING

ISOPROPYL ALCOHOL IS AN AGENT THAT IS FLAMMABLE, EXPLOSIVE, AND POISONOUS. MAKE SURE ALL PERSONS OBEY ALL OF THE PRECAUTIONS WHEN ISOPROPYL ALCOHOL IS USED.

- GAS/AIR MIXTURES MORE THAN THE LOWER EXPLOSIVE LIMIT (LEL) CAN CAUSE AN EXPLOSION IF HIGH HEAT, SPARKS, OR FLAMES SUPPLY IGNITION.
- USE IN AN AREA OPEN TO THE AIR.
- CLOSE THE CONTAINER WHEN NOT USED.
- DO NOT GET ISOPROPYL ALCOHOL IN THE EYES, ON THE SKIN, OR ON YOUR CLOTHES.
- DO NOT BREATHE THE GAS.

(2) Put on rubber glove, STD-1086.

SUBTASK 26-16-01-160-003

(3) Remove the dirt from the exterior of the smoke detector with the cotton wiper, G00034, soaked with alcohol, B00065.

SUBTASK 26-16-01-210-001

CAUTION

BLOW THE AIR LIGHTLY INTO THE SENSOR ASSEMBLY. TOO MUCH FORCE CAN CAUSE STATIC ELECTRICITY ON THE ELECTRODES WHICH CAN CAUSE DAMAGE TO THE SMOKE SENSOR.

(4) Use soft bristle brush, STD-123, and compressed nitrogen at a maximum pressure of 15 psig (103 kPa), or with the 0 to 10 psig air source, STD-3939, lightly blow any dirt or particles out of the sensor assembly.

NOTE: Scheduled cleaning is not necessary, but the smoke detector labyrinth must be kept clean to operate accurately.

(a) Clean the connector.

(b) Clean the dust screens.

(c) Clean the inner area of the labyrinth.

1) Remove the smoke detector cover.

NOTE: Refer to the applicable CMM for the instructions. It is not necessary to fully disassemble the smoke detector to clean the inner area of the labyrinth.

2) Clean unwanted material from the inner area of the labyrinth with a soft bristle brush, STD-123, and nitrogen (or clean, dry air).

3) Install the smoke detector cover.

SUBTASK 26-16-01-160-004

(5) Clean the electrical connector pins with alcohol, B50073.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-16-01-860-005

(6) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-16-01-710-002

(7) Do the operational test of the cargo bay smoke detection (TASK 26-16-00-710-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO ELECTRONIC UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the Cargo Electrical Unit (CEU).

(2) Installation of the CEU.

B. The CEU is installed in the ceiling of each cargo bay.

TASK 26-16-02-000-801

2. Cargo Electronic Unit Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Cargo Electronic Unit (CEU).

B. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
141	Aft Cargo Compartment - Left

C. Prepare for the Removal

SUBTASK 26-16-02-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

SUBTASK 26-16-02-020-001

(2) To get access to the CEUs, remove the screws [4] that attach the cover [7] to the cargo bay ceiling.

D. CEU Removal

SUBTASK 26-16-02-860-002

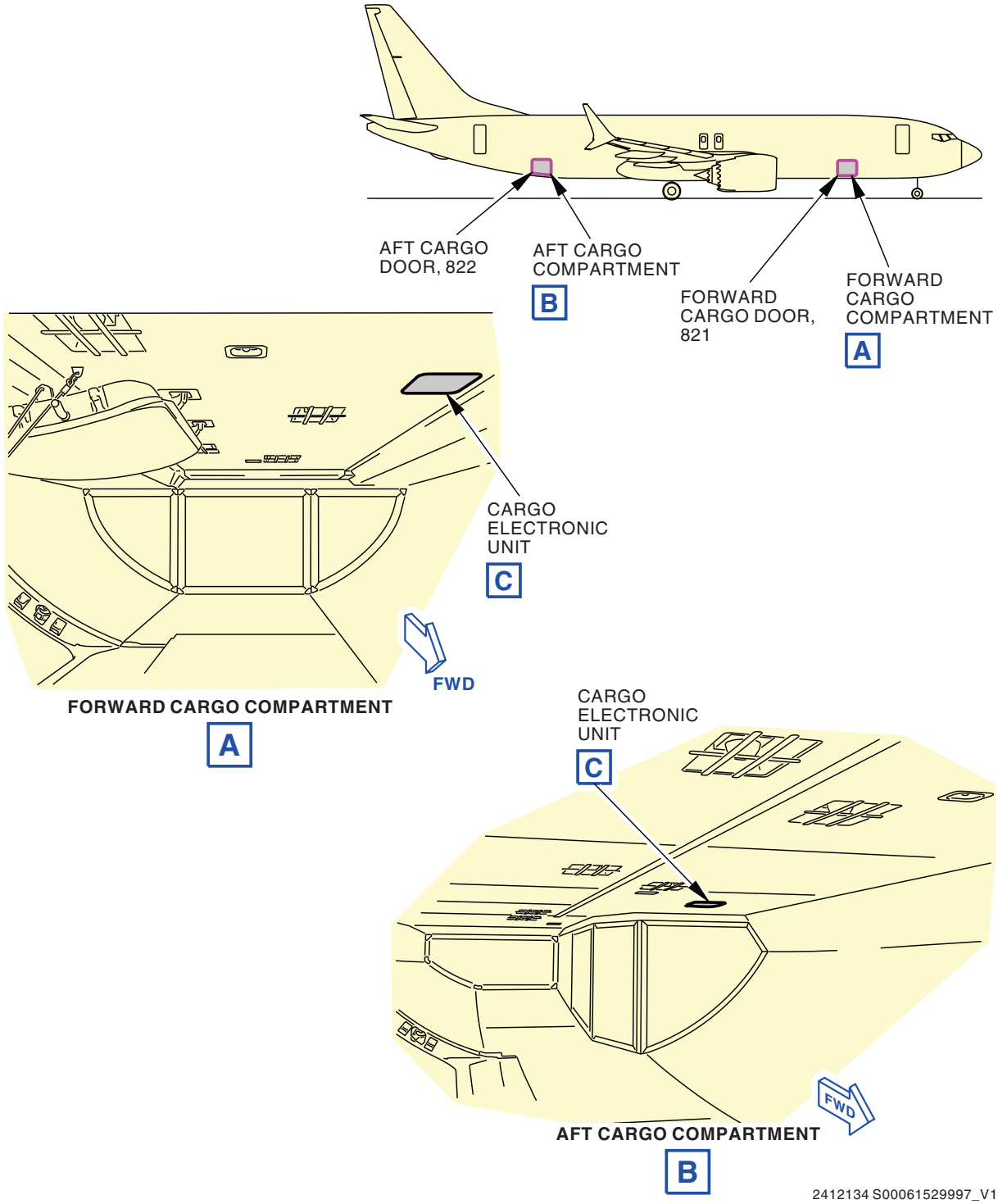
(1) Remove the bolts [2] that attach the cargo electronic unit [1] to the airplane.

SUBTASK 26-16-02-020-002

(2) Disconnect the electrical connectors from the cargo electronic unit [1].

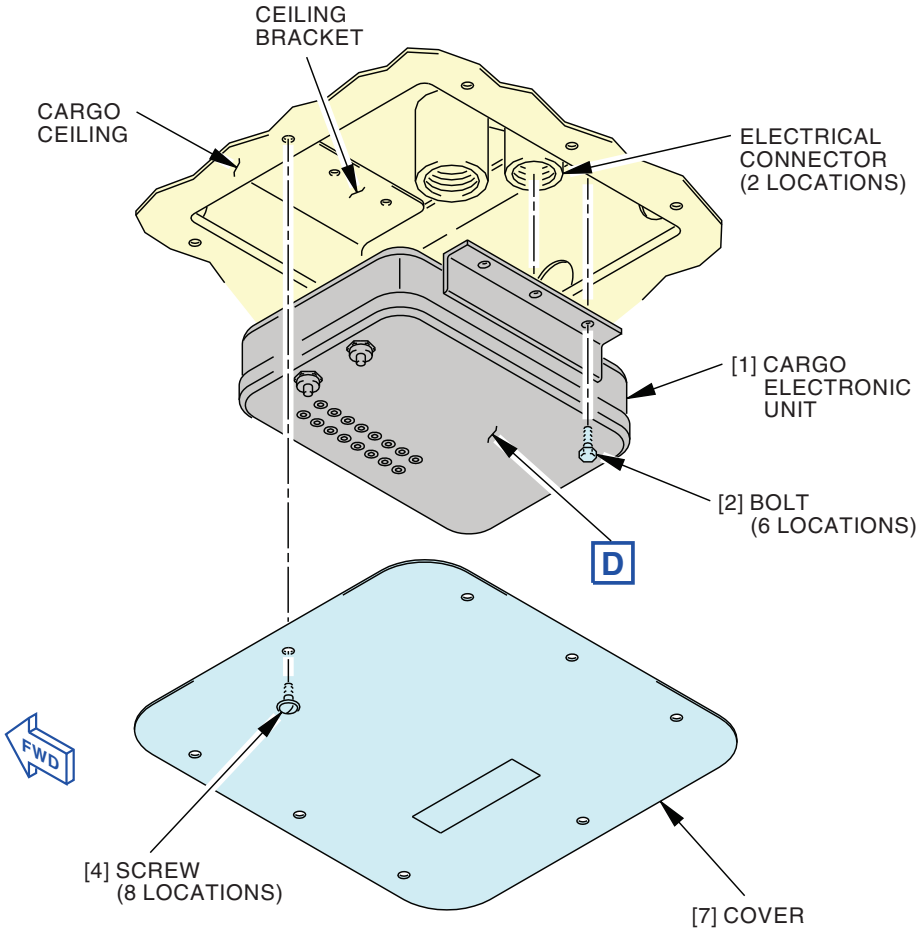
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cargo Electronic Unit Installation
Figure 401/26-16-02-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



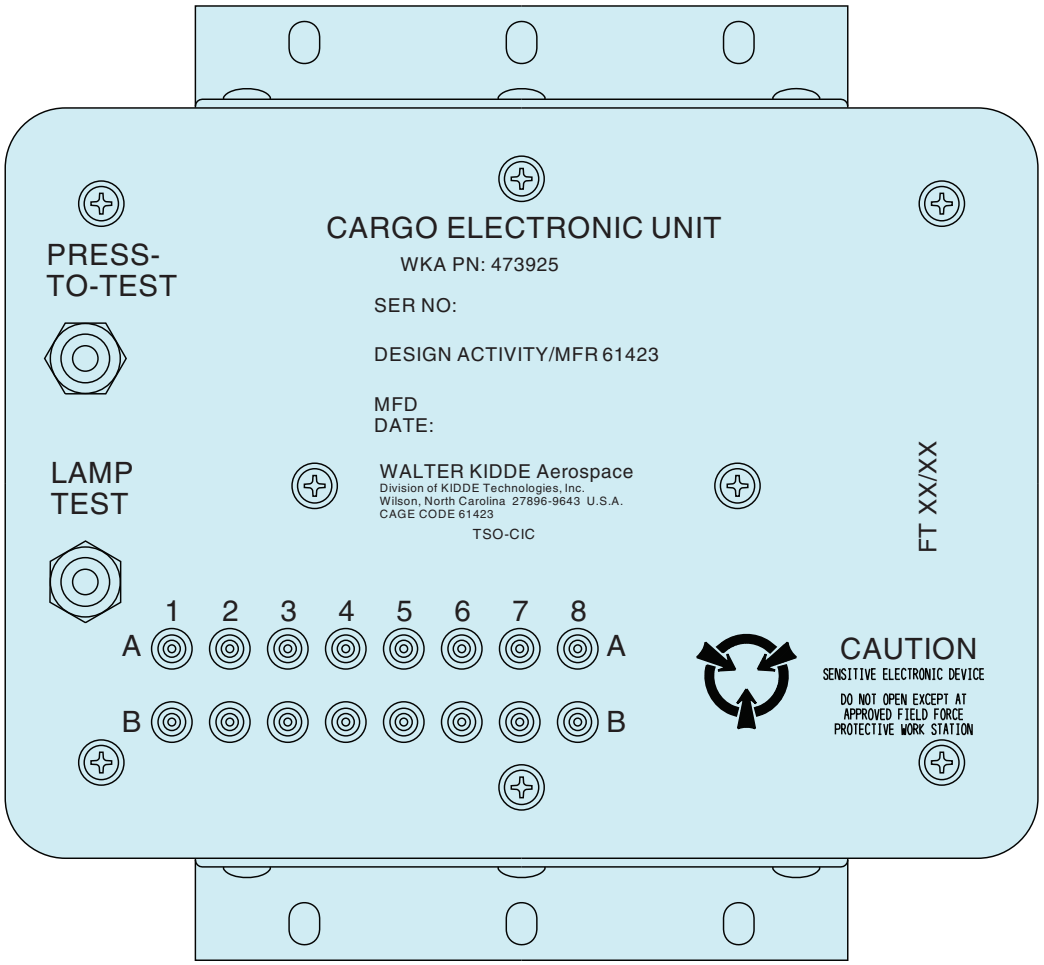
CARGO ELECTRONIC UNIT (EXAMPLE)

C

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Cargo Electronic Unit Installation
Figure 401/26-16-02-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



D

2412136 S00061529999_V1

Cargo Electronic Unit Installation
Figure 401/26-16-02-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-16-02-400-801

3. Cargo Electronic Unit Installation

(Figure 401)

A. General

(1) This task gives instructions for the installation of the forward or aft Cargo Electronic Unit (CEU).

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cargo electronic unit	26-16-02-01-050	PTF ALL
		26-16-02-10-050	PTF ALL

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
141	Aft Cargo Compartment - Left

D. CEU Installation

SUBTASK 26-16-02-420-001

(1) Connect the airplane electrical connectors to the electrical connectors on the cargo electronic unit [1].

SUBTASK 26-16-02-420-002

(2) Put the cargo electronic unit [1] in position on the airplane bracket and attach it with bolts [2].

E. Put the Airplane Back to Its Usual Condition

SUBTASK 26-16-02-860-004

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

SUBTASK 26-16-02-710-005

(2) Do this task: Cargo Electronic Unit Installation Test, TASK 26-16-02-710-801.

——— END OF TASK ———

TASK 26-16-02-710-801

4. Cargo Electronic Unit Installation Test

(Figure 401)

A. Prepare for the Test

SUBTASK 26-16-02-020-003

(1) If they are installed, remove the screws [4] that attach the cover [7] to the cargo bay ceiling.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Test

SUBTASK 26-16-02-710-001

(1) Push and hold the LAMP TEST switch on the CEU.

(a) Make sure all A and B loop smoke detector indicators on the CEU come on.

SUBTASK 26-16-02-710-002

(2) Release the LAMP TEST switch.

(a) Make sure all A and B loop smoke detector indicators on the CEU come off.

SUBTASK 26-16-02-710-003

(3) Push and hold the PRESS TO TEST switch on the CEU for a minimum of five seconds.

(a) Make sure all A and B loop smoke detector indicators on the CEU come on.

NOTE: Properly functioning smoke detectors and unused positions respond by lighting up their corresponding indicator. Unlit indicators represent non-functioning smoke detectors.

(b) Make sure that the flight compartment fire bell in the Aural Warning Module, M315, does not come on.

SUBTASK 26-16-02-710-004

(4) Release the PRESS TO TEST switch.

C. Put the Airplane Back to Its Usual Condition

SUBTASK 26-16-02-420-003

(1) Put the cover [7] in position on the cargo ceiling and attach it with screws [4].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WHEEL WELL, WING AND LOWER AFT BODY OVERHEAT DETECTION SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Wheel well, wing and lower aft body overheat detection system - deactivation.

(2) Wheel well, wing and lower aft body overheat detection system - activation.

TASK 26-18-00-040-801

2. Wheel Well, Wing and Lower Aft Body Overheat Detection System - Deactivation

A. General

(1) This task will deactivate the wheel well, wing and lower aft body overheat detection system.

B. Location Zones

Zone	Area
129	Keel Beam (Part) - Section 43
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
149	Keel Beam (Part) Body Station 727.00 to Body Station 743.95
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box
311	Area Aft of Pressure Bulkhead - Left
431	Engine 1 - Forward Strut Fairing
441	Engine 2 - Forward Strut Fairing
511	Left Wing - Leading Edge To Front Spar
521	Left Wing - Leading Edge to Front Spar
611	Right Wing - Leading Edge to Front Spar
621	Right Wing - Leading Edge to Front Spar

C. Procedure

SUBTASK 26-18-00-020-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

D. Wheel Well, Wing and Lower Aft Body Overheat Detection System - Tryout

NOTE: This tryout is to make sure that the power to the wheel well, wing and lower aft body overheat detection system is in zero energy state.

SUBTASK 26-18-00-410-001

(1) Push and hold the OVHT TEST switch on the P5-10 panel for a minimum of 5 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that neither of the two WING-BODY OVERHEAT lights comes on.

(b) On the P7 panel, make sure that the AIR COND light does not come on.

(c) On the P7 panel, make sure that the captain's and the first officer's MASTER CAUTION lights do not come on.

SUBTASK 26-18-00-020-002

(2) Release the OVHT TEST switch.

SUBTASK 26-18-00-410-002

(3) Put and hold the TEST switch to the OVHT/FIRE position on the P8-1 panel.

(a) Make sure that the WHEEL WELL light does not come on.

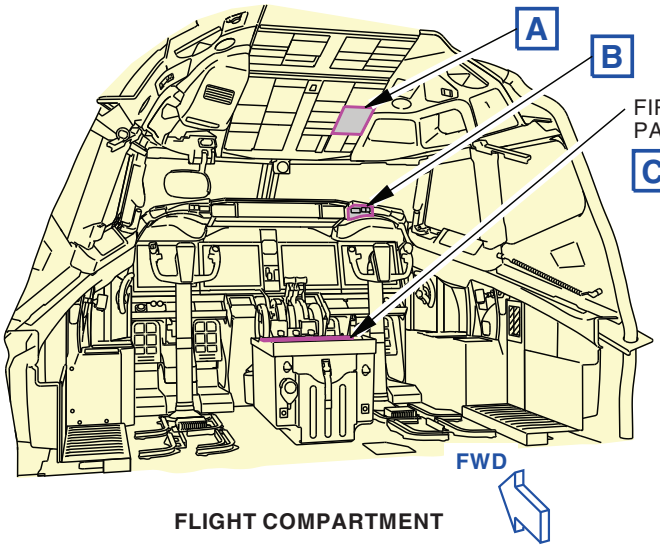
(b) Make sure that the flight compartment fire bell does not come on.

SUBTASK 26-18-00-020-003

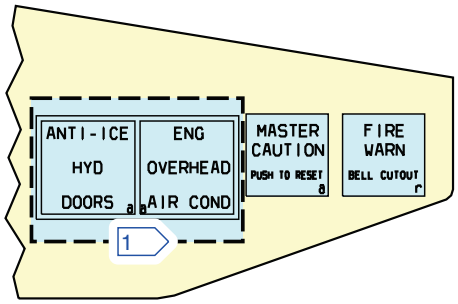
(4) Release the TEST switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

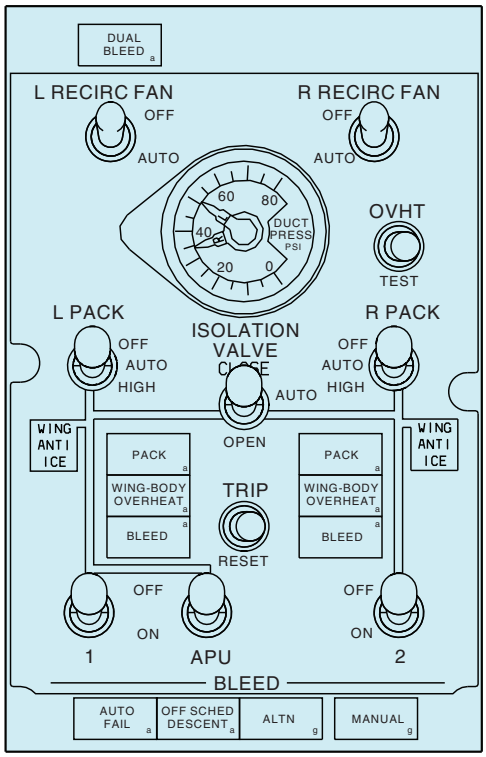


FLIGHT COMPARTMENT



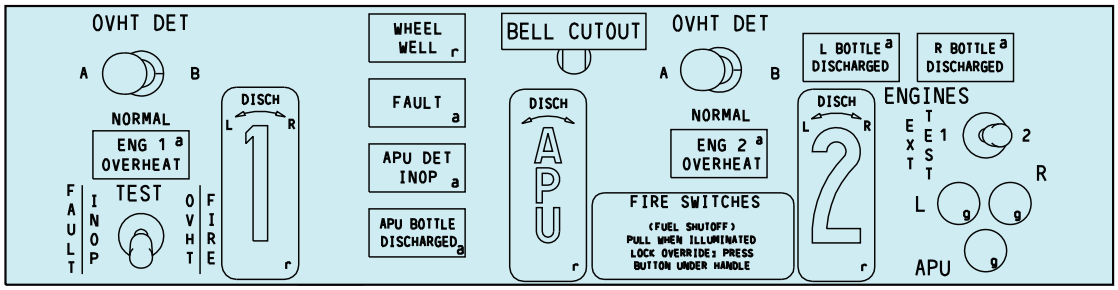
FIRST OFFICER'S WARNING LIGHTS (P7)
(CAPTAIN'S FIRE WARN AND MASTER CAUTION LIGHTS ARE OPPOSITE)

B



BLEED AIR CONTROL MODULE (P5)

A



FIRE CONTROL PANEL (P8-1)

C

1 FIRST OFFICER'S

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Wheel Well, Wing and Lower Aft Body Overheat Detection System
Figure 201/26-18-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-18-00-440-801

3. Wheel Well, Wing and Lower Aft Body Overheat Detection System - Activation

(Figure 201)

A. General

(1) This task will activate the wheel well, wing and lower aft body overheat detection system.

B. Location Zones

Zone	Area
129	Keel Beam (Part) - Section 43
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
149	Keel Beam (Part) Body Station 727.00 to Body Station 743.95
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box
311	Area Aft of Pressure Bulkhead - Left
431	Engine 1 - Forward Strut Fairing
441	Engine 2 - Forward Strut Fairing
511	Left Wing - Leading Edge To Front Spar
521	Left Wing - Leading Edge to Front Spar
611	Right Wing - Leading Edge to Front Spar
621	Right Wing - Leading Edge to Front Spar

C. Procedure

SUBTASK 26-18-00-420-001

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WHEEL WELL, WING AND LOWER AFT BODY OVERHEAT DETECTION SYSTEM - ADJUSTMENT/TEST

1. General

A. The procedures that follow contain the tests for the wheel well, wing and lower aft body overheat detection system, which will be referred to as the overheat detection system throughout the procedures.

B. This procedure consists of this task:

(1) An operational test of the overheat detection system.

TASK 26-18-00-710-801

2. Wheel Well and Lower Aft Body Overheat Detection System - Operational Test

(Figure 501)

A. General

(1) This procedure does a check of the Overheat Fire Detection System without disturbing the detection system and it only uses the equipment that is on the airplane.

(2) This procedure can be performed on the flight deck, or in the E/E bay.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Operational Test

SUBTASK 26-18-00-860-002

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	12	C00132	MASTER CAUTION ANNUNCIATOR BUS 1
B	13	C00131	MASTER CAUTION ANNUNCIATOR BAT
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
D	12	C00310	INDICATOR MASTER DIM BAT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
E	11	C00313	INDICATOR MASTER DIM SECT 1
E	12	C00314	INDICATOR MASTER DIM SECT 2
E	13	C00315	INDICATOR MASTER DIM SECT 3
E	14	C00316	INDICATOR MASTER DIM SECT 4
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6
F	13	C01179	INDICATOR MASTER DIM SECT 7

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(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	14	C01180	INDICATOR MASTER DIM SECT 8

SUBTASK 26-18-00-860-003

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

D. Overheat Detection System Operational Test - Flight Compartment

SUBTASK 26-18-00-710-001

(1) Move and hold the TEST switch on the Engine and APU Fire Control Panel to the OVHT/FIRE position.

(a) Make sure that the WHEEL WELL light on the Engine and APU Fire Control Panel comes on.

(b) Make sure that the Captain's and First Officer's FIRE WARN lights, on the P7 Pilot's Glareshield, come on.

(c) Make sure that the flight compartment fire bell in the Aural Warning Module, M315, comes on.

SUBTASK 26-18-00-710-012

(2) Push the BELL CUTOUT switch on the P8-1 panel.

(a) Make sure that the fire bell goes off immediately.

(b) Make sure that both FIRE WARN lights go off.

(c) Make sure that the WHEEL WELL light, on the P8-1 panel, remains on.

SUBTASK 26-18-00-710-013

(3) Set the LIGHTS switch, on the P1 pilots main panel, to the DIM position.

(a) Make sure that the WHEEL WELL light dims.

SUBTASK 26-18-00-710-014

(4) Set the LIGHTS switch, on the P1 pilots main panel, to the BRT position.

(a) Make sure that the WHEEL WELL light returns to its original brightness.

SUBTASK 26-18-00-710-002

(5) Release the TEST switch on the Engine and APU Fire Control Panel.

(a) Make sure that the WHEEL WELL light on the Engine and APU Fire Control Panel goes off.

(b) Make sure that the Captain's and First Officer's FIRE WARN lights on the Pilot's Glareshield remains off.

(c) Make sure that the flight compartment fire bell in the Aural Warning Module, M315, remains off.

SUBTASK 26-18-00-710-016

(6) Push and hold the OVHT TEST switch, on the P5-10 air conditioning panel, for at least 5 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5-10 air conditioning panel, make sure that these lights come on:

1) The left WING-BODY OVERHEAT light.

2) The right WING-BODY OVERHEAT light.

(b) On the P7 Pilot's Glareshield, make sure that these lights come on:

1) The AIR COND light.

2) The Captain's and First Officer's MASTER CAUTION lights.

SUBTASK 26-18-00-710-017

(7) Release the OVHT TEST switch on the P5-10 air conditioning panel.

(a) On the P5-10 air conditioning panel, make sure that these lights go off:

1) The left WING-BODY OVERHEAT light.

2) The right WING-BODY OVERHEAT light.

(b) On the P7 Pilot's Glareshield, make sure that these lights go off:

1) The AIR COND light.

2) The Captain's and First Officer's MASTER CAUTION lights.

E. Overheat Detection System Operational Test - E/E Bay

SUBTASK 26-18-00-710-005

(1) Push and release the DISPLAY TEST switch on the compartment overheat detection module, M237.

(a) Make sure that all LED's come on, then go off.

SUBTASK 26-18-00-710-006

(2) Push and release the LOCAL TEST switch on the compartment overheat detection module, M237.

(a) Make sure that CONFIG MAX1 comes on while the test is in progress, then the TEST COMPLETE comes on to show there are no faults.

SUBTASK 26-18-00-710-007

(3) Push and release the MEMORY READ switch on the compartment overheat detection module, M237.

(a) Make sure that END OF MEMORY comes on to show the test is complete.

SUBTASK 26-18-00-710-008

(4) Push and hold momentarily the MEMORY CLEAR switch on the compartment overheat detection module, M237.

(a) Make sure that MEMORY CLEAR is displayed.

(b) Make sure that the MAINT ADV light stays off.

F. Put the Airplane Back to Its Usual Condition.

SUBTASK 26-18-00-860-004

(1) Remove the safety tags and close these circuit breakers:

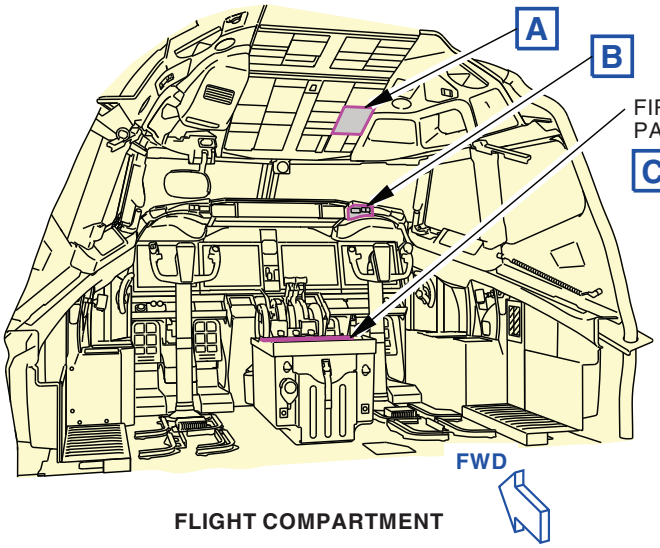
F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

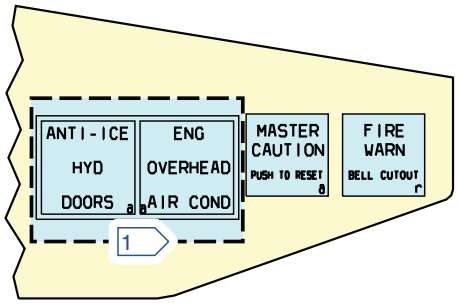
ECCN 9E991 BOEING PROPRIETARY - See title page for details

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

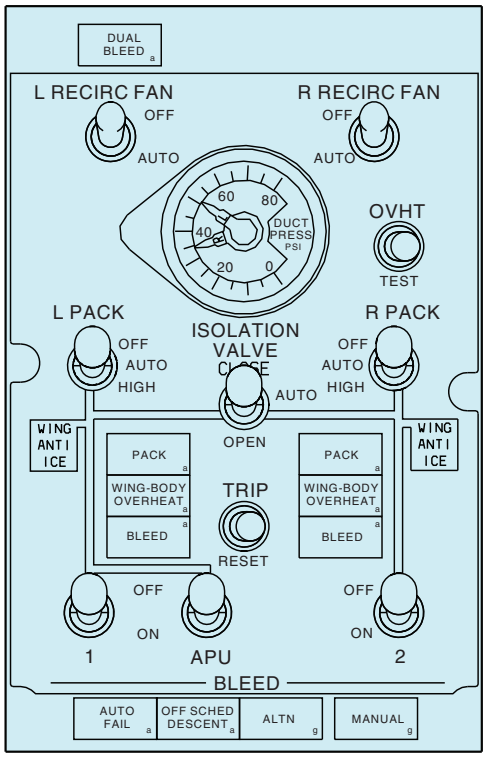


FLIGHT COMPARTMENT



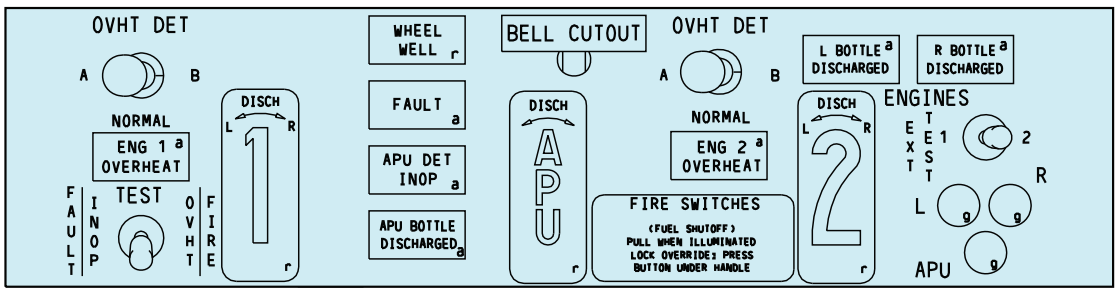
FIRST OFFICER'S WARNING LIGHTS (P7)
(CAPTAIN'S FIRE WARN AND MASTER CAUTION LIGHTS ARE OPPOSITE)

B



BLEED AIR CONTROL MODULE (P5)

A



FIRE CONTROL PANEL (P8-1)

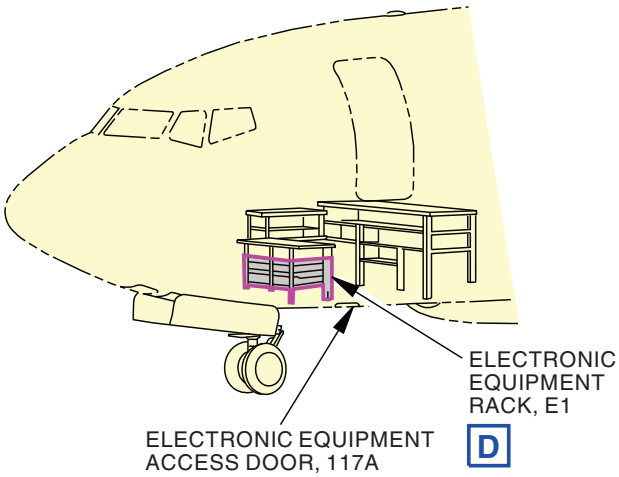
C

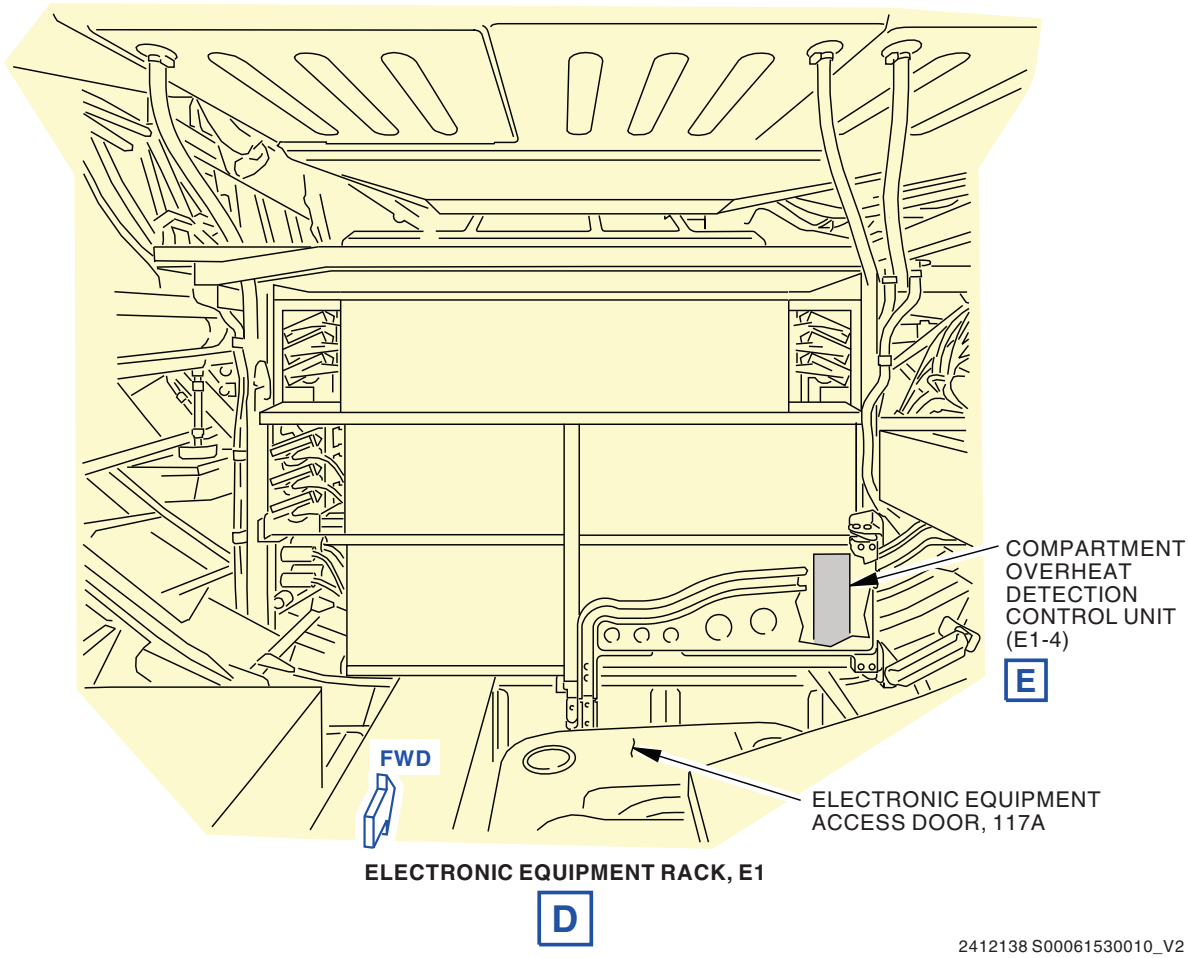
1 FIRST OFFICER'S

2412137 S00061530006_V1

Wheel Well, Wing and Lower Aft Body Overheat Detection System Operational Test
Figure 501/26-18-00-990-802 (Sheet 1 of 3)

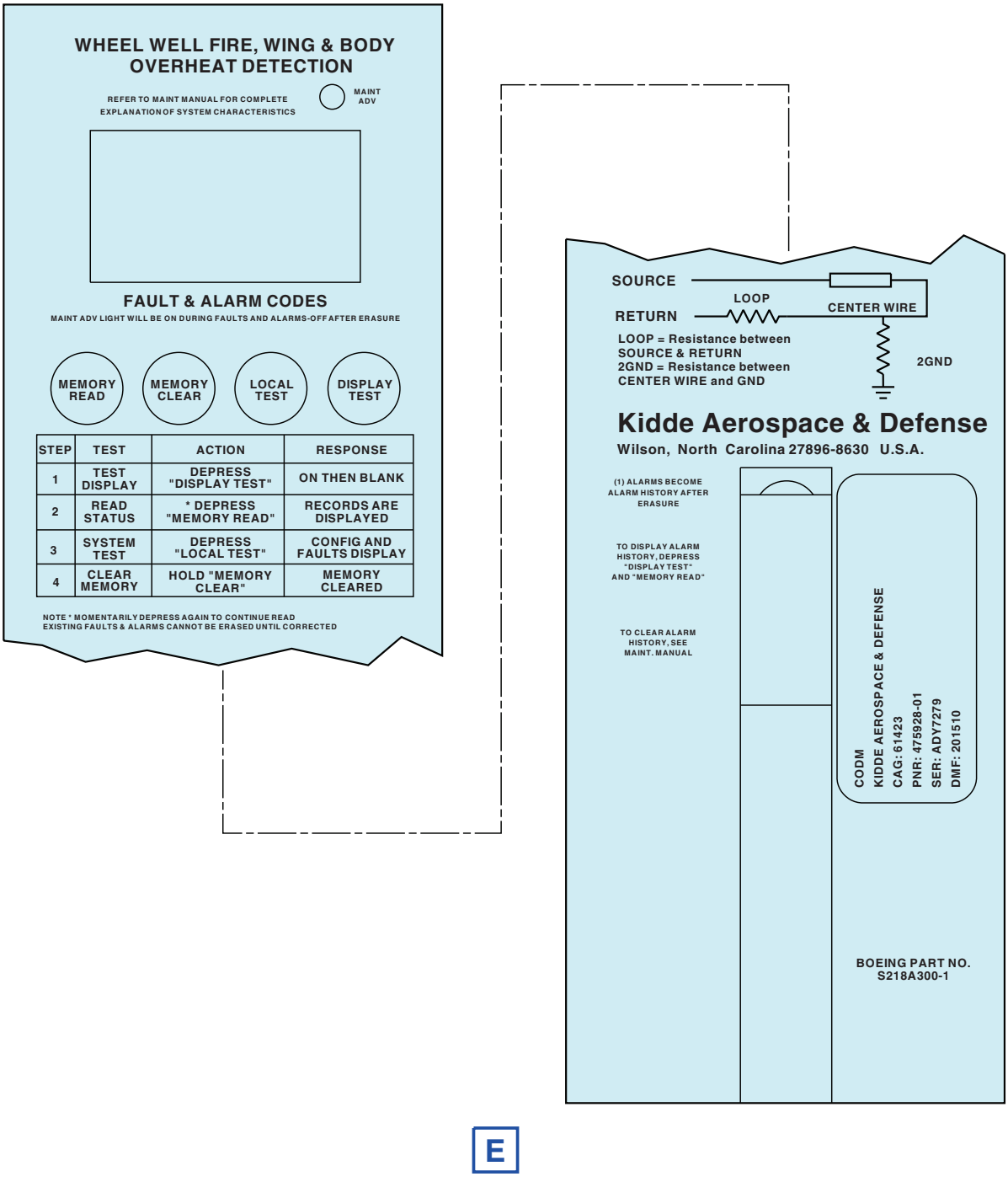
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Wheel Well, Wing and Lower Aft Body Overheat Detection System Operational Test
Figure 501/26-18-00-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412139 S00061530011_V1

Wheel Well, Wing and Lower Aft Body Overheat Detection System Operational Test
Figure 501/26-18-00-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-18-00-710-802

3. Wing and Lower Aft Body Overheat Detection System - Operational Test

(Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The Operational Test does a check of the Overheat Fire Detection System without disturbing the detection system and it only uses the equipment that is on the airplane.

(2) The operational test can be performed on the flight deck.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Operational Test

SUBTASK 26-18-00-860-005

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	12	C00132	MASTER CAUTION ANNUNCIATOR BUS 1
B	13	C00131	MASTER CAUTION ANNUNCIATOR BAT
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
D	12	C00310	INDICATOR MASTER DIM BAT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
E	11	C00313	INDICATOR MASTER DIM SECT 1
E	12	C00314	INDICATOR MASTER DIM SECT 2
E	13	C00315	INDICATOR MASTER DIM SECT 3
E	14	C00316	INDICATOR MASTER DIM SECT 4
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

SUBTASK 26-18-00-860-006

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Overheat Detection System Operational Test - Flight Compartment

SUBTASK 26-18-00-710-009

(1) Push and hold the OVHT TEST switch, on the P5-10 air conditioning panel, for at least 5 seconds.

(a) On the P5-10 air conditioning panel, make sure that these lights come on:

1) The left WING-BODY OVERHEAT light.

2) The right WING-BODY OVERHEAT light.

(b) On the P7 Pilot's Glareshield, make sure that these lights come on:

1) The AIR COND light.

2) The Captain's and First Officer's MASTER CAUTION lights.

SUBTASK 26-18-00-710-010

(2) Release the OVHT TEST switch on the P5-10 air conditioning panel.

(a) On the P5-10 air conditioning panel, make sure that these lights go off:

1) The left WING-BODY OVERHEAT light.

2) The right WING-BODY OVERHEAT light.

(b) On the P7 Pilot's Glareshield, make sure that these lights go off:

1) The AIR COND light.

2) The Captain's and First Officer's MASTER CAUTION lights.

E. Put the Airplane Back to Its Usual Condition.

SUBTASK 26-18-00-860-007

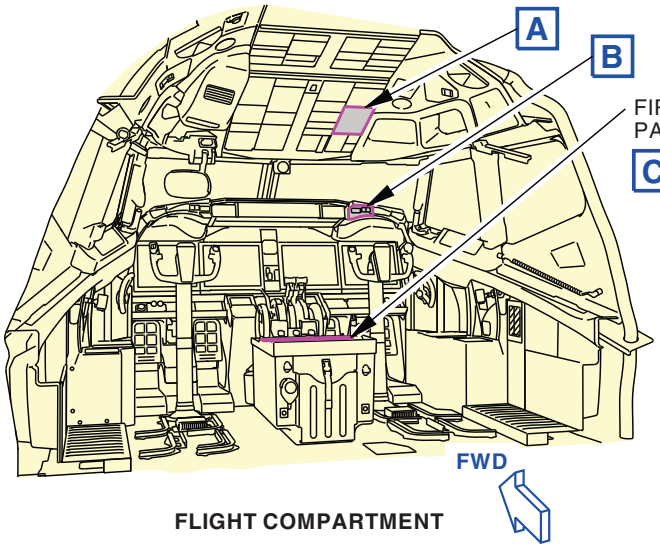
(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

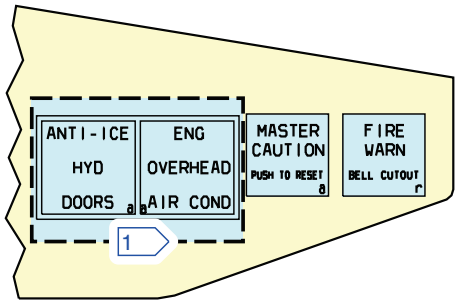
Row	Col	Number	Name
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

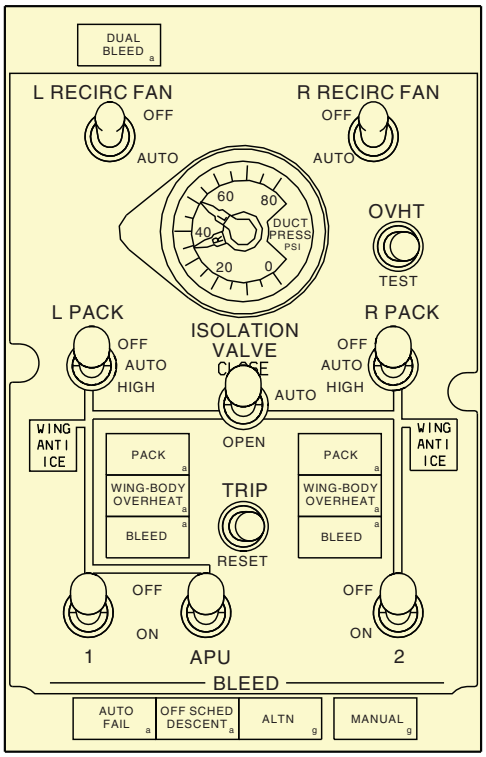


FLIGHT COMPARTMENT



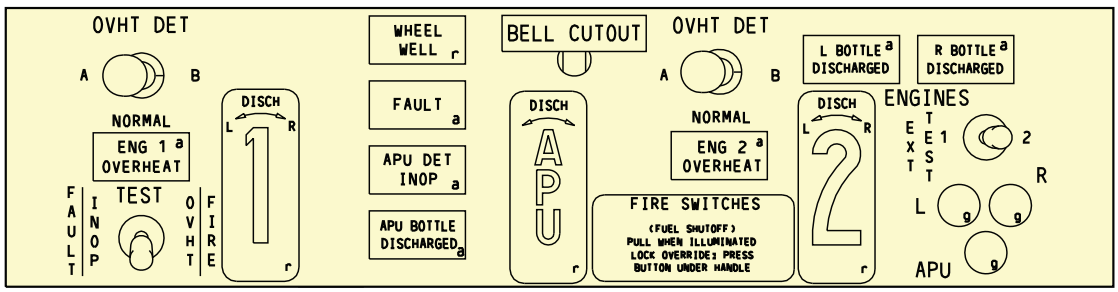
FIRST OFFICER'S WARNING LIGHTS (P7)
(CAPTAIN'S FIRE WARN AND MASTER CAUTION LIGHTS ARE OPPOSITE)

B



BLEED AIR CONTROL MODULE (P5)

A



FIRE CONTROL PANEL (P8-1)

C

1 FIRST OFFICER'S

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Wing and Lower Aft Body Overheat Detection System Operational Test
Figure 502/26-18-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COMPARTMENT OVERHEAT DETECTION CONTROL UNIT, M237 - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the Compartment Overheat Detection Control Unit (M237), referred to as the control unit.

(2) An installation of the Compartment Overheat Detection Control Unit.

B. The Compartment Overheat Detection Control Unit is installed on the E1-4 shelf in the electronic equipment compartment.

TASK 26-18-01-000-801

2. Compartment Overheat Detection Control Unit Removal

(Figure 401)

A. General

(1) This task gives instructions for the removal of the compartment overheat detection control unit.

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 26-18-01-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

SUBTASK 26-18-01-010-001

(2) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

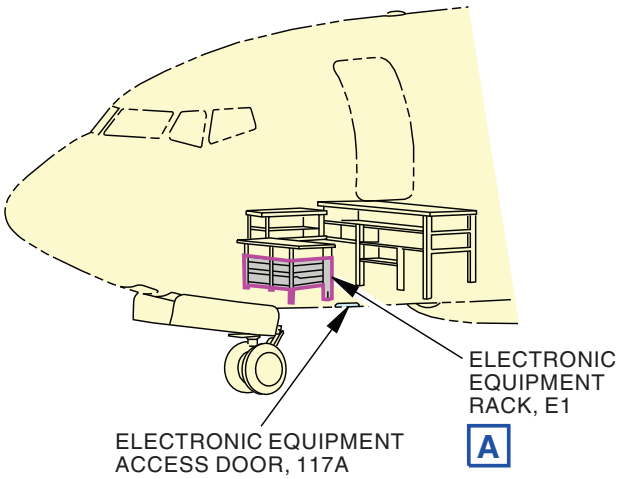
F. Compartment Overheat Detection Control Unit Removal

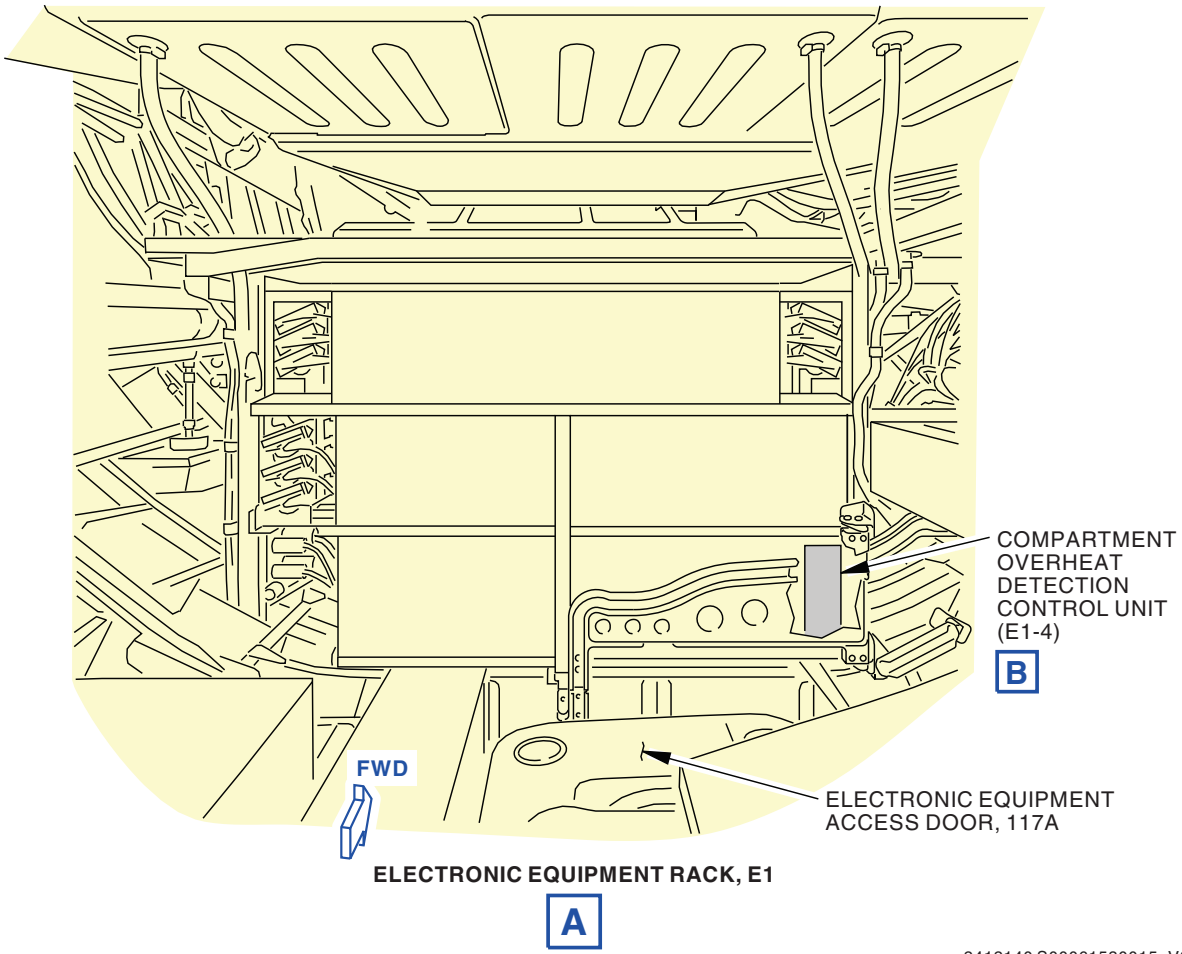
SUBTASK 26-18-01-020-001

(1) Remove the compartment overheat detection control unit [1] (TASK 20-10-07-000-801).

——— END OF TASK ———

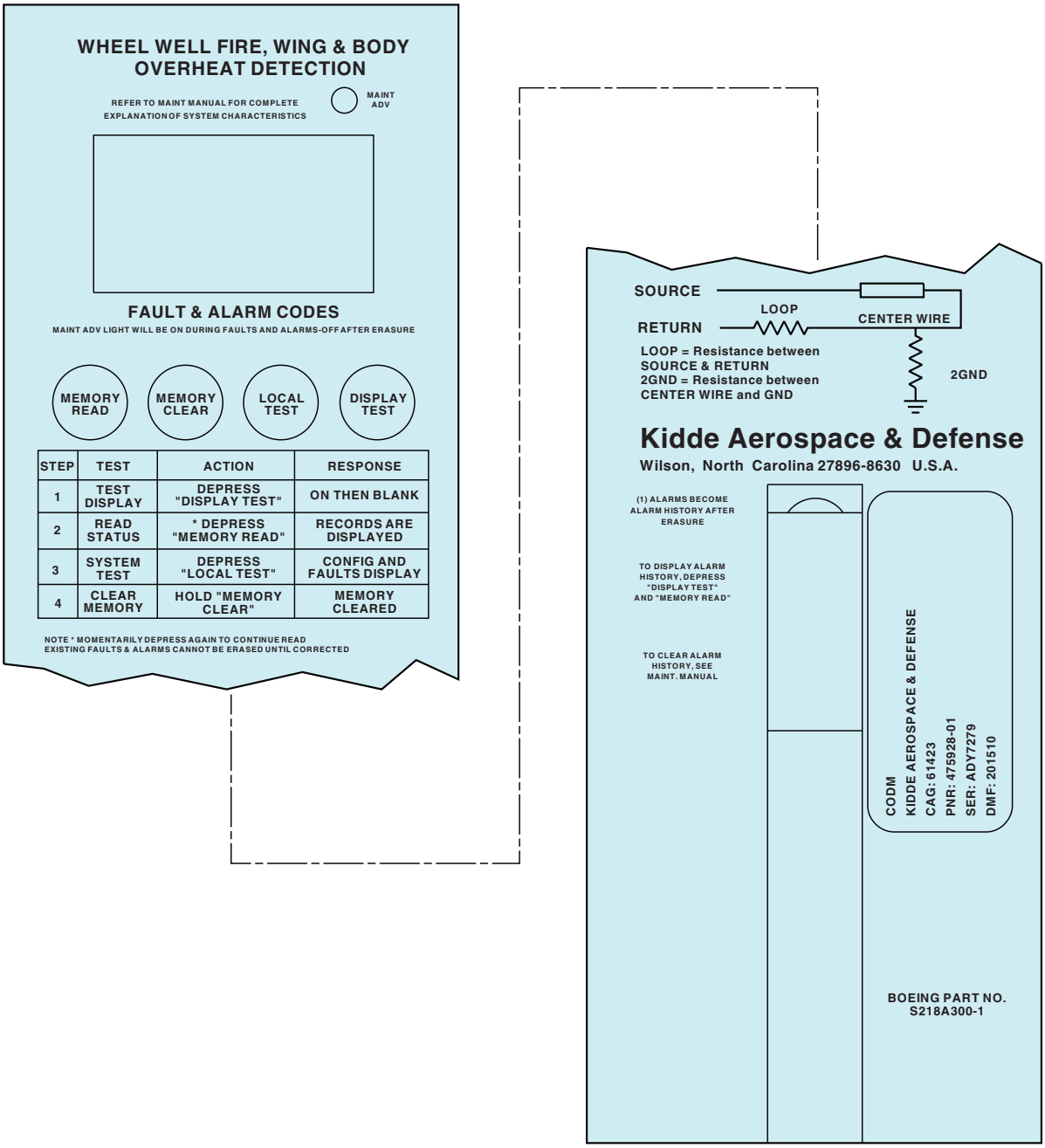
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Compartment Overheat Detection Control Unit Installation
Figure 401/26-18-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



COMPARTMENT OVERHEAT DETECTION CONTROL UNIT (E1-4)

B

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Compartment Overheat Detection Control Unit Installation
Figure 401/26-18-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-18-01-400-801

3. Compartment Overheat Detection Control Unit Installation

(Figure 401)

A. General

(1) This task gives instructions for the installation of the compartment overhead detection control unit.

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
26-18-00-710-801	Wheel Well and Lower Aft Body Overheat Detection System - Operational Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Compartment overheat detection control unit	26-18-01-01-010	PTF ALL

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Installation

SUBTASK 26-18-01-860-002

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

SUBTASK 26-18-01-010-002

(2) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

G. Compartment Overheat Detection Control Unit Installation

SUBTASK 26-18-01-020-002

(1) Install the compartment overheat detection control unit [1] (TASK 20-10-07-400-801).

H. Compartment Overheat Detection Control Unit Installation Test

SUBTASK 26-18-01-860-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-18-01-710-001

(2) Do this task: Wheel Well and Lower Aft Body Overheat Detection System - Operational Test, TASK 26-18-00-710-801.

NOTE: Do the portion of the test for the E/E bay.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 26-18-01-010-003

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COMPARTMENT OVERHEAT DETECTION CONTROL UNIT, M237 - ADJUSTMENT/TEST

1. General

A. The Compartment Overheat Detection Control Unit, M237 is installed on the E1-4 shelf in the electronic equipment compartment.

B. This procedure has this task:

(1) The Compartment Overheat Detection Control Unit (M237), BITE Test.

C. Do not hold the control module switches in the CLOSED position for more than 10 seconds or the software could stop the test.

D. Unless the test specifies differently, continue from one step to the subsequent step in less than 60 seconds.

TASK 26-18-01-740-801

2. Compartment Overheat Detection Control Unit, M237 - BITE Test

Figure 501

A. General

(1) The wheel well fire, wing & body compartment overheat detection control unit, M237 is located on the E1-4 shelf in the electronic equipment (EE) compartment. The wheel well fire, wing & body overheat detection control unit will be referred to as the control module throughout this procedure. Access the front of panel of the control module to do the BITE test.

(2) This task gives instructions to performed these procedures with the control module:

(a) Display Test Procedure.

(b) Local Test Procedure.

(c) Memory Read Procedure.

(d) Memory Clear Procedure.

(e) Zone Resistance Test.

(3) Do not hold the control module switches in the CLOSED position for more than 10 seconds or the software could stop the test.

(4) Unless the test specifies differently, continue from one step to the subsequent step in less than 60 seconds.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. BITE Procedure

SUBTASK 26-18-01-740-001

(1) Do these steps to do the display test procedure:

(a) Push and release the DISPLAY TEST switch on the control module.

1) Make sure that all LED's come on, then go off.

SUBTASK 26-18-01-740-002

(2) Do these steps to do the local test procedure for the control module:

(a) Push and release the LOCAL TEST switch on the control module.

1) The CONFIG MAX1 comes on while the test is in progress, then the TEST COMPLETE comes on to show that there are no faults.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If an alarm/fault condition is found, the display will show a fault code.

SUBTASK 26-18-01-740-003

(3) Do these steps to do the memory read procedure:

(a) Push and release the MEMORY READ switch on the control module.

1) Make sure that END OF MEMORY comes on to show the test is complete.

2) Push and release the MEMORY READ switch again until all the alarm/faults have been read.

NOTE: A minimum of 10 LRU failures can be stored in non-volatile memory.

3) Push and release the MEMORY READ switch again so the display shows no data.

SUBTASK 26-18-01-740-004

(4) Do these steps to do the memory clear procedure:

(a) Push and momentarily hold the MEMORY CLEAR switch on the control module.

(b) Make sure that MEMORY CLEAR is displayed.

(c) Make sure that the MAINT ADV light stays off.

NOTE: If there are fault codes in the memory, the display will show the last code that was recorded. If there are no fault codes in the memory, the display will show no data.

(d) Push and release the MEMORY CLEAR switch to clear the displayed code.

1) The display shows no data.

(e) Push and release the MEMORY READ switch again to display the next code.

1) Repeat the two steps above until no data shows.

NOTE: Alarm/fault codes can not be removed until the condition is corrected.

(f) Push the MEMORY READ switch again so the display shows no data and the MAINT ADV light is off.

SUBTASK 26-18-01-740-005

(5) Do these steps to do the Zone Resistance test (CODM BITE):

(a) Push and release the LOCAL TEST button on the front panel of the CODM.

1) The front panel of the CODM displays CONFIG MAX1 followed by L ENGINE STR.

(b) Continue to push and release LOCAL TEST. This will cycle through the different zones followed by LOOP ___ which shows the resistance value in the loop or 2GND_____ which shows the resistance value to ground.

NOTE: If the loops are functioning properly all zones will have a 2GND value of 5000.

(c) Make sure that the resistance value agree with the Table 501 below:

Table 501/26-18-01-993-801 Zone Resistance Test Data

KIDDE ALARM/FAULT CODES	RESISTANCE VALUE or FAULT MESSAGE	SENSORS
L ENGINE STR	< 15 Ohms	M1761, M1762, M3057
L ENGINE STR OPEN	OPEN	
L ENGINE STR OVHT	LOW	
L ENGINE STR SHRT	0	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 501/26-18-01-993-801 Zone Resistance Test Data (Continued)

KIDDE ALARM/FAULT CODES	RESISTANCE VALUE or FAULT MESSAGE	SENSORS
L WING LE L WING LE OPEN L WING LE OVHT L WING LE SHRT	< 15 Ohms OPEN LOW 0	M268, M370
L PACK BAY L PACK BAY OPEN L PACK BAY OVHT L PACK BAY SHRT	< 15 Ohms OPEN LOW 0	M1911, M1909, M355
NITROGEN GEN NITROGEN GEN OPEN NITROGEN GEN OVHT NITROGEN GEN SHRT	< 15 Ohms OPEN LOW 0	M3067
KEELBEAM KEELBEAM OPEN KEELBEAM OVHT KEELBEAM SHRT	< 15 Ohms OPEN LOW 0	M272, M273
AFT CARGO AFT CARGO OPEN AFT CARGO OVHT AFT CARGO SHRT	< 20 Ohms OPEN LOW 0	M275, M347, M348, M1147
AFT BODY AFT BODY OPEN AFT BODY OVHT AFT BODY SHRT	< 15 Ohms OPEN LOW 0	M276
MWW LOOP A MWW LOOP A OPEN MWW LOOP A FIRE MWW LOOP A SHRT	< 15 Ohms OPEN LOW 0	M270
MWW LOOP B MWW LOOP B OPEN MWW LOOP B FIRE MWW LOOP B SHRT	< 15 Ohms OPEN LOW 0	M3068
R ENGINE STR R ENGINE STR OPEN R ENGINE STR OVHT R ENGINE STR SHRT	< 15 Ohms OPEN LOW 0	M1763, M1764, M3058
R WING LE R WING LE OPEN R WING LE OVHT R WING LE SHRT	< 15 Ohms OPEN LOW 0	M269, M371

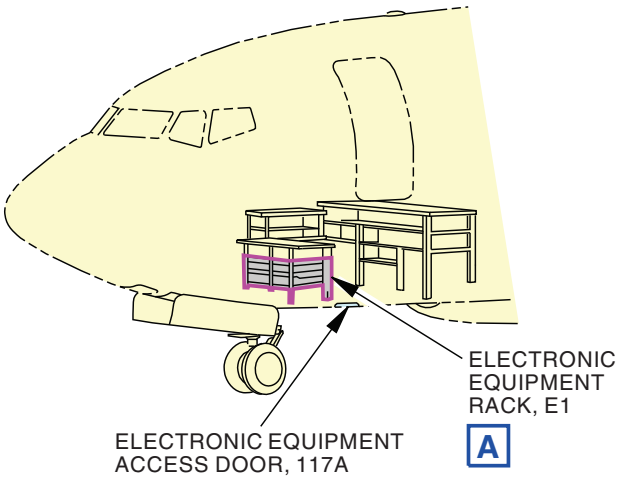
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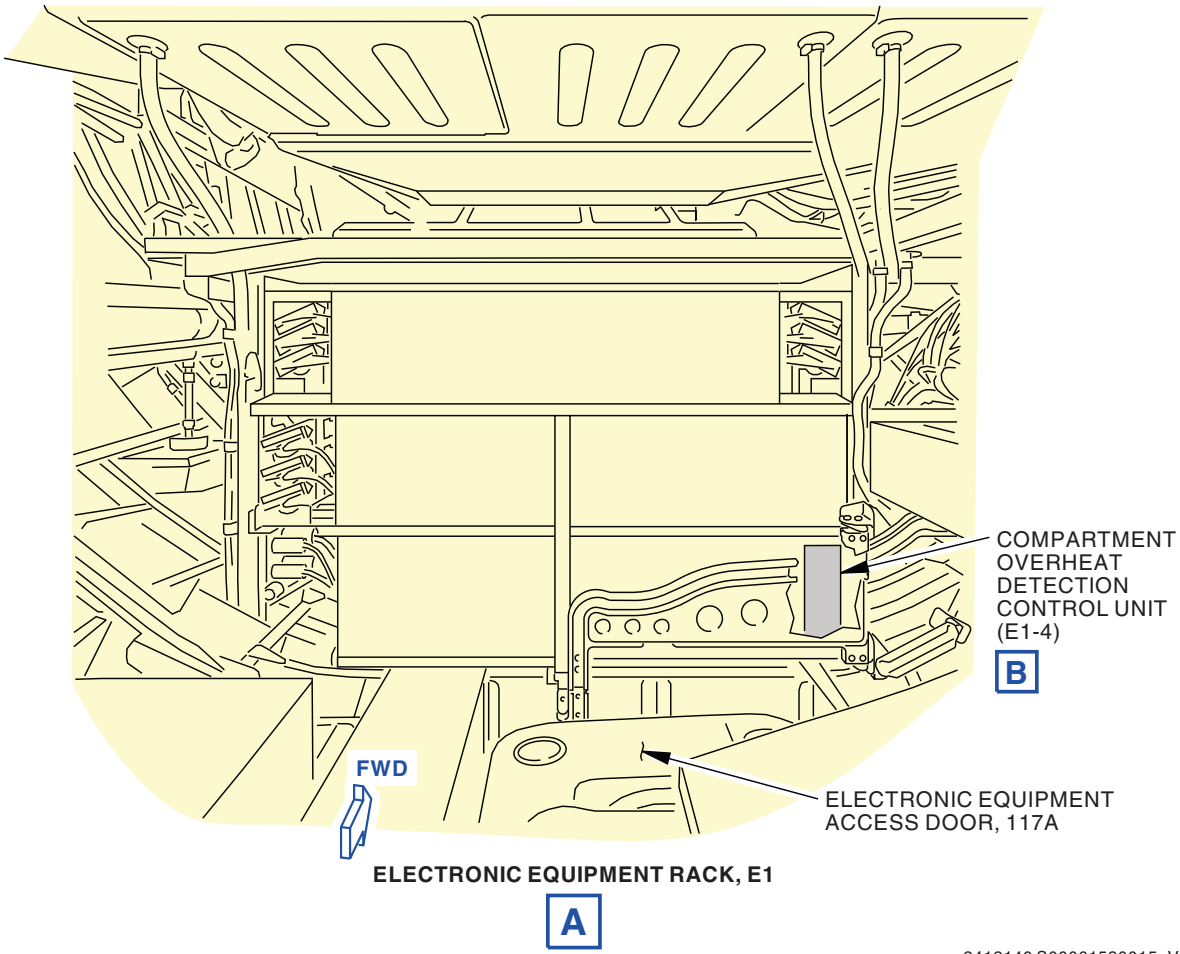
Table 501/26-18-01-993-801 Zone Resistance Test Data (Continued)

KIDDE ALARM/FAULT CODES	RESISTANCE VALUE or FAULT MESSAGE	SENSORS
R PACK BAY R PACK BAY OPEN R PACK BAY OVHT R PACK BAY SHRT	< 15 Ohms OPEN LOW 0	M356, M1910, M1912

——— END OF TASK ———

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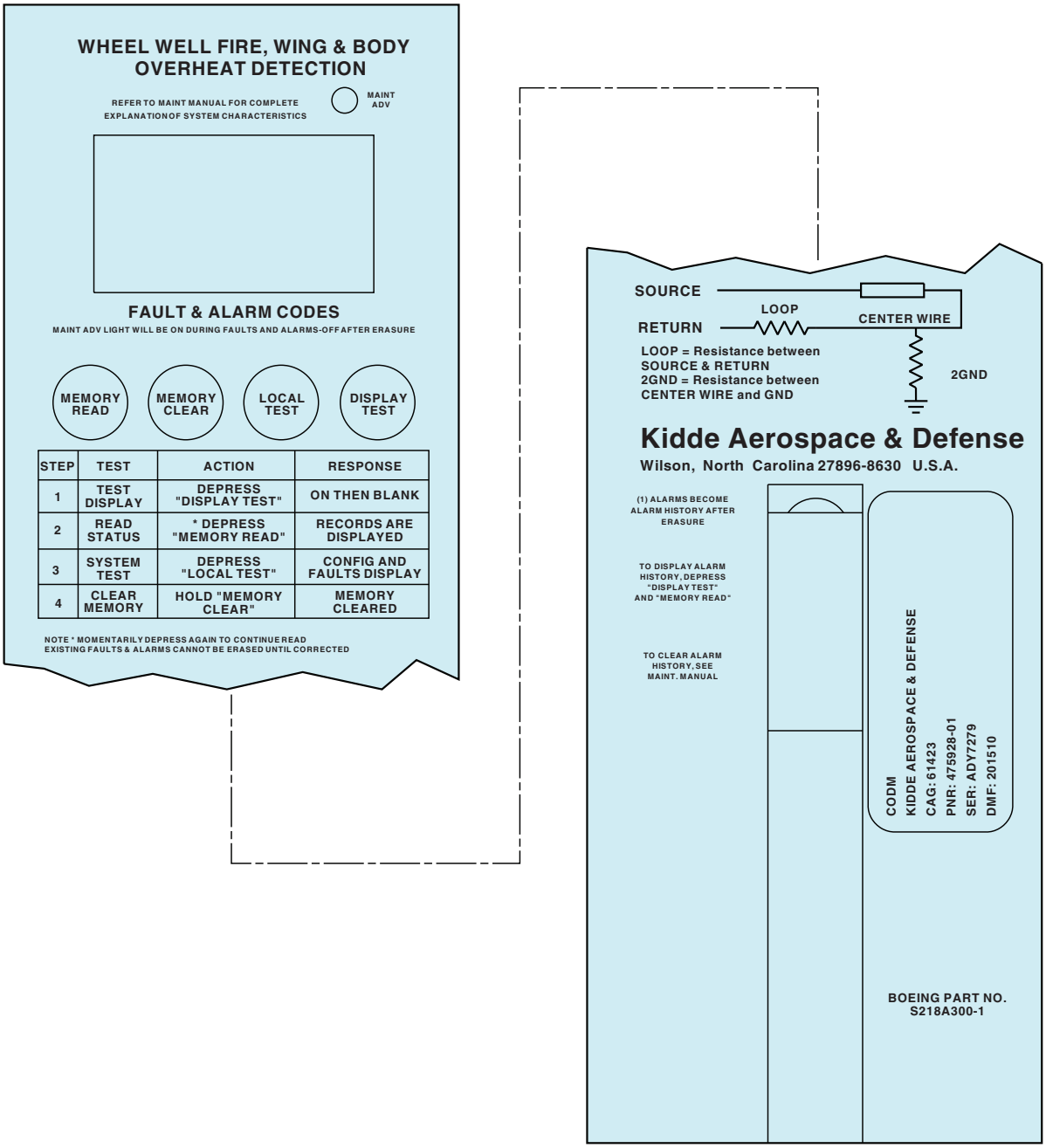




2412140 S00061530015_V2

Compartment Overheat Detection Control Unit M237
Figure 501/26-18-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



COMPARTMENT OVERHEAT DETECTION CONTROL UNIT (E1-4)

B

2412141 S00061530016_V1

Compartment Overheat Detection Control Unit M237
Figure 501/26-18-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-18-01-700-801

3. Compartment Overheat System Automated Test with WiN MS Aero FLi Test Kit

Figure 502

A. General

This task provides instructions on how to attach, test, and diagnose the Compartment Overheat System utilizing the WiN MS Aero FLi test set.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
26-18-01-000-801	Compartment Overheat Detection Control Unit Removal (P/B 401)
26-18-01-400-801	Compartment Overheat Detection Control Unit Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-23026	Tester - Overheat Detection Loop, Aero Fli Part #: 411001 Supplier: FB3T5

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

E. Prepare for the Compartment Overheat System Automated Test

SUBTASK 26-18-01-860-004

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

NOTE: Failure to remove power from the aircraft will result in inaccurate test results.

SUBTASK 26-18-01-020-003

(2) Do this task: Compartment Overheat Detection Control Unit Removal, TASK 26-18-01-000-801.

SUBTASK 26-18-01-480-001

(3) Install the aero fli overheat detection loop tester, SPL-23026:

(a) Install the interface module on the Compartment Overheat Detection Control Unit rack.

(b) Connect the cable to the interface tablet and the interface module.

(c) Turn on the interface module.

(d) Turn on the interface tablet.

F. Compartment Overheat System Automated Test

SUBTASK 26-18-01-700-001

(1) Perform the automated test:

NOTE: For detailed instructions please refer to equipment user manual.

(a) Start the AERO FLi (or OHDL Studio) App.

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(b) Enter the applicable Operator, Station, and Aircraft ID.

NOTE: This will be used for PDF report.

(c) Select the applicable Program.

(d) Select Loop: [All].

(e) Select Static Test.

(f) Select the applicable loops that are to be scanned.

(g) Select Launch Scan.

1) If the scan takes longer than two minutes, restart the hardware.

(h) When scan is finished, select PDF Report.

(i) Review the PDF report to identify potential problems among the reported items.

1) If problems are found, the report will show the location of the problem.

2) Correct the problems and perform the automated test again.

3) When the problems have been corrected, there will be no problems listed in the report.

G. Put the Aircraft Back to its Usual Condition

SUBTASK 26-18-01-080-001

(1) Remove the Aero FLi interface module:

(a) Turn off the interface module.

(b) Turn off the interface tablet.

(c) Disconnect the interface cable.

(d) Remove the module from the rack.

SUBTASK 26-18-01-020-004

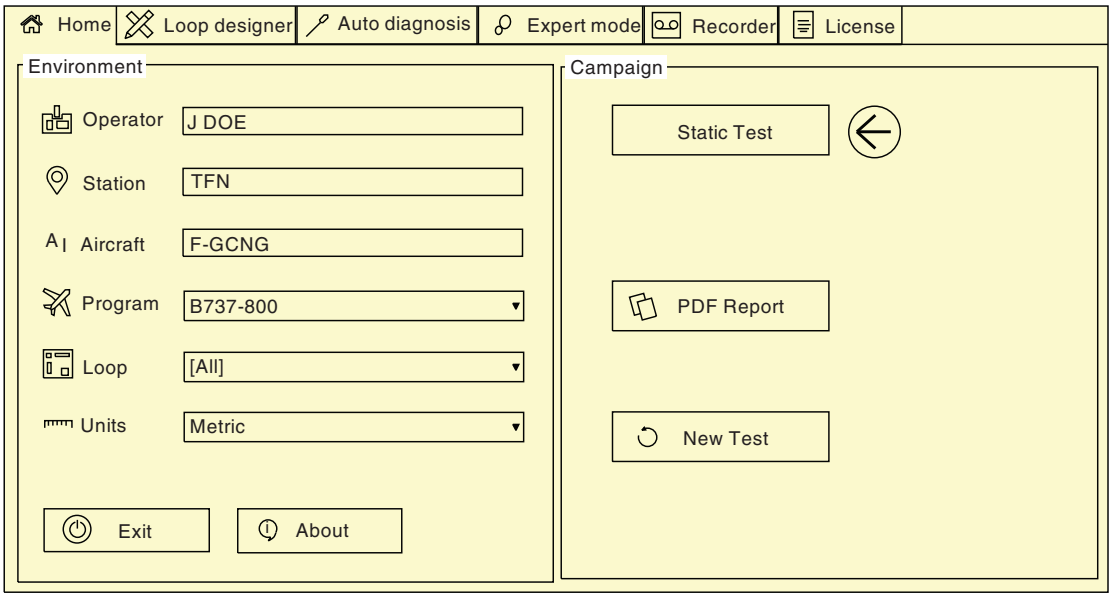
(2) Do this task: Compartment Overheat Detection Control Unit Installation, TASK 26-18-01-400-801.

SUBTASK 26-18-01-860-005

(3) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

——— END OF TASK ———

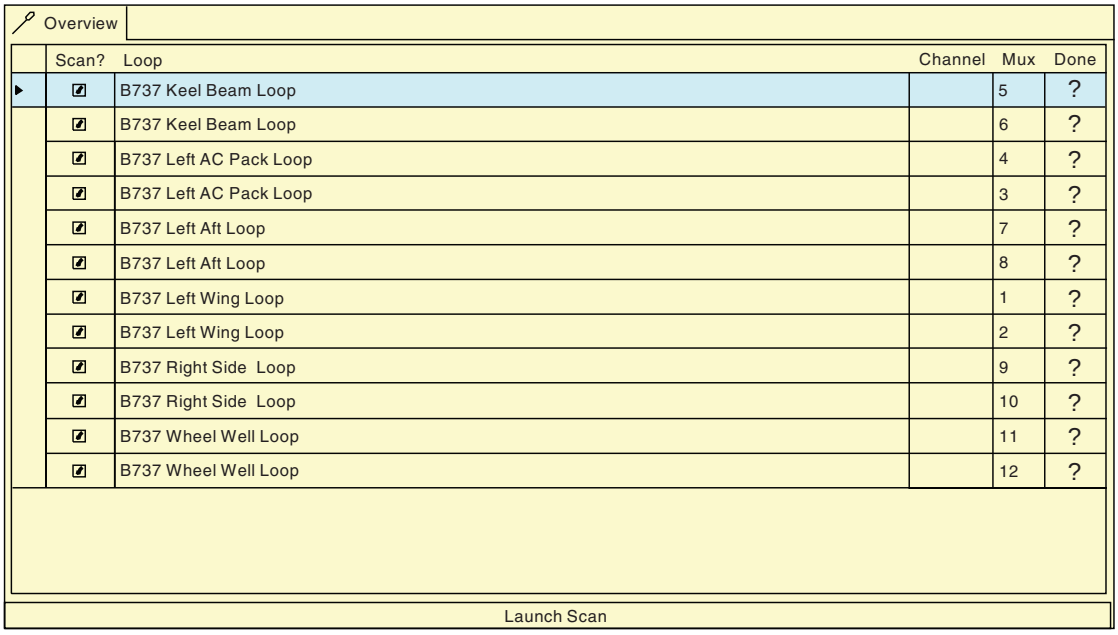
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Compartment Overheat System Automated Test with WiN MS Aero FLi Test Kit
Figure 502/26-18-01-990-803 (Sheet 1 of 3)

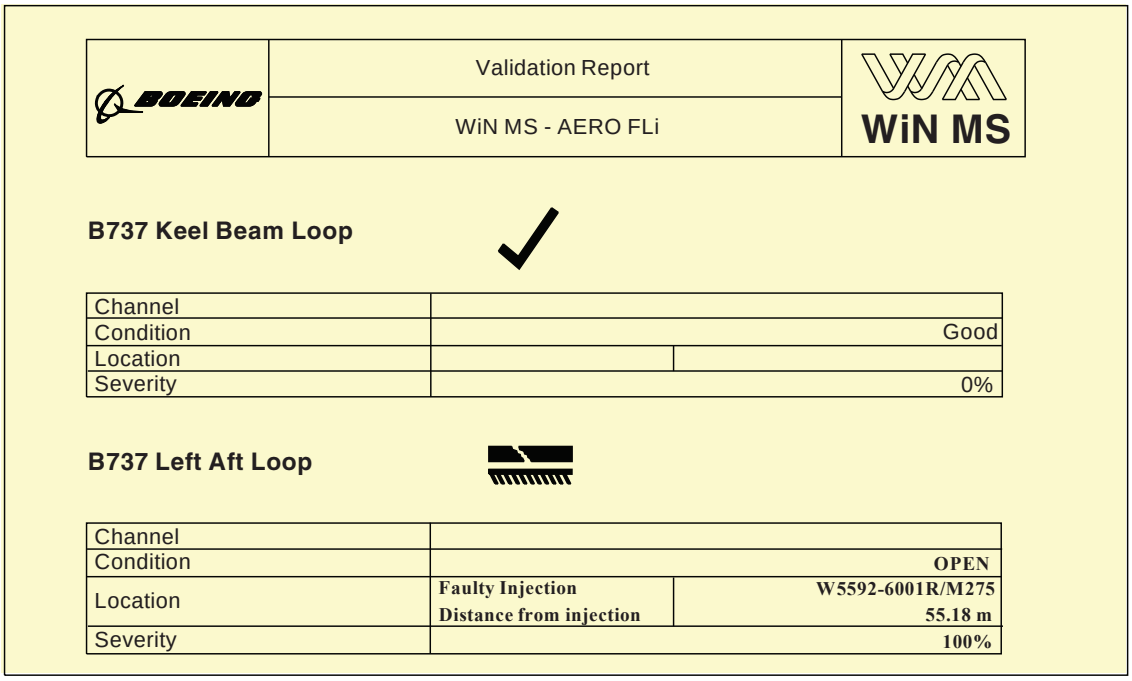
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3027724 S0000798888_V1

Compartment Overheat System Automated Test with WiN MS Aero FLi Test Kit
Figure 502/26-18-01-990-803 (Sheet 2 of 3)

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3027725 S0000798889_V1

Compartment Overheat System Automated Test with WiN MS Aero FLi Test Kit
Figure 502/26-18-01-990-803 (Sheet 3 of 3)

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WING, WHEEL WELL, AND LOWER AFT BODY OVERHEAT DETECTION SENSING ELEMENT - REMOVAL/INSTALLATION

1. General

A. This section contains procedures to remove and install the long sensor elements, and the looped sensor elements.

(1) The long sensor elements are located in the wing area, the forward Air Conditioning (A/C) Pack Bay, the Wheel Well, and the Aft Body Area.

(2) The looped sensor elements are located in the mid and aft A/C pack, and along the strut.

B. The minimum element bend radius is 1 in. (25.4 mm). The best bend radius is three inches.

C. The sensor element must be straight and away from the connector for one inch before bending.

D. A list of all the compartment overheat detection sensing elements is shown in Table 401. Each loop is made up of one or more sensing elements that make a complete circuit with the Compartment Overheat Detection Module, M237. The loop is defined by the two pins in the electrical connector D742 that complete the circuit. The pins that define each loop are shown in the column "D742 PINS" of Table 401.

TASK 26-18-02-000-801

2. Wheel Well, Wing and Lower Aft Body Overheat Sensor Element - Removal

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405, Figure 406, Figure 407, Figure 408)

A. General

(1) This task gives instructions to remove the Wheel Well, Wing and Lower Aft Body Overheat Sensor Element.

B. References

Reference	Title
20-10-44-000-801	Lockwire, Cotter Pins, and Lockrings - Removal (P/B 401)
21-51-03-000-803	Heat Exchanger and Plenum/Diffuser Assembly - Removal (P/B 401)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-803	Leading Edge Flaps and Slats Extension (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)

C. Location Zones

Zone	Area
129	Keel Beam (Part) - Section 43
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
149	Keel Beam (Part) Body Station 727.00 to Body Station 743.95
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box

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(Continued)

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
431	Engine 1 - Forward Strut Fairing
441	Engine 2 - Forward Strut Fairing
511	Left Wing - Leading Edge To Front Spar
521	Left Wing - Leading Edge to Front Spar
611	Right Wing - Leading Edge to Front Spar
621	Right Wing - Leading Edge to Front Spar

D. Prepare for the Removal

SUBTASK 26-18-02-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

SUBTASK 26-18-02-010-001

(2) Open the applicable access doors and panels (Table 401).

Table 401/26-18-02-993-801

LOOP	D742 PINS	EQPT NO.	Sensor Element	Figure	DETECTOR	OPEN/REMOVE FOR ACCESS
1	42, 41	M1912	element [13]	Figure 403	R AFT A/C PACK	ECS ACCESS DOOR, 192CR *[1]
		M1910	element [14]	Figure 403	R MID A/C PACK	ECS ACCESS DOOR, 192CR *[1]
		M356	element [15]	Figure 402	R FWD A/C PACK	ECS ACCESS DOOR, 192CR *[1] A/C PNEUMATIC TUBES
2	30, 35	M371	element [16]	Figure 401	R WING INBOARD	LE SLATS - EXTEND FULLY AND DEACTIVATE (TASK 27-81-00-860-803 and TASK 27-81-00-040-801) - ACTIVATE AND RETRACT (TASK 27-81-00-860-804 and TASK 27-81-00-440-801) ECS ACCESS DOOR, 192CR *[1] RAM AIR INLET LIP PANEL, 191HR FORWARD WING TO BODY FAIRING PANEL, 191FR INBOARD LEADING EDGE, STRAKELET UPPER ACCESS PANEL, 611AT

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Table 401/26-18-02-993-801 (Continued)

LOOP	D742 PINS	EQPT NO.	Sensor Element	Figure	DETECTOR	OPEN/REMOVE FOR ACCESS
		M269	element [17]	Figure 401	R WING OUTBOARD	LE SLATS - EXTEND FULLY AND DEACTIVATE (TASK 27-81-00-860-803 and TASK 27-81-00-040-801) - ACTIVATE AND RETRACT (TASK 27-81-00-860-804 and TASK 27-81-00-440-801) ECS ACCESS DOOR, 192CR *[1] RAM AIR INLET LIP PANEL, 191HR FORWARD WING TO BODY FAIRING PANEL, 191FR INBOARD LEADING EDGE, REMOVABLE ACCESS PANEL, 611AB
3	11, 10	M1763	element [18]	Figure 404	R FWD STRUT	FORWARD STRUT FAIRING, 441BT
		M3058	element [18]	Figure 404	R MID STRUT	FORWARD STRUT FAIRING, 441BR
		M1764	element [18]	Figure 404	R AFT STRUT	OUTBOARD LEADING EDGE BLOWOUT ACCESS DOOR, 621AB
4	36, 37	M276	element [19]	Figure 406	AFT, SECTION 48	STAB TRIM ACCESS
5	13, 27	M275	element [20]	Figure 406	AFT, SECTION 47	AFT CARGO LINING
		M347	element [21]	Figure 406	AFT, SECTION 46	AFT CARGO LINING
		M1147	element [22]	Figure 406	AFT, SECTION 46	AFT CARGO LINING
		M348	element [23]	Figure 406	AFT, SECTION 46	AFT CARGO LINING
6	29, 9	M273	element [24]	Figure 407	AFT KEEL BEAM	PANELS BELOW BEAM LANDING GEAR TRANSFER VALVE HYDRAULIC LINES COVERS TO FORWARD BULKHEAD COVERS TO AFT BULKHEAD
		M272	element [25]	Figure 407	FWD KEELBEAM	
7	8, 2	M1911	element [26]	Figure 403	L AFT A/C PACK	ECS ACCESS DOOR, 192CL
		M1909	element [27]	Figure 403	L MID A/C PACK	ECS ACCESS DOOR, 192CL

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Table 401/26-18-02-993-801 (Continued)

LOOP	D742 PINS	EQPT NO.	Sensor Element	Figure	DETECTOR	OPEN/REMOVE FOR ACCESS
		M355	element [28]	Figure 402	L FWD A/C PACK	ECS ACCESS DOOR, 192CL A/C PNEUMATIC TUBES
8	39, 40	M370	element [16]	Figure 401	L WING INBOARD	LE SLATS - EXTEND FULLY AND DEACTIVATE (TASK 27-81-00-860-803 and TASK 27-81-00-040-801) - ACTIVATE AND RETRACT (TASK 27-81-00-860-804 and TASK 27-81-00-440-801) ECS ACCESS DOOR, 192CL RAM AIR INLET LIP PANEL, 191HL FORWARD WING TO BODY FAIRING PANEL, 191FL INBOARD LEADING EDGE, STRAKELET UPPER PANEL, 511AT
		M268	element [17]	Figure 401	L WING OUTBOARD	LE SLATS - EXTEND FULLY AND DEACTIVATE (TASK 27-81-00-860-803 and TASK 27-81-00-040-801) - ACTIVATE AND RETRACT (TASK 27-81-00-860-804 and TASK 27-81-00-440-801) ECS ACCESS DOOR, 192CL RAM AIR INLET LIP PANEL, 191HL FORWARD WING TO BODY FAIRING PANEL, 191FL INBOARD LEADING EDGE, REMOVABLE ACCESS PANEL, 511AB
9	4, 12	M1761	element [18]	Figure 404	L STRUT FWD	FORWARD STRUT FAIRING, 431BT
		M3057	element [18]	Figure 404	L MID STRUT	FORWARD STRUT FAIRING, 431BR
		M1762	element [18]	Figure 404	L STRUT AFT	OUTBOARD LEADING EDGE BLOWOUT ACCESS DOOR, 521AB
10	21, 24	M270	detector [29]	Figure 405	WHEEL WELL LOOP A	LANDING GEAR DOWN
11	31, 34	M3068	detector [29]	Figure 405	WHEEL WELL LOOP B	LANDING GEAR DOWN

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Table 401/26-18-02-993-801 (Continued)

LOOP	D742 PINS	EQPT NO.	Sensor Element	Figure	DETECTOR	OPEN/REMOVE FOR ACCESS
12	32, 33	M3067	element [41]	Figure 408	NITROGEN GEN	HEAT EXCHANGER INLET DUCT - Heat Exchanger and Plenum/Diffuser Assembly - Removal, TASK 21-51-03-000-803

*[1] Do not open the ECS access door, 192CR, until you open the ECS high pressure access door, 192DR.

SUBTASK 26-18-02-010-002

(3) Remove any airplane parts that are in the way (Table 401).

E. Long Sensor Element Removal

NOTE: This procedure only applies to the following sensors:

element [15], M356, R FWD A/C PACK
element [16], M370, L WING INBOARD
element [16], M371, R WING INBOARD
element [17], M268, L WING OUTBOARD
element [17], M269, R WING OUTBOARD
element [19], M276, AFT - SECTION 48
element [20], M275, AFT - SECTION 47
element [21], M347, AFT - SECTION 46
element [22], M1147, AFT - SECTION 46
element [23], M348, AFT - SECTION 46
element [24], M273, AFT KEELBEAM
element [25], M272, FWD KEELBEAM
element [28], M355, L FWD A/C PACK
detector [29], M270, WHEEL WELL LOOP A
detector [29], M3068, WHEEL WELL LOOP B.

SUBTASK 26-18-02-020-001

(1) Loosen all the bolts [6] securing the mounting clips [4] that hold the element in its position.

SUBTASK 26-18-02-020-002

(2) Remove the element from all the mounting clips [4].

SUBTASK 26-18-02-020-011

(3) Remove the grommet [5] from the element.

NOTE: Keep the grommet to reuse during installation procedure.

SUBTASK 26-18-02-030-002

(4) Do these steps at both ends of the applicable sensor element on the wing and lower aft body:

(a) Remove the lockwires [1] from the jamnut [2].

(b) If it is necessary, disconnect the coupling element connector or electrical connector and remove the jamnut [2].

(c) Disconnect the connector [3].

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(d) Put covers on the open end of the connector [3] with tape or caps to prevent damage.

SUBTASK 26-18-02-020-012

(5) Do these steps for sensor elements in the main wheel well:

(a) Remove the nut [43], two washers [44], and screw [45] from the terminal lug [10].

(b) Remove the lockwires [1] from the connector hex nut [42].

(c) Loosen the connector hex nut [42].

(d) Put covers on the open end of the connector [3] with tape or caps to prevent damage.

SUBTASK 26-18-02-020-003

(6) Hold the element carefully to keep it in its position and to prevent bends.

SUBTASK 26-18-02-020-004

(7) Do these steps for sensor elements on the aft keel beam:

(a) Tie a cord to the forward end of the element.

1) Make sure that the cord is longer than the element.

(b) Slowly pull the element aft through the keel beam.

1) Use the access holes below the keel beam to help guide the element through the holes in the structure.

(c) Remove the element from the cord and leave the cord in its position inside the keel beam.

SUBTASK 26-18-02-550-001

(8) Carefully roll the element into a 6 in. (15.2 cm) diameter coil.

(a) Use tape if it is necessary.

F. Looped Sensor Element Removal

NOTE: This procedure only applies to the following sensors:

element [13], M1912, R AFT A/C PACK
element [14], M1910, R MID A/C PACK
element [18], M1763, R FWD STRUT
element [18], M3058, R MID STRUT
element [18], M1764, R AFT STRUT
element [18], M1761, L FWD STRUT
element [18], M3057, L MID STRUT
element [18], M1762, L AFT STRUT
element [26], M1911, L AFT A/C PACK
element [27], M1909, L MID A/C PACK
element [41], M3067, NITROGEN GEN.

SUBTASK 26-18-02-020-005

(1) Remove the nuts [8], washers [9] and bolts [11] that secure each lead to the terminal lugs [10].

SUBTASK 26-18-02-020-009

(2) Remove the jumper [30] from the bracket [12].

(a) For the Air Conditioning (A/C) pack locations remove the screw [31] and washer [32].

(b) For the Strut locations remove the nut [33] and washer [34].

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SUBTASK 26-18-02-020-006

(3) For the Strut locations release the elements from the lower quick release clamps [35].

(a) Open the quick release clamps [35] to free the element.

(b) Remove the grommets [5] from the element.

NOTE: Keep the grommet to reuse during installation procedure.

SUBTASK 26-18-02-020-010

(4) Remove the bracket [12] with the element.

(a) For the A/C pack locations remove the screws [36] and washers [37].

(b) For the Strut locations remove the bolts [38] and washers [39].

SUBTASK 26-18-02-010-005

(5) Place the bracket [12] with the element on a clean work surface.

SUBTASK 26-18-02-010-006

(6) Release the element from the quick release clamp [35] on the bracket [12].

(a) Open the quick release clamp [35] to free the element.

(b) Remove the grommet [5] from the element.

NOTE: Keep the grommet to reuse during installation procedure.

SUBTASK 26-18-02-020-007

(7) Remove the lockwire [1] from the jamnuts [2] (TASK 20-10-44-000-801).

SUBTASK 26-18-02-010-007

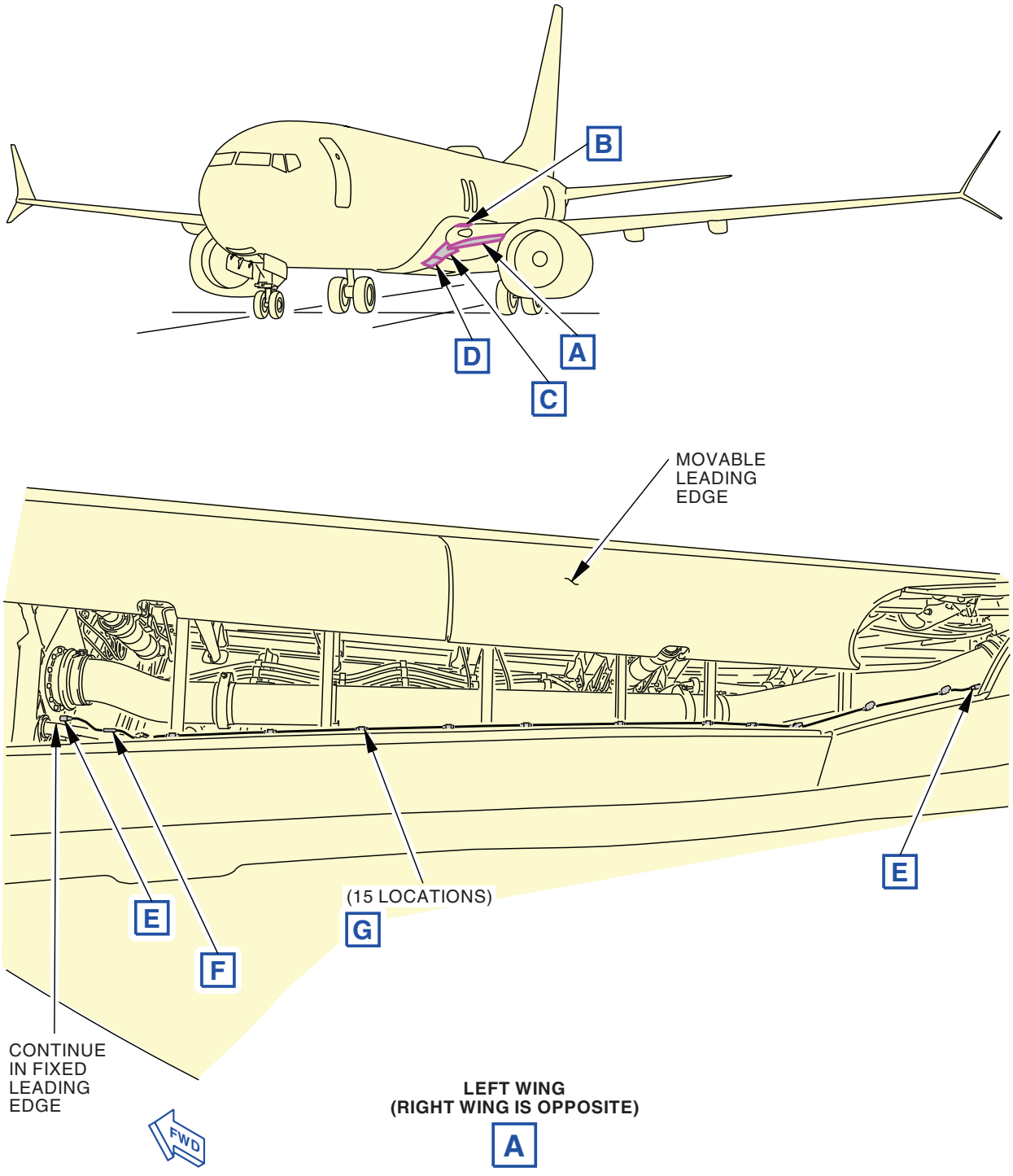
(8) Remove the jamnuts [2] which secure each connector [3] to the bracket [12].

SUBTASK 26-18-02-020-008

(9) Remove the element from the bracket [12].

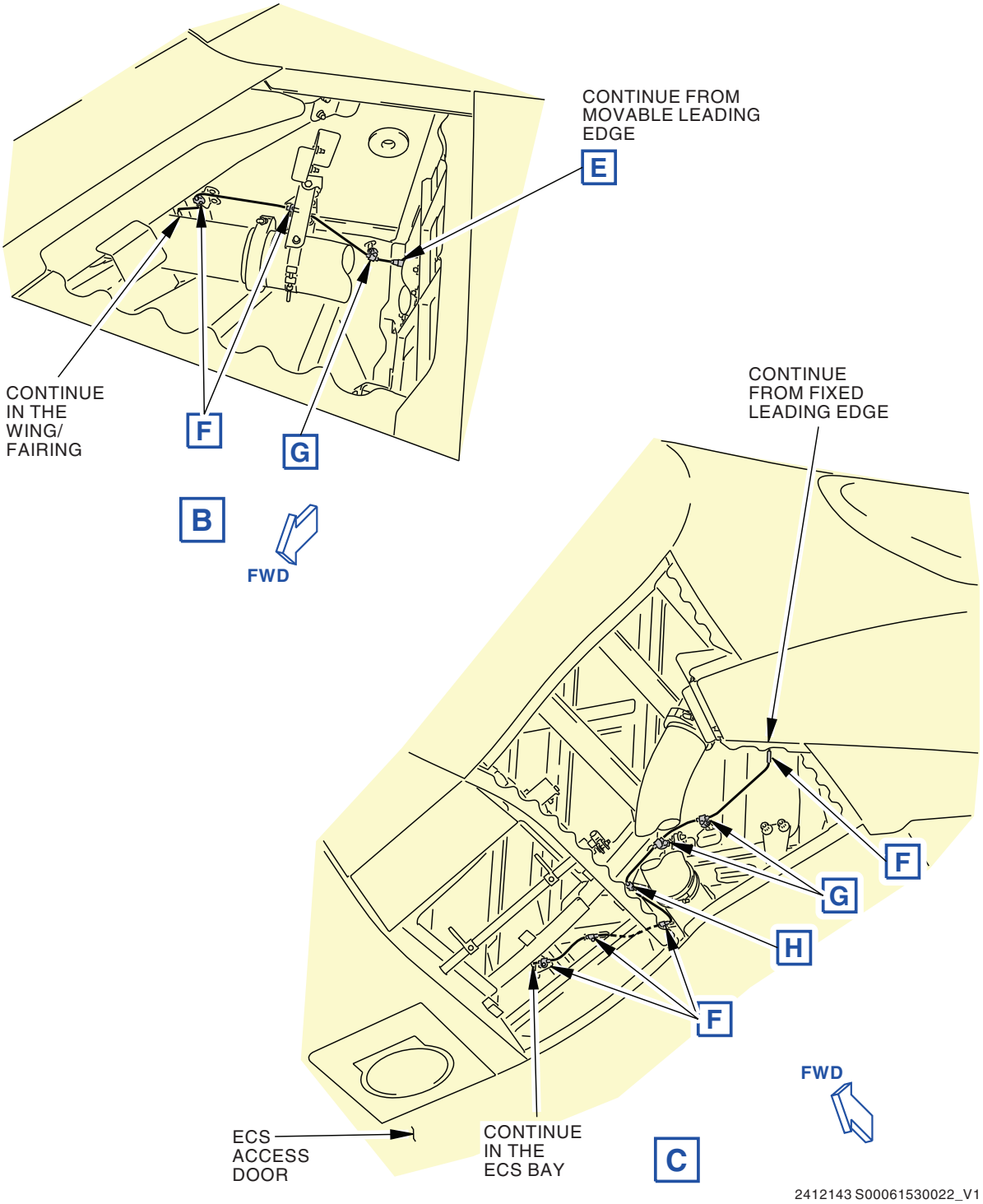
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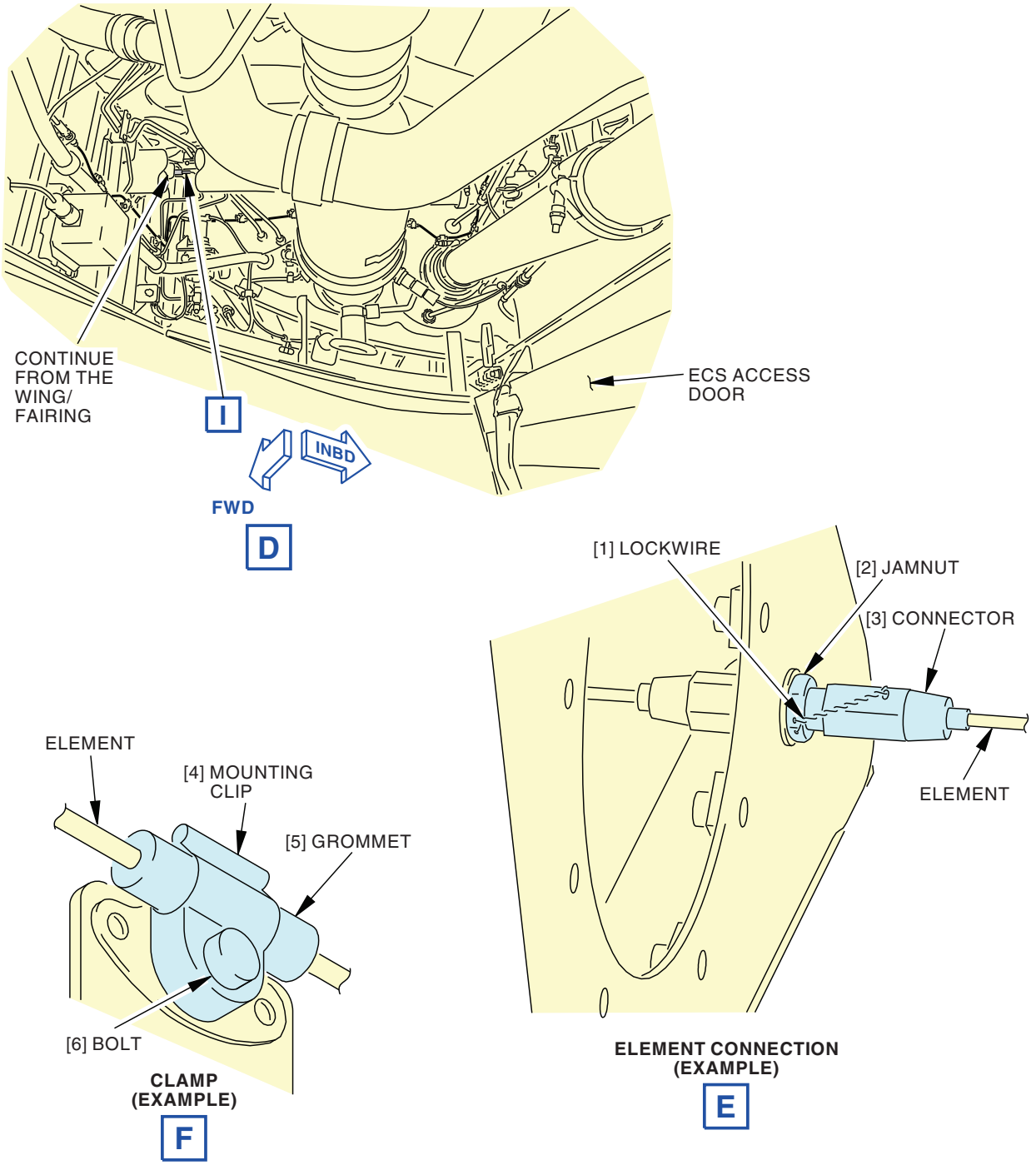
Wing and Lower Aft Body Overheat Detectors Installation
Figure 401/26-18-02-990-801 (Sheet 1 of 4)

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Wing and Lower Aft Body Overheat Detectors Installation
Figure 401/26-18-02-990-801 (Sheet 2 of 4)

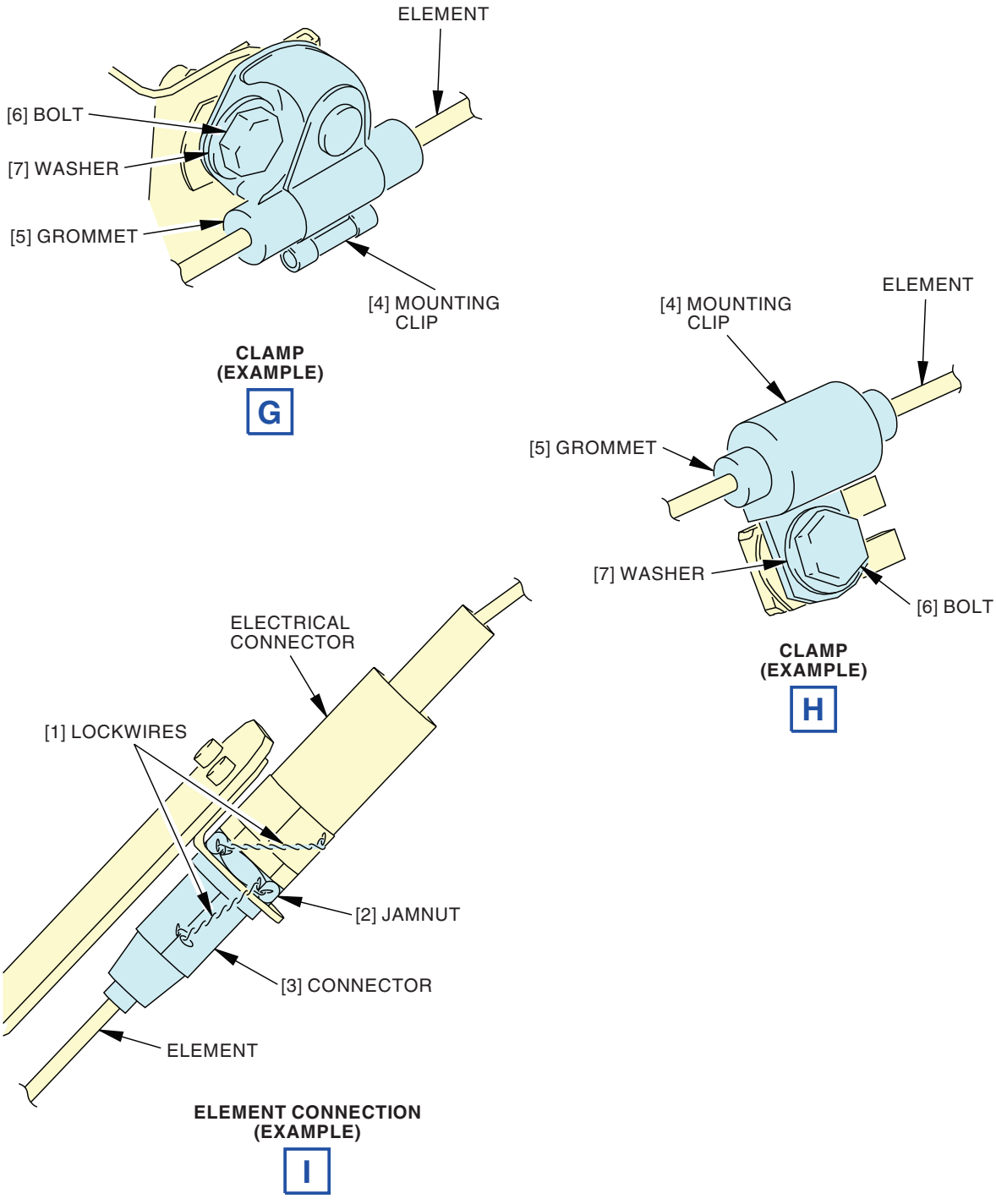
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Wing and Lower Aft Body Overheat Detectors Installation
Figure 401/26-18-02-990-801 (Sheet 3 of 4)

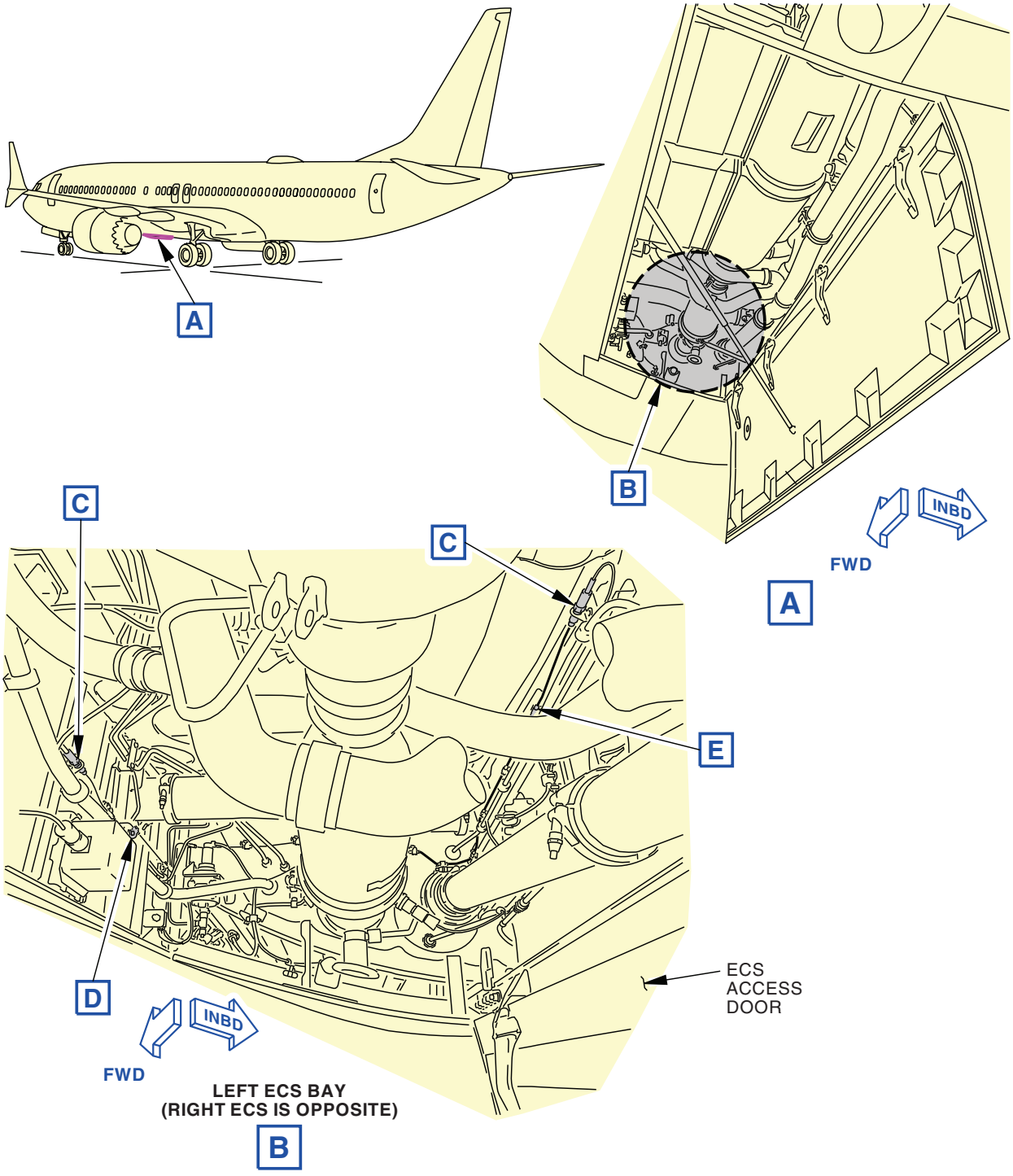
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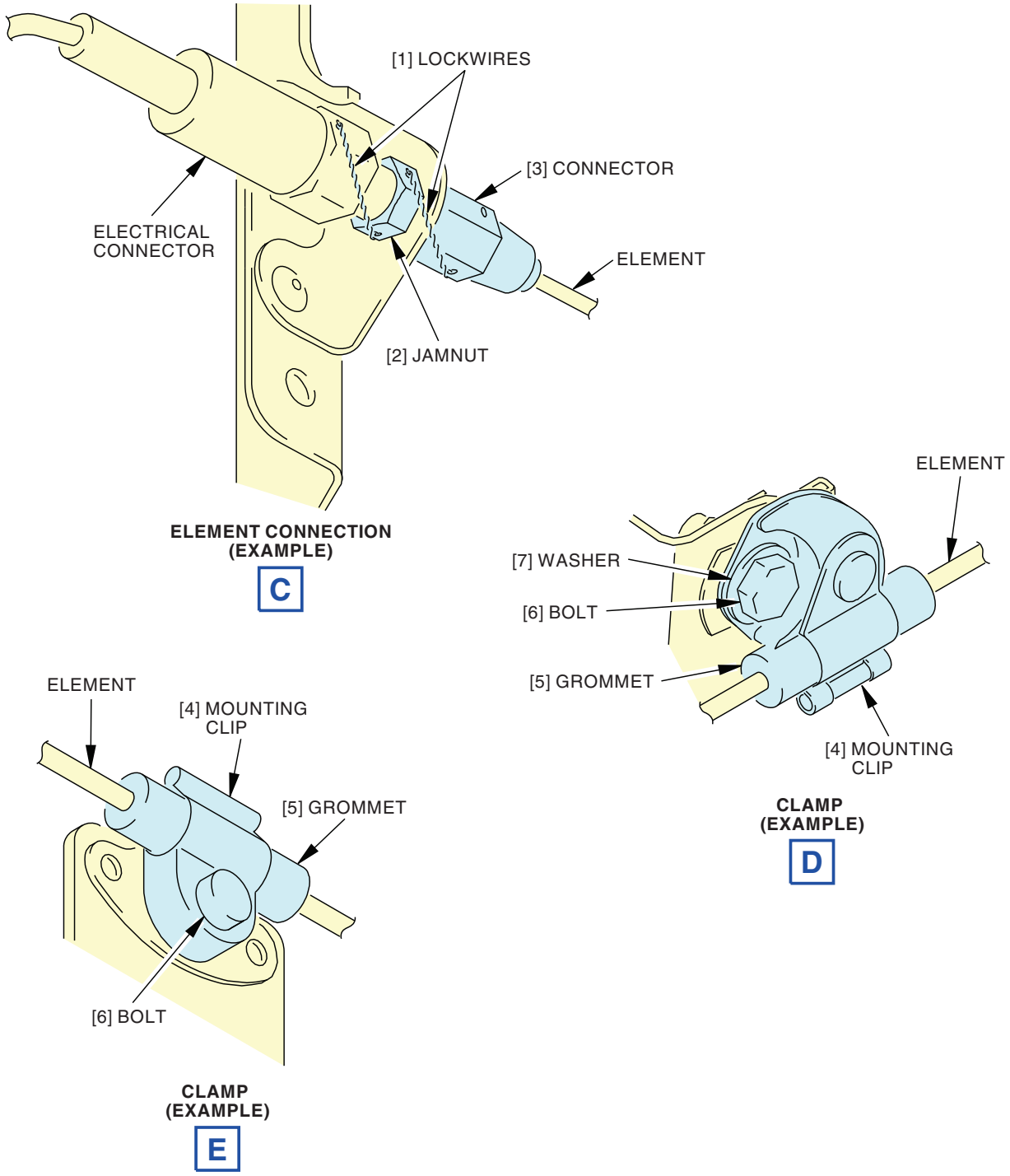
Wing and Lower Aft Body Overheat Detectors Installation
Figure 401/26-18-02-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A/C Pack Overheat Detectors Installation
Figure 402/26-18-02-990-802 (Sheet 1 of 2)

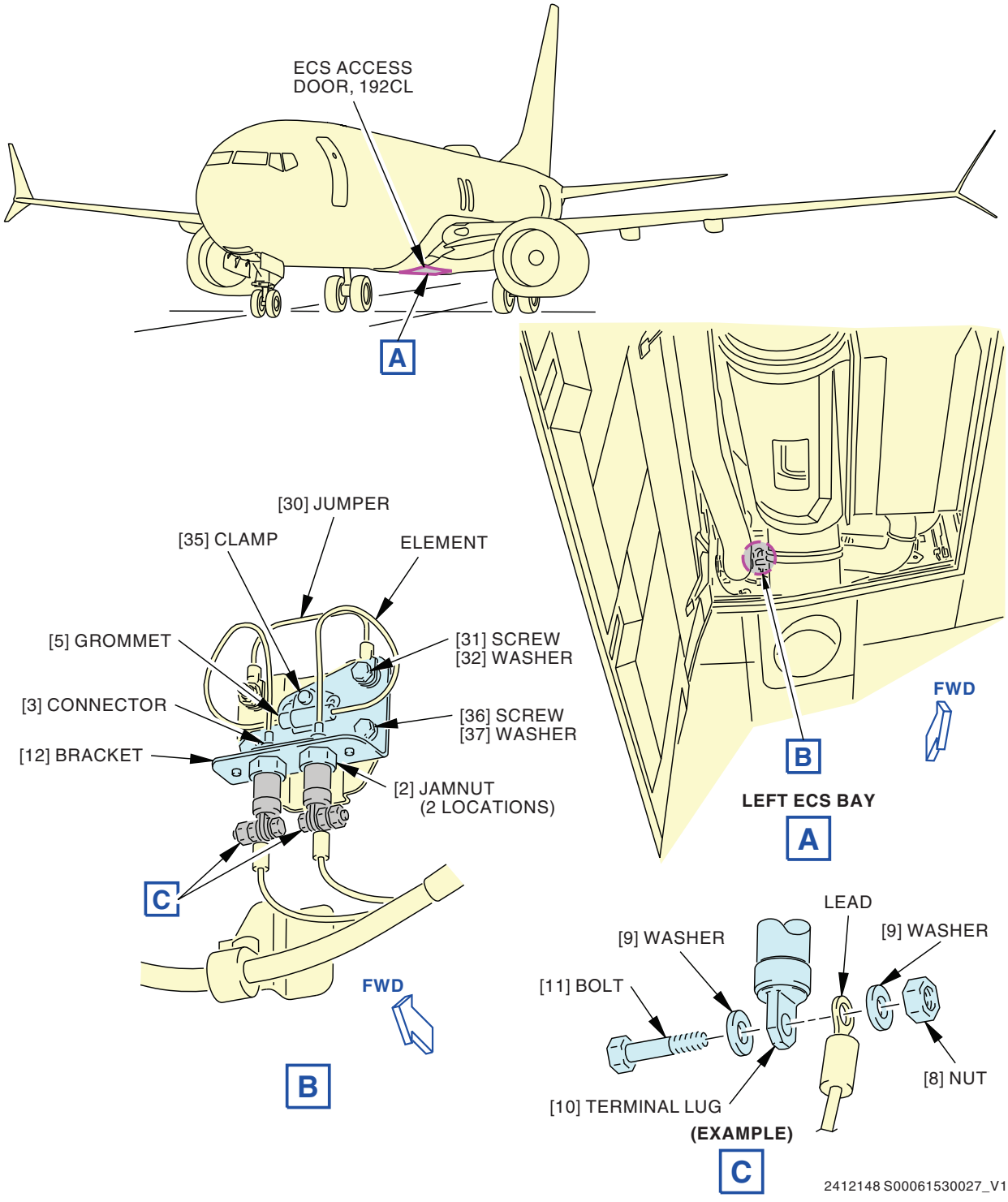
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412147 S00061530026_V2

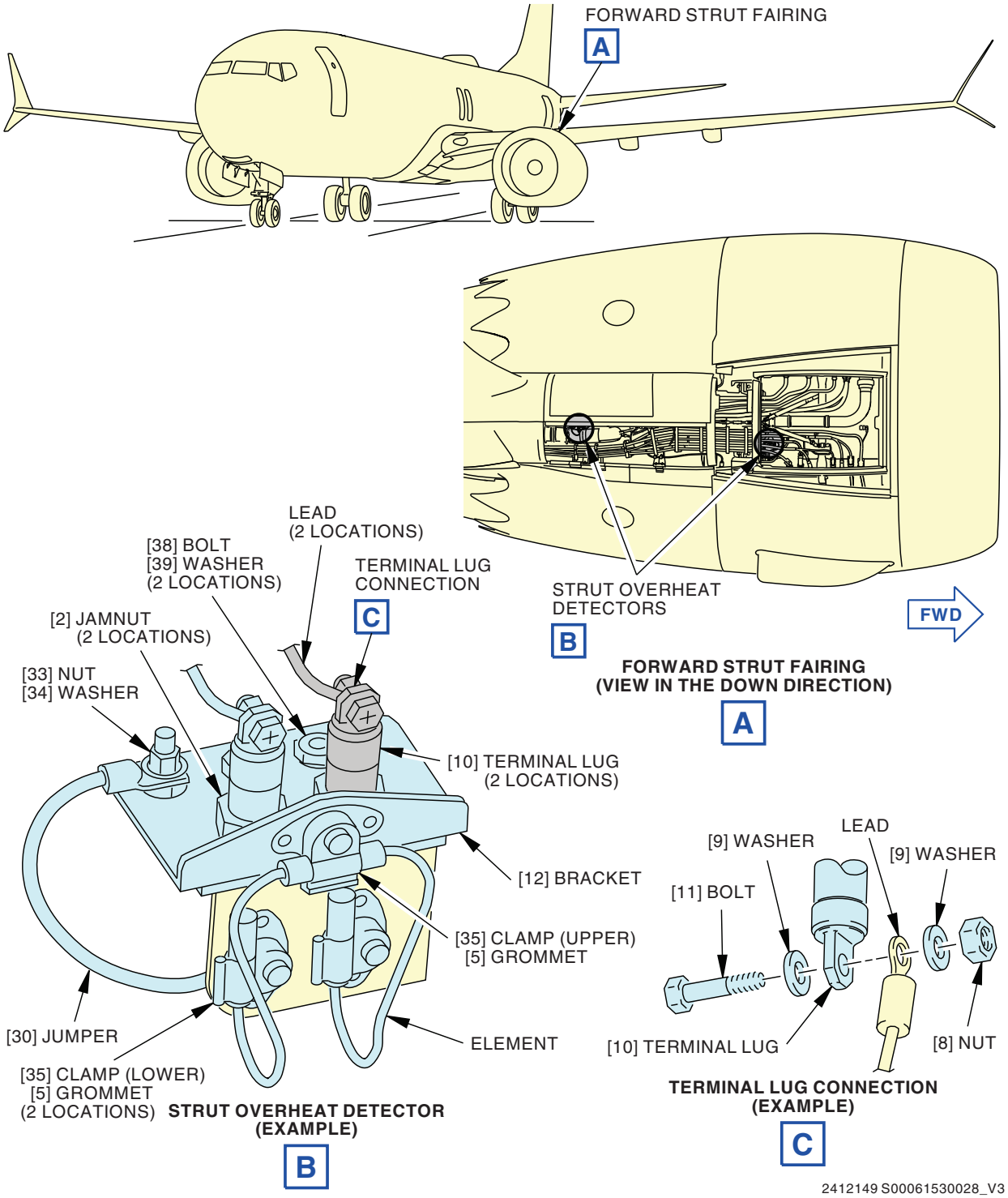
A/C Pack Overheat Detectors Installation
Figure 402/26-18-02-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



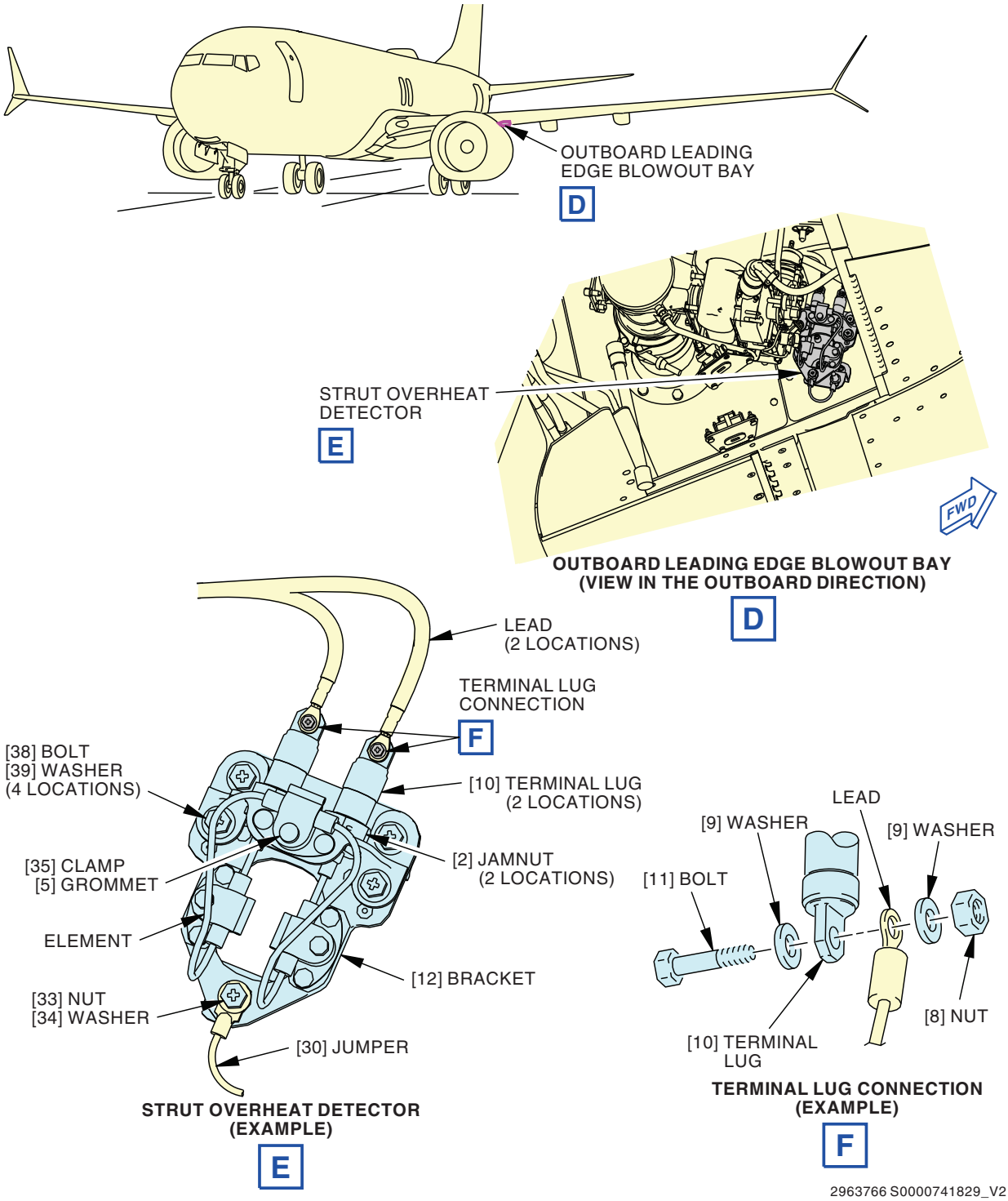
Mid and Aft A/C Pack Overheat Detectors Installation
Figure 403/26-18-02-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details



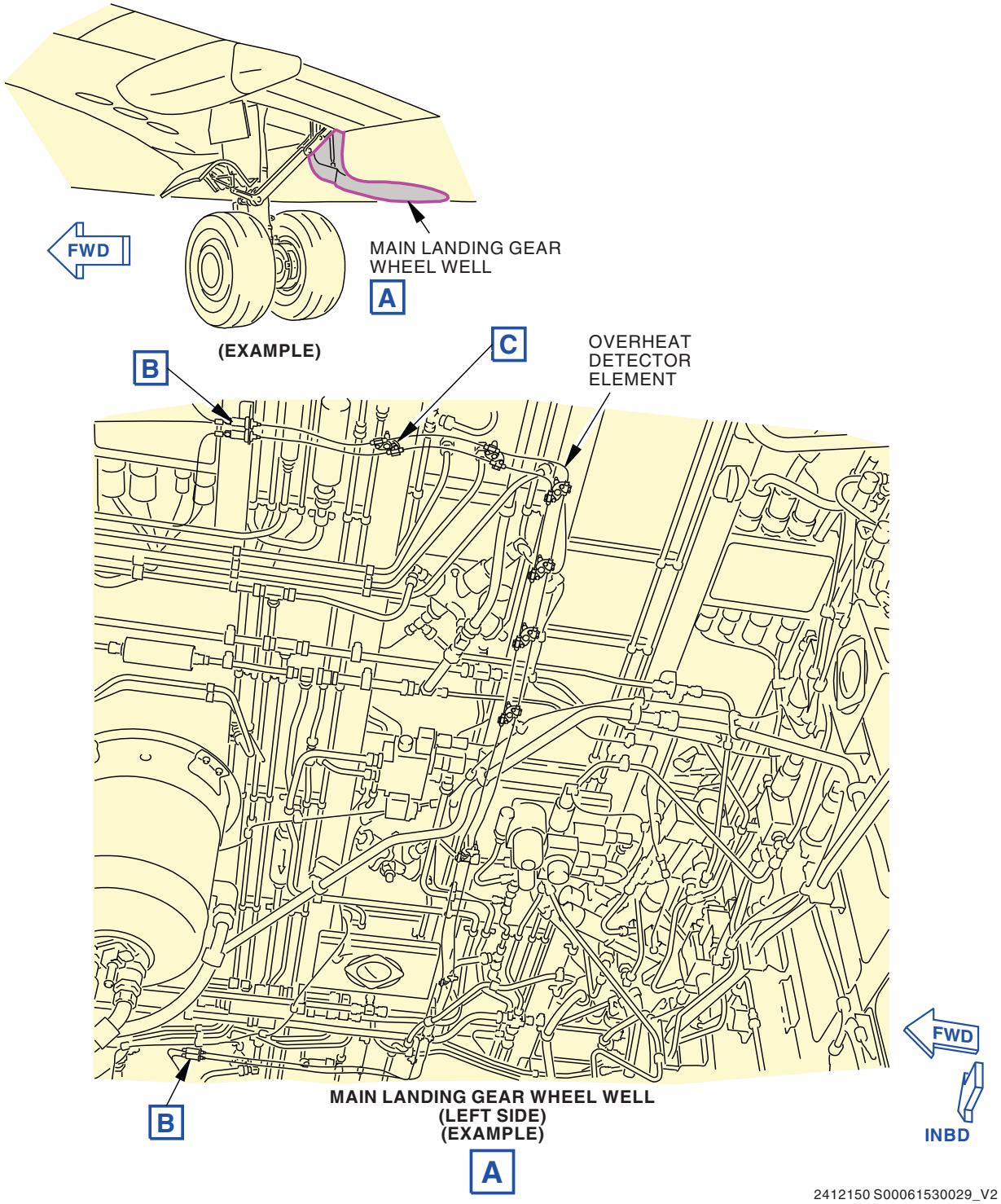
Strut Overheat Detectors Installation
Figure 404/26-18-02-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



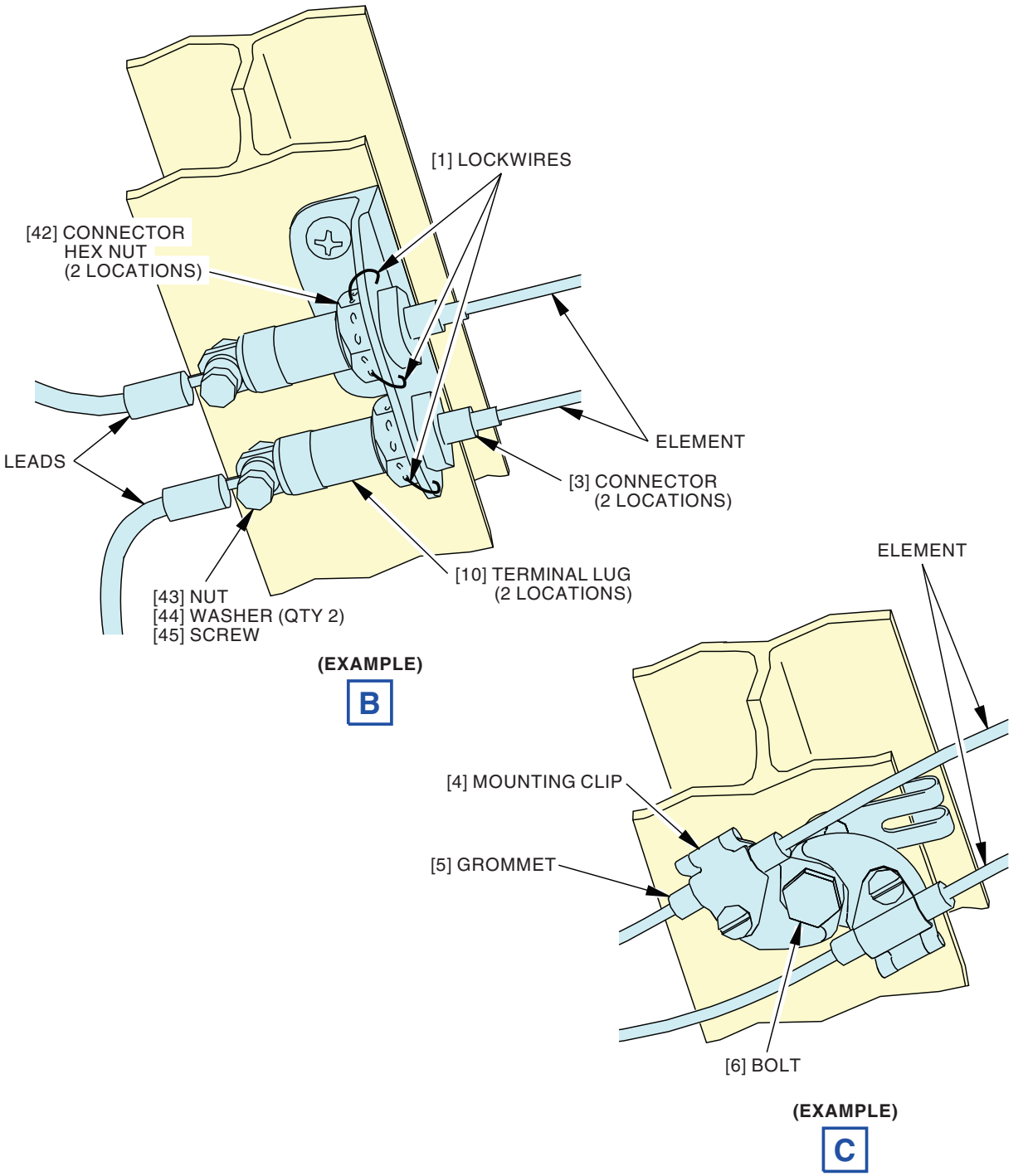
Strut Overheat Detectors Installation
Figure 404/26-18-02-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



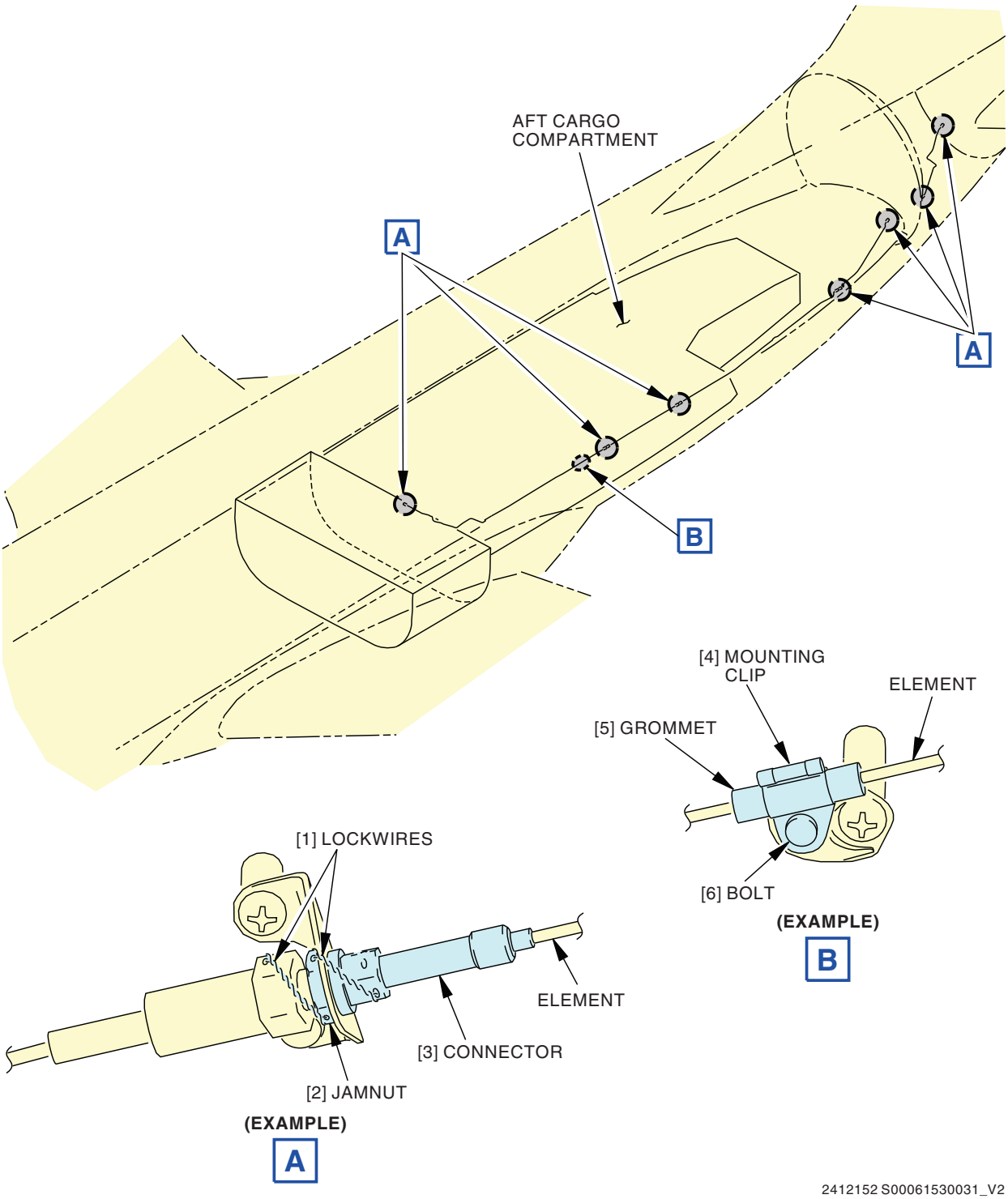
Wheel Well Overheat Detection Sensing Element Installation
Figure 405/26-18-02-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



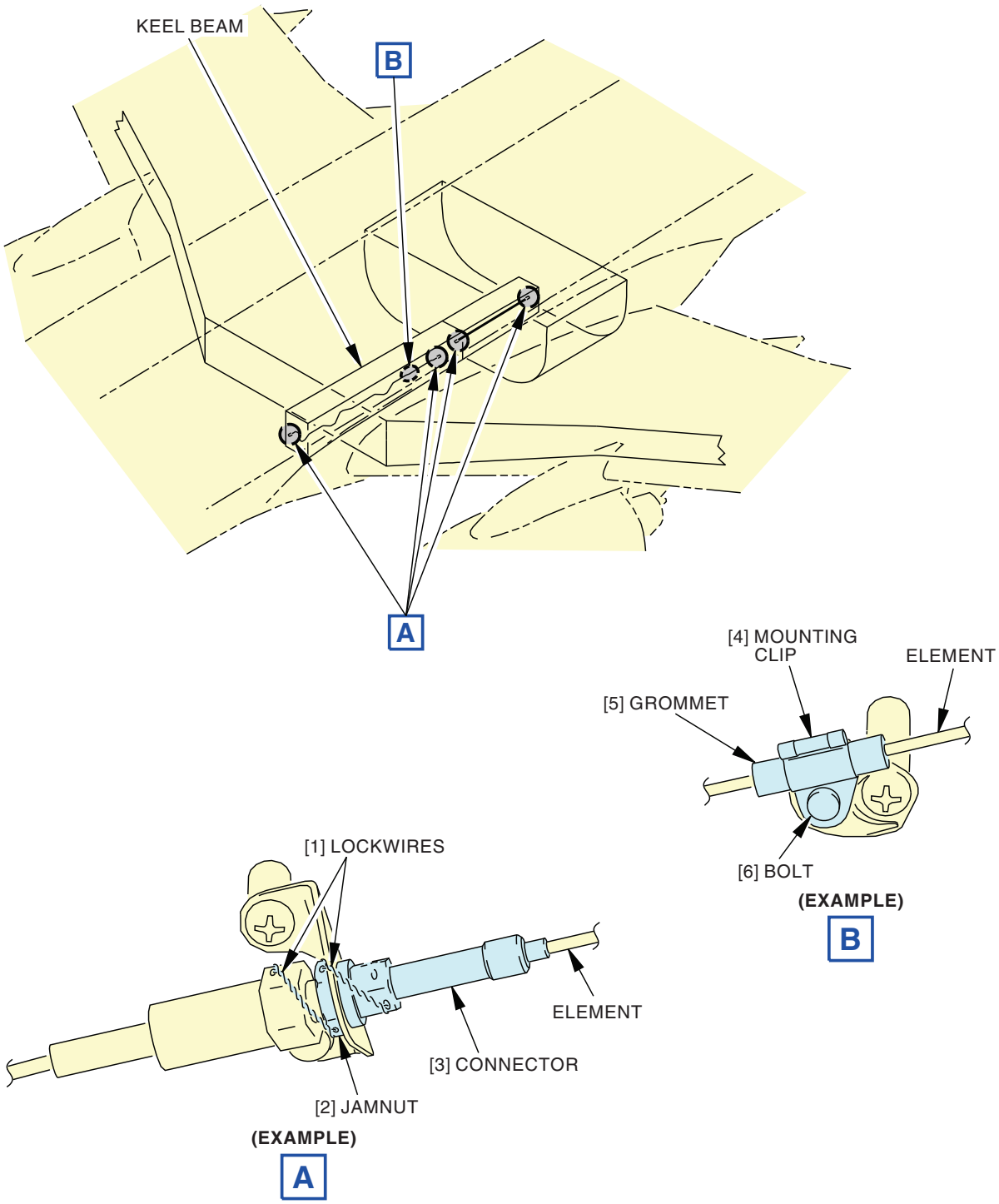
Wheel Well Overheat Detection Sensing Element Installation
Figure 405/26-18-02-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



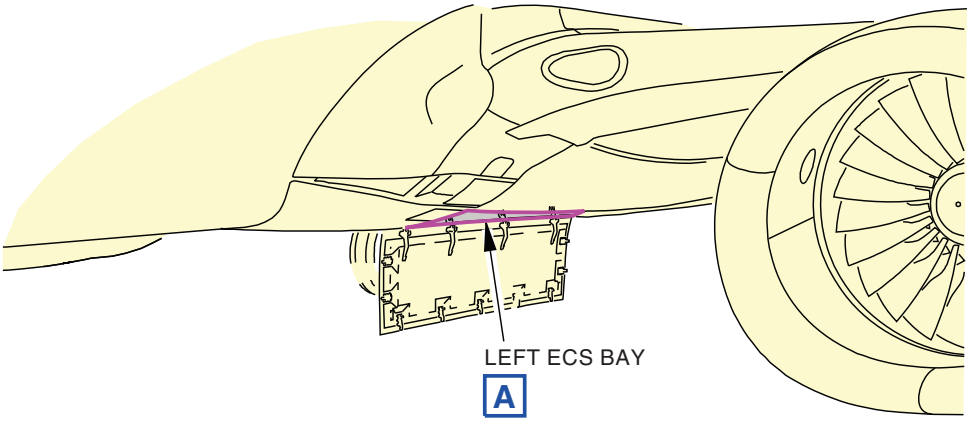
Aft Cargo Overheat Detectors Installation
Figure 406/26-18-02-990-806

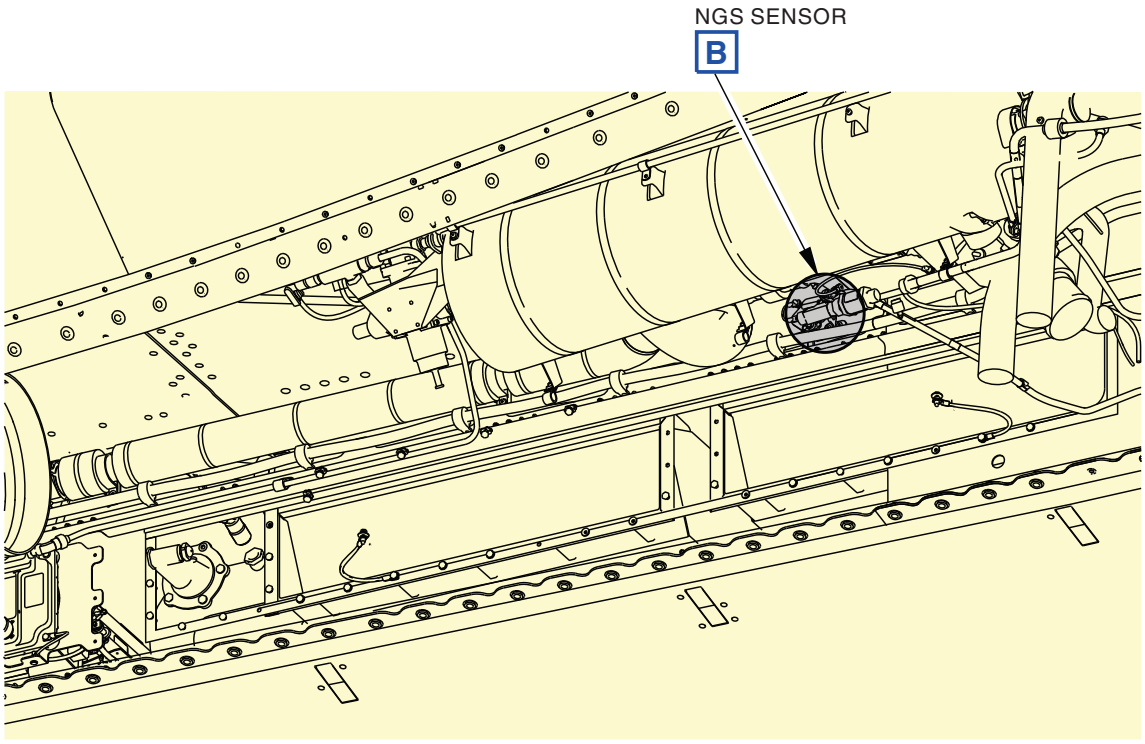
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Keel Beam Overheat Detectors Installation
Figure 407/26-18-02-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details



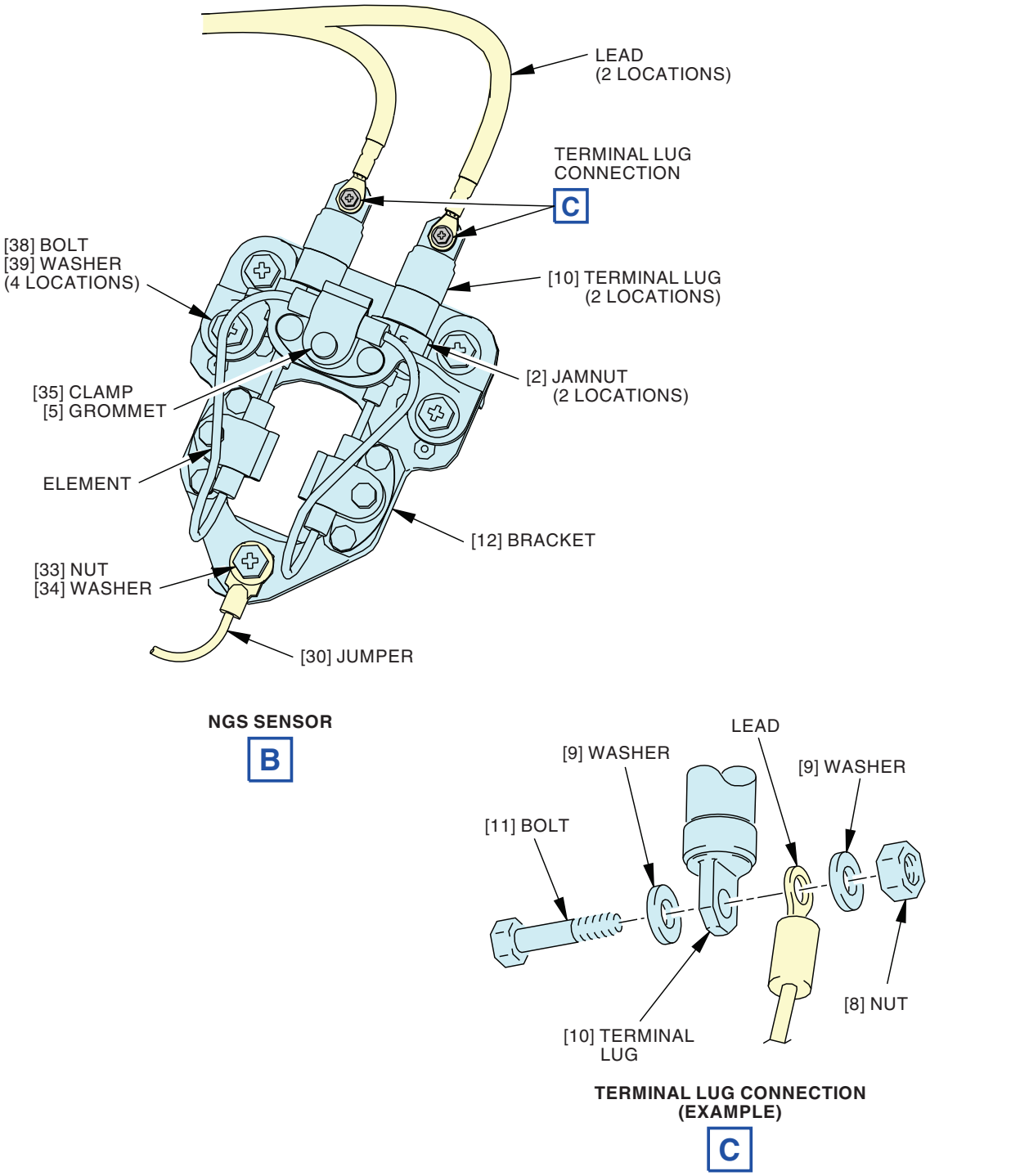


LEFT ECS BAY
(HEAT EXCHANGER INLET DUCT REMOVED FOR CLARITY)
A

2963767 S0000741830_V2

NGS Overheat Detector Installation
Figure 408/26-18-02-990-809 (Sheet 1 of 2)

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NGS Overheat Detector Installation
Figure 408/26-18-02-990-809 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-18-02-400-801

3. Wheel Well, Wing and Lower Aft Body Overheat Sensor Element - Installation

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405, Figure 406, Figure 407, Figure 408)

A. General

(1) This task gives instructions to install the wheel well, wing, and lower aft body overheat sensor element.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
20-40-11-760-801	Electrical Bonding (P/B 201)
26-18-00-710-801	Wheel Well and Lower Aft Body Overheat Detection System - Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

D. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
A00230	Compound - Electrical Insulating Coating	BMS5-37
A00767	Sealant - Fuel Tank	BMS5-45
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00130	Alcohol - Isopropyl	TT-I-735
B00316	Solvent - Aliphatic Naphtha (For Organic Coatings)	TT-N-95 Type I, ASTM D-3735 Type I
B50078	Solvent - Aliphatic Naphtha (For Organic Coatings)	TT-N-95 Type I (Supersedes BMS3-2 Type I)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
B50079	Solvent - Aliphatic Naphtha (For Acrylic Plastics)	TT-N-95 Type II (Supersedes BMS3-2 Type II)
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50262	Wiper - Cleaning	BMS15-5
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50470	Compound - Electrical Insulating, Molykote 4 (formerly Dow Corning 4)	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	Element	26-18-02-12-050	PTF ALL
14	Element	26-18-02-11-050	PTF ALL
15	Element	26-18-02-13A-045	PTF ALL
16	Element	26-18-02-05-055	PTF ALL
17	Element	26-18-02-10B-045	PTF ALL
18	Element	26-18-02-15-100	PTF ALL
		26-18-02-16-100	PTF ALL
		26-18-02-20-060	PTF ALL
19	Element	26-18-02-04-025	PTF ALL
20	Element	26-18-02-01-010	PTF ALL
21	Element	26-18-02-01-015	PTF ALL
22	Element	26-18-02-03A-070	PTF ALL
23	Element	26-18-02-03A-075	PTF ALL
24	Element	26-18-02-09A-025	PTF ALL
25	Element	26-18-02-07A-030	PTF ALL
26	Element	26-18-02-12-050	PTF ALL
27	Element	26-18-02-11-050	PTF ALL
28	Element	26-18-02-13-045	PTF ALL
29	Detector	26-18-02-06-110	PTF ALL
41	Element	26-18-02-21-060	PTF ALL

F. Location Zones

Zone	Area
129	Keel Beam (Part) - Section 43
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
149	Keel Beam (Part) Body Station 727.00 to Body Station 743.95
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box
311	Area Aft of Pressure Bulkhead - Left
431	Engine 1 - Forward Strut Fairing
441	Engine 2 - Forward Strut Fairing
511	Left Wing - Leading Edge To Front Spar
521	Left Wing - Leading Edge to Front Spar
611	Right Wing - Leading Edge to Front Spar
621	Right Wing - Leading Edge to Front Spar

G. Prepare for the Installation

SUBTASK 26-18-02-860-002

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

SUBTASK 26-18-02-010-003

(2) Make sure that the applicable access doors and panels are open (Table 401) (TASK 26-18-02-000-801).

SUBTASK 26-18-02-010-004

(3) Make sure that you remove the airplane parts that are in the way (Table 401) (TASK 26-18-02-000-801).

H. Long Sensor Element Installation

NOTE: This procedure only applies to the following sensors:

element [15], M356, R FWD A/C PACK
element [16], M370, L WING INBOARD
element [16], M371, R WING INBOARD
element [17], M268, L WING OUTBOARD
element [17], M269, R WING OUTBOARD
element [19], M276, AFT - SECTION 48
element [20], M275, AFT - SECTION 47
element [21], M347, AFT - SECTION 46
element [22], M1147, AFT - SECTION 46
element [23], M348, AFT - SECTION 46
element [24], M273, AFT KEELBEAM
element [25], M272, FWD KEELBEAM
element [28], M355, L FWD A/C PACK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

detector [29], M270, WHEEL WELL LOOP A

detector [29], M3068, WHEEL WELL LOOP B.

SUBTASK 26-18-02-420-001

CAUTION

WHEN YOU INSTALL THE ELEMENT, DO NOT BEND THE ELEMENT TO A RADIUS THAT IS LESS THAN 1 IN. (25.4 MM.). DO NOT LET A BEND IN 1 IN. (25.4 MM.) DISTANCE FROM THE EDGE OF THE QUICK-RELEASE CLAMP OR THE FLANGE ASSEMBLY. CONTOUR THE ELEMENT TO CLEAR ADJACENT PARTS AND STRUCTURES. HAVE THE ELEMENT LENGTH EQUALLY BETWEEN CLAMPS. IF THE BEND RADIUS IS LESS THAN ONE INCH, DAMAGE TO THE ELEMENT CAN OCCUR.

CAUTION

DO NOT TURN, PULL, OR ATTACH THE SENSING ELEMENT WITH A CLAMP. DO NOT MAKE SERVICEABLE KINKS, BENDS, OR DENTS STRAIGHT. IF YOU MAKE SERVICEABLE KINKS, BENDS, OR DENTS STRAIGHT, DAMAGE TO THE SENSING ELEMENT CAN OCCUR.

(1) Carefully put the element into position.

NOTE: This step will require two people.

(a) Carefully unwind the element.

1) Make sure that there are no kinks.

(b) Carefully bend the element to fit into its approximate position.

(c) Carefully put and hold the element in its position.

(d) Do these steps for sensor elements on the aft keel beam:

1) Tie the existing cord to the aft end of the new element.

2) Slowly pull the forward through the keel beam.

a) Use the access holes below the keel beam to help guide the element through the holes in the structure.

3) Remove the cord from the element.

SUBTASK 26-18-02-420-015

(2) Remove the caps from the end fittings of the electrical connectors [3].

SUBTASK 26-18-02-420-016

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(3) Examine the electrical connector [3] at each end of the heat detector for corrosion.

(a) If corrosion is present, replace the element.

SUBTASK 26-18-02-420-017

(4) Put covers on the open end of the electrical connectors [3] with caps to prevent damage.

SUBTASK 26-18-02-420-002

(5) Install the element into each of the mounting clips [4].

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(a) Install the grommets [5] around the elements so that each grommet [5] is in the center of the mounting clip [4].

(b) Install the grommets [5] into the mounting clips [4] so that the slit is at 90 degrees from the clip hinge except at bends where the slit is facing outside of the nearest bend.

1) Make sure that the mounting clips [4] and grommets [5] are attached to the element tightly.

2) If the used sensor element is in its position as a reference, remove it.

(c) Locate element in approximate installation position and temporarily support as necessary to maintain position.

(d) Tighten the bolts [6] to secure the mounting clips [4].

(e) For the Wing-to-body zone, make sure that the minimum clearance is as follows:

1) 0.5 in. (12.7 mm) between the element and structure, except at mounting clips [4].

2) 0.5 in. (12.7 mm) between the element and other systems, except at mounting clips [4].

3) 1.65 in. (41.91 mm) between the element and duct.

(f) For the Wing Leading Edge zone, make sure that the minimum clearance is as follows:

1) 0.25 in. (6.4 mm) between the element and structure, except at mounting clips [4].

2) 0.15 in. (3.81 mm) between the element and other systems, except at mounting clips [4].

3) 1.5 in. (38.10 mm) between the element and duct.

(g) For the Bulkhead zone, make sure that the minimum clearance is as follows:

1) 0.25 in. (6.4 mm) between the element and structure, except at mounting clips [4].

2) 0.5 in. (12.7 mm) between the element and other systems, except at mounting clips [4].

3) 1.5 in. (38.10 mm) between the element and duct.

(h) For all the other zones, make sure that the minimum clearance is as follows:

1) 0.25 in. (6.4 mm) between the element and structure, except at mounting clips [4].

2) 0.5 in. (12.7 mm) between the element and other systems, except at mounting clips [4].

3) 3 in. (76.2 mm) between the element and duct.

(i) Make sure that the minimum clearance between the elements is 0.5 in. (12.7 mm).

SUBTASK 26-18-02-420-003

(6) Remove the caps from the end fittings to connect the connectors [3].

SUBTASK 26-18-02-210-001

(7) Make sure that the connectors [3] have ceramic inserts on both ends.

SUBTASK 26-18-02-210-002

(8) If it is applicable, make sure that the pins on the connectors [3] are centered, not bent, recessed, or damaged.

SUBTASK 26-18-02-100-001

(9) Clean the mating surfaces of the connectors [3], element, and structure (SWPM 20-20-00).

(a) Use a cotton wiper, G00034, made moist with alcohol, B00130.

(b) Immediately dry the surfaces with a new cotton wiper, G00034.

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SUBTASK 26-18-02-390-001

(10) Apply Compound, G50470, to the internal components of the connectors [3] and ends of the element.

SUBTASK 26-18-02-420-005

(11) Do these steps to connect the connector [3]:

(a) Attach the connector [3].

(b) If it was removed, install the jamnut [2] or connector hex nut [42] and torque to 50 in-lb (5.6 N·m) - 60 in-lb (6.8 N·m).

(c) If it was disconnected, connect the coupling element connector or electrical connector.

(d) Install lockwires [1] on the jamnut [2] or connector hex nut [42] as follows:

NOTE: Lockwire diameter must fill between 1/3 and 3/4 of the hole diameter, 0.032 in. (0.813 mm) diameter minimum except for lockwire 0.020 in. (0.508 mm) in diameter can be used if the lockwire hole is 0.045 in. (1.143 mm) diameter or smaller, or the spacing between parts is less than 2 in. (51 mm) and lockwire hole diameter is between 0.045 in. (1.143 mm) and 0.062 in. (1.575 mm) diameter.

1) Install MS20995NC32 lockwire, G50347, on the connector hex nut [42] or jamnut [2] and connector [3].

2) If it is necessary, install MS20995NC32 lockwire, G50347, on the jamnut [2] and coupling element connector or electrical connector.

3) To obtain adequate tension without overstressing the lockwire [1], apply the number of twists per inch of lockwire as in Table 402.

Table 402/26-18-02-993-802 TWISTS PER INCH

RANGE OF WIRE DIAMETERS (IN)	LESS THAN 0.019	0.019 TO 0.026	0.027 TO 0.042	0.043 TO 0.065	MORE THAN 0.065
Twists per inch	11 to 14	9 to 12	7 to 10	5 to 8	4 to 7

SUBTASK 26-18-02-860-009

(12) Measure the electrical bonding resistance between airplane structure and overheat sensor elements.

(a) Use a non-intrinsically safe bonding meter, COM-614.

(b) Make sure that the electrical bonding resistance is 0.005 ohms (5 milliohm) or less.

SUBTASK 26-18-02-390-002

(13) Apply a fillet seal to all edges of the mating surfaces between the bracket and M270, M3068 sensors with sealant, A50337, sealant, A50231, or sealant, A50359 (SWPM 20-20-00).

SUBTASK 26-18-02-420-006

(14) Do these steps for sensing elements in the main wheel well:

(a) Clean the terminal lug [10], nut [43], washers [44], and screw [45] with aliphatic naphtha, B50079, or aliphatic naphtha, B50078, or solvent, B00316, and wiper, G50262.

(b) Attach the terminal lug [10] with the nut [43], two washers [44], and screw [45].

(c) Tighten the screw [45] to the specified torque of 25 in-lb (2.8 N·m) - 35 in-lb (4.0 N·m).

(d) Apply compound, A00230, to the terminal of the long sensor element.

1) Make sure that the sealant covers all exposed metal from the insulator to the wiring.

NOTE: The sealant should not bridge the insulator.

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SUBTASK 26-18-02-860-003

(15) Only the M270 and M3068 sensors require application of sealant, A50337, sealant, A50231, or sealant, A50359.

I. Looped Sensor Element Installation

NOTE: This procedure only applies to the following sensors:

element [13], M1912, R AFT A/C PACK
element [14], M1910, R MID A/C PACK
element [18], M1763, R FWD STRUT
element [18], M3058, R MID STRUT
element [18], M1764, R AFT STRUT
element [18], M1761, L STRUT FWD
element [18], M3057, L MID STRUT
element [18], M1762, L STRUT AFT
element [26], M1911, L AFT A/C PACK
element [27], M1909, L MID A/C PACK
element [41], M3067, NITROGEN GEN.

SUBTASK 26-18-02-100-002

(1) On a clean work surface clean the mating surfaces of the bracket [12] and element (SWPM 20-20-00).

(a) Use cotton wiper, G00034, made moist with alcohol, B00130.

(b) Immediately dry the surfaces with a new cotton wiper, G00034.

SUBTASK 26-18-02-420-007

CAUTION

WHEN YOU INSTALL THE ELEMENT, DO NOT BEND THE ELEMENT TO A RADIUS THAT IS LESS THAN 1 IN. (25.4 MM.). DO NOT LET A BEND IN 1 IN. (25.4 MM.) DISTANCE FROM THE EDGE OF THE QUICK-RELEASE CLAMP OR THE FLANGE ASSEMBLY. CONTOUR THE ELEMENT TO CLEAR ADJACENT PARTS AND STRUCTURES. HAVE THE ELEMENT LENGTH EQUALLY BETWEEN CLAMPS. IF THE BEND RADIUS IS LESS THAN ONE INCH, DAMAGE TO THE ELEMENT CAN OCCUR.

CAUTION

DO NOT TURN, PULL, OR ATTACH THE SENSING ELEMENT WITH A CLAMP. DO NOT MAKE SERVICEABLE KINKS, BENDS, OR DENTS STRAIGHT. IF YOU MAKE SERVICEABLE KINKS, BENDS, OR DENTS STRAIGHT, DAMAGE TO THE SENSING ELEMENT CAN OCCUR.

(2) Position the element in the bracket [12] and quick release clamp [35].

NOTE: Make sure that the element is routed straight inside the clamps and minimum clearance is maintained as follow: 0.25 inches between the detector and adjacent structure or other systems; 1.5 inches between the detector and ducts.

SUBTASK 26-18-02-390-009

(3) For M1762 and M1764, apply a fay seal between the bracket and sensors with sealant, A50359, sealant, A50337, or sealant, A50231 (SWPM 20-20-00).

SUBTASK 26-18-02-420-008

(4) Secure each terminal lug [10] with jamnut [2].

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(a) Do this for sensor elements M1909, M1910, M1911, M1912, M1762, M1764, and M3067:

1) Torque jamnuts [2] to 80 in-lb (9.0 N·m) - 100 in-lb (11.3 N·m).

(b) Do this for sensor elements M1761, M1763, M3058 and M3057:

1) Torque jamnuts [2] to 50 in-lb (5.6 N·m) - 60 in-lb (6.8 N·m).

(c) Measure the electrical bonding resistance between the detector element sensors and bracket for M1909, M1910, M1911, M1912, M1762, and M1764 sensors (TASK 20-40-11-760-801).

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.005 ohms (5 milliohms) or less.

(d) Measure the electrical bonding resistance between the detector element sensors and bracket for M1761, M1763, M3057, and M3058 sensors (TASK 20-40-11-760-801).

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.01 ohm (10 milliohms) or less.

(e) Install the lockwire [1] on the jamnuts [2], do this task: Lockwire, Cotter Pins, and Lockrings - Installation, TASK 20-10-44-400-801.

SUBTASK 26-18-02-420-009

(5) Install the element in the quick release clamp [35] on the bracket [12].

(a) Install the grommet [5] on the element.

(b) Position the grommet [5] in the center of the quick release clamp [35].

(c) Secure the quick release clamp [35].

SUBTASK 26-18-02-420-012

(6) Place the assembled bracket [12] and element in position.

(a) For the A/C pack locations secure with screws [36] and washers [37].

(b) For the strut locations secure with bolts [38] and washers [39].

SUBTASK 26-18-02-420-013

(7) For the strut locations secure the element to the lower quick release clamps [35].

(a) Install the grommets [5] on the element.

(b) Position the grommets [5] in the center of the quick release clamps [35].

(c) Secure the quick release clamps [35].

SUBTASK 26-18-02-420-014

(8) Attach the jumper [30] to the bracket [12].

(a) For the A/C pack locations secure with screw [31] and washer [32].

(b) For the strut locations secure with nut [33] and washer [34].

SUBTASK 26-18-02-420-010

(9) Attach each terminal lug [10] to its corresponding lead and secure with a bolt [11], washers [9], and nut [8].

(a) Tighten nut [8].

SUBTASK 26-18-02-420-018

(10) Attach the jumper [30] to the structure.

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SUBTASK 26-18-02-860-017

(11) Measure the electrical bonding resistance between airplane structure and M1909, M1910, M1911, M1912, M1762, M1764, M1761, M1763, M3057, M3058, and M3068 sensors.

(a) Use a non-intrinsically safe bonding meter, COM-614.

(b) Make sure that the electrical bonding resistance is 0.010 ohms (10 milliohms) or less.

SUBTASK 26-18-02-914-001

(12) Apply compound, A00230, or sealant, A00767, to the terminal of the overheat loop sensor.

(a) Make sure that the sealant covers all exposed metal from the insulator to the wiring.

NOTE: The sealant should not bridge the insulator.

SUBTASK 26-18-02-390-003

(13) Apply a fillet seal between the bracket [12] and jamnut [2] for M1761, M1763, M3057, and M3058 sensors with sealant, A00160 (SWPM 20-20-00).

J. Wheel Well, Wing and Lower Aft Body Overheat Sensor Element Installation Test

SUBTASK 26-18-02-860-004

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

SUBTASK 26-18-02-420-011

(2) Do this task: Wheel Well and Lower Aft Body Overheat Detection System - Operational Test, TASK 26-18-00-710-801.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 26-18-02-410-001

(1) Install any airplane parts that were removed (Table 401) (TASK 26-18-02-000-801).

SUBTASK 26-18-02-410-002

(2) Close the applicable access doors and panels (Table 401) (TASK 26-18-02-000-801).

——— END OF TASK ———

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WING, WHEEL WELL, AND LOWER AFT BODY OVERHEAT DETECTION SENSING ELEMENT - ADJUSTMENT/TEST

1. General

A. This procedure provides the instructions to test the sensing elements for the wing, wheel well and lower aft body overheat detection system.

TASK 26-18-02-720-801

2. Wing, Wheel Well and Lower Aft Body Overheat Sensing Element Test

(Figure 501)

A. General

(1) Do a test of each sensing loop only when all of the units in the system are stable at ambient temperature. If you cannot complete the tests that follow on an element, replace the element.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
26-18-00-710-801	Wheel Well and Lower Aft Body Overheat Detection System - Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1741	Meter - LCR (Inductance, Capacitance, Resistance; Use COM-20443 for OH bleed loop testing) Part #: 878B Supplier: 08098 Part #: 879B Supplier: 08098 Part #: 880 Supplier: 08098 Part #: U1733P Supplier: 1LQK8 Opt Part #: 875B Supplier: 08098 Opt Part #: 878A Supplier: 08098 Opt Part #: 879 Supplier: 08098
COM-20443	Meter - LCR and Z (Inductance, Capacitance, Resistance and Impedance) Part #: 880 Supplier: 08098

D. Location Zones

Zone	Area
129	Keel Beam (Part) - Section 43
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
149	Keel Beam (Part) Body Station 727.00 to Body Station 743.95

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(Continued)

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box
311	Area Aft of Pressure Bulkhead - Left
431	Engine 1 - Forward Strut Fairing
441	Engine 2 - Forward Strut Fairing
511	Left Wing - Leading Edge To Front Spar
521	Left Wing - Leading Edge to Front Spar
611	Right Wing - Leading Edge to Front Spar
621	Right Wing - Leading Edge to Front Spar

E. Sensing Element Continuity Test (Resistance Measurement)

SUBTASK 26-18-02-865-001

(1) Disconnect the connectors from the loop.

SUBTASK 26-18-02-860-013

(2) Connect the Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443, or LCR meter (Inductance, Capacitance, Resistance), COM-1741, to the center of the conductor at each end of the loop.

NOTE: This procedure does a test of center conductor continuity.

NOTE: A standard digital voltmeter can be used for the center conductor resistance.

(a) If the Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443, or LCR meter (Inductance, Capacitance, Resistance), COM-1741, is used, make sure that the meter is in DCR (DC Resistance) mode.

SUBTASK 26-18-02-860-014

(3) Set the Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443, or LCR meter (Inductance, Capacitance, Resistance), COM-1741, to measure continuity through DCR mode.

SUBTASK 26-18-02-860-015

(4) Allow the reading to stabilize and record the reading.

SUBTASK 26-18-02-865-002

(5) Do the check for an open detector or circuit:

(a) Compare the values from the reading to the values from Table 501.

Table 501/26-18-02-993-803 Sensing Loop Test Data

KIDDE ALARM/FAULT CODES	LOOP CONNECTOR/PINS		LOOP CENTER CONDUCTOR RESISTANCE (< 130 ° F)	SENSORS
L ENGINE STR OPEN L ENGINE STR OVHT L ENGINE STR SHRT	D742, pin 4	D742, pin 12	< 15 ohms	M1761, M1762, M3057
L PACK BAY OPEN L PACK BAY OVHT L PACK BAY SHRT	D742, pin 2	D742, pin 8	< 15 ohms	M1911, M1909, M355

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 501/26-18-02-993-803 Sensing Loop Test Data (Continued)

KIDDE ALARM/FAULT CODES	LOOP CONNECTOR/PINS		LOOP CENTER CONDUCTOR RESISTANCE (< 130 ° F)	SENSORS
KEELBEAM OPEN KEELBEAM OVHT KEELBEAM SHRT	D742, pin 9	D742, pin 29	< 25 ohms	M272, M273
AFT CARGO APU OPEN AFT CARGO APU OVHT AFT CARGO APU SHRT	D742, pin 13	D742, pin 27	< 25 ohms	M275, M347, M348, M1147
R ENGINE STR OPEN R ENGINE STR OVHT R ENGINE STR SHRT	D742, pin 10	D742, pin 11	< 15 ohms	M1763, M1764, M3058
MWW LOOP A FIRE MWW LOOP A OPEN MWW LOOP A SHRT	D742, pin 21	D742, pin 24	< 15 ohms	M270
AFT BODY APU OPEN AFT BODY APU OVHT AFT BODY APU SHRT	D742, pin 36	D742, pin 37	< 25 ohms	M276
L WING LE OPEN L WING LE OVHT L WING LE SHRT	D742, pin 39	D742, pin 40	< 15 ohms	M268, M370
MWW LOOP B FIRE MWW LOOP B OPEN MWW LOOP B SHRT	D742, pin 31	D742, pin 34	< 15 ohms	M3068
NITROGEN GEN OPEN NITROGEN GEN OVHT NITROGEN GEN SHRT	D742, pin 32	D742, pin 33	< 15 ohms	M3067
R WING LE OPEN R WING LE OVHT R WING LE SHRT	D742, pin 30	D742, pin 35	< 15 ohms	M269, M371
R PACK BAY OPEN R PACK BAY OVHT R PACK BAY SHRT	D742, pin 41	D742, pin 42	< 15 ohms	M356, M1910, M1912

(b) If the reading is greater than the values shown in Table 501, do the continuity checks of the applicable loop segments to isolate to the failed loop or wiring.

(c) Replace the faulty detector or repair the wiring, as needed.

F. Sensing Element Pin to Case Test (Resistance Measurement)

SUBTASK 26-18-02-860-006

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

SUBTASK 26-18-02-720-001

(2) Disconnect the connectors from both ends of the sensors.

SUBTASK 26-18-02-860-016

(3) Connect the Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443, or LCR meter (Inductance, Capacitance, Resistance), COM-1741, to the center of the conductor and to the sheath of the sensor tube.

SUBTASK 26-18-02-720-002

(4) Set the Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443, or LCR meter (Inductance, Capacitance, Resistance), COM-1741, to R-mode (AC Resistance), parallel mode, and 1KHz.

NOTE: Do not use a Digital Multi Meter (DMM) for center conductor to outer case resistance checks on individual sensors. The data below is for LCR only.

NOTE: For individual sensing elements, continuity (center conductor to center conductor) can be measured using a standard digital voltmeter, or using DCR mode on Inductance, Capacitance, Resistance and Impedance (LCRZ) meter, COM-20443, or LCR meter (Inductance, Capacitance, Resistance), COM-1741.

(a) Measure the resistance from the center of the conductor to the case ground (outer side of the element) (Table 502).

NOTE: Conductance equals 1/Resistance.

Table 502/26-18-02-993-804 Sensing Element Conductance and Resistance Center Conductor to Case Test Data

SENSOR	SENSOR PART NO.	MAXIMUM CONDUCTANCE, MICRO-SIEMENS*[1]	MINIMUM RESISTANCE, (MEGOHMS) *[1]	MAXIMUM CONTINUITY, (MILLIOHMS)
M371	35574-4-255	0.74	1.35	563.00
M370	35574-4-255	0.74	1.35	563.00
M356	35594-4-255	0.94	1.06	703.00
M355	35594-4-255	0.94	1.06	703.00
M348	35658-4-255	1.58	0.63	1151.00
M347	35646-2-255	1.46	0.68	1067.00
M276	35555-4-255	0.55	1.82	423.00
M275	35575-2-255	0.75	1.33	570.00
	35599-2-255	0.99	1.01	738.00
M273	35555-4-255	0.55	1.82	423.00
M272	35626-4-255	1.26	0.79	927.00

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Table 502/26-18-02-993-804 Sensing Element Conductance and Resistance Center Conductor to Case Test Data (Continued)

SENSOR	SENSOR PART NO.	MAXIMUM CONDUCTANCE, MICRO-SIEMENS*[1]	MINIMUM RESISTANCE, (MEGOHMS) *[1]	MAXIMUM CONTINUITY, (MILLIOHMS)
M270	04-90010-110D	1.10	0.91	660.00
	35610-4-400	1.10	0.909	815.00
	35614-4-400	1.14	0.877	843.00
M269	35599-2-255	0.99	1.01	738.00
M268	35599-2-255	0.99	1.01	738.00
M1912	35712-80 (255)	1.20	0.833	885.00
M1911	35712-80 (255)	1.20	0.833	885.00
M1910	35712-80 (255)	1.20	0.833	885.00
M1909	35712-80 (255)	1.20	0.833	885.00
M1764	35712-75 (310)	2.24	0.417	820.00
	35712-79 (310)	0.60	1.67	820.00
M1763	35712-79 (310)	0.60	1.67	820.00
M1762	35712-75 (310)	2.24	0.417	820.00
	35712-79 (310)	0.60	1.67	820.00
M1761	35712-79 (310)	0.60	1.67	820.00
M1147	35675-2-255	1.75	0.571	1270.00
M3057	35712-79 (310)	0.60	1.67	820.00
M3058	35712-79 (310)	0.60	1.67	820.00
M3067	1060011-031A	1.20	0.833	885.00
M3068	04-90010-110D	1.10	0.91	660.00
	35610-4-400	1.10	0.91	815.00
	35614-4-400	1.14	0.88	843.00

*[1] from center conductor to ground

SUBTASK 26-18-02-720-003

(5) Connect the connectors to the sensor elements.

SUBTASK 26-18-02-720-004

(6) Tighten the jamnuts to 50 in-lb (5.6 N·m) - 60 in-lb (6.8 N·m).

SUBTASK 26-18-02-720-005

(7) Attach the lockwires to the sensor element connectors (TASK 20-10-44-400-801).

SUBTASK 26-18-02-720-006

(8) Repeat the sensing element continuity and sensing element pin to case test for each remaining loop.

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G. Zone Resistance Test (CODM BITE)

SUBTASK 26-18-02-740-004

(1) Push and release the LOCAL TEST button on the front panel of the Compartment Overheat Detection Module (CODM) (Figure 501).

(a) The front panel of the CODM displays CONFIG MAX1 followed by L ENGINE STR.

SUBTASK 26-18-02-740-005

(2) Continue to push and release LOCAL TEST.

NOTE: This will cycle through the different zones followed by LOOP ___ that shows the resistance value in the loop or 2GND_____ that shows the resistance value to ground.

NOTE: If the loops are functioning properly all zones will have a 2GND value of 5000.

SUBTASK 26-18-02-740-003

(3) Make sure that the resistance value agree with the Table 503 below:

Table 503/26-18-02-993-805 Zone Resistance Test Data

KIDDE ALARM/FAULT CODES	RESISTANCE VALUE or FAULT MESSAGE	SENSORS
L ENGINE STR L ENGINE STR OPEN L ENGINE STR OVHT L ENGINE STR SHRT	< 15 Ohms OPEN LOW 0	M1761, M1762, M3057
L WING LE L WING LE OPEN L WING LE OVHT L WING LE SHRT	< 15 Ohms OPEN LOW 0	M268, M370
L PACK BAY L PACK BAY OPEN L PACK BAY OVHT L PACK BAY SHRT	< 15 Ohms OPEN LOW 0	M1911, M1909, M355
NITROGEN GEN NITROGEN GEN OPEN NITROGEN GEN OVHT NITROGEN GEN SHRT	< 15 Ohms OPEN LOW 0	M3067
KEELBEAM KEELBEAM OPEN KEELBEAM OVHT KEELBEAM SHRT	< 25 Ohms OPEN LOW 0	M272, M273
AFT CARGO APU AFT CARGO APU OPEN AFT CARGO APU OVHT AFT CARGO APU SHRT	< 25 Ohms OPEN LOW 0	M275, M347, M348, M1147
AFT BODY AFT BODY OPEN AFT BODY OVHT AFT BODY SHRT	< 25 Ohms OPEN LOW 0	M276

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Table 503/26-18-02-993-805 Zone Resistance Test Data (Continued)

KIDDE ALARM/FAULT CODES	RESISTANCE VALUE or FAULT MESSAGE	SENSORS
MWW LOOP A MWW LOOP A OPEN MWW LOOP A FIRE MWW LOOP A SHRT	< 15 Ohms OPEN LOW 0	M270
MWW LOOP B MWW LOOP B OPEN MWW LOOP B FIRE MWW LOOP B SHRT	< 15 Ohms OPEN LOW 0	M3068
R ENGINE STR R ENGINE STR OPEN R ENGINE STR OVHT R ENGINE STR SHRT	< 15 Ohms OPEN LOW 0	M1763, M1764, M3058
R WING LE R WING LE OPEN R WING LE OVHT R WING LE SHRT	< 15 Ohms OPEN LOW 0	M269, M371
R PACK BAY R PACK BAY OPEN R PACK BAY OVHT R PACK BAY SHRT	< 15 Ohms OPEN LOW 0	M356, M1910, M1912

H. Put the Airplane Back to Its Usual Condition

NOTE: Do the following only if a resistance measurement was performed.

SUBTASK 26-18-02-720-007

(1) Remove the wire and fittings from the loop connectors.

SUBTASK 26-18-02-720-008

(2) Tighten the jamnuts to 50 in-lb (5.6 N·m) - 60 in-lb (6.8 N·m).

SUBTASK 26-18-02-720-009

(3) Attach the lockwire to the element connectors (TASK 20-10-44-400-801).

SUBTASK 26-18-02-860-008

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

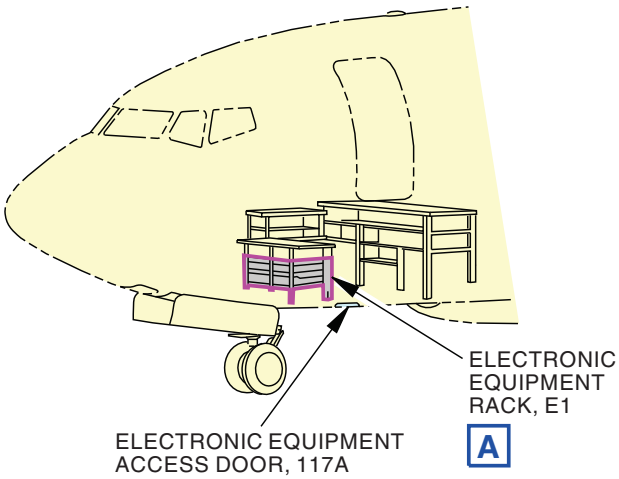
Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT

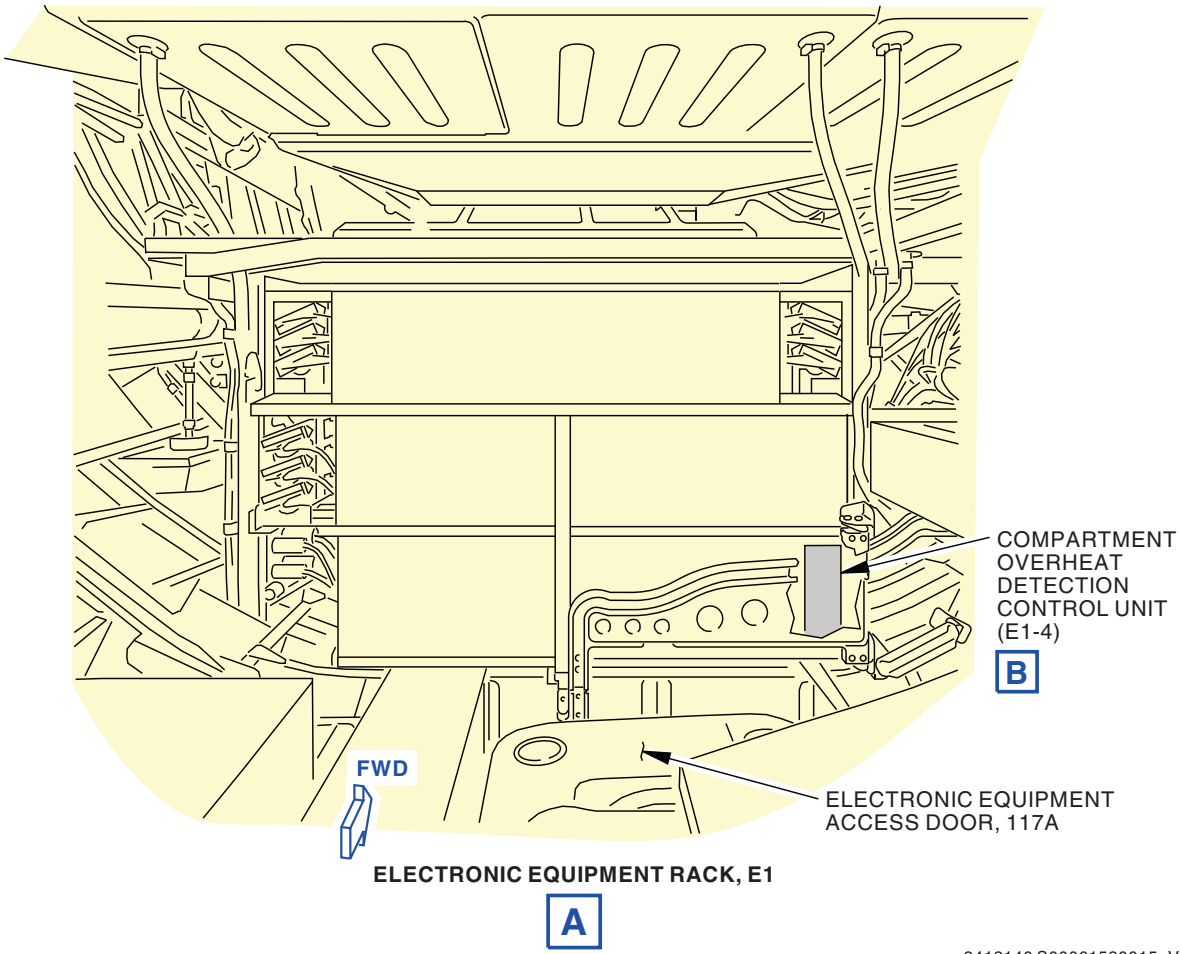
SUBTASK 26-18-02-710-001

(5) Do this task: Wheel Well and Lower Aft Body Overheat Detection System - Operational Test, TASK 26-18-00-710-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

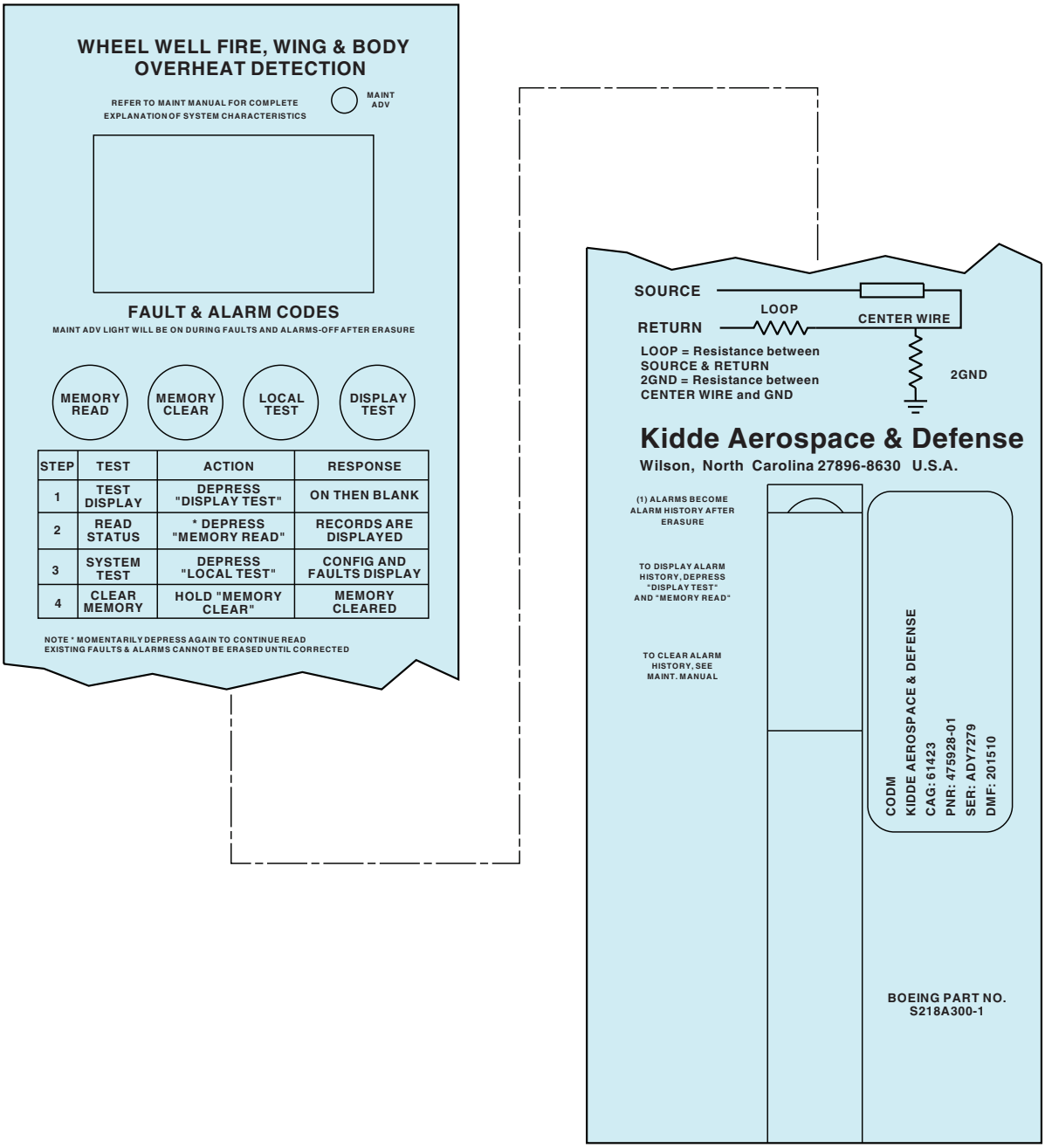




2412140 S00061530015_V2

Compartment Overheat Detection Control Unit M237
Figure 501/26-18-02-990-808 (Sheet 1 of 2)

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COMPARTMENT OVERHEAT DETECTION CONTROL UNIT (E1-4)

B

2412141 S00061530016_V1

Compartment Overheat Detection Control Unit M237
Figure 501/26-18-02-990-808 (Sheet 2 of 2)

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E/E COOLING SMOKE DETECTION - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) E/E Cooling Smoke Detection - Operational Test

(2) E/E Cooling Smoke Detector - Operational Test

TASK 26-19-00-710-801

2. E/E Cooling Smoke Detection - Operational Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The operational test makes sure that the E/E Cooling Smoke Detection system operates correctly. The operational test uses only equipment installed on the airplane.

B. References

Reference	Title
21-25-02 P/B 401	RECIRCULATION FAN - REMOVAL/INSTALLATION
21-27-02 P/B 401	EQUIPMENT COOLING SUPPLY FAN - REMOVAL/INSTALLATION
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. E/E Cooling Smoke Detection Operational Test

SUBTASK 26-19-00-710-001

(1) Do these steps to do the EQUIP COOL SMOKE DETECTOR SUPPLY TEST:

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	13	C04013	EQPT COOL SMOKE DET EXHAUST

(b) Make sure that the S870 Supply Equipment Cooling Switch is set to NORMAL position.

(c) Make sure that these switches are in the correct positions:

SWITCH	POSITION
L RECIRC FAN	AUTO
R RECIRC FAN	AUTO
L PACK	AUTO
R PACK	AUTO

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(d) Make sure that the airplane is in GROUND mode.

(e) Push and hold the S1204 Equip Cooling Smoke Test button for 2 to 3 seconds (Figure 501).

NOTE: Timing for subsequent verification steps start when the test button is pushed.

(f) Release the S1204 Equip Cooling Smoke Test button.

(g) Make sure that the L1293 Equip Cool Supl Smk Det Light, on the J23 panel in the E/E bay, comes on.

(h) Make sure that 2 seconds after the S1204 EQUIP COOL SMK DET TEST button is pushed, the L1554 EQUIP COOLING EQUIP SMOKE detection light, on the P5 Equipment Cooling panel, in the flight deck comes on (Figure 501).

(i) Make sure that 2 seconds after the S1204 Equip Cooling Smoke Test button is pushed, the M1322 Normal Supply Fan turns off (PAGEBLOCK 21-27-02/401).

(j) Make sure that 2 seconds after the S1204 Equip Cooling Smoke Test button is pushed, the M1440 L Cabin Air Recirc Fan shuts off (PAGEBLOCK 21-25-02/401).

(k) Make sure that 2 seconds after the S1204 Equip Cooling Smoke Test button is pushed, the M1441 R Cabin Air Recirc Fan shuts off (PAGEBLOCK 21-25-02/401).

(l) Make sure that approximately 302 seconds ± 30 seconds after the S1204 Equip Cooling Smoke Test button is pushed, the M1322 Normal Supply Fan turns back on.

(m) Push and hold the S1204 Equip Cooling Smoke Test button for 10 to 20 seconds immediately after the M1322 Normal Supply fan turns back on.

NOTE: This 10 second window after the supply fan turns back on is the system purge time, where recurrence of supply smoke will have no effect on the system state.

(n) Make sure that during the 10 second purge time window, the L1554 EQUIP COOL EQUIP SMOKE Det Light does not come on.

(o) Make sure that after the 10 second purge time window, the L1554 EQUIP COOL EQUIP SMOKE Det Light comes on.

(p) Release the S1204 Equip Cooling Smoke Test button after the L1554 EQUIP COOL EQUIP SMOKE Det Light comes on.

(q) Wait 330 seconds or longer before the next step.

1) Make sure that the L and R Cabin Recirc Fans turn back on.

(r) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	13	C04013	EQPT COOL SMOKE DET EXHAUST

SUBTASK 26-19-00-710-003

(2) Do these steps to do the EQUIP COOL SMOKE DET EXHAUST TEST:

WARNING

OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	4	C00700	HEATERS DRAIN MAST AIR

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	16	C01356	LANDING GEAR AIR/GND SYS 1

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
A	11	C04014	EQPT COOL SMOKE DET SUPPLY

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE AWAY FROM THE FLIGHT CONTROL SURFACES. THE HYDRAULIC PRESSURE THAT STAYS IN THE SYSTEM CAN CAUSE THE SLATS OR FLAPS TO MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(c) If trailing edge flaps are extended, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

(d) Make sure that these switches are in the correct positions:

SWITCH	POSITION
L RECIRC FAN	AUTO
R RECIRC FAN	AUTO
L PACK	AUTO
R PACK	AUTO

(e) Make sure that the airplane is in GROUND mode.

(f) Make sure that the LEFT FMCS BITE screen shows on the CDU-1 screen.

1) Push the INIT REF key on the CDU keyboard.
2) Push these Line Select Key (LSK):
   a) INDEX
   b) MAINT
   c) FMCS
   d) BITE.

(g) Make sure that the RIGHT FMCS BITE screen shows on the CDU-2 screen.

1) Push the INIT REF key on the CDU keyboard.
2) Push these LSK:
   a) INDEX
   b) MAINT
   c) FMCS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

d) BITE.

(h) Make sure that the FMC Transfer Switch is set to NORMAL position.

(i) Push LSK 4L (DISCRETES) on both CDUs.

NOTE: The status of LEFT and RIGHT discretes depend on cockpit and airplane conditions.

NOTE: If an analog discrete status is highlighted, a mismatch exists between the FMCs, and must be corrected before further testing can be accomplished. A momentary highlight after the change of a switch state is not considered a mismatch.

(j) Set the engine 1 BLEED switch or engine 2 BLEED switch to ON position.

(k) Make sure that on both CDUs that the LEFT ECS PACK H/L status shows LO, and the RIGHT ECS PACK H/L status shows LO.

(l) Push and hold the S1204 Equip Cooling Smoke Test button for 2 to 3 seconds (Figure 501).

NOTE: Timing for subsequent verification steps start when the test button is pushed.

(m) Release the S1204 Equip Cooling Smoke Test button.

1) Make sure that the L1294 Equip Cool Exhaust Smk Det Light comes on.

2) Make sure that after 2 seconds, the L1554 EQUIP COOL EQUIP SMOKE Det Light comes on (Figure 501).

3) Make sure that after 2 seconds, the M1322 Normal Supply Fan turns off (PAGEBLOCK 21-27-02/401).

4) Make sure that after 2 seconds, the M1440 L Cabin Air Recirc Fan shuts off (PAGEBLOCK 21-25-02/401).

5) Make sure that after 2 seconds, the M1441 R Cabin Air Recirc Fan shuts off (PAGEBLOCK 21-25-02/401).

6) Make sure that after 2 seconds, the Master Caution operates.

7) Make sure that after 2 seconds, the OVERHEAD light on the right system annunciator panel comes on.

(n) Make sure that on the two CDUs the LEFT ECS PACK H/L and the RIGHT ECS PACK H/L condition show HI.

(o) After you release the S1204 test switch, for a maximum of 30 seconds, do this step:

1) Hold the P8-75 Cargo Fire Control module TEST switch while the L1554 EQUIP COOL EQUIP SMOKE Det Light is on (Figure 501).

(p) Make sure that the L1554 EQUIP COOL EQUIP SMOKE Det Light is suppressed and goes off.

(q) Release the P8-75 Cargo Fire Control module TEST switch.

(r) Make sure that after approximately 302 seconds ± 30 seconds, the M1322 Normal Supply Fan turns back on.

(s) Make sure that the engine 1 BLEED switch and engine 2 BLEED switch are set to OFF position.

(t) Wait 330 seconds or longer before the next step.

1) Make sure that the L and R Cabin Recirc Fans turn back on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(u) If trailing edge flaps are extended, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

(v) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	4	C00700	HEATERS DRAIN MAST AIR

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	16	C01356	LANDING GEAR AIR/GND SYS 1

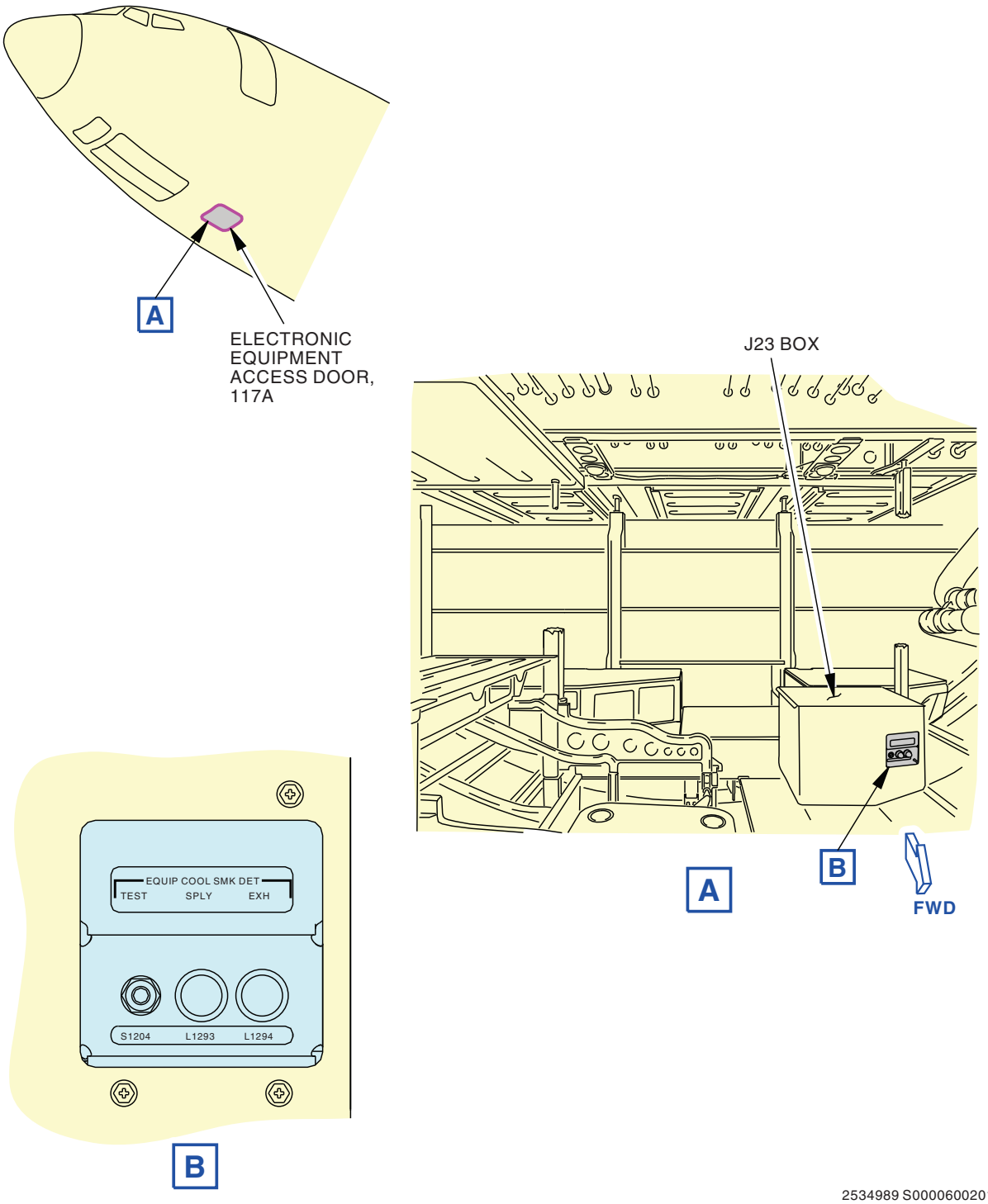
F/O Electrical System Panel, P6-5

Row	Col	Number	Name
A	11	C04014	EQPT COOL SMOKE DET SUPPLY

(w) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

——— END OF TASK ———

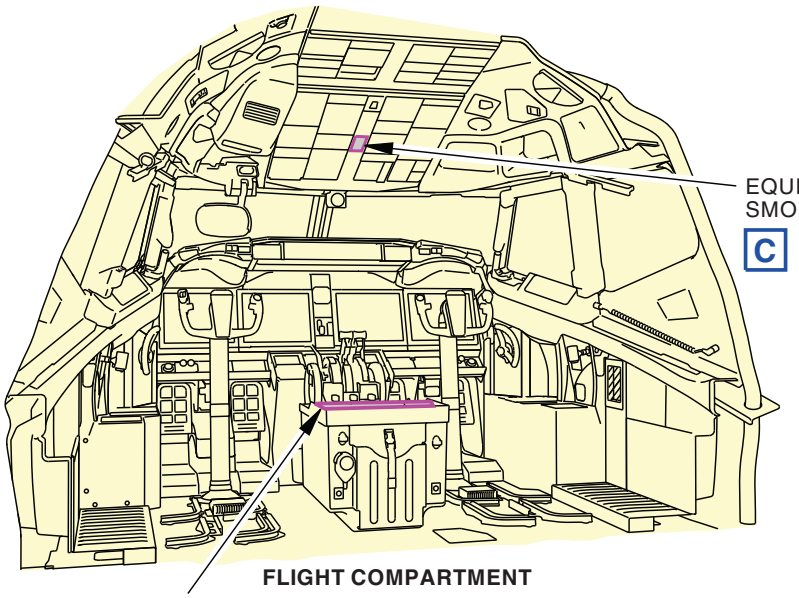
ECCN 9E991 BOEING PROPRIETARY - See title page for details

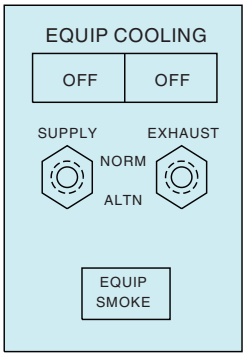


2534989 S0000600201_V1

E/E Cooling Smoke Detection - Operational Test
Figure 501/26-19-00-990-801 (Sheet 1 of 2)

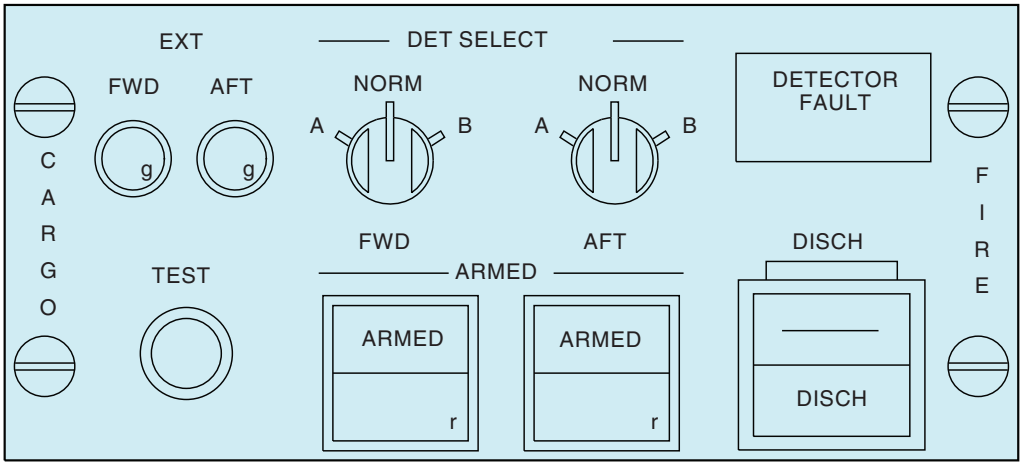
ECCN 9E991 BOEING PROPRIETARY - See title page for details





EQUIP COOLING EQUIP SMOKE DETECTION (P5)

C



CARGO FIRE CONTROL MODULE (P8-75)

D

2534992 S0000600203_V1

E/E Cooling Smoke Detection - Operational Test
Figure 501/26-19-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-19-00-710-802

3. E/E Cooling Smoke Detector - Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

B. Prepare for Operational Test

SUBTASK 26-19-00-860-002

(1) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	13	C04013	EQPT COOL SMOKE DET EXHAUST

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
A	11	C04014	EQPT COOL SMOKE DET SUPPLY

C. E/E Cooling Smoke Detector Operational Test

SUBTASK 26-19-00-710-002

(1) On the J23 panel in the E/E bay, do these steps to do the EQUIP COOL SMOKE DETECTOR SUPPLY TEST:

(a) Push then release the S1204 Equip Cooling Smoke Test button (Figure 501).

NOTE: If the Test button is held for more than 2 seconds, the cooling fans will go off. The cooling fans will come on again on after approximately 5 minutes.

(b) Make sure that the L1293 EQUIP COOL SPLY SMK DET light comes on.

(c) Make sure that the L1294 EQUIP COOL EXH SMK DET light comes on.

(d) Make sure that the L1293 EQUIP COOL SPLY SMK DET and L1294 EQUIP COOL EXH SMK DET lights go off immediately after you release the test button.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Equipment Supply Smoke Detector - Removal/Installation

1. General

A. This procedure has these tasks:

(1) Removal of the equipment supply smoke detector.

(2) Installation of the equipment supply smoke detector.

TASK 26-19-01-000-801

2. Equipment Supply Smoke Detector Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Equipment Supply Smoke Detector.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 26-19-01-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
A	11	C04014	EQPT COOL SMOKE DET SUPPLY

SUBTASK 26-19-01-010-001

(2) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Equipment Supply Smoke Detector Removal

SUBTASK 26-19-01-020-001

(1) Remove and discard the tiewrap from the tubing [2] and disconnect the tubing [2].

SUBTASK 26-19-01-020-005

(2) Install protective caps on the equipment supply smoke detector [1] tubing assemblies.

SUBTASK 26-19-01-020-002

(3) Disconnect the electrical connector [3].

SUBTASK 26-19-01-020-003

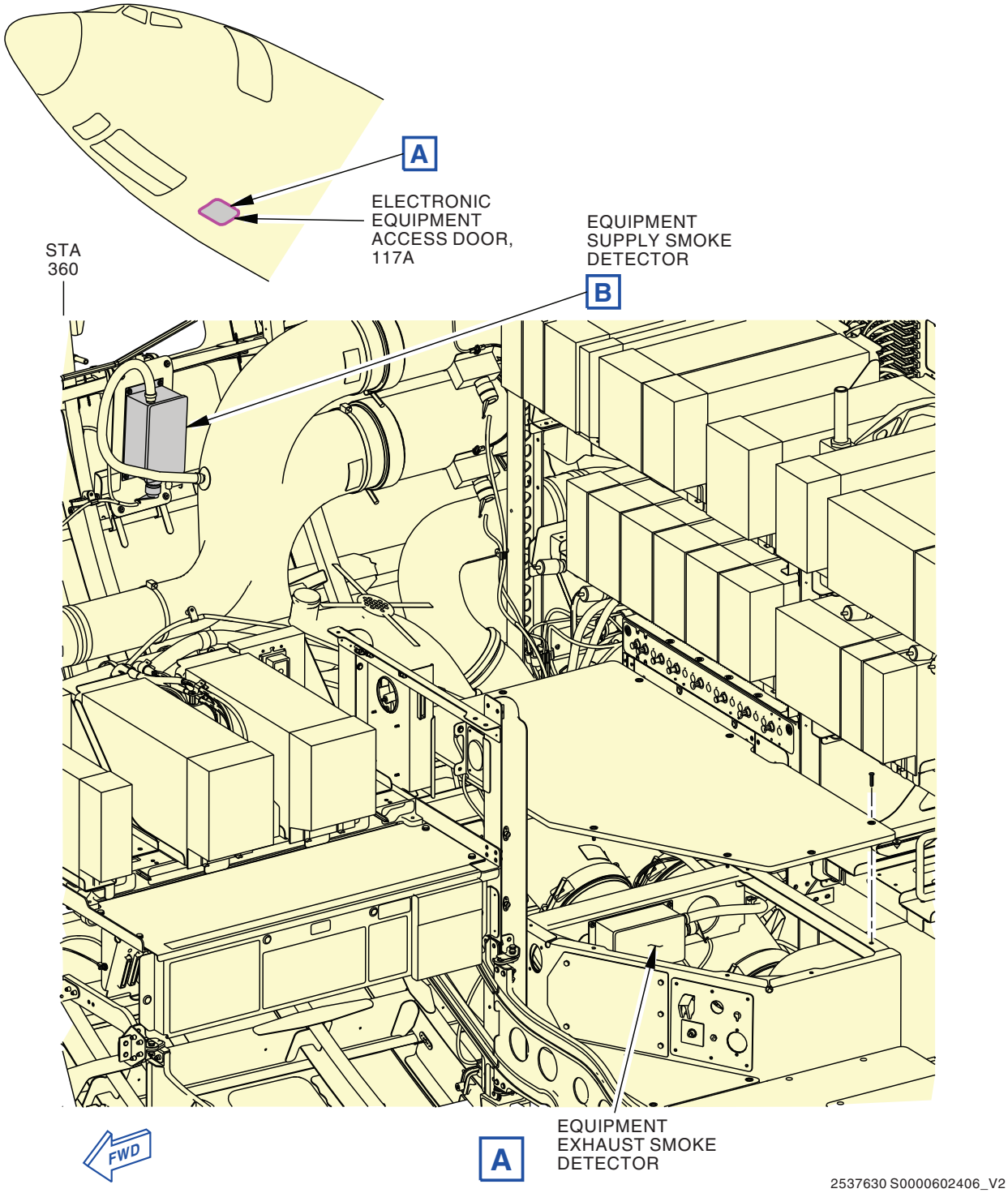
(4) Remove the four screws [4].

SUBTASK 26-19-01-020-004

(5) Remove the equipment supply smoke detector [1].

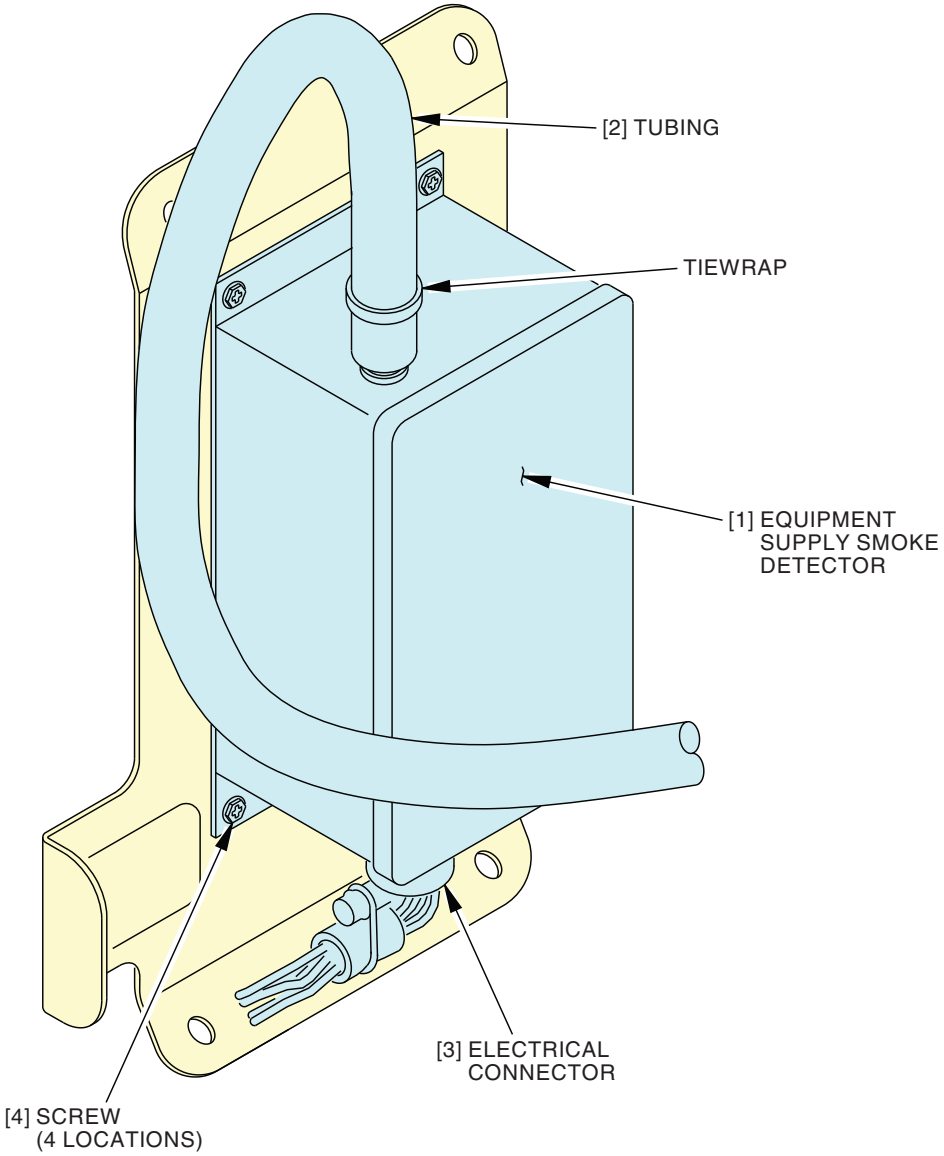
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Equipment Supply Smoke Detector Installation
Figure 401/26-19-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EQUIPMENT SUPPLY SMOKE DETECTOR

B

2537639 S0000602410_V1

Equipment Supply Smoke Detector Installation
Figure 401/26-19-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-19-01-400-801

3. Equipment Supply Smoke Detector Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Equipment Supply Smoke Detector.

B. References

Reference	Title
26-19-00-710-802	E/E Cooling Smoke Detector - Operational Test (P/B 501)

C. Consumable Materials

Reference	Description	Specification
G51414	Strap - Plastic, Adjustable, Self-locking, 7.30 Inches (185.42 mm) Long	BACS38K1

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Equipment supply smoke detector	26-19-01-01-005	PTF ALL

E. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. Prepare for the Installation

SUBTASK 26-19-01-860-002

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
A	11	C04014	EQPT COOL SMOKE DET SUPPLY

SUBTASK 26-19-01-010-002

(2) Make sure that this access panel is open:

Number	Name/Location
117A	Electronic Equipment Access Door

H. Equipment Supply Smoke Detector Installation

SUBTASK 26-19-01-420-001

(1) Install the equipment supply smoke detector [1].

SUBTASK 26-19-01-420-002

(2) Install the four screws [4].

SUBTASK 26-19-01-420-003

(3) Connect the electrical connector [3].

SUBTASK 26-19-01-420-004

(4) Remove the protective caps from the equipment supply smoke detector [1] tubing assemblies.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-19-01-420-005

(5) Connect the tubing [2] to the top tubing assembly and install new adjustable plastic strap, G51414, to the tubing [2].

SUBTASK 26-19-01-860-003

(6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
A	11	C04014	EQPT COOL SMOKE DET SUPPLY

SUBTASK 26-19-01-710-001

(7) Do this task: E/E Cooling Smoke Detector - Operational Test, TASK 26-19-00-710-802

I. Put the Airplane Back to Its Usual Condition

SUBTASK 26-19-01-410-001

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Equipment Exhaust Smoke Detector - Removal/Installation

1. General

A. This procedure has these tasks:

(1) Removal of the equipment exhaust smoke detector.

(2) Installation of the equipment exhaust smoke detector.

TASK 26-19-02-000-801

2. Equipment Exhaust Smoke Detector Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Equipment Exhaust Smoke Detector.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 26-19-02-860-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	13	C04013	EQPT COOL SMOKE DET EXHAUST

SUBTASK 26-19-02-010-001

(2) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Equipment Exhaust Smoke Detector Removal

SUBTASK 26-19-02-010-003

(1) Remove the eight screws [5] and the panel [6] to access the equipment exhaust smoke detector [1]

SUBTASK 26-19-02-020-001

(2) Remove and discard the tiewrap from the two tubings [2] and disconnect the two tubings [2].

SUBTASK 26-19-02-020-005

(3) Install protective caps on the equipment exhaust smoke detector [1] tubing assemblies.

SUBTASK 26-19-02-020-002

(4) Disconnect the electrical connector [3].

SUBTASK 26-19-02-020-003

(5) Remove the four screws [4].

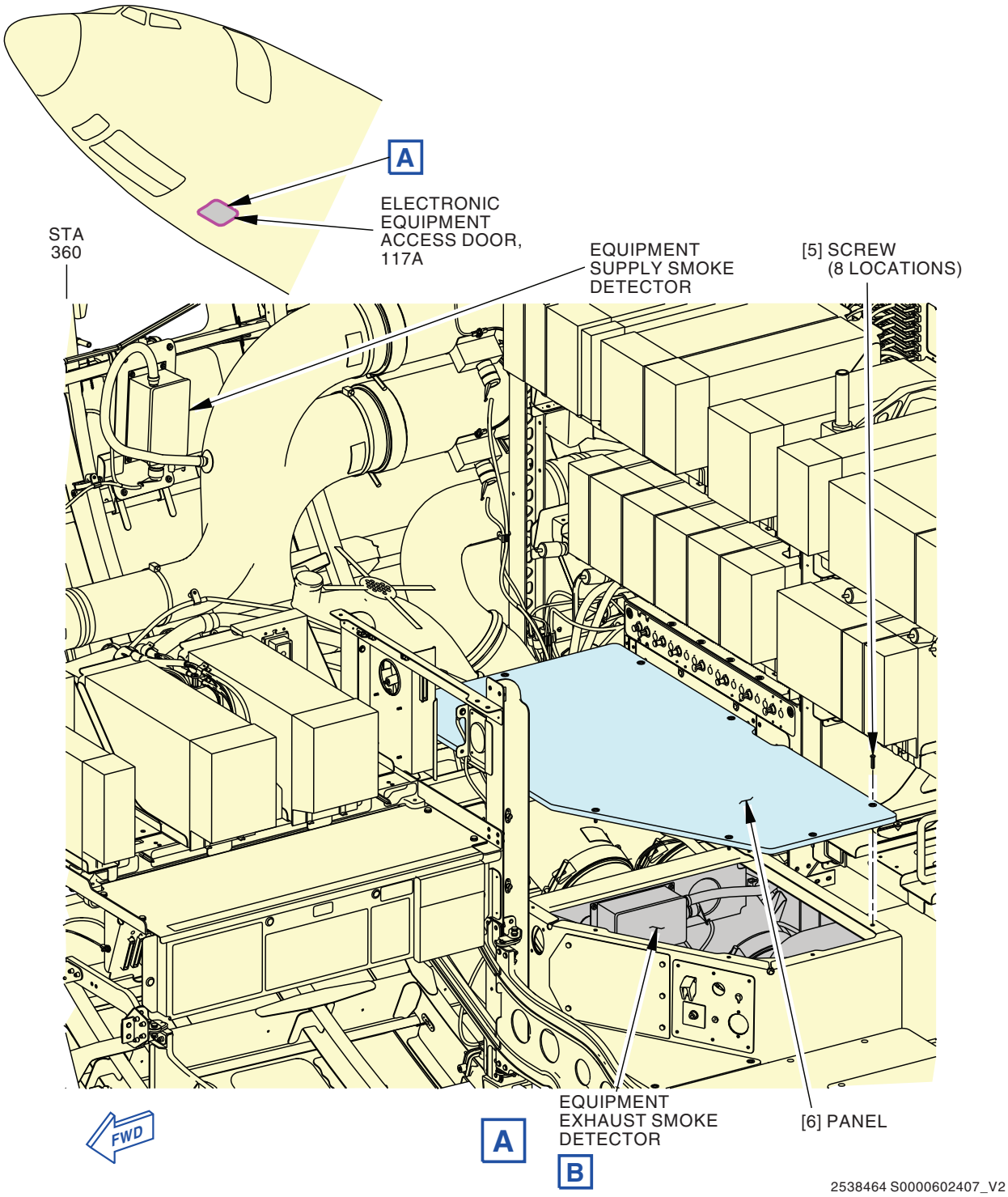
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-19-02-020-004

(6) Remove the equipment exhaust smoke detector [1].

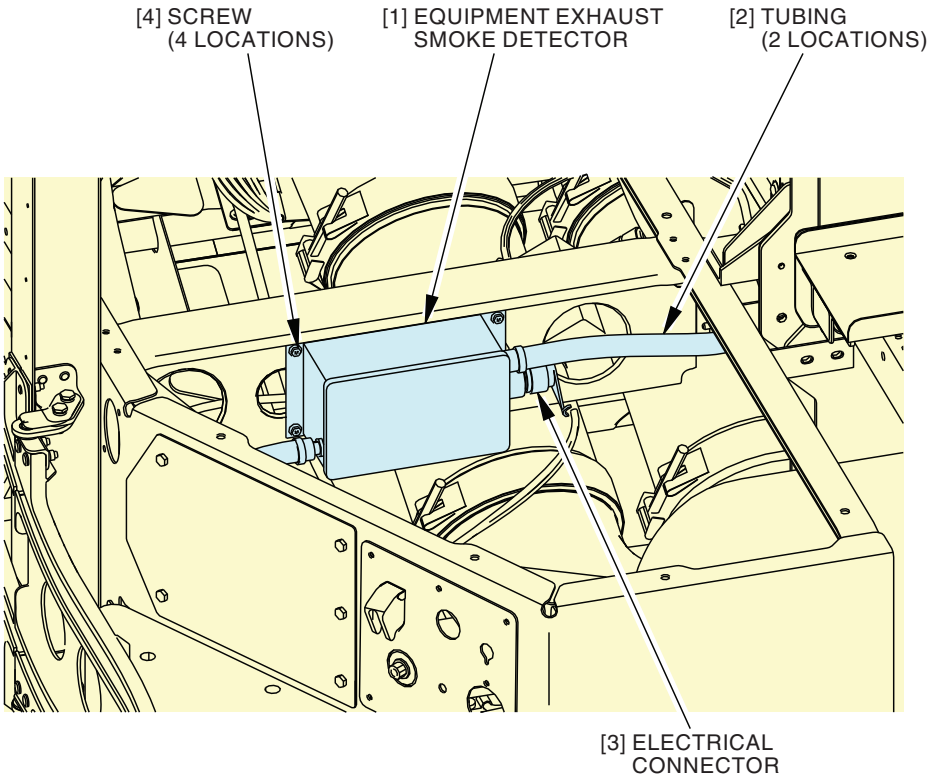
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Equipment Exhaust Smoke Detector Installation
Figure 401/26-19-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EQUIPMENT EXHAUST SMOKE DETECTOR

B

2538480 S0000602453_V2

Equipment Exhaust Smoke Detector Installation
Figure 401/26-19-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-19-02-400-801

3. Equipment Exhaust Smoke Detector Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Equipment Exhaust Smoke Detector.

B. References

Reference	Title
26-19-00-710-802	E/E Cooling Smoke Detector - Operational Test (P/B 501)

C. Consumable Materials

Reference	Description	Specification
G51414	Strap - Plastic, Adjustable, Self-locking, 7.30 Inches (185.42 mm) Long	BACS38K1

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Equipment exhaust smoke detector	26-19-02-01-005	PTF ALL

E. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. Prepare for the Installation

SUBTASK 26-19-02-860-002

(1) Make sure that this circuit breaker is open and has safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	13	C04013	EQPT COOL SMOKE DET EXHAUST

SUBTASK 26-19-02-010-002

(2) Make sure that this access panel is open:

Number	Name/Location
117A	Electronic Equipment Access Door

H. Equipment Exhaust Smoke Detector Installation

SUBTASK 26-19-02-420-001

(1) Install the equipment exhaust smoke detector [1].

SUBTASK 26-19-02-420-002

(2) Install the four screws [4].

SUBTASK 26-19-02-410-001

(3) Connect the electrical connector [3].

SUBTASK 26-19-02-410-002

(4) Remove the protective caps from the equipment exhaust smoke detector [1] tubing assemblies.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-19-02-420-003

(5) Connect the two tubings [2] and install new adjustable plastic straps, G51414.

SUBTASK 26-19-02-860-003

(6) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	13	C04013	EQPT COOL SMOKE DET EXHAUST

SUBTASK 26-19-02-710-001

(7) Do this task: E/E Cooling Smoke Detector - Operational Test, TASK 26-19-00-710-802.

SUBTASK 26-19-02-410-004

(8) Install the panel [6] and the eight screws [5].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 26-19-02-410-003

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FIRE EXTINGUISHING BOTTLE PRESSURE GAUGE - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) A check of the engine fire extinguishing bottle

(2) A check of the Auxiliary Power Unit (APU) fire extinguishing bottle.

TASK 26-20-00-210-801

2. Engine Fire Extinguishing Bottle Check

(Figure 601 and Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
26-21-01 P/B 401	ENGINE FIRE EXTINGUISHING BOTTLE - REMOVAL/INSTALLATION

B. Tools/Equipment

Reference	Description
STD-1336	Thermometer - Digital, 0-150 +/- 2 Degrees F, with a Probe, Thermocouple or Equivalent

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
311	Area Aft of Pressure Bulkhead - Left

D. Fire Extinguishing Bottle Check

NOTE: This procedure is for fire extinguishing bottles with pressure gauges. No action is required if a pressure gauge is not installed.

SUBTASK 26-20-00-860-001

(1) Allow the fire bottle to reach ambient temperature.

NOTE: It may take several hours with the airplanes on the ground for the halon in the bottles to reach ambient temperature.

SUBTASK 26-20-00-010-001

(2) Get access to the engine fire bottles that are located on the aft bulkhead in the left main landing gear wheel well.

SUBTASK 26-20-00-970-001

CAUTION

KEEP MERCURY THERMOMETERS AWAY FROM THE AIRPLANE. IF THE THERMOMETER BREAKS, MERCURY CAN CAUSE DAMAGE TO THE AIRPLANE COMPONENTS.

(3) Use a 0-150 +/- 2 degrees F digital thermometer, STD-1336, to get the ambient air temperature by the fire bottle.

SUBTASK 26-20-00-210-001

(4) Find the acceptable pressure range for the current temperature in the temperature curve (Figure 602).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-20-00-210-002

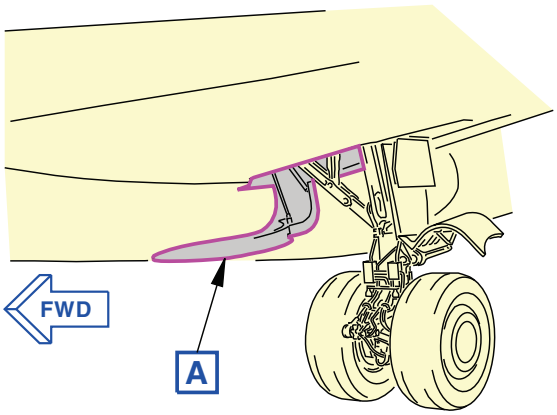
(5) Read the pressure gauge on the fire bottles.

(a) Make sure that the pressure on the gauge falls within the pressure limits.

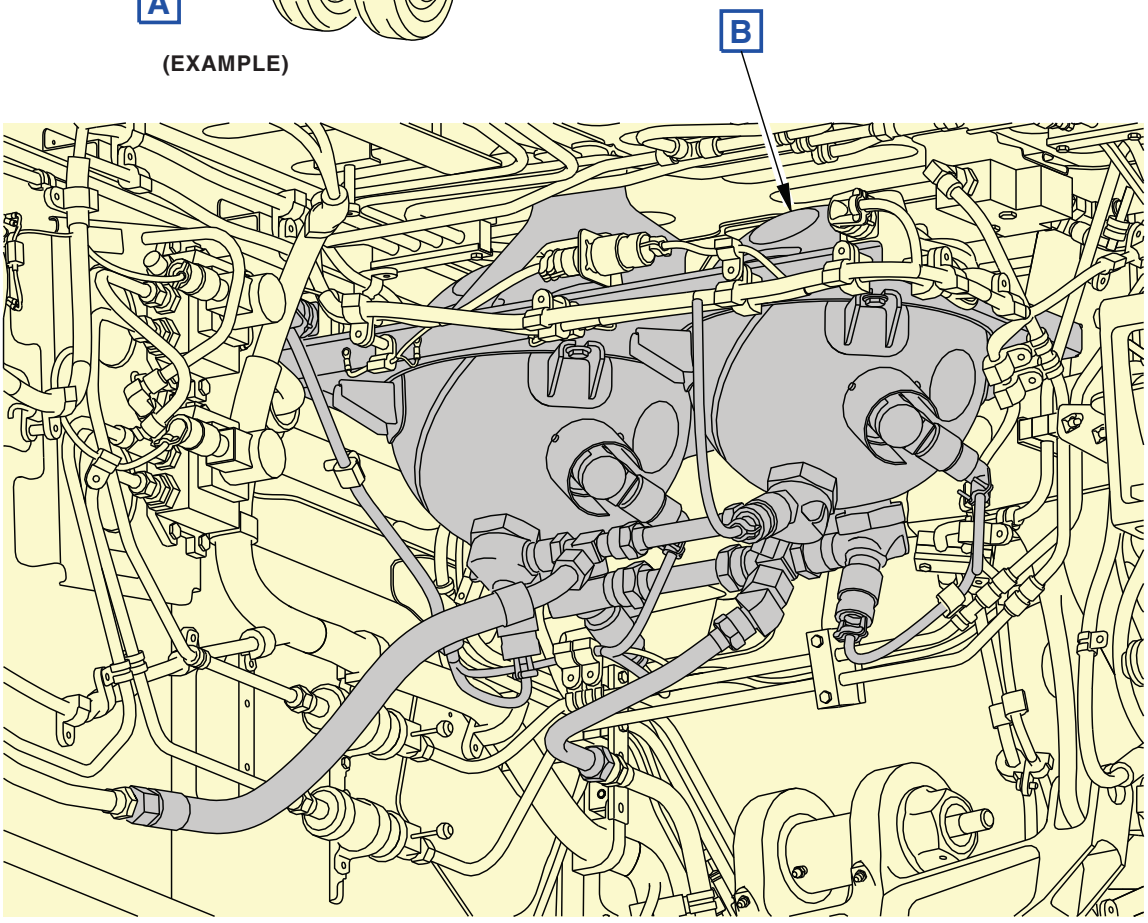
1) If the pressure on the gauge is not within the limits shown, replace the fire bottle with a serviceable fire bottle (PAGEBLOCK 26-21-01/401).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)



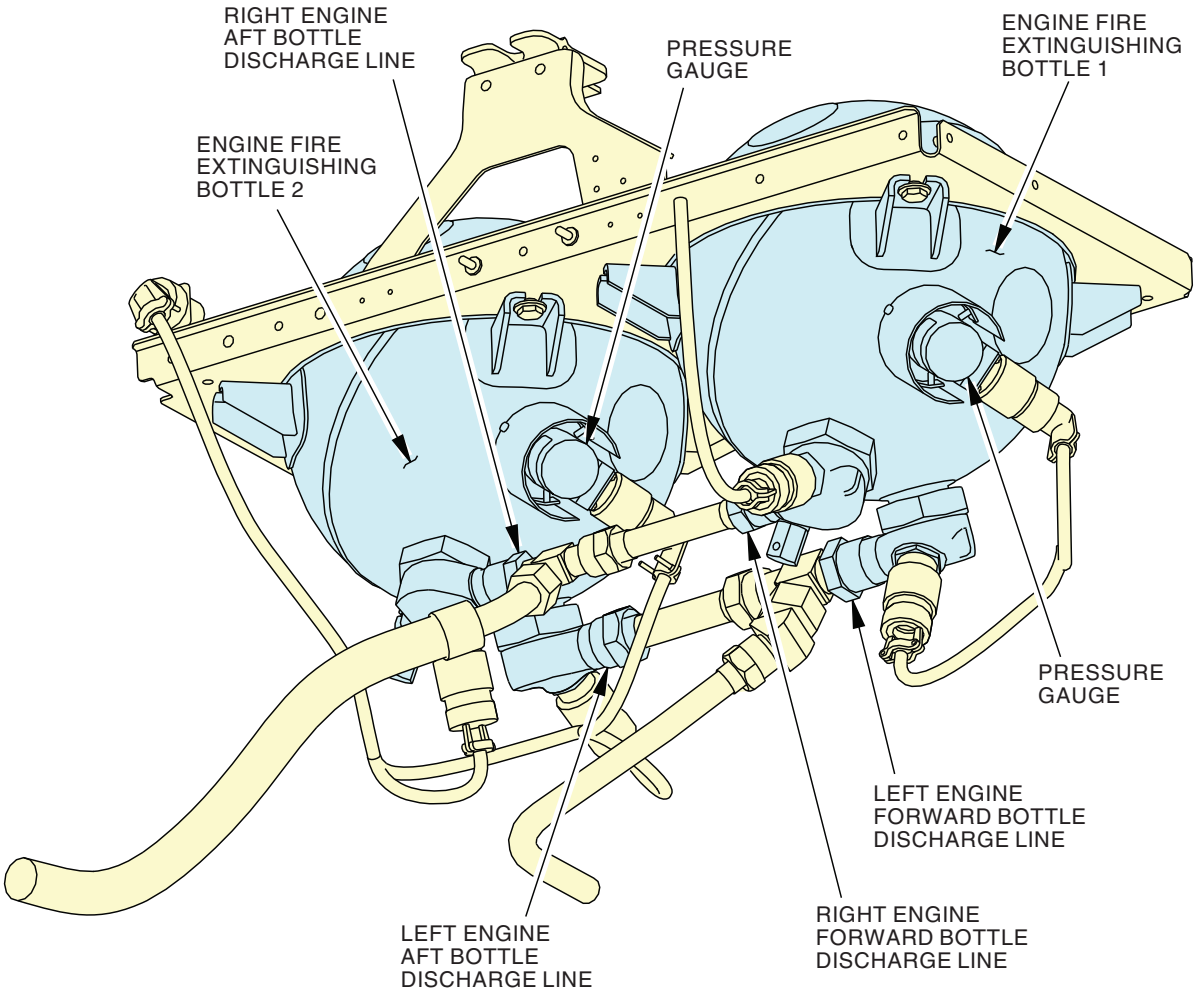
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)
(EXAMPLE)



2412155 S00061530041_V3

Engine Fire Extinguishing Bottle Inspection
Figure 601/26-20-00-990-801 (Sheet 1 of 2)

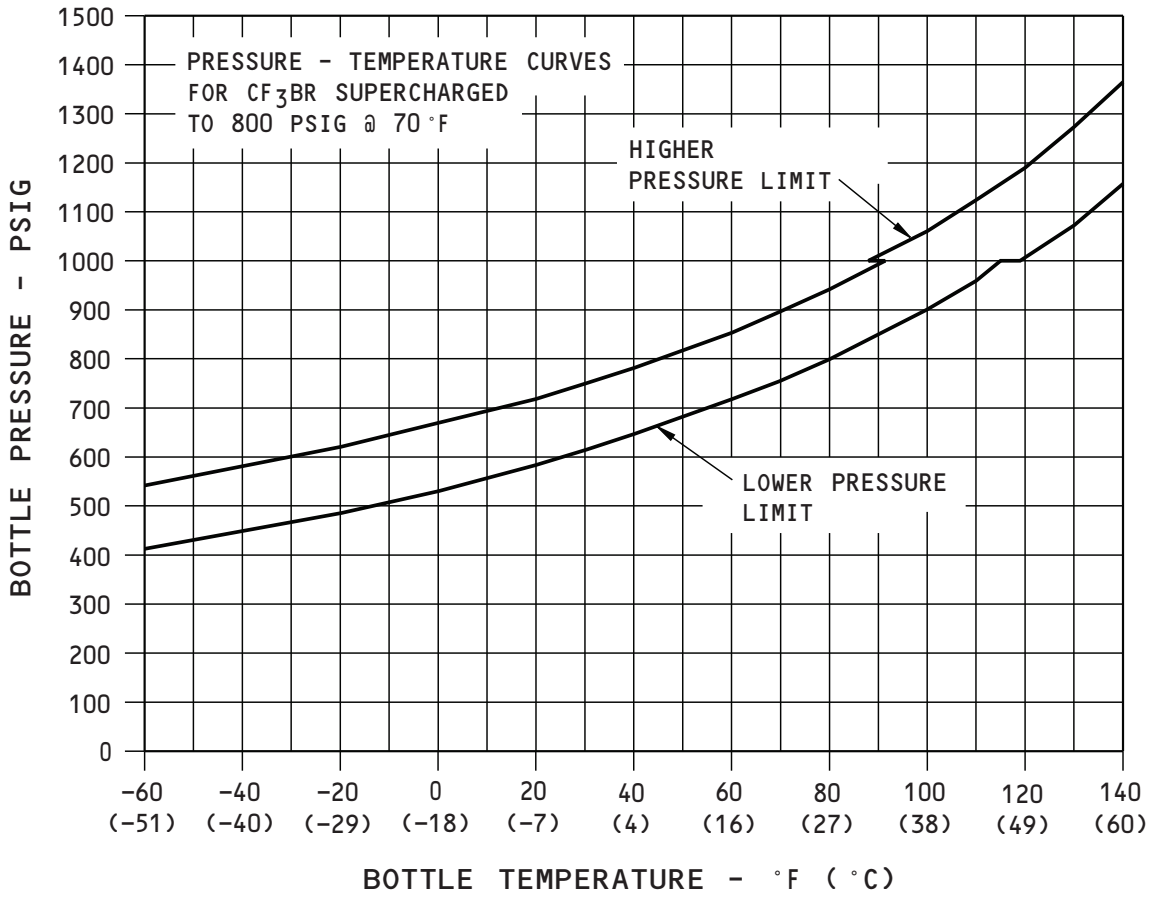
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412156 S00061530042_V2

Engine Fire Extinguishing Bottle Inspection
Figure 601/26-20-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412157 S00061530043_V1

Fire Extinguisher Bottle Pressure - Temperature Curve
Figure 602/26-20-00-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-20-00-210-802

3. APU Fire Extinguishing Bottle Check

(Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is applicable to fire extinguishing bottles with pressure gauges. This procedure is not required if a pressure gauge is not installed.

B. References

Reference	Title
26-22-01 P/B 401	APU FIRE EXTINGUISHING BOTTLE - REMOVAL/INSTALLATION

C. Tools/Equipment

Reference	Description
STD-1336	Thermometer - Digital, 0-150 +/- 2 Degrees F, with a Probe, Thermocouple or Equivalent

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
311	Area Aft of Pressure Bulkhead - Left

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Fire Extinguishing Bottle Check

SUBTASK 26-20-00-860-002

(1) Allow the fire bottle to reach ambient temperature.

NOTE: It will take several hours with the airplane on the ground for the halon in the bottles to reach ambient temperature.

SUBTASK 26-20-00-010-002

(2) To access the APU fire bottle, do this step:

Open this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 26-20-00-970-002

CAUTION

KEEP MERCURY THERMOMETERS AWAY FROM THE AIRPLANE. IF THE THERMOMETER BREAKS, MERCURY CAN CAUSE DAMAGE TO THE AIRPLANE COMPONENTS.

(3) Use a 0-150 +/- 2 degrees F digital thermometer, STD-1336, to get the ambient air temperature by the fire bottle.

(a) Do not use the total temperature indication from the airplane for the ambient air temperature.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-20-00-210-003

(4) Find the acceptable pressure range for the current temperature in the temperature curve (Figure 604).

SUBTASK 26-20-00-210-004

(5) Do these steps:

(a) Read the pressure gauge on the fire bottle.

(b) Make sure that the pressure on the gauge falls within the pressure limits.

1) If the pressure on the gauge is not within the limits shown, then replace the fire bottle with a serviceable fire bottle (PAGEBLOCK 26-22-01/401).

G. Put the Airplane Back to Its Usual Condition

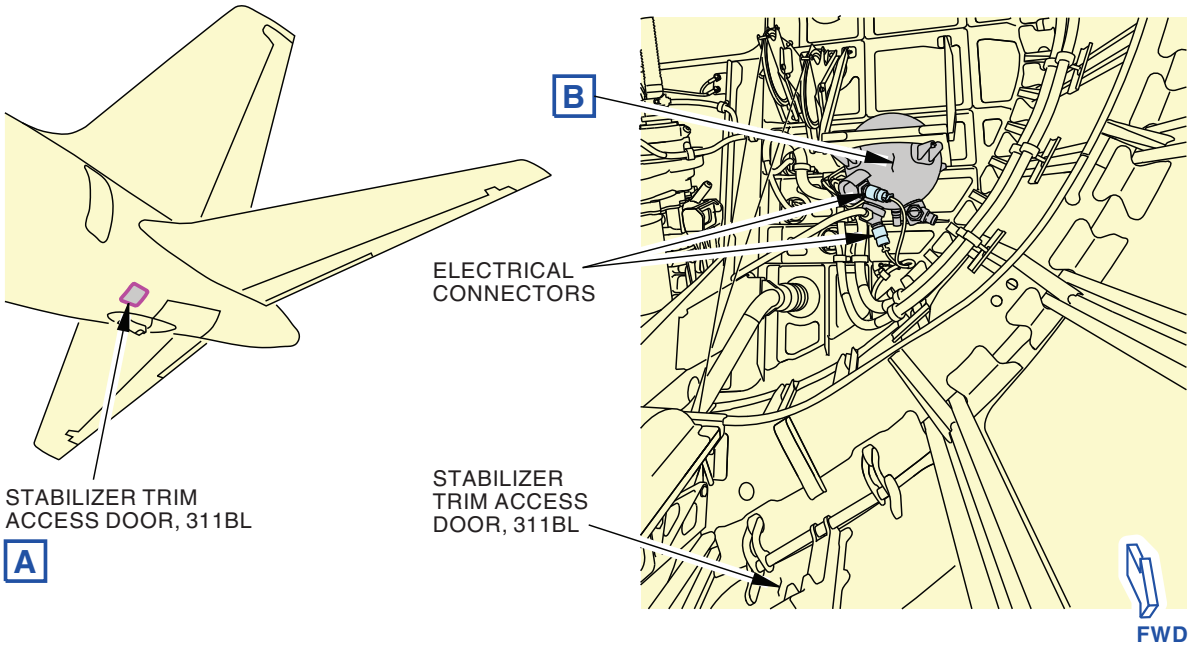
SUBTASK 26-20-00-410-001

(1) Close this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

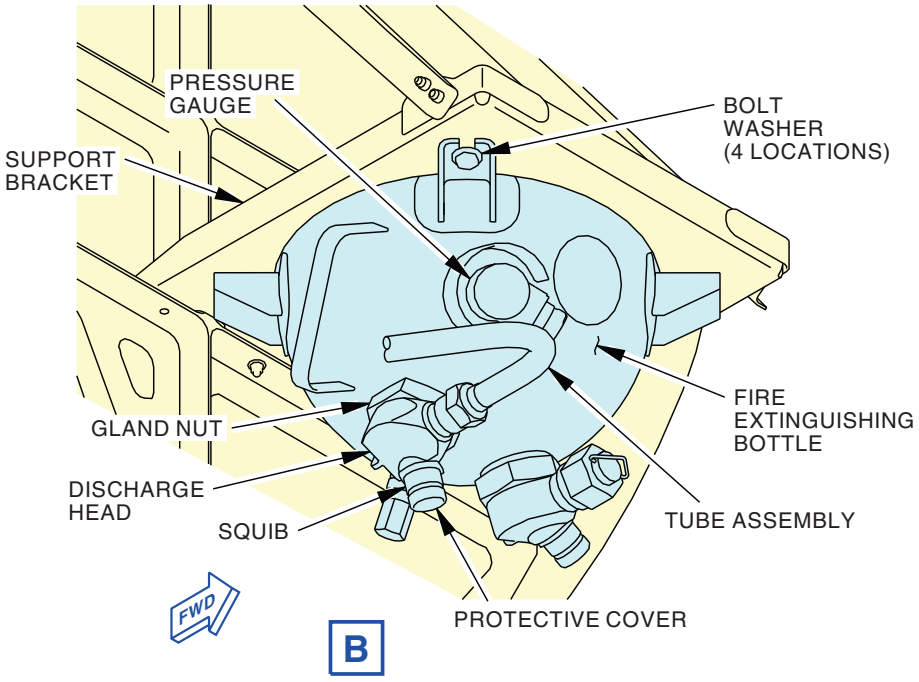
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACCESS DOOR, 311BL

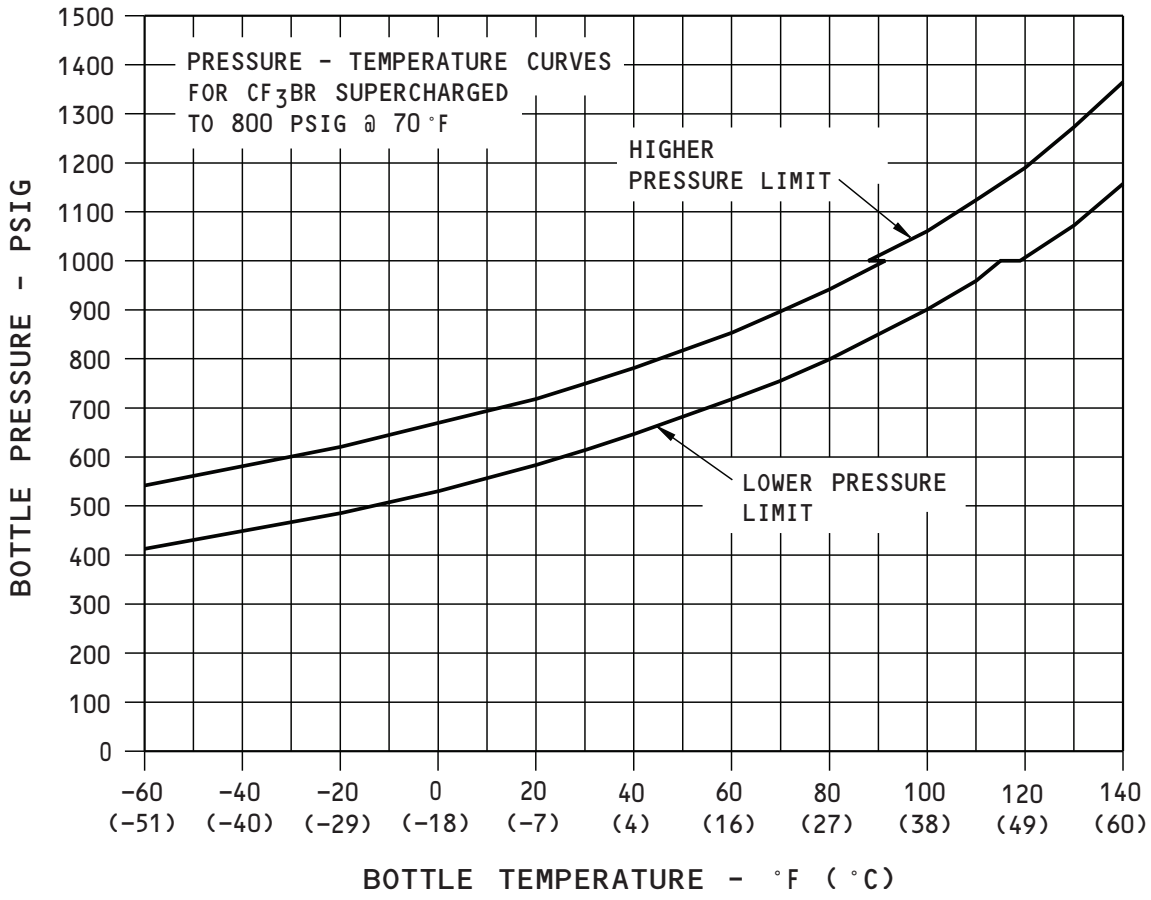
A



2412158 S00061530045_V1

APU Fire Extinguishing Bottle Inspection
Figure 603/26-20-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412157 S00061530043_V1

Fire Extinguisher Bottle Pressure - Temperature Curve
Figure 604/26-20-00-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE EXTINGUISHING - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Engine fire extinguishing system - deactivation.

(2) Engine fire extinguishing system - activation.

TASK 26-21-00-040-801

2. Engine Fire Extinguishing System - Deactivation

A. General

(1) This task will deactivate the engine fire extinguishing system.

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
315	APU Compartment - Left
316	APU Compartment - Right
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Procedure

SUBTASK 26-21-00-020-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-00-420-001

(2) Put DO NOT OPERATE tags, STD-858 on the ENGINE 1 and ENGINE 2 fire handles on the P8-1 panel.

E. Engine Fire Extinguishing System - Tryout

NOTE: This tryout is to make sure that the power to the engine fire extinguishing system is in a zero energy state.

SUBTASK 26-21-00-020-002

(1) Remove the DO NOT OPERATE tags, STD-858 from the ENGINE 1 and ENGINE 2 fire handles on the P8-1 panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-00-210-001

(2) Push the manual override button and pull the ENGINE 1 and ENGINE 2 fire extinguishing handles on the P8-1 panel.

SUBTASK 26-21-00-210-002

(3) Turn the ENGINE 1 handle counterclockwise.

(a) Make sure that the L BOTTLE DISCHARGED light on the P8-1 panel stays off.

SUBTASK 26-21-00-210-003

(4) Turn the ENGINE 1 handle back to its initial position.

SUBTASK 26-21-00-210-004

(5) Turn the ENGINE 1 handle clockwise.

(a) Make sure that the R BOTTLE DISCHARGED light on the P8-1 panel stays off.

SUBTASK 26-21-00-210-005

(6) Turn the ENGINE 1 handle back to its initial position.

SUBTASK 26-21-00-210-006

(7) Turn the ENGINE 2 handle counterclockwise.

(a) Make sure that the L BOTTLE DISCHARGED light on the P8-1 panel stays off.

SUBTASK 26-21-00-210-007

(8) Turn the ENGINE 2 handle back to its initial position.

SUBTASK 26-21-00-210-008

(9) Turn the ENGINE 2 handle clockwise.

(a) Make sure that the R BOTTLE DISCHARGED light on the P8-1 panel stays off.

SUBTASK 26-21-00-210-009

(10) Turn the ENGINE 2 handle back to its initial position.

SUBTASK 26-21-00-210-010

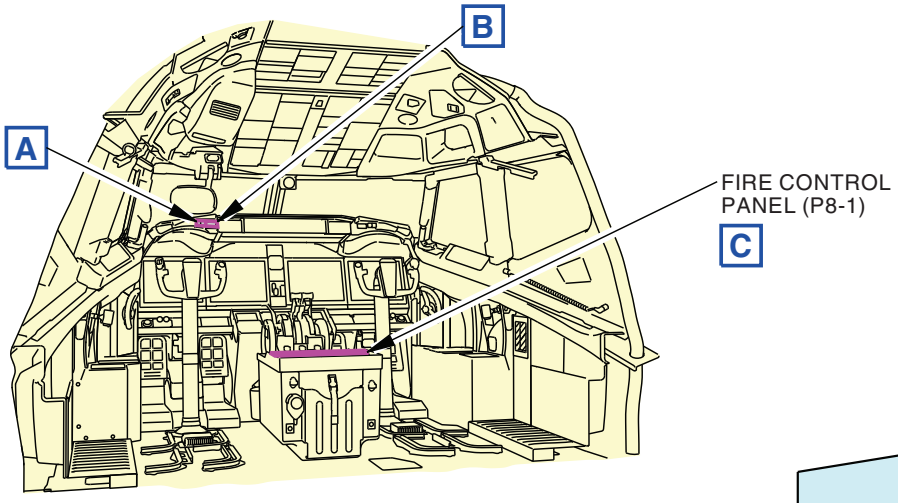
(11) Set the ENGINE 1 and ENGINE 2 fire handles to their usual positions.

SUBTASK 26-21-00-420-002

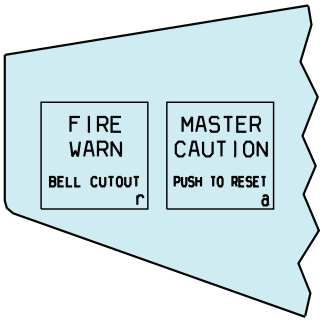
(12) Put DO NOT OPERATE tags, STD-858 on the ENGINE 1 and ENGINE 2 fire handles on the P8-1 panel.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

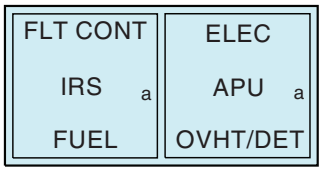


FLIGHT COMPARTMENT



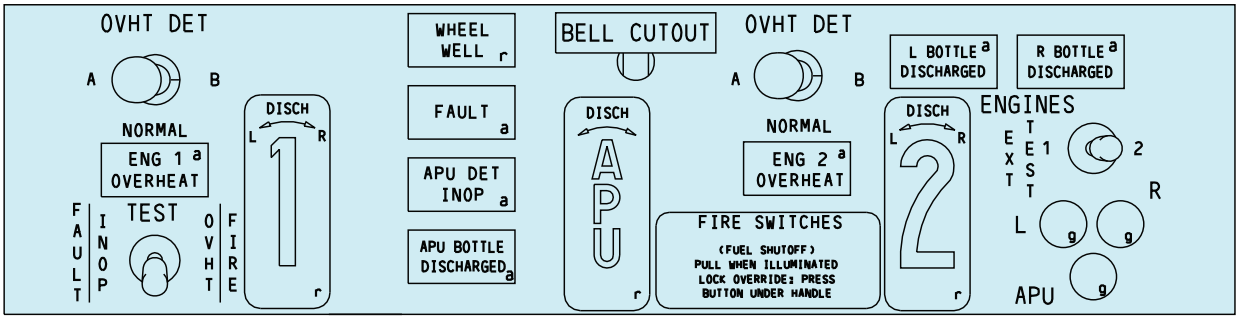
CAPTAIN'S FIRE WARNING SWITCH (P7)
(FIRST OFFICER'S IS OPPOSITE)

A



CAPTAIN'S WARNING
LIGHTS (P7)

B



FIRE CONTROL PANEL (P8-1)

C

2412159 S00061530050_V3

Engine Fire Extinguishing System
Figure 201/26-21-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-21-00-440-801

3. Engine Fire Extinguishing System - Activation

(Figure 201)

A. General

(1) This task will activate the engine fire extinguishing system.

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
315	APU Compartment - Left
316	APU Compartment - Right
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Procedure

SUBTASK 26-21-00-420-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-00-020-003

(2) Remove the DO NOT OPERATE tags, STD-858, from the ENGINE 1 and ENGINE 2 fire handles on the P8-1 panel.

SUBTASK 26-21-00-710-001

(3) Put and hold the EXT TEST switch on the P8-1 panel to 1.

(a) Make sure the L and R lights come on.

NOTE: The APU light will also come on.

SUBTASK 26-21-00-710-002

(4) Put and hold the EXT TEST switch on the P8-1 panel to 2.

(a) Make sure the L and R lights come on.

NOTE: The APU light will also come on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-00-710-003

(5) Release the EXT TEST switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE EXTINGUISHING - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) A test of the engine fire bottle squib circuit.

(2) A test of the engine fire bottle pressure switch circuit.

(3) An airflow test through the discharge tubes.

(4) A pressure test of the engine fire bottle discharge tubes.

(5) A shutdown test of the engine systems.

TASK 26-21-00-730-801

2. Engine Fire Extinguishing Bottle Squib Circuit Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-1663	Test Equipment - Fire Extinguishing System Part #: F80229-69 Supplier: 81205 Opt Part #: F80229-15 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
SPL-10039	Cable Assy - Fire Extinguishing System Test Equipment Part #: F80229-20 Supplier: 81205 Part #: F80229-48 Supplier: 81205 Part #: F80229-59 Supplier: 81205 Part #: F80229-61 Supplier: 81205
SPL-10040	Shorting Plug - Fire Extinguishing System Test Equipment Part #: F80229-52 Supplier: 81205
STD-1079	Resistor - 10K Ohm or Greater

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 26-21-00-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-00-020-004

(2) Disconnect the electrical connectors from the squibs.



WARNING DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(a) Before squibs are touched, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

(b) Disconnect D582 and D584 to test the engine 1 circuits.

(c) Disconnect D1322 and D1324 to test the engine 2 circuits.

SUBTASK 26-21-00-020-005



WARNING MAKE SURE THAT YOU PUT A SHORTING PLUG ON THE SQUIB. IF YOU DO NOT PUT A SHORTING PLUG ON THE SQUIB, THE FIRE EXTINGUISHING BOTTLE CAN RELEASE ITS CONTENTS SUDDENLY. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(3) Install a shorting plug, SPL-10040, on each squib.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-00-860-002

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-00-480-001

(5) Set the toggle switch on the Fire System Test Equipment, SPL-1663 (test box), to the ON position.

SUBTASK 26-21-00-760-001

(6) Measure the resistance (RL) of the test box with a digital/analog multimeter, COM-1793, connected to the voltmeter jacks.

(a) Write down the value of RL in the Table 502 or, in the Table 503 as applicable for the adapter cable.

SUBTASK 26-21-00-480-002

(7) Set the toggle switch on the Fire System Test Equipment, SPL-1663, to the OFF position.

SUBTASK 26-21-00-480-003

(8) Set the multimeter to read DC volts.

SUBTASK 26-21-00-480-004

(9) Connect the applicable cable assembly, SPL-10039, for the primary circuit (A) to the connector on the Fire System Test Equipment, SPL-1663.

NOTE: The primary circuit (A) is between pins 4 and 3 of each squib connector.

NOTE: Look at the placard on the test box lid to determine the applicable adapter.

NOTE: Each squib has two circuits. The primary circuit (A) is between pins 4 and 3; the alternate circuit (B) is between pins 6 and 7. Either circuit can fire the squib. There are different adapter cables in the test set to test each circuit.

E. Engine Fire Extinguishing Bottle Squib Circuit Test

SUBTASK 26-21-00-860-003

(1) Push the manual override button and pull the ENGINE 1 and ENGINE 2 fire handles on the P8-1 panel.

SUBTASK 26-21-00-480-005

(2) Connect the cable assembly, SPL-10039, for the primary circuit (A) to one of the squib connectors.

NOTE: These connectors are listed in the Table 501.

Table 501/26-21-00-993-801 Engine Fire Bottle Squib Circuit Test Data

CONNECTOR	SQUIB LOCATION	ENGINE	DIRECTION
D582	L (forward) bottle, squib 1A squib 1B	1	CCW
D1322	L (forward) bottle, squib 2A squib 2B	2	CCW
D584	R (aft) bottle, squib 1A squib 1B	1	CW
D1324	R (aft) bottle, squib 2A squib 2B	2	CW

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-00-860-004

(3) Turn the applicable fire handle and hold it in the direction specified in the Table 501.

(a) Make sure that the BOTTLE DISCHARGE light, on the Fire System Test Equipment, SPL-1663, comes on.

(b) Write down the voltage shown on the multimeter as (V1) in the Table 502 or, in the Table 503 as applicable for the adapter cable.

(c) Set the switch on the test box to the ON position.

(d) Write down the voltage shown on the multimeter as (V2) in the Table 502 or, in the Table 503 as applicable for the adapter cable.

SUBTASK 26-21-00-730-001

(4) Release the fire handle.

(a) Make sure that the handle moves toward the center 10 degrees.

(b) Make sure that the multimeter shows 0 (± 2) volts.

SUBTASK 26-21-00-760-002

(5) Calculate the resistance Rc of the squib discharge circuit using this formula:

Rc = RL (V1/V2) - RL

(a) Make sure Rc is less than or equal to 3 ohms.

NOTE: If Rc is more than 3 ohms, the circuit can not supply enough current to fire the squib cartridge.

Table 502/26-21-00-993-802 Primary (A) Circuit Engine Fire Bottle Squib Test Result

CONNECTOR	RL (ohm)	V1 (volt)	V2 (volt)	Rc (ohm)
D582				
D1322				
D584				
D1324				

Table 503/26-21-00-993-803 Alternate (B) Circuit Engine Fire Bottle Squib Test Result

CONNECTOR	RL (ohm)	V1 (volt)	V2 (volt)	Rc (ohm)
D582				
D1322				
D584				
D1324				

SUBTASK 26-21-00-480-006

(6) Disconnect the adapter cable from the squib connector, and repeat the procedure until all the connectors listed in the Table 501 are tested.

SUBTASK 26-21-00-080-009

(7) Disconnect the adapter cable for the primary circuit (A) from the squib connector.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-00-480-018

(8) Do the procedure again with the adapter cable for the Alternate circuit (B) until all the connectors listed in the Table 501 are tested.

NOTE: Each squib has two circuits. The primary circuit (A) is between pins 4 and 3; the alternate circuit (B) is between pins 6 and 7. Either circuit can fire the squib. There are different adapter cables in the test set to test each circuit.

SUBTASK 26-21-00-080-010

(9) Disconnect the adapter cable from the squib connector when you have completed the procedure with both adapter cables.

SUBTASK 26-21-00-860-005

(10) Push the fire handles to their usual position.

SUBTASK 26-21-00-860-006

(11) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-00-700-001

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(12) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-21-00-080-011

(13) Remove the shorting plug, SPL-10040, from the squibs.

SUBTASK 26-21-00-420-004

(14) Connect the electrical connectors to the squibs.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(a) Before squibs are touched, do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

(b) Connect D582 and D584 to Squib 1 on the left and right bottle.

(c) Connect D1322 and D1324 to Squib 2 on the left and right bottle.

SUBTASK 26-21-00-860-007

(15) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-00-710-004

(16) Set the EXT TEST switch to the 1 position.

(a) Make sure that the L and R lights come on.

NOTE: The Auxiliary Power Unit (APU) light should also come on.

SUBTASK 26-21-00-710-005

(17) Set the EXT TEST switch to the 2 position.

(a) Make sure that the L and R lights come on.

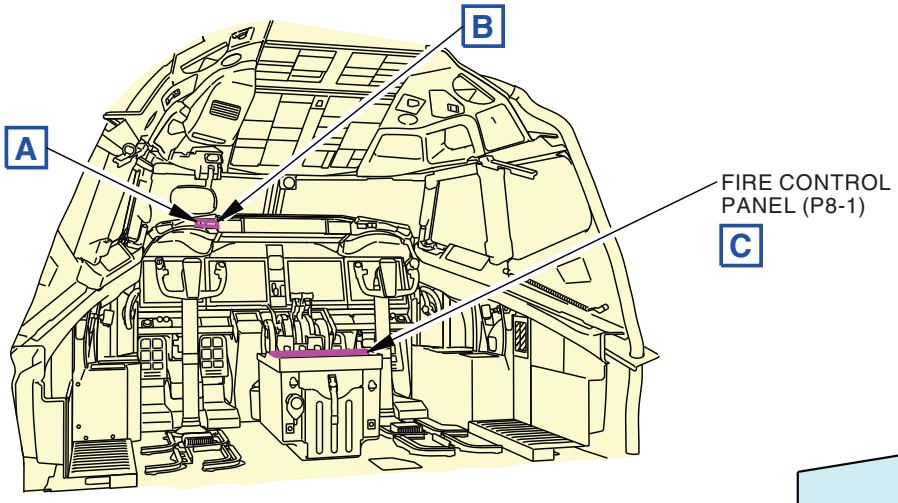
NOTE: The APU light should also come on.

SUBTASK 26-21-00-710-006

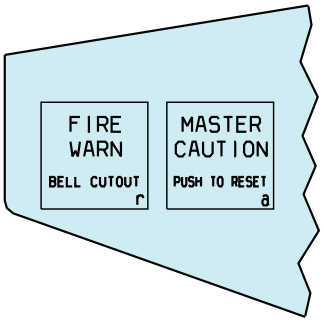
(18) Release the EXT TEST switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

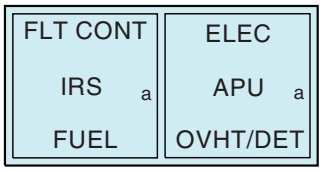


FLIGHT COMPARTMENT



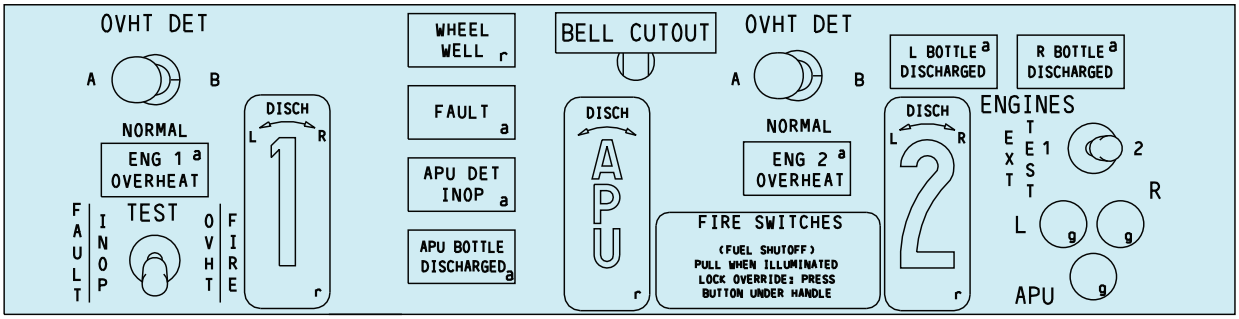
CAPTAIN'S FIRE WARNING SWITCH (P7)
(FIRST OFFICER'S IS OPPOSITE)

A



CAPTAIN'S WARNING
LIGHTS (P7)

B



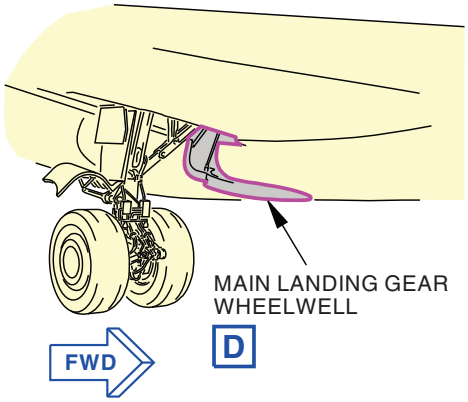
FIRE CONTROL PANEL (P8-1)

C

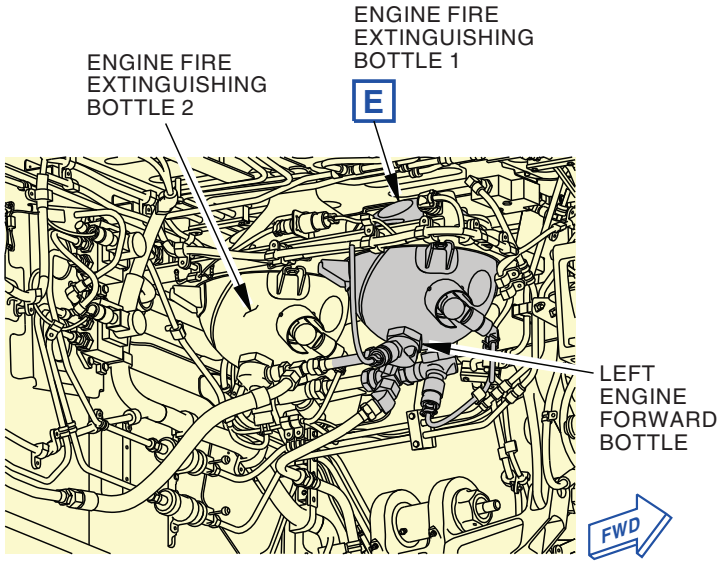
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Engine Fire Detection Test
Figure 501/26-21-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

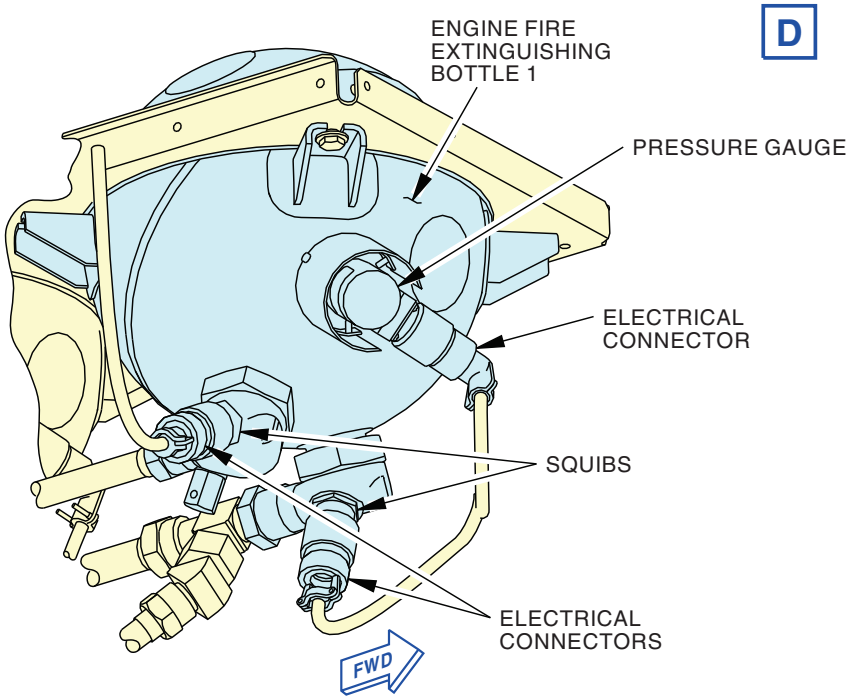


(EXAMPLE)



MAIN LANDING GEAR WHEELWELL
(EXAMPLE)

D



ENGINE FIRE EXTINGUISHING BOTTLE 1
(ENGINE FIRE EXTINGUISHING BOTTLE 2 IS EQUIVALENT)

E

2412160 S00061530054_V4

Engine Fire Detection Test
Figure 501/26-21-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-21-00-730-802

3. Engine Fire Extinguishing Bottle Pressure Switch Circuit Test

(Figure 501)

A. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Engine Fire Extinguishing Bottle Pressure Switch Test

SUBTASK 26-21-00-020-006

(1) Disconnect D586 from the pressure switch on the left (forward) fire bottle.

SUBTASK 26-21-00-020-007

(2) Disconnect D588 from the pressure switch on the right (aft) fire bottle.

SUBTASK 26-21-00-730-002

(3) Do a test of the engine fire bottle pressure switch circuit.

(a) Connect a jumper wire between pins 1 and 2 on D586.

1) Make sure the L BOTTLE DISCHARGED light on the Engine and APU Fire Control Module comes on.

(b) Remove the jumper from connector D586.

(c) Connect a jumper wire between pins 1 and 2 on D588.

1) Make sure the R BOTTLE DISCHARGED light on the Engine and APU Fire Control Module comes on.

(d) Remove the jumper from connector D588.

SUBTASK 26-21-00-420-005

(4) Connect D586 to the pressure switch on the left (forward) fire bottle.

SUBTASK 26-21-00-420-006

(5) Connect D588 to the pressure switch on the right (aft) fire bottle.

——— END OF TASK ———

TASK 26-21-00-720-801

4. Engine Fire Extinguishing Discharge Line Flow Test

(Figure 502, Figure 503)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task checks the operation of the check valve and makes sure that there is nothing blocking the discharge lines.

B. References

Reference	Title
26-21-01-000-801	Engine Fire Extinguishing Bottle Removal (P/B 401)
26-21-01-400-801	Engine Fire Extinguishing Bottle Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

Reference	Description
STD-1325	Source - Pressure, Capable of Supplying 35 +/- 5 PSIG

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Engine Fire Extinguishing Discharge Line Flow Test

SUBTASK 26-21-00-010-001

WARNING

DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSER: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THIS SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 26-21-00-020-012

(2) If necessary, remove one engine fire extinguishing bottle to get clearance for the pressure source. Do this task: Engine Fire Extinguishing Bottle Removal, TASK 26-21-01-000-801.

SUBTASK 26-21-00-020-008

(3) Remove the discharge tube from the discharge outlets on both engine fire bottles at positions 1 through 4.

NOTE: Loosen the hard tube tee assembly and other fittings as necessary to connect the test equipment.

SUBTASK 26-21-00-480-007

(4) Connect a pressure source, STD-1325, to the L fire bottle discharge line at position 2.

(a) When connecting the test equipment to the discharge lines, do not let the test equipment hang from the discharge lines.

(b) Give sufficient support to the test equipment during the flow test.

(c) Make sure that air is flowing through all the nozzles on the right engine.

(d) Make sure that little or no air is flowing through the R fire bottle discharge line at position 4.

SUBTASK 26-21-00-080-002

(5) Remove the pressure source from the L fire bottle discharge line at position 2.

SUBTASK 26-21-00-480-008

(6) Connect a pressure source, STD-1325, to the R fire bottle discharge line at position 4.

(a) Make sure that air is flowing through all the nozzles on the right engine.

(b) Make sure that little or no air is flowing through the L fire bottle discharge line at position 2.

SUBTASK 26-21-00-080-003

(7) Remove the pressure source from the R fire bottle discharge line at position 4.

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SUBTASK 26-21-00-480-009

(8) Connect a pressure source, STD-1325, to the L fire bottle discharge line at position 1.

(a) Make sure that air is flowing through all the nozzles on the left engine.

(b) Make sure that little or no air is flowing through the R fire bottle discharge line at position 3.

SUBTASK 26-21-00-080-004

(9) Remove the pressure source from the L fire bottle discharge line at position 1.

SUBTASK 26-21-00-480-010

(10) Connect a pressure source, STD-1325, to the R fire bottle discharge line at position 3.

(a) Make sure that air is flowing through all the nozzles on the left engine.

(b) Make sure that little or no air is flowing through the L fire bottle discharge line at position 1.

SUBTASK 26-21-00-080-005

(11) Remove the pressure source from the R fire bottle discharge line at position 3.

SUBTASK 26-21-00-420-010

(12) If necessary, install the engine fire extinguishing bottle that was removed. Do this task: Engine Fire Extinguishing Bottle Installation, TASK 26-21-01-400-801.

SUBTASK 26-21-00-420-014

(13) Make sure the discharge tubes at positions 1 through 4 are connected (TASK 26-21-01-400-801).

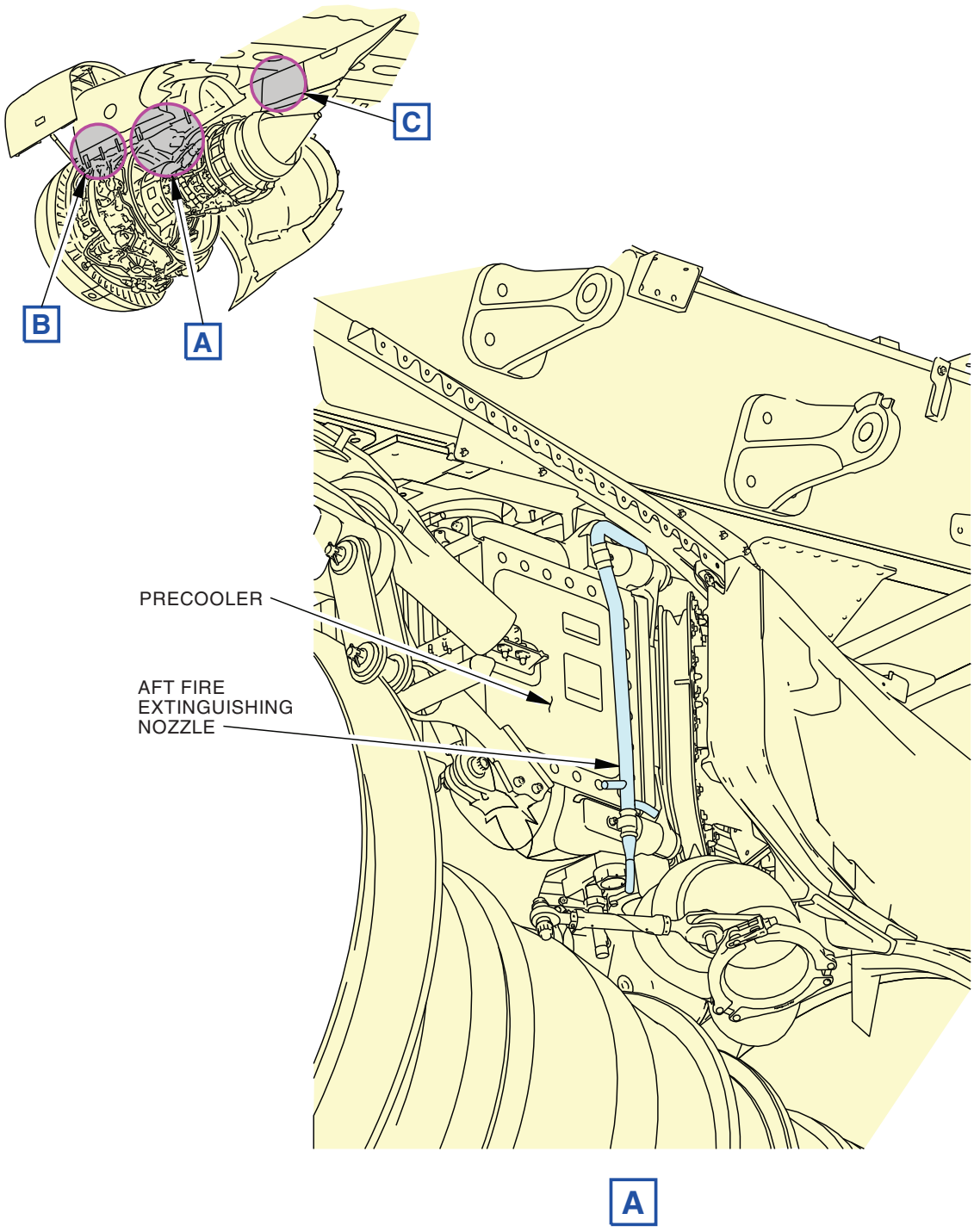
SUBTASK 26-21-00-010-002

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURES TO OPEN AND CLOSE THE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(14) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

END OF TASK

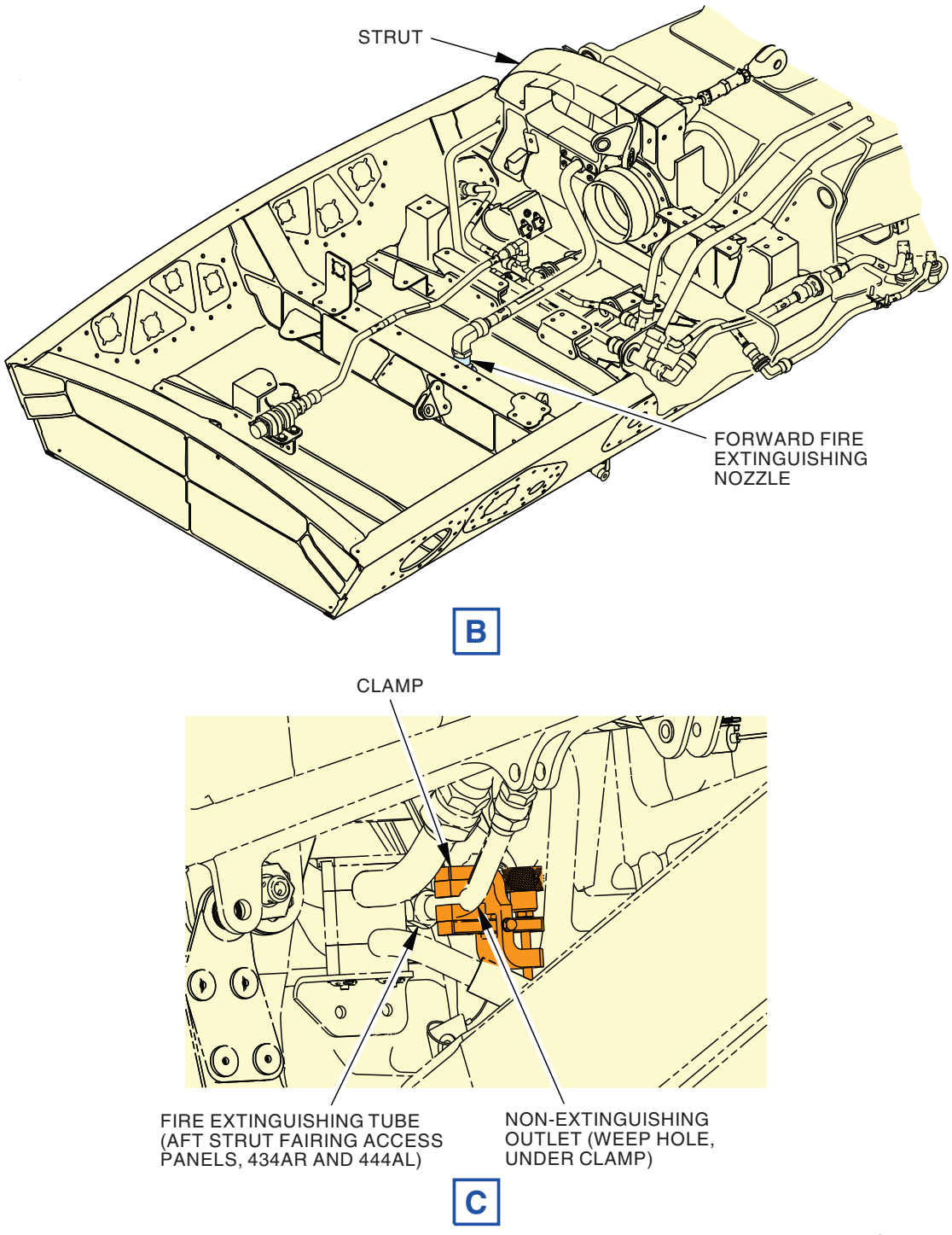
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412161 S00061530058_V2

Engine Fire Extinguishing Test
Figure 502/26-21-00-990-807 (Sheet 1 of 2)

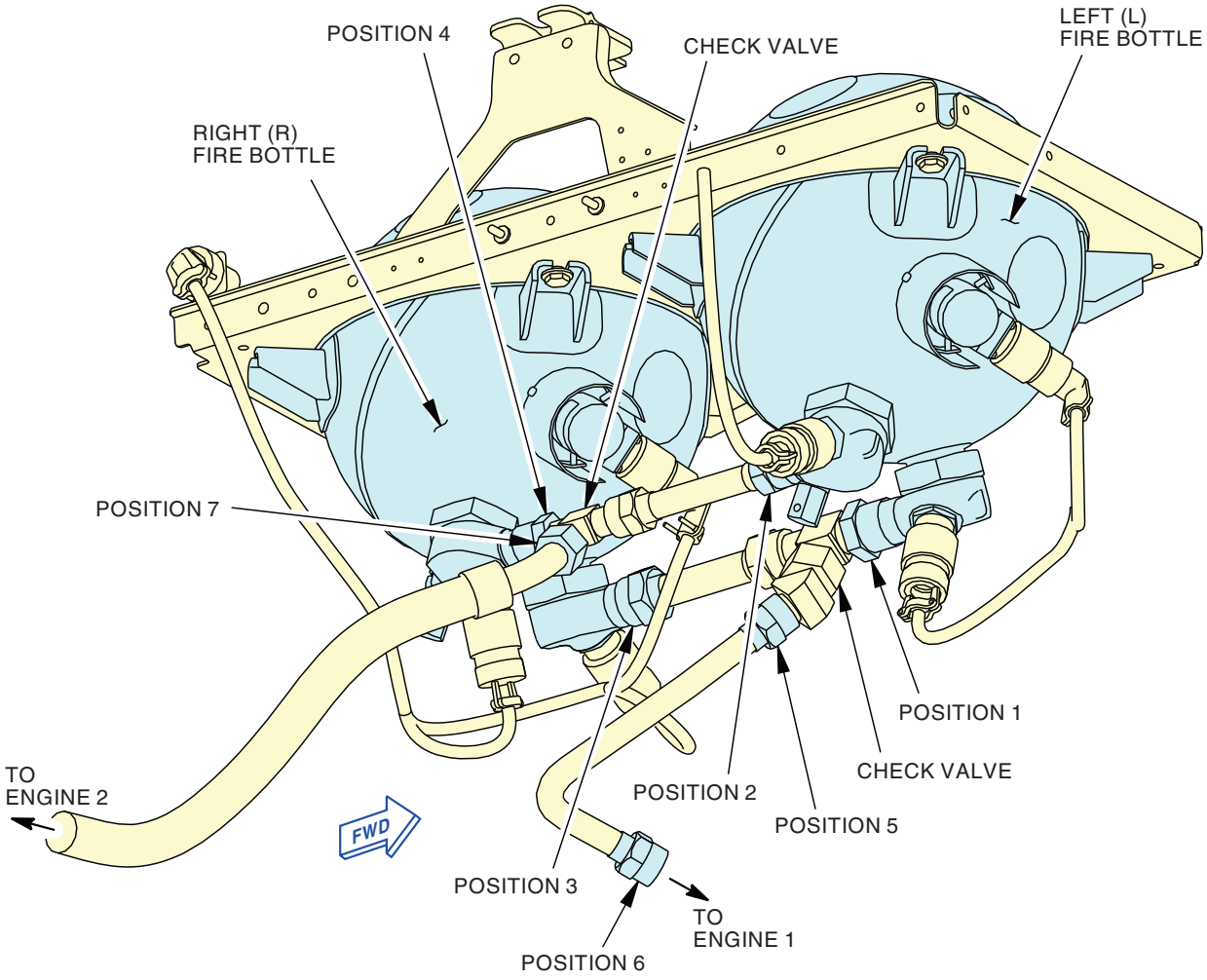
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412162 S00061530059_V3

Engine Fire Extinguishing Test
Figure 502/26-21-00-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

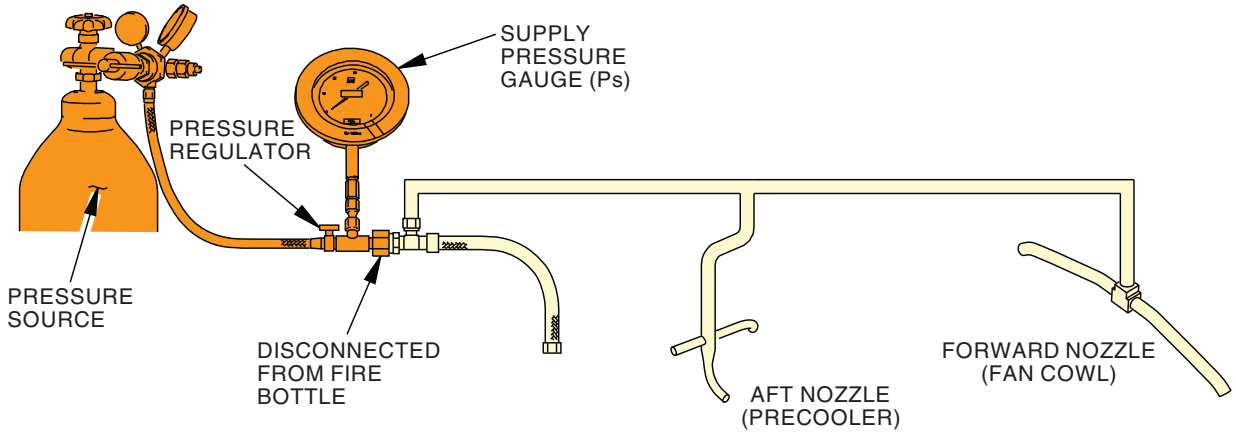


ENGINE FIRE EXTINGUISHING BOTTLES
(MAIN LANDING GEAR WHEEL WELL, LEFT SIDE)

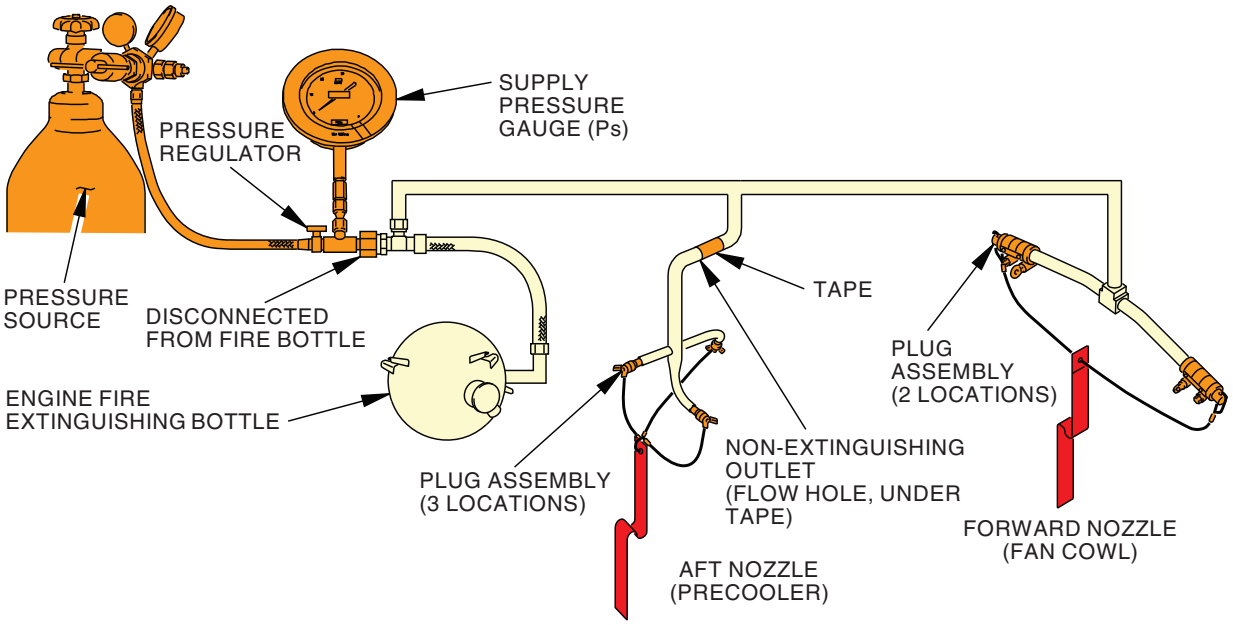
2412163 S00061530060_V3

Test Setup
Figure 503/26-21-00-990-808 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLOW TEST



PRESSURE TEST

2412164 S00061530061_V3

Test Setup
Figure 503/26-21-00-990-808 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-21-00-730-803

5. Engine Fire Extinguishing Discharge Line Pressure Test

(Figure 502, Figure 503)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task makes sure that there are no leaks in the discharge tubes. The test is the same for both engines.

B. References

Reference	Title
26-21-01-000-801	Engine Fire Extinguishing Bottle Removal (P/B 401)
26-21-01-400-801	Engine Fire Extinguishing Bottle Installation (P/B 401)
54-52-06-000-801	Aft Fairing Access Panel Removal (P/B 401)
54-52-06-400-801	Aft Fairing Access Panel Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1647	Test Equipment - Engine Fire Extinguishing System Part #: C26004-83 Supplier: 81205 Part #: C26004-84 Supplier: 81205
SPL-15147	Assembly - 0.625 inch Diameter Clamp Part #: C26004-101 Supplier: 81205
STD-1325	Source - Pressure, Capable of Supplying 35 +/- 5 PSIG
STD-12129	Pressure Regulator - Adjustable

D. Consumable Materials

Reference	Description	Specification
A50464	Tape - Vinyl, Adhesive Coated	BMS5-179
G51367	Strap - Plastic, Adjustable, Self-locking, 3.71 Inches (94.23 mm) Long	BACS38K4

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
434AR	Aft Strut Fairing, Right Panel, Strut 1
444AL	Aft Strut Fairing, Left Panel, Strut 2

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G. Engine Fire Extinguishing Discharge Line Pressure Test

SUBTASK 26-21-00-010-003

WARNING

DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSERS: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THE ABOVE SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 26-21-00-020-013

(2) If necessary, remove one engine fire extinguishing bottle to get clearance for the pressure source. Do this task: Engine Fire Extinguishing Bottle Removal, TASK 26-21-01-000-801.

SUBTASK 26-21-00-020-009

(3) Disconnect the discharge outlet tube from the tee fitting for the engine 1 at position 5.

SUBTASK 26-21-00-480-011

(4) Connect the pressure source, STD-1325, and adjustable pressure regulator, STD-12129, to the engine 1 discharge line at position 5.

(a) Adjust the position of the discharge outlet tube for the engine 1.

1) Loosen the nut at position 6.

2) Adjust the position of the tube to allow connection with the test equipment.

3) Re-tighten the nut at position 6 to 360 ±18 in-lb (41 ±2 N·m).

(b) When connecting the test equipment to the discharge lines, do not let the test equipment hang from the discharge lines.

(c) Give sufficient support to the test equipment during the pressure test.

SUBTASK 26-21-00-480-012

(5) Install plugs on the fire extinguishing nozzles at the left engine (engine 1).

NOTE: The equipment, SPL-1647, or other suitable plug may be used. If it is necessary to use C26004–78 and C26004–79, you must also use 0.625 inch diameter clamp assembly, SPL-15147.

SUBTASK 26-21-00-010-007

(6) Open the applicable aft fairing access panels:

(TASK 54-52-06-000-801)

Number	Name/Location
434AR	Aft Strut Fairing, Right Panel, Strut 1
444AL	Aft Strut Fairing, Left Panel, Strut 2

SUBTASK 26-21-00-480-013

(7) Install the clamp and tape on the non-extinguishing outlets for the applicable engine.

NOTE: This is to allow pressurization of tubing.

(a) Install a 0.625 inch diameter clamp assembly, SPL-15147, on the weep hole (View C, Figure 502).

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(b) Install a tape, A50464, and adjustable plastic strap, G51367 (or equivalent), on the flow hole (Figure 503).

NOTE: The flow hole is not on the personnel view side.

NOTE: The flow hole is a 0.125 in. (3.175 mm) dia hole on the 0.625 in. (15.875 mm) dia tube.

SUBTASK 26-21-00-780-001

(8) Pressurize the discharge tubing that goes from L bottle to engine 1 to a pressure of 35 ±5 psi (241 ±35 kPa).

(a) Wait at least 3 minutes for the air pressure to stabilize.

SUBTASK 26-21-00-780-002

(9) Turn off the pressure source.

(a) Record the stabilized air pressure.

(b) Wait 2 minutes and check the pressure.

1) Make sure that the pressure loss during the 2 minute waiting period is not more than 1 psi.

2) If the pressure loss is more than 1 psi, do a check of the pressure source connections and plugs that used to seal the fire extinguishing nozzles.

3) If they do not show any indication of leakage, do a check of all fittings, seal surfaces, valves, and tubing in the discharge line.

a) Repair the problems found.

SUBTASK 26-21-00-780-003

(10) Bleed the fire extinguishing plumbing to 0 psi.

(a) Disconnect the test equipment from position 5.

SUBTASK 26-21-00-420-011

(11) Connect the discharge outlet tube to the tee fitting for the engine 1 at position 5.

(a) Loosen the nut at position 6.

(b) Restore the position of the tube.

(c) Tighten the nuts at position 5 and 6 to 360 ±18 in-lb (41 ±2 N·m).

SUBTASK 26-21-00-020-010

(12) Disconnect the discharge outlet hose from the tee fitting for the engine 2 at position 7.

SUBTASK 26-21-00-480-014

(13) Connect the test equipment to the engine 2 discharge line at position 7.

SUBTASK 26-21-00-480-015

(14) Install plugs on the fire extinguishing nozzles at the right engine (engine 2).

NOTE: The equipment, SPL-1647, or other suitable plug may be used. If it is necessary to use C26004–78 and C26004–79, you must also use 0.625 inch diameter clamp assembly, SPL-15147.

SUBTASK 26-21-00-780-004

(15) Pressurize the discharge tubing that goes to engine 2 to a pressure of 35 ±5 psi (241 ±35 kPa).

(a) Wait at least 3 minutes for the air pressure to stabilize.

SUBTASK 26-21-00-780-005

(16) Turn off the pressure source.

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(a) Record the stabilized air pressure.

(b) Wait 2 minutes and check the pressure.

1) Make sure that the pressure loss during the 2 minute waiting period is not more than 1 psi.

2) If the pressure loss is more than 1 psi, do a check of the pressure source connections and plugs that used to seal the fire extinguishing nozzles.

3) If they do not show any indication of leakage, then do a check of all fittings, seal surfaces, valves, and tubing in the discharge line.

a) Repair the problems found.

SUBTASK 26-21-00-780-006

(17) Bleed the fire extinguishing plumbing to 0 psi.

(a) Disconnect the pressure source at position 7.

SUBTASK 26-21-00-420-012

(18) Connect the discharge outlet hose to the tee fitting for the engine 2 at position 7.

(a) Make sure that the minimum clearance between the hose and electrical harness is 0.25 in. (6 mm).

(b) Tighten the nut at position 7 to 360 ±18 in-lb (41 ±2 N·m).

SUBTASK 26-21-00-080-007

CAUTION

MAKE SURE THAT YOU REMOVE THE PLUGS FROM THE FIRE EXTINGUISHING NOZZLES AT THE TWO ENGINES. IF YOU DO NOT REMOVE THE PLUGS, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(19) Remove plugs on the fire extinguishing nozzles at the right engine (engine 2).

SUBTASK 26-21-00-420-013

(20) If necessary, install the engine fire extinguishing bottle that was removed. Do this task: Engine Fire Extinguishing Bottle Installation, TASK 26-21-01-400-801.

SUBTASK 26-21-00-080-008

CAUTION

MAKE SURE THAT YOU REMOVE THE PLUGS FROM THE FIRE EXTINGUISHING NOZZLES AT THE TWO ENGINES. IF YOU DO NOT REMOVE THE PLUGS, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(21) Remove plugs on the fire extinguishing nozzles at the left engine (engine 1).

SUBTASK 26-21-00-080-006

(22) Remove the clamps, tapes, and test equipment.

SUBTASK 26-21-00-010-004

WARNING

OBEY THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE PROCEDURE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(23) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 26-21-00-410-001

(24) Close these access panels:

(TASK 54-52-06-400-801)

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Number	Name/Location
434AR	Aft Strut Fairing, Right Panel, Strut 1
444AL	Aft Strut Fairing, Left Panel, Strut 2

——— END OF TASK ———

TASK 26-21-00-720-802

6. Engine Fire Switch System Shutdown Test

(Figure 501, Figure 504)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task makes sure that the engine is isolated from the fuel, hydraulic, and electrical systems, and that the thrust reverser is disabled when the fire handle is pulled. It should be done when the Engine and Auxiliary Power Unit (APU) Fire Control Panel is replaced.

(2) This test is the same for both engines.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
410	Subzone - Engine 1
420	Subzone - Engine 2

D. Prepare for the Test

SUBTASK 26-21-00-010-005

(1) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00

SUBTASK 26-21-00-860-009

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

E. Engine Fire Switch System Shutdown Test

SUBTASK 26-21-00-860-011

(1) Make sure that the START SWITCH for each engine is in the OFF position.

SUBTASK 26-21-00-860-012

(2) Set the engine start lever to the IDLE position.

(a) Make sure that the engine SPAR VALVE CLOSED light, on the P5 panel, comes on bright, then goes out.

(b) Make sure that the engine ENG VALVE CLOSED light, on the P5 panel, comes on bright.

SUBTASK 26-21-00-860-013

(3) Pull the engine fire handle on the P8 panel.

(a) Make sure that the fire handle does not extend.

SUBTASK 26-21-00-860-014

(4) Push the manual override button and pull the engine fire handle on the P8 panel.

(a) Make sure that the engine SPAR VALVE CLOSED light, on the P5 panel, comes on bright while the valve is in transition, then goes dim.

(b) Make sure that the engine ENG VALVE CLOSED light, on the P5 panel, goes dim.

SUBTASK 26-21-00-860-015

(5) Push in the engine fire handle, on the P8 panel, to the normal position.

SUBTASK 26-21-00-860-016

(6) Set the engine start lever to the CUTOFF position.

SUBTASK 26-21-00-860-017

(7) Push the manual override button and pull the engine fire handle on the P8 panel.

(a) Make sure that the engine hydraulic system A or B fluid shutoff valve, V33 or V34 moves to the down position (Figure 504).

NOTE: The indicator arm in the down position shows that the valve is closed.

SUBTASK 26-21-00-860-018

(8) Push in the engine fire handle, on the P8 panel, to the normal position.

(a) Make sure that the engine hydraulic system A or B fluid shutoff valve, V33 or V34 moves to the up position.

NOTE: The indicator arm in the up position shows that the valve is open.

SUBTASK 26-21-00-860-041

(9) Open these circuit breakers and install safety tags for the applicable engine:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	8	C01286	GENERATOR DISC 1
F	9	C01287	GENERATOR DISC 2
F	10	C01283	GENERATOR CONT UNIT 1
F	11	C01284	GENERATOR CONT UNIT 2

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SUBTASK 26-21-00-020-011

(10) Disconnect the electrical connector P103 from the engine Integrated Drive Generator (IDG).

SUBTASK 26-21-00-860-043

(11) Remove the safety tags and close these circuit breakers for the applicable engine:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	8	C01286	GENERATOR DISC 1
F	9	C01287	GENERATOR DISC 2
F	10	C01283	GENERATOR CONT UNIT 1
F	11	C01284	GENERATOR CONT UNIT 2

SUBTASK 26-21-00-860-024

(12) Set the GEN CONT switch, on the P5-4 panel, to the OFF position.

(a) Make sure that the resistance between pins 3 (LO) and 2 (HI) on P103 is less than 10 ohms.

SUBTASK 26-21-00-860-025

(13) Set the GEN CONT switch, on the P5-4 panel, to the ON position.

(a) Make sure that the resistance between pins 3 (LO) and 2 (HI) on P103 is at least 230 kilohms.

SUBTASK 26-21-00-860-026

(14) Push the manual override button and pull the engine fire handle on the P8 panel.

(a) Make sure that the resistance between pins 3 (LO) and 2 (HI) on P103 is less than 10 ohms after 5 seconds.

SUBTASK 26-21-00-860-027

(15) Push in the engine fire handle, on the P8 panel, to the normal position.

SUBTASK 26-21-00-860-044

(16) Open these circuit breakers and install safety tags for the applicable engine:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	8	C01286	GENERATOR DISC 1
F	9	C01287	GENERATOR DISC 2
F	10	C01283	GENERATOR CONT UNIT 1
F	11	C01284	GENERATOR CONT UNIT 2

SUBTASK 26-21-00-420-009

(17) Connect the electrical connector P103 to the engine IDG.

SUBTASK 26-21-00-860-042

(18) Remove the safety tags and close these circuit breakers for the applicable engine:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	8	C01286	GENERATOR DISC 1
F	9	C01287	GENERATOR DISC 2
F	10	C01283	GENERATOR CONT UNIT 1
F	11	C01284	GENERATOR CONT UNIT 2

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SUBTASK 26-21-00-010-006

(19) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00

SUBTASK 26-21-00-860-028

(20) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 26-21-00-860-029

(21) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 26-21-00-860-030

(22) Set the applicable EEC MAINT POWER switch, on the MISC SYSTEM CTRLS page, on the Multi Function Display (MFD), to the TEST position.

(a) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.

(b) Wait for a minimum of 30 seconds to let the Electronic Engine Control (EEC) complete the test.

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER switch to the TEST position.

SUBTASK 26-21-00-860-040

(23) Make sure that the thrust reverser is in the retracted position.

SUBTASK 26-21-00-750-001

(24) Start the thrust reverser special function (TASK 71-00-00-700-816-G00).

SUBTASK 26-21-00-860-031

(25) Push the manual override button and pull the engine fire handle on the P8 panel.

SUBTASK 26-21-00-860-032

WARNING KEEP ALL PERSONNEL AWAY FROM THE THRUST REVERSER SLEEVE. THE THRUST REVERSER SLEEVE WILL RETRACT WHEN YOU SUPPLY HYDRAULIC PRESSURE. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(26) Move the engine reverse thrust lever up and aft to the extend position.

(a) Make sure that the thrust reverser does not extend.

SUBTASK 26-21-00-860-033

(27) Move the engine reverse thrust lever down and forward to the retract position.

SUBTASK 26-21-00-860-034

(28) Push in the engine fire handle, on the P8 panel, to the normal position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-00-860-035

WARNING KEEP ALL PERSONNEL AWAY FROM THE THRUST REVERSER SLEEVE. THE THRUST REVERSER SLEEVE WILL RETRACT WHEN YOU SUPPLY HYDRAULIC PRESSURE. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(29) Move the engine reverse thrust lever up and aft to the extend position.

(a) Make sure that the thrust reverser extends.

SUBTASK 26-21-00-860-036

(30) Move the engine reverse thrust lever down and forward to the retract position.

(a) Make sure that the thrust reverser retracts.

SUBTASK 26-21-00-750-002

(31) Stop the thrust reverser special function (TASK 71-00-00-700-816-G00).

SUBTASK 26-21-00-860-037

(32) Set the ENG START switch to the OFF position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 26-21-00-860-038

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 26-21-00-860-039

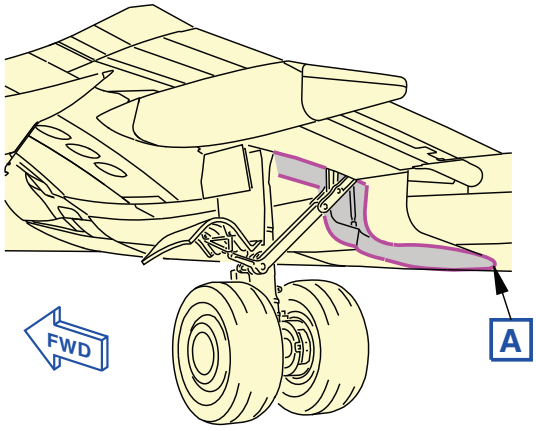
(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

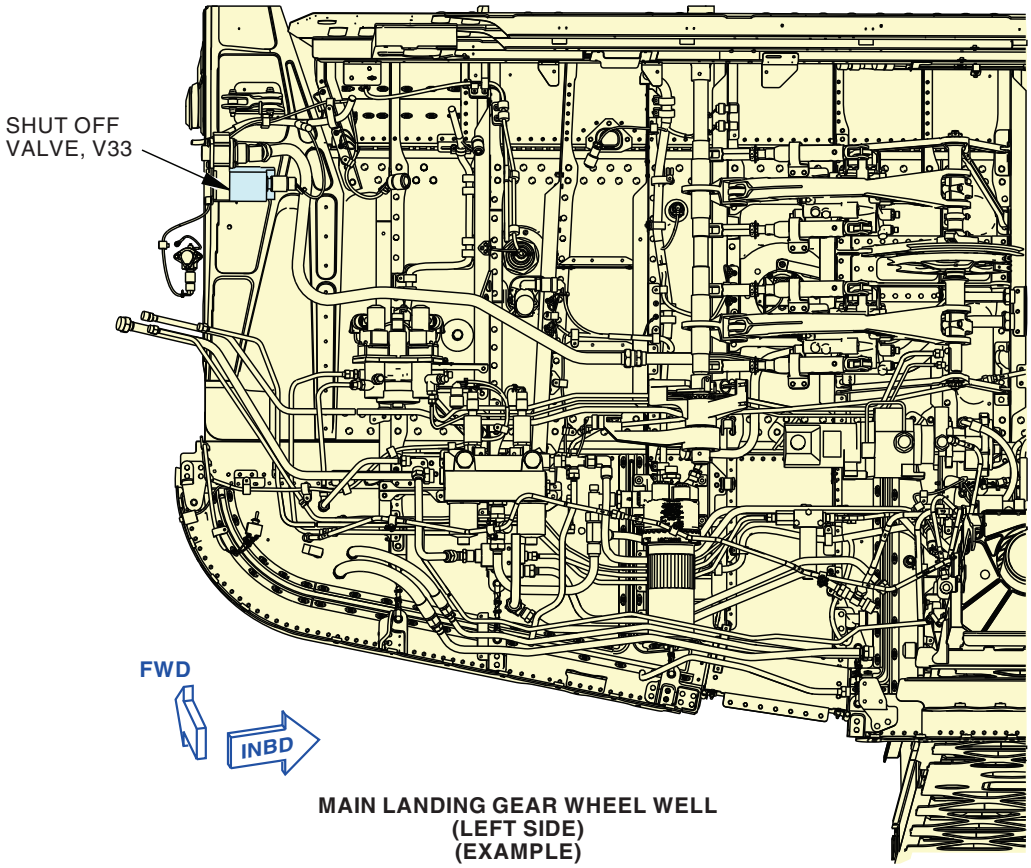
Row	Col	Number	Name
A	19	C00388	FIRE PROTECTION DET OVHT WW WING BODY
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)



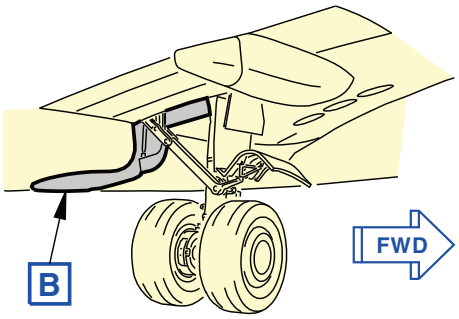
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)
(EXAMPLE)

A

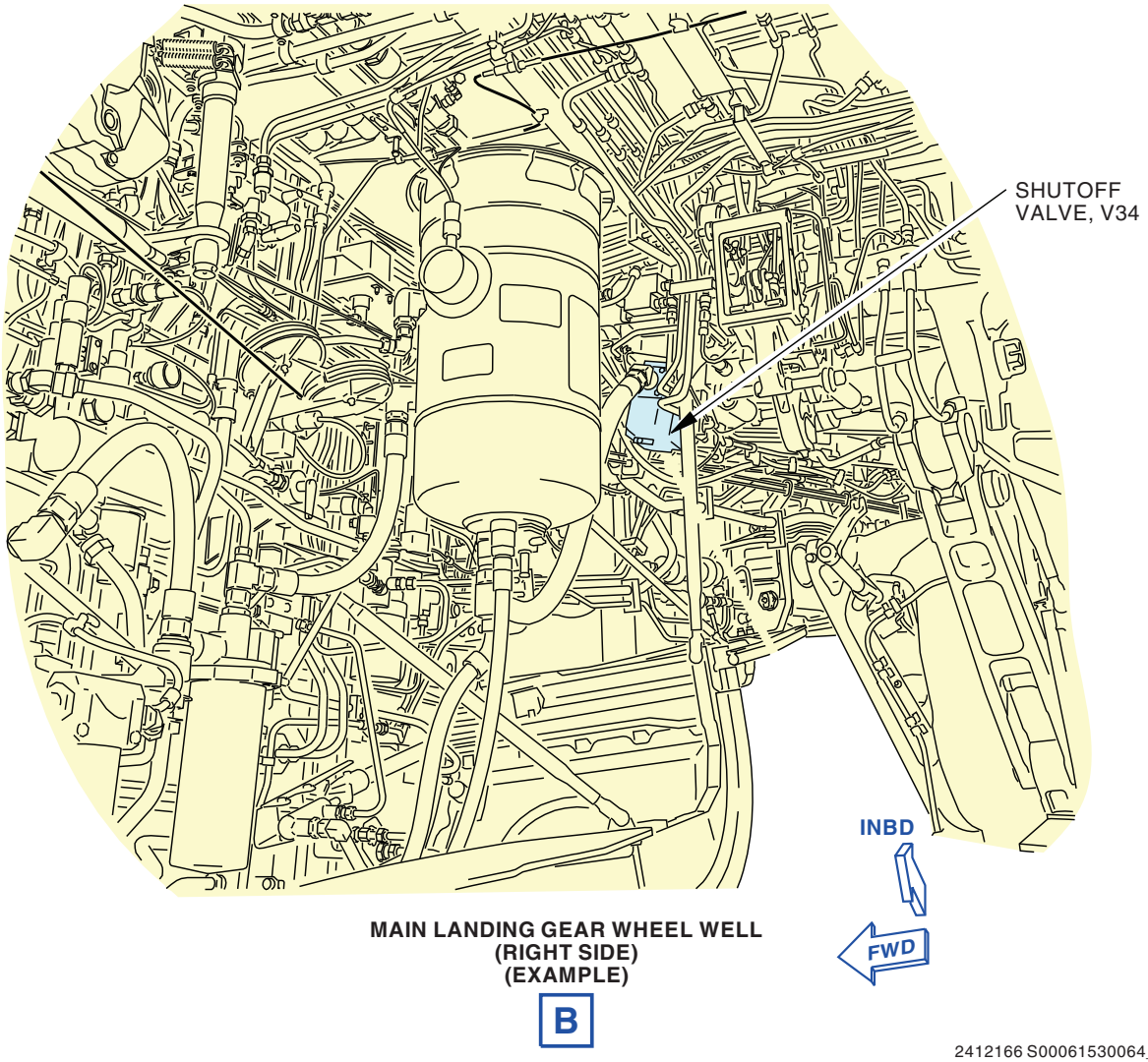
2412165 S00061530063_V2

Engine EDP Shutoff Valve Test
Figure 504/26-21-00-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)



MAIN LANDING GEAR WHEEL WELL
(RIGHT SIDE)
(EXAMPLE)

2412166 S00061530064_V2

Engine EDP Shutoff Valve Test
Figure 504/26-21-00-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE EXTINGUISHING BOTTLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of an engine fire extinguishing bottle.

(2) An installation of an engine fire extinguishing bottle.

TASK 26-21-01-000-801

2. Engine Fire Extinguishing Bottle Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to remove the engine fire extinguishing bottle [1].

(2) The engine fire extinguishing bottles [1] are installed in the main landing gear wheel well. The removal procedure is the same for each bottle.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1168	Cap - Shorting or Faraday Cap

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Engine Fire Extinguishing Bottle Removal

SUBTASK 26-21-01-940-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 26-21-01-860-001

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-01-020-001

WARNING

DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSON AND DAMAGE EQUIPMENT.

(3) Disconnect the electrical connectors [6] from the squibs [7].

(a) Before you touch the squib [7], do the procedure for devices that are sensitive to electrostatic discharge.

1) These are the tasks:

- ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.
- ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-21-01-020-002

WARNING

DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION

DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(4) Install the shorting or faraday cap, STD-1168, on each squib [7].

NOTE: The shorting caps are attached to the fire extinguishing bottles. These shorting caps prevent the squibs from possible activation due to static electricity.

SUBTASK 26-21-01-020-003

(5) Disconnect the electrical connector [5] from the pressure switch [4].

SUBTASK 26-21-01-020-004

(6) Disconnect the discharge line from both fire bottle discharge outlets.

NOTE: Loosen the hard tube tee assembly and other fittings as necessary to move the discharge lines out of the way.

SUBTASK 26-21-01-420-001

(7) Install the protective caps on each of the discharge outlets.

SUBTASK 26-21-01-420-005

(8) Install the protective caps on each of the discharge lines.

SUBTASK 26-21-01-020-008

(9) Remove the bolts [2] and washers [3].

NOTE: Loosen two opposite bolts at a time for easier removal.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-01-020-006

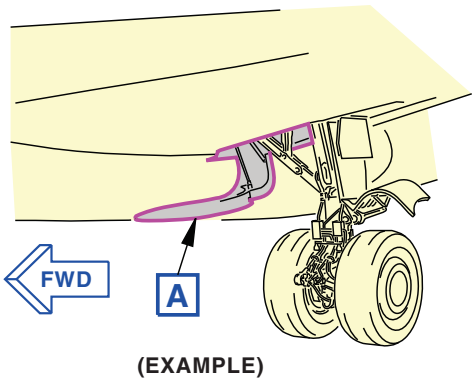
WARNING

BE CAREFUL WHEN YOU MOVE THE FIRE EXTINGUISHING BOTTLE. THE FIRE EXTINGUISHING BOTTLE IS PRESSURIZED AND HAS AN EXPLOSIVE CARTRIDGE AS A COMPONENT. IF THE FIRE EXTINGUISHING BOTTLE RELEASES ITS CONTENTS, IT CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

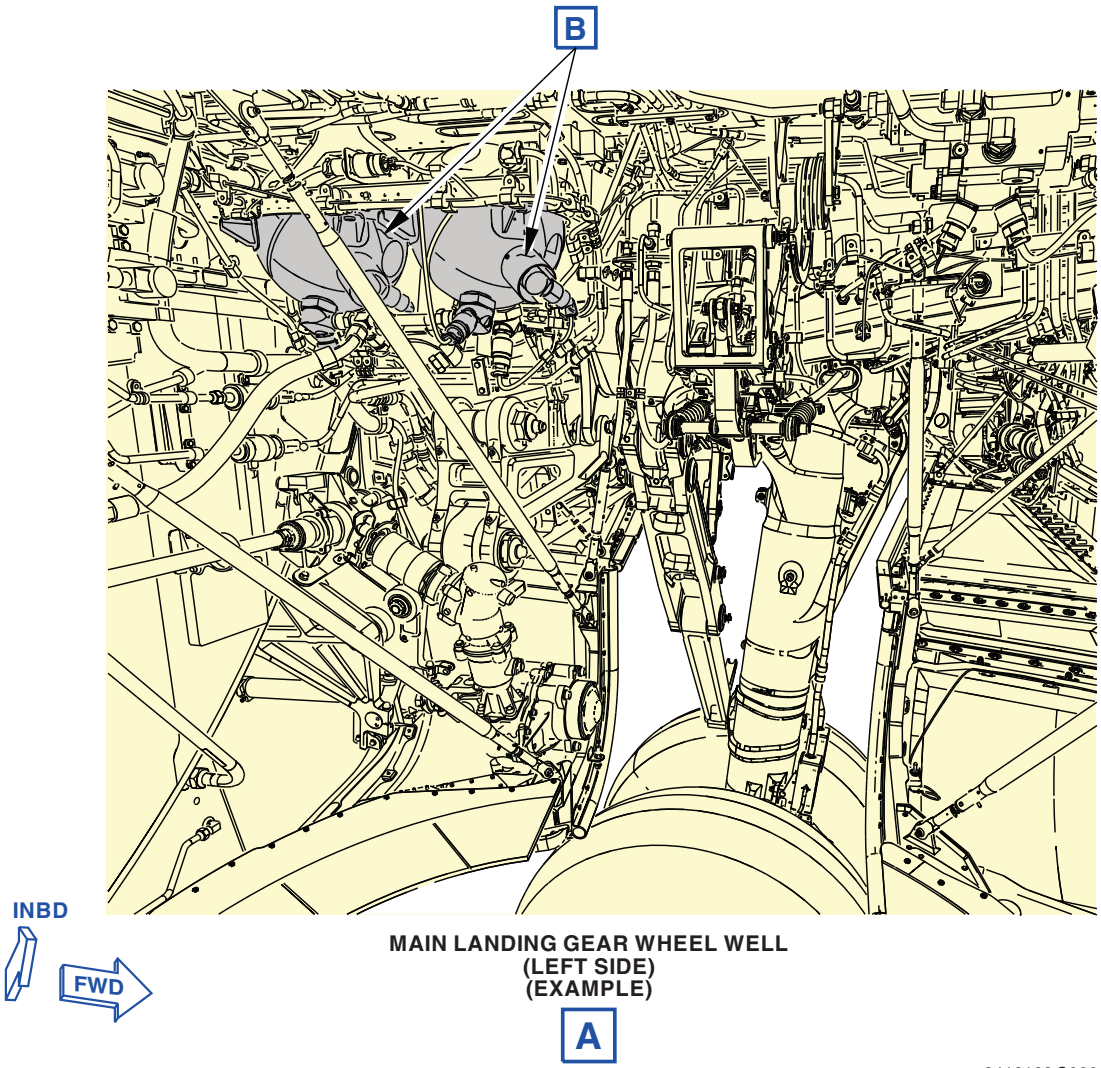
(10) Remove the fire extinguishing bottle [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

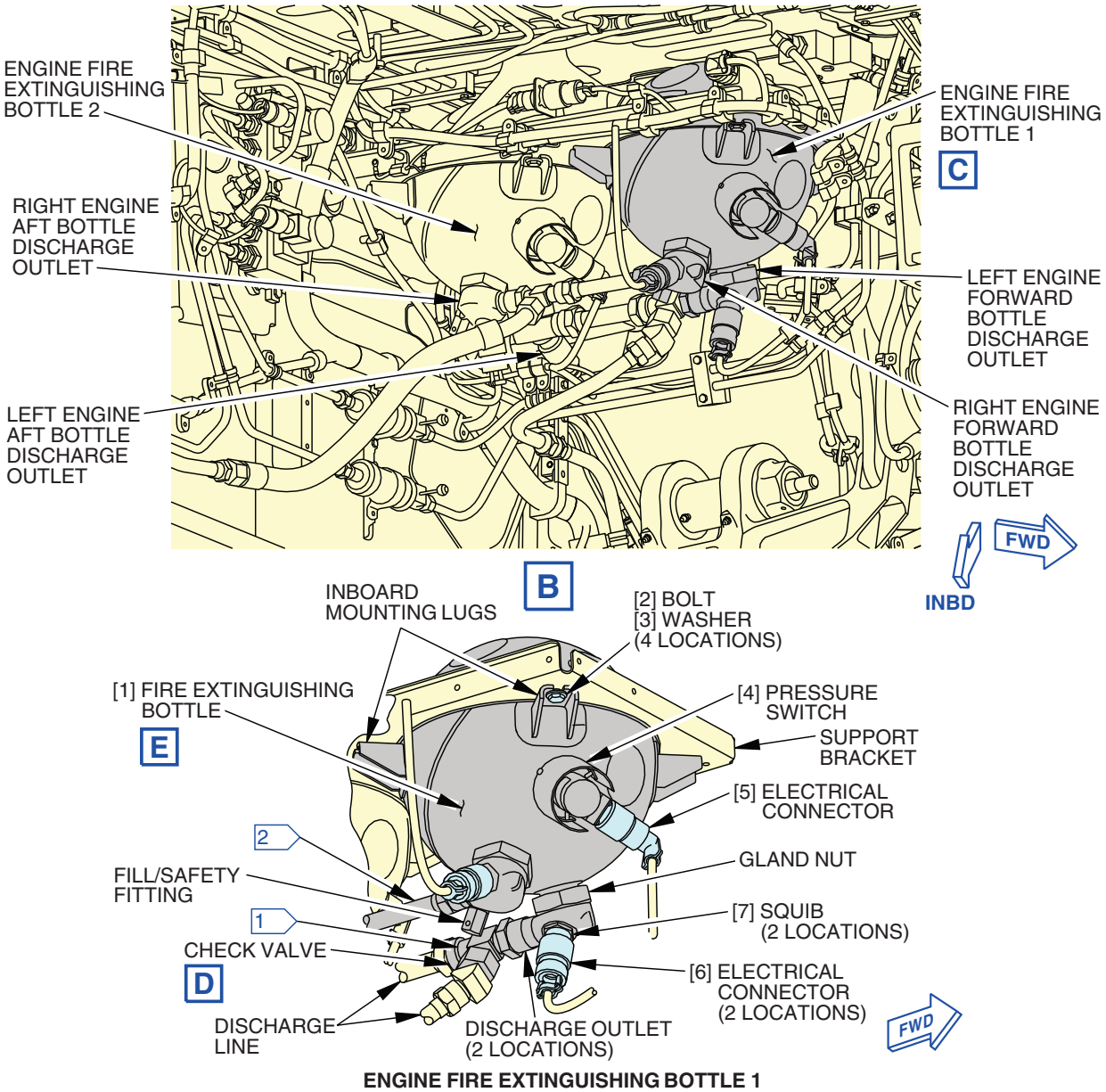


MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)
(EXAMPLE)

2412168 S00061530069_V3

Engine Fire Extinguishing Bottle Installation
Figure 401/26-21-01-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ENGINE FIRE EXTINGUISHING BOTTLE 1
(ENGINE FIRE EXTINGUISHING BOTTLE 2 IS EQUIVALENT)

C

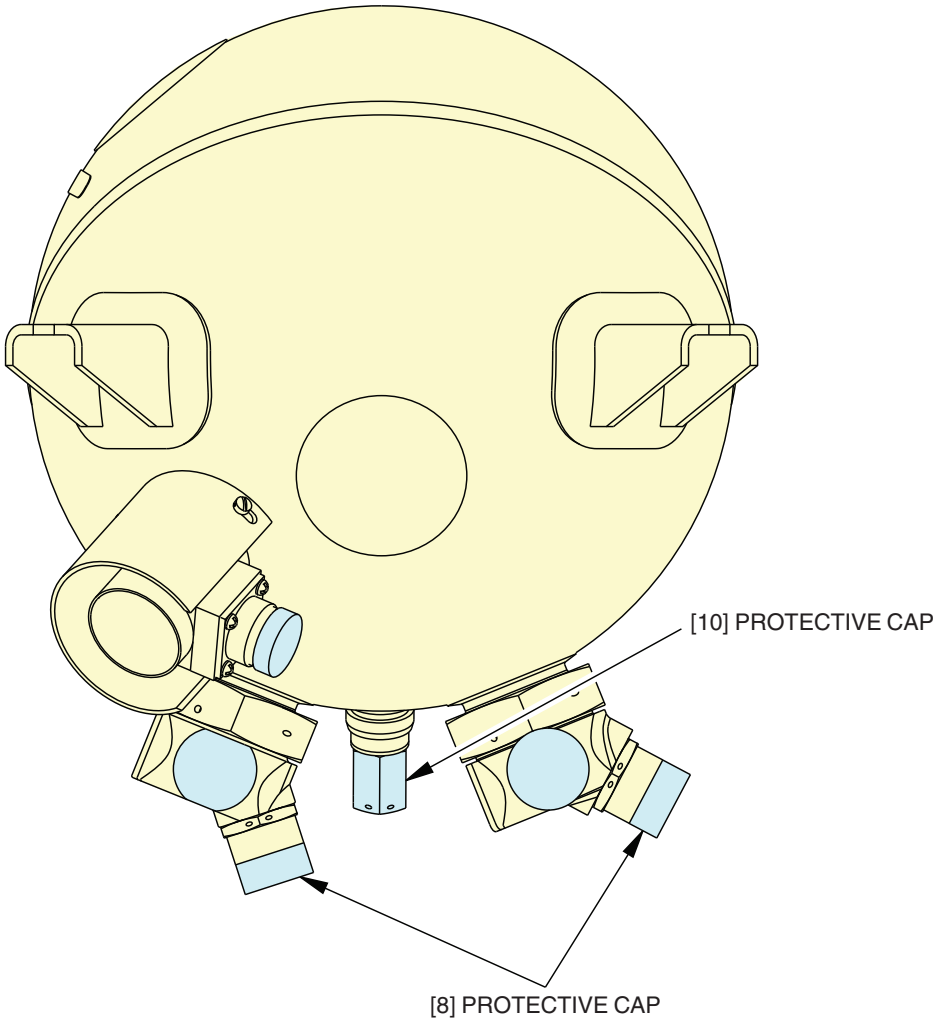
1 POSITION #1: T-FITTING CONNECTION TO THE ENGINE FIRE EXTINGUISHING BOTTLE 1 (POSITION #4: ENGINE FIRE EXTINGUISHING BOTTLE 2 IS EQUIVALENT)

2 POSITION #2: HOSE ASSEMBLY CONNECTION TO THE ENGINE FIRE EXTINGUISHING BOTTLE 1 (POSITION #3: ENGINE FIRE EXTINGUISHING BOTTLE 2 IS EQUIVALENT)

3117797 S0000854089_V2

Engine Fire Extinguishing Bottle Installation
Figure 401/26-21-01-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



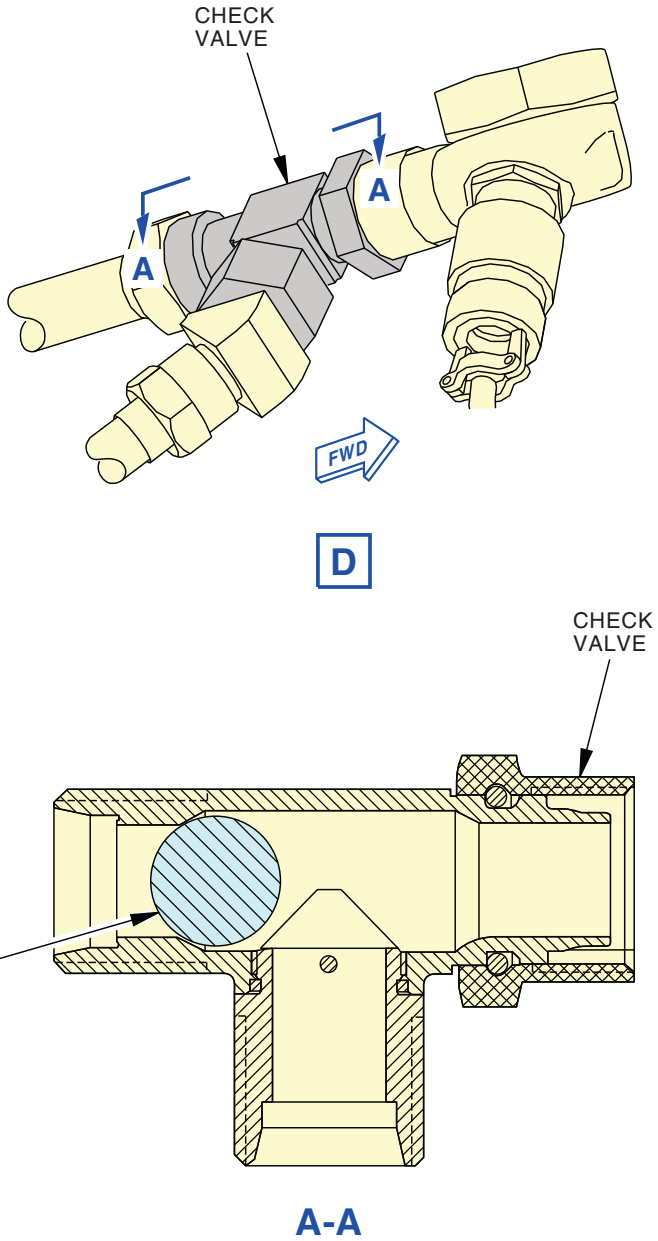
ENGINE FIRE EXTINGUISHING BOTTLE

E

3036530 S0000806021_V2

Engine Fire Extinguishing Bottle Installation
Figure 401/26-21-01-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2540267 S0000603665_V2

Engine Fire Extinguishing Bottle Installation
Figure 401/26-21-01-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-21-01-400-801

3. Engine Fire Extinguishing Bottle Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to install the engine fire extinguishing bottles [1].

(2) The engine fire extinguishing bottles [1] are installed in the left main wheel well. The installation procedure is the same for each bottle.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
26-20-00-210-801	Engine Fire Extinguishing Bottle Check (P/B 601)
26-21-00-720-801	Engine Fire Extinguishing Discharge Line Flow Test (P/B 501)
26-21-00-730-802	Engine Fire Extinguishing Bottle Pressure Switch Circuit Test (P/B 501)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
COM-4907	Force Gauge, 10 Lbs. with Resolution of 0.1 lb, +/-0.3% or Better Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: LG-5KG Supplier: 92456 Part #: PS-10 Supplier: 1GHM7 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-14896	Alignment Equipment - Fire Suppression Bottle Nozzle Part #: C26011-1 Supplier: 81205
STD-1079	Resistor - 10K Ohm or Greater
STD-1168	Cap - Shorting or Faraday Cap

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
D50004	Compound - Antiseize	BMS3-28
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50077	Abrasive - Aluminum Oxide Paper, 240 grit or finer	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fire extinguishing bottle	26-21-01-01-090	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

G. Engine Fire Extinguishing Bottle Installation

SUBTASK 26-21-01-940-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 26-21-01-280-001

(2) Use the force gauge, COM-4907, to weigh the fire extinguishing bottle [1].

NOTE: The bottle assembly includes the charged bottle, discharge heads, squibs, and shorting or faraday caps, STD-1168, on the squibs.

NOTE: The squibs are also known as cartridges.

(a) Weigh the shorting or faraday caps, STD-1168, including its caps and record as "Value A".

(b) Record the weight marked on the fire extinguishing bottle [1] as "Value B".

(c) Replace the squib protective caps [8] on the squibs [7] with the shorting or faraday caps, STD-1168.

(d) Make sure that the shorting or faraday caps, STD-1168, are installed on the squibs [7] during handling, weighing and installation of the fire extinguishing bottle [1].

(e) Remove the remaining protective caps from the fire extinguishing bottle [1].

(f) Weigh the assembly, including the charged bottle, discharge heads, squibs [7], shorting or faraday caps, STD-1168 (including shorting plugs' caps), and the loose squib protective caps [8], and record as “Value C”.

(g) Find the measured weight of the fire extinguishing bottle [1] by subtracting “Value A” from “Value C” and record as “Value D” (Value D = Value C - Value A).

SUBTASK 26-21-01-280-002

(3) Replace the fire extinguishing bottle [1] if the measured weight (“Value D”) deviates from the weight marked on the fire extinguishing bottle [1] (“Value B”) by more than 0.00 +0.12 / -0.06 lb (0.00 +0.05 / -0.03 kg).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(4) Install the protective caps to the discharge outlets on the fire extinguishing bottle [1].

SUBTASK 26-21-01-820-001

(5) If it is necessary, on a clean work surface, use fire suppression bottle nozzle alignment equipment, SPL-14896, to check and align each discharge head on the fire extinguishing bottle [1].

(a) Remove the protective cap from the discharge outlet.

(b) Loosen the gland nut to align the discharge head.

(c) Tighten the gland nut to 540 in-lb (61.0 N·m) - 660 in-lb (74.6 N·m).

(d) Install MS20995C32 lockwire, G01048, between the gland nuts on the discharge heads.

(e) Install the protective cap on the discharge outlet.

(f) Make sure that MS20995C32 lockwire, G01048, is installed between the cartridges and the discharge outlets.

SUBTASK 26-21-01-200-001

(6) If the previous fire extinguishing bottle [1] was discharged, do this step:

(a) Do a check of the discharge line for contamination.

NOTE: Pay particular attention to the discharge outlets (nozzles) at the engine.

SUBTASK 26-21-01-200-002

(7) Examine the check valve to make sure that the ball moves freely.

NOTE: It may be necessary to further disconnect the discharge lines to get access to the check valve.

(8) Do this task: Engine Fire Extinguishing Discharge Line Flow Test, TASK 26-21-00-720-801.

SUBTASK 26-21-01-860-002

(9) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-01-420-002

(10) Do these steps to install the fire extinguishing bottle [1]:

(a) Use 240 grit or finer abrasive paper, G50077, on the inboard support bracket mounting surfaces to remove all surface finishes until a bright aluminum bonding surface is visible.

(b) Use new cotton wiper, G00034, moist in solvent, B00184, to clean the washers [3], inboard mounting lugs, and mounting surfaces of the support bracket.

(c) Apply a fay surface seal with sealant, A50110, sealant, A50155, sealant, A50231, or sealant, A50296 to the inboard mounting surfaces.

(d) Install the bolt [2] and washer [3] into the support bracket finger tight.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING BE CAREFUL WHEN YOU MOVE THE FIRE EXTINGUISHING BOTTLE. THE FIRE EXTINGUISHING BOTTLE IS PRESSURIZED AND HAS AN EXPLOSIVE CARTRIDGE AS A COMPONENT. IF THE FIRE EXTINGUISHING BOTTLE RELEASES ITS CONTENTS, IT CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(e) Put the fire extinguishing bottle [1] in its correct position on the support bracket.

1) Slide the bottle lug between previously installed bolt [2], washer [3] and support bracket.

(f) Install a second bolt [2] and washer [3] finger tight cross corner from the first.

1) Do not install the remaining bolts [2] and washers [3] at this time.

2) Make sure that the bolts [2] are fully engaged in the fire bottle lugs.

NOTE: Movement of the bottle is allowed for installation of the discharge lines.

(g) If the discharge head does not align with the hose or tube assembly, do this step:

1) Loosen the discharge outlet gland nut and rotate the discharge outlet so it aligns with the tube assembly.

(h) Tighten the gland nut to 540 in-lb (61.0 N·m) - 660 in-lb (74.6 N·m).

NOTE: Even if the discharge head is already aligned, it could be necessary to torque the gland nut.

NOTE: If the gland nut is torqued and the lockwire is installed, it is not necessary to remove the lockwire and torque the gland nut.

(i) Install MS20995C32 lockwire, G01048, between the gland nuts on the discharge heads.

(j) Remove the protective caps from each of the discharge outlets.

(k) Remove the protective caps from each of the discharge lines.

WARNING YOU MUST WEAR THE APPLICABLE GLOVES WHEN YOU USE HIGH-TEMPERATURE ANTISEIZE COMPOUND. IF YOU GET THE HIGH-TEMPERATURE ANTISEIZE COMPOUND ON YOUR SKIN, REMOVE IT WITH SOAP AND WATER. IF THE HIGH-TEMPERATURE ANTISEIZE COMPOUND GOES THROUGH YOUR SKIN, INJURY TO PERSONNEL CAN OCCUR.

(l) Apply a layer of compound, D50004, to the external threads of the discharge heads, and to the shoulders of the sleeves that contact the coupling nuts on the discharge lines.

NOTE: Do not allow antiseize to enter the discharge heads or the discharge lines. Do not apply antiseize to the sealing surface of the sleeves on the discharge lines.

(m) Connect the discharge lines to the discharge heads.

1) Begin with outboard connections first, then inboard.

NOTE: Yellow is used to show the plumbing that supplies extinguishing agent to engine number 1. Blue is used to show the plumbing that supplies extinguishing agent to engine number 2.

NOTE: Loosely connect the hard tube tee and discharge line before you tighten the discharge line.

(n) Remove all unwanted compound.

I (o) Install the protective cap [10] to the fill fitting.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Secure the protective cap [10] with MS20995C32 lockwire, G01048.

(p) Install the remaining bolts [2] and washers [3] that attach the fire extinguishing bottle [1].

1) Tighten the bolts [2] snug, then back off 1/2 turn.

2) Make sure that the discharge lines do not bind.

(q) At positions 1 and 3, tighten the nuts on the tube assemblies to 450 ±23 in-lb (50.8 ±2.6 N·m).

NOTE: Position 1 connects the T-fitting to the engine fire extinguishing bottle 1.

NOTE: Position 3 connects the tube assembly to the engine fire extinguishing bottle 2.

(r) At positions 2 and 4, tighten the nuts on the tube assemblies to 360 ±18 in-lb (40.7 ±2.0 N·m).

NOTE: Position 2 connects the tube assembly to the engine fire extinguishing bottle 1.

NOTE: Position 4 connects the T-fitting to the engine fire extinguishing bottle 2.

(s) Tighten the bolts [2] to 150 in-lb (16.9 N·m).

(t) Measure the electrical bonding resistance between the fire extinguishing bottle [1] and support bracket (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the resistance is 0.0025 ohm (2.5 milliohms) or less.

3) If the resistance is more than 0.0025 ohm (2.5 milliohms), clean the bonding surfaces between the inboard mounting lugs and support bracket with solvent, B00083, and retest.

SUBTASK 26-21-01-710-001

(11) Do these steps before you connect the electrical connectors [6] to the squibs [7].

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Before you touch the squibs [7], do the procedure for devices that are sensitive to electrostatic discharge.

1) These are the tasks:

- ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.
- ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-21-01-700-001

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(12) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-21-01-600-001

(13) Do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

SUBTASK 26-21-01-420-007

(14) Remove the shorting or faraday cap, STD-1168, from the squibs [7].

SUBTASK 26-21-01-420-003

(15) Connect the electrical connectors [6] to the squibs [7].

(a) Make sure that the pins fully engage the electrical connectors [6].

SUBTASK 26-21-01-420-004

(16) Do a test of the electrical wiring to the engine fire bottle pressure switch [4] (TASK 26-21-00-730-802).

SUBTASK 26-21-01-420-006

(17) Connect the electrical connector [5] to the pressure switch [4].

SUBTASK 26-21-01-210-004

(18) Do a visual check of the pressure gauge (TASK 26-20-00-210-801).

H. Engine Fire Extinguishing Bottle Installation Test

SUBTASK 26-21-01-860-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-01-741-001

(2) Set the Engines EXT TEST switch, on the P8-1 panel, to the 1 position.

(a) Make sure that the L and R lights come on.

NOTE: The APU light will also come on.

SUBTASK 26-21-01-741-002

(3) Set the Engines EXT TEST switch, on the P8-1 panel, to the 2 position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the L and R lights come on.

NOTE: The APU light will also come on.

SUBTASK 26-21-01-741-003

(4) Release the Engines EXT TEST switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE EXTINGUISHING BOTTLE - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) Engine Fire Extinguisher Distribution System Inspection.

TASK 26-21-01-210-801

2. Engine Fire Extinguisher Distribution System Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

C. Prepare for Inspection

SUBTASK 26-21-01-010-001

WARNING

DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSERS: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THE ABOVE SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

D. Engine Fire Extinguisher Distribution System Inspection

SUBTASK 26-21-01-210-001

(1) Make sure the engine fire extinguishing bottle 1 and 2 are not damaged.

SUBTASK 26-21-01-210-002

(2) Make sure the discharge lines are not damaged.

SUBTASK 26-21-01-210-003

(3) Make sure the forward and aft fire extinguishing nozzles are free of obstructions.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Put the Airplane Back to Its Usual Condition

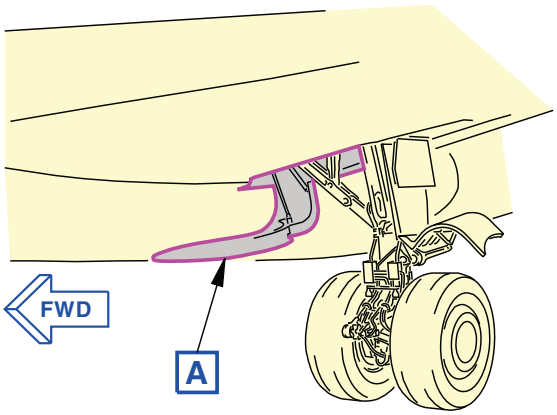
SUBTASK 26-21-01-010-002

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS WHEN YOU CLOSE THE THRUST REVERSERS, INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

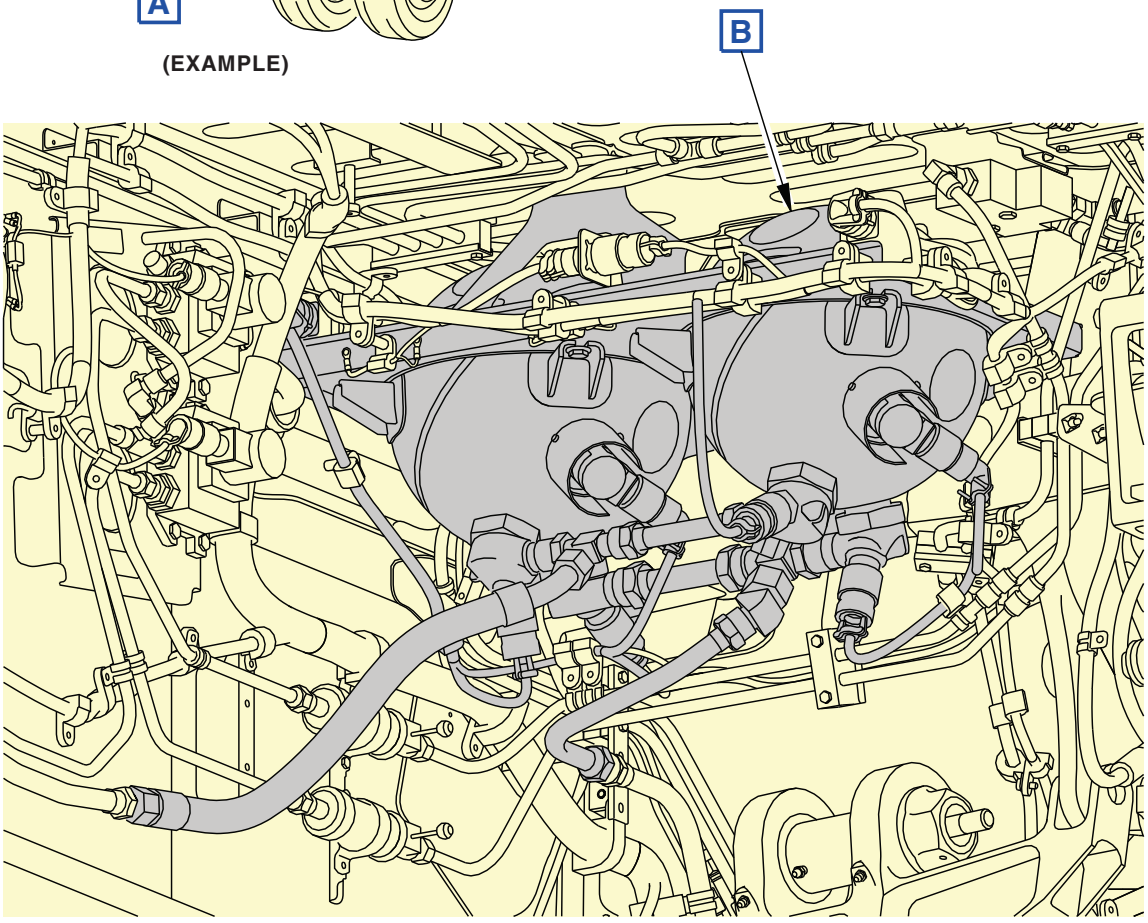
(1) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

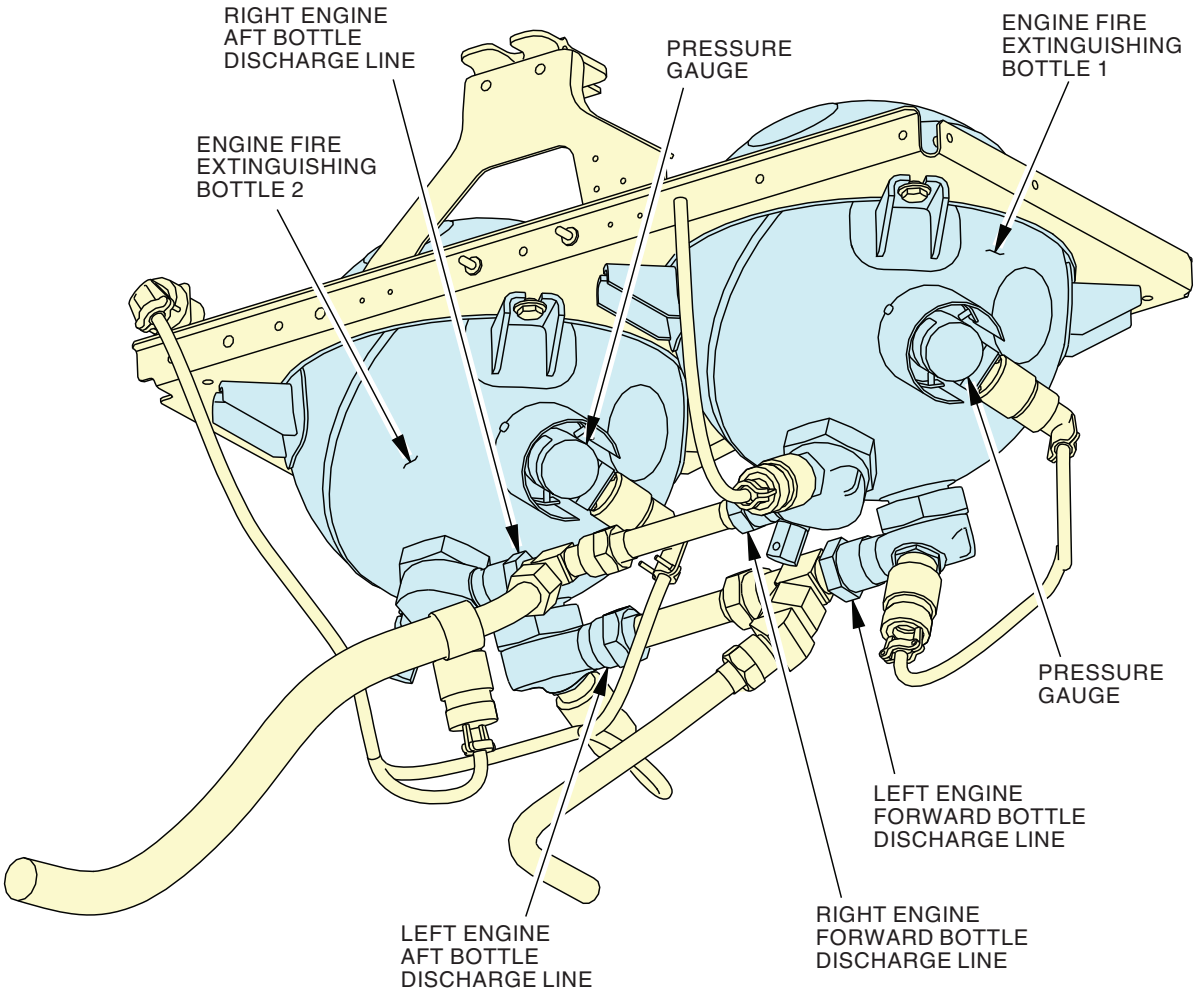


MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)
(EXAMPLE)

2412155 S00061530041_V3

Engine Fire Extinguishing Bottle Inspection
Figure 601/26-21-01-990-802 (Sheet 1 of 4)

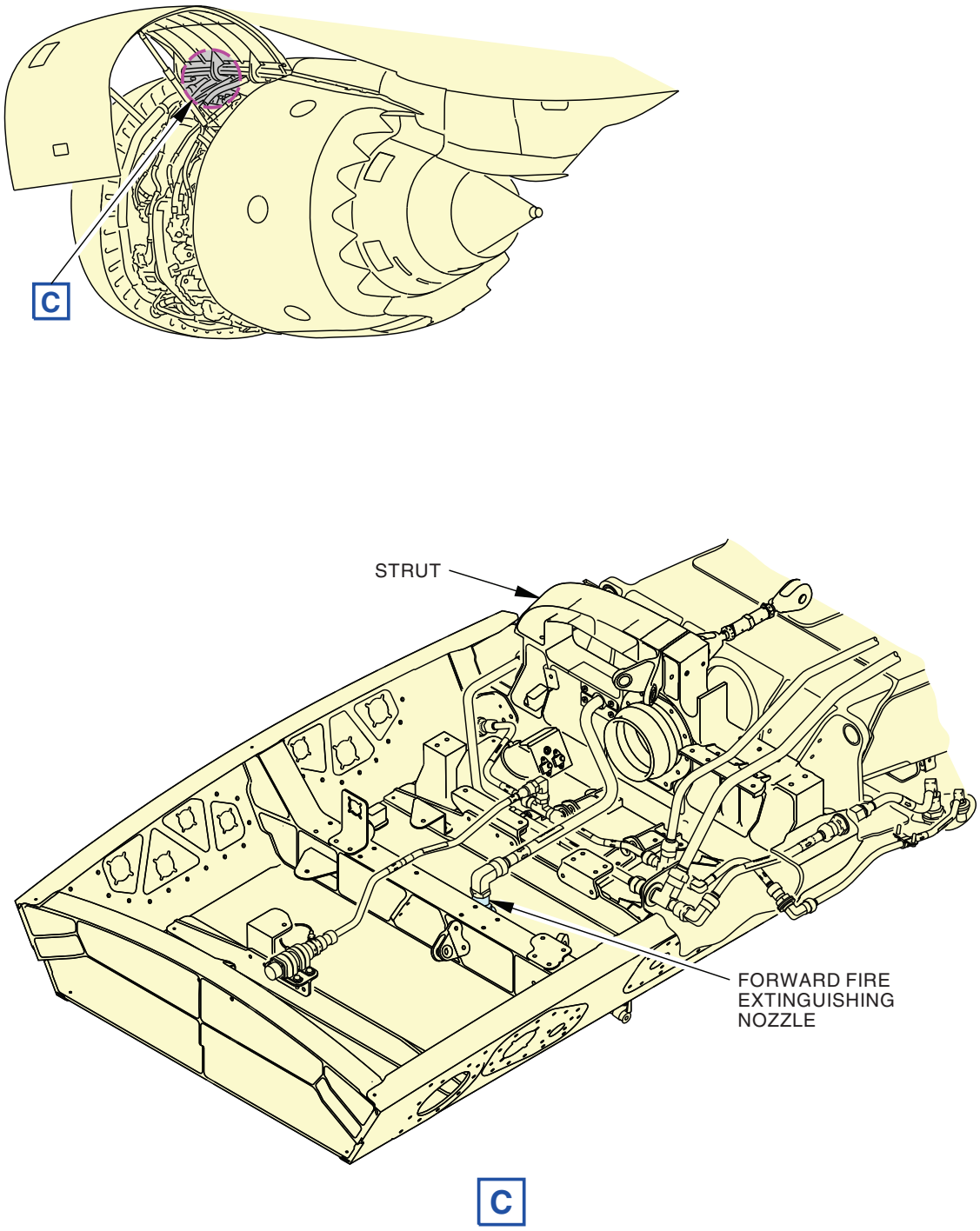
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412156 S00061530042_V2

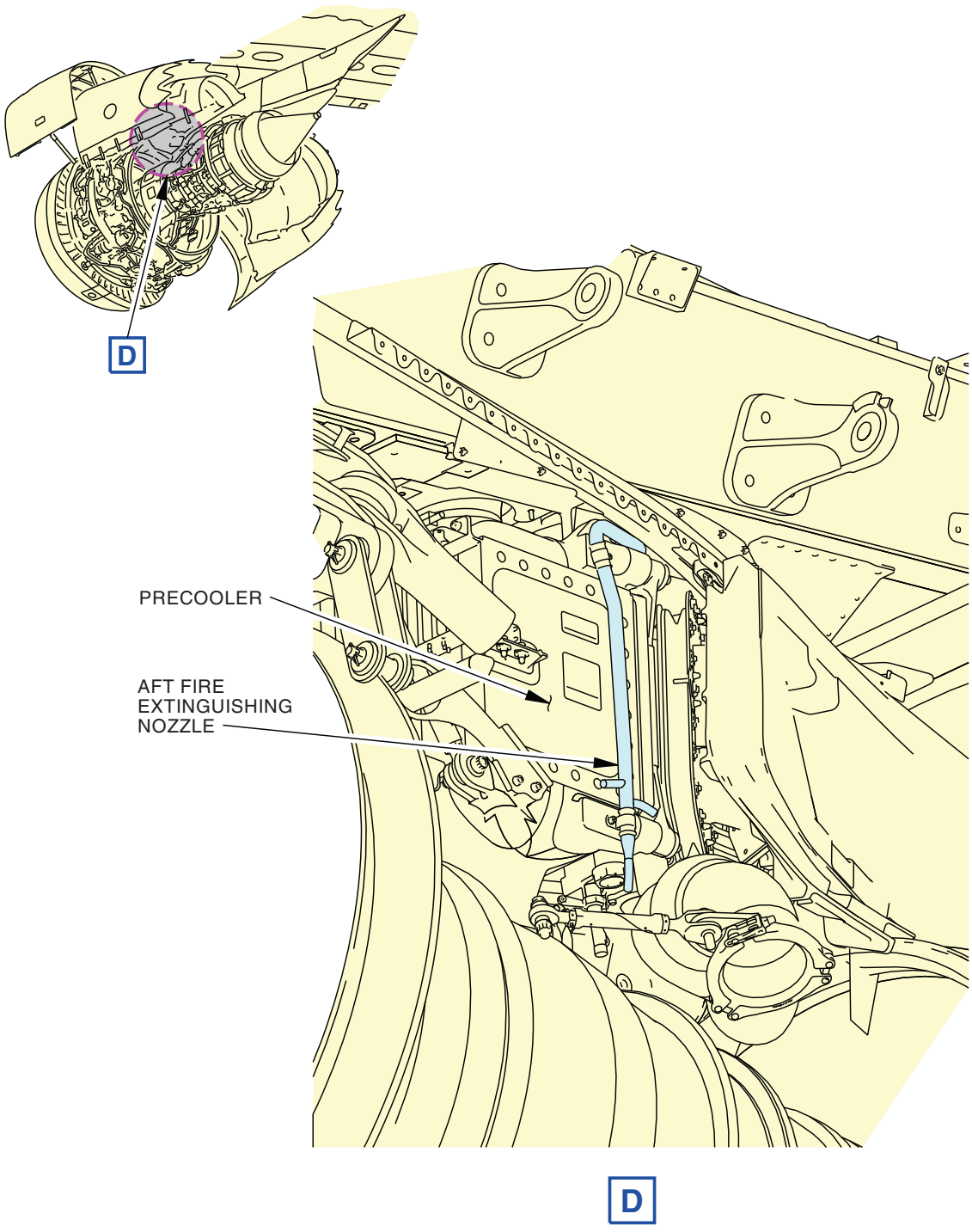
Engine Fire Extinguishing Bottle Inspection
Figure 601/26-21-01-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Fire Extinguishing Bottle Inspection
Figure 601/26-21-01-990-802 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2524344 S0000595728_V1

Engine Fire Extinguishing Bottle Inspection
Figure 601/26-21-01-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE FIRE EXTINGUISHER BOTTLE SQUIB - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Engine fire extinguisher bottle squib removal

(2) Engine fire extinguisher bottle squib installation.

TASK 26-21-02-000-801

2. Engine Fire Extinguisher Bottle Squib Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The squibs are installed on the engine fire extinguishing bottles, located in the main landing gear wheel well. The removal procedure is the same for each squib, but each squib has a unique location and they are not interchangeable.

(2) This task gives instructions to remove the engine fire extinguisher bottle squib.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1168	Cap - Shorting or Faraday Cap

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Removal

SUBTASK 26-21-02-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-02-860-005

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Before the squib [2] or squib [3] is touched, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

F. Engine Fire Extinguisher Bottle Squib Removal

SUBTASK 26-21-02-020-005

(1) Disconnect the applicable electrical connector [1] from squib [2] or squib [3].

NOTE: The squibs are not interchangeable and each squib has a unique location on the discharge port.

SUBTASK 26-21-02-020-003

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(2) Install the shorting or faraday cap, STD-1168, on the squib [2] or squib [3].

NOTE: The shorting caps are attached to the fire extinguishing bottles. These shorting caps prevent the squibs from possible activation due to static electricity.

SUBTASK 26-21-02-020-002

(3) Remove the lockwire between the squib [2] or squib [3] and discharge outlet.

SUBTASK 26-21-02-020-004

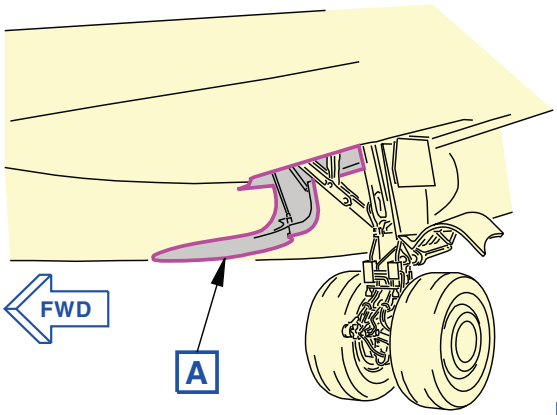
(4) Remove the squib [2] or squib [3] from the discharge head.

SUBTASK 26-21-02-910-001

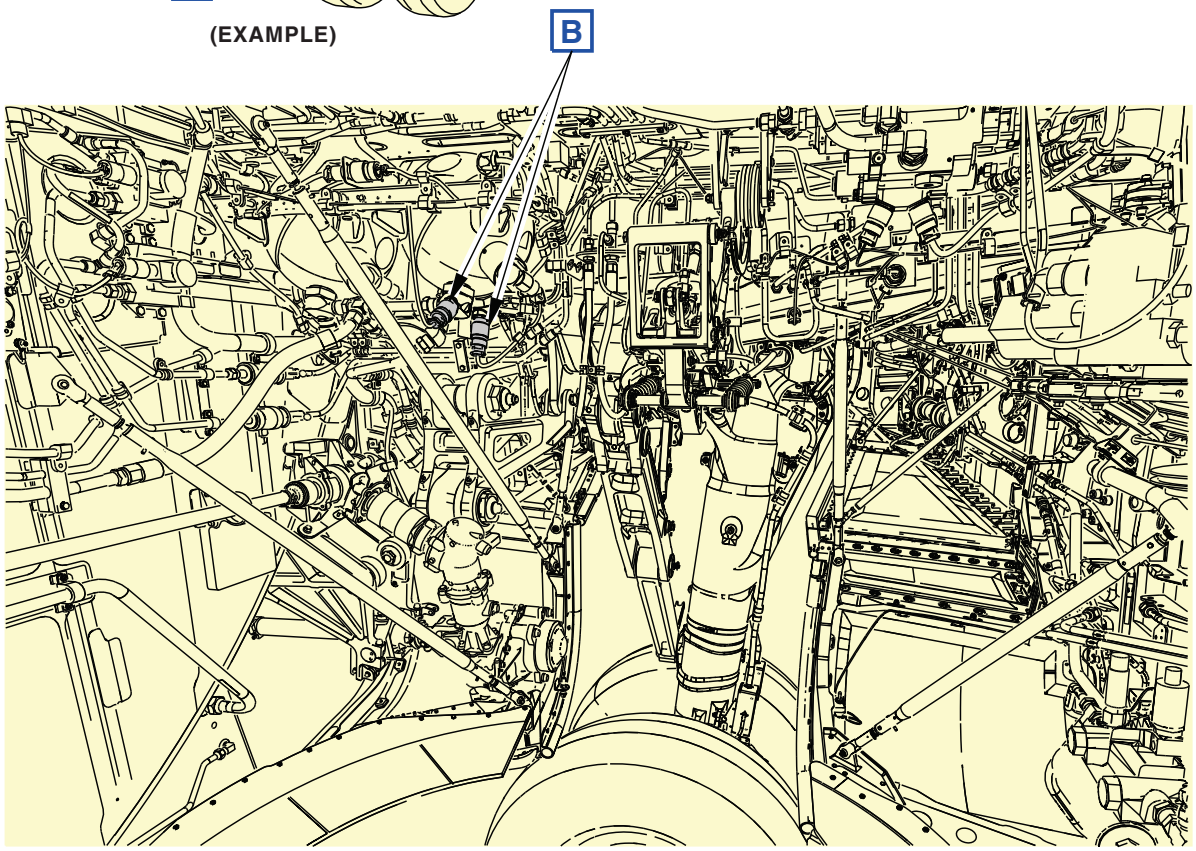
(5) Discard the packing [4] from the removed squibs.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)



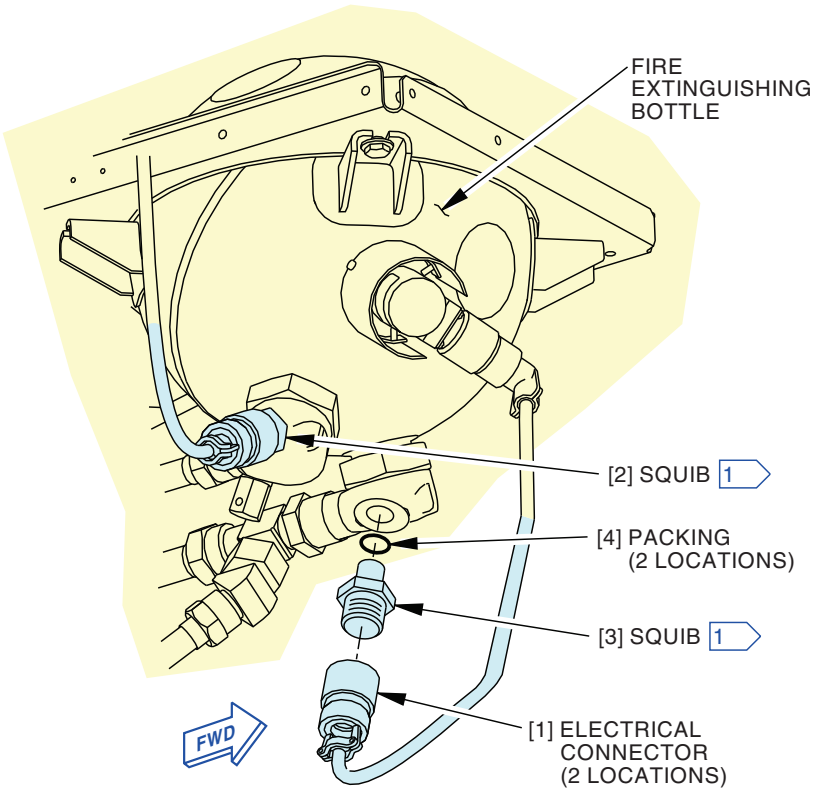
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)
(EXAMPLE)

A

2412171 S00061530078_V4

Engine Fire Extinguishing Bottle Squib Installation
Figure 401/26-21-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ENGINE FIRE EXTINGUISHING BOTTLE (EXAMPLE)

B

1 EACH SQUIB HAS A UNIQUE LOCATION AND THEY ARE NOT INTERCHANGEABLE.

2412172 S00061530079_V3

Engine Fire Extinguishing Bottle Squib Installation
Figure 401/26-21-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-21-02-400-801

3. Engine Fire Extinguisher Bottle Squib Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The squibs are installed on the engine fire extinguishing bottles, found in the left main wheel well. The installation procedure is the same for each squib, but each squib has a unique location and they are not interchangeable.

(2) This task gives instructions to install the engine fire extinguisher bottle squib.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
STD-1079	Resistor - 10K Ohm or Greater
STD-1168	Cap - Shorting or Faraday Cap

D. Consumable Materials

Reference	Description	Specification
D00652	Lubricant - O-Ring - Parker Super-O-Lube	
D50287	Lubricant - O-Ring, Drilube 822 (MMS-N306A)	
G02392 [P05-262]	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8121 mm) Diameter	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Squib	26-21-01-01-100	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Squib	26-21-01-01-095	PTF ALL
4	Packing	26-21-01-01-115	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

G. Prepare for the Installation

SUBTASK 26-21-02-860-002

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

H. Engine Fire Extinguisher Bottle Squib Installation

SUBTASK 26-21-02-420-001

(1) Do these steps to install squib [2] or squib [3] into the discharge port of the fire extinguishing bottle:

NOTE: The squibs are not interchangeable and each squib has a unique location on the discharge port.

WARNING

DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Before the squib [2] or squib [3] is touched, do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

(b) Lubricate the new packing [4] with O-ring lubricant, D50287, or Parker Super-O-Lube lubricant, D00652.

(c) Put the packing [4] on the squib [2] or squib [3].

(d) Install the squib [2] or squib [3] into the fire bottle.

1) Tighten the squib [2] or squib [3] to 90 in-lb (10.2 N·m) - 100 in-lb (11.3 N·m).

2) Install MS20995C32 lockwire, G02392 [P05-262], between the squib [2] or squib [3] and discharge outlet.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-02-700-001

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(2) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-21-02-420-003

(3) Remove the shorting or faraday cap, STD-1168, from the squib [2] or squib [3].

SUBTASK 26-21-02-420-002

CAUTION MAKE SURE THAT THE SQUIB PINS CORRECTLY ENGAGE THE ELECTRICAL CONNECTOR. IF NOT, DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(4) Connect the electrical connector [1] to the squib [2] or squib [3].

I. Engine Fire Extinguisher Squib Installation Test

SUBTASK 26-21-02-860-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	20	C00297	FIRE PROTECTION EXTINGUISHERS RIGHT
B	22	C00296	FIRE PROTECTION EXTINGUISHERS LEFT
B	23	C01022	FIRE PROTECTION EXTINGUISHERS ALTN R
B	24	C01021	FIRE PROTECTION EXTINGUISHERS ALTN L

SUBTASK 26-21-02-740-001

(2) Put and hold the EXT TEST switch, on the P8-1 panel, to the 1 position.

(a) Make sure that the L and R lights come on.

NOTE: The APU light will also come on.

SUBTASK 26-21-02-740-002

(3) Put and hold the EXT TEST switch, on the P8-1 panel, to the 2 position.

(a) Make sure that the L and R lights come on.

NOTE: The APU light will also come on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-21-02-710-001

(4) Release the EXT TEST switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FIRE EXTINGUISHING - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) A test of the APU fire extinguishing bottle squib circuit

(2) A test of the APU fire extinguishing bottle pressure switch circuit

(3) A shutdown test of the APU fire switch system.

TASK 26-22-00-730-801

2. APU Fire Extinguishing Bottle Squib Circuit Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-1663	Test Equipment - Fire Extinguishing System Part #: F80229-69 Supplier: 81205 Opt Part #: F80229-15 Supplier: 81205
SPL-10039	Cable Assy - Fire Extinguishing System Test Equipment Part #: F80229-20 Supplier: 81205 Part #: F80229-48 Supplier: 81205 Part #: F80229-59 Supplier: 81205 Part #: F80229-61 Supplier: 81205

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(Continued)

Reference	Description
SPL-10040	Shorting Plug - Fire Extinguishing System Test Equipment Part #: F80229-52 Supplier: 81205
STD-1079	Resistor - 10K Ohm or Greater

C. Consumable Materials

Reference	Description	Specification
G50414	Lockwire - MS20995CY15, Copper - 0.015 Inch (0.381 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 26-22-00-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

SUBTASK 26-22-00-860-002

(2) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 26-22-00-020-004

WARNING

DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Before you touch the squib, do these tasks for devices that are sensitive to electrostatic discharge:

- ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802
- ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-22-00-020-007

(4) Disconnect electrical connector D594 from the squib.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-22-00-020-005

WARNING MAKE SURE THAT YOU PUT A SHORTING PLUG ON THE SQUIB. IF YOU DO NOT PUT A SHORTING PLUG ON THE SQUIB, THE FIRE EXTINGUISHING BOTTLE CAN RELEASE ITS CONTENTS SUDDENLY. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(5) Install a shorting plug, SPL-10040, to the squib.

SUBTASK 26-22-00-860-003

(6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

SUBTASK 26-22-00-480-001

(7) Set the toggle switch on Fire System Test Equipment, SPL-1663, to the ON position.

SUBTASK 26-22-00-760-001

(8) Measure the resistance (RL) of the test box with a digital/analog multimeter, COM-1793, connected to the voltmeter jacks.

(a) Write down the value of RL in Table 501, Table 502, and Table 503.

SUBTASK 26-22-00-480-002

(9) Set the toggle switch, on the test box, to the OFF position.

SUBTASK 26-22-00-480-003

(10) Set the multimeter to read DC volts.

SUBTASK 26-22-00-480-004

(11) Connect the applicable cable assembly, SPL-10039, to the connector on the test box.

NOTE: Look at the placard on the test box lid to determine the applicable adapter.

G. APU Fire Extinguishing Bottle Squib Circuit Test

SUBTASK 26-22-00-730-001

(1) Push the manual override button and pull the Auxiliary Power Unit (APU) fire handle, on the P8-1 panel.

SUBTASK 26-22-00-480-005

(2) Connect the adapter cable to the squib electrical connector D594.

SUBTASK 26-22-00-860-004

(3) Turn the APU fire handle and hold it in the CW (Clockwise) direction.

(a) Make sure that the BOTTLE DISCHARGE light on the test box comes on.

(b) Write down the voltage shown on the multimeter as (V_1) in Table 501.

(c) Set the switch on the test box to the ON position.

(d) Write down the voltage shown on the multimeter as (V_2) in Table 501.

SUBTASK 26-22-00-860-005

(4) Release the fire handle.

(a) Make sure that the handle moves toward the center 10 degrees (0.2 rad).

(b) Make sure that the multimeter shows 0 (±2) V (Volts).

(c) Set the switch on the test box to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-22-00-760-002

(5) Calculate the resistance R_C of the squib discharge circuit using this formula:

$R_C = RL (V_1/V_2) - RL$.

(a) Make sure that R_C is less than or equal to 5 ohms.

NOTE: If R_C is more than 5 ohms, the circuit can not supply enough current to fire the squib cartridge.

Table 501/26-22-00-993-801 APU Fire Bottle Squib Circuit Test Result

CONNECTOR	RL	V1	V2	Rc
D594				

SUBTASK 26-22-00-860-006

(6) Turn the APU fire handle and hold it in the CCW (Counterclockwise) direction.

(a) Make sure that the BOTTLE DISCHARGE light on the test box comes on.

(b) Write down the voltage shown on the multimeter as (V_1) in Table 502.

(c) Set the switch on the test box to the ON position.

(d) Write down the voltage shown on the multimeter as (V_2) in Table 502.

SUBTASK 26-22-00-860-007

(7) Release the fire handle.

(a) Make sure that the handle moves toward the center 10 degrees (0.2 rad).

(b) Make sure that the multimeter shows 0 (±2) V.

(c) Set the switch on the test box to the OFF position.

SUBTASK 26-22-00-760-003

(8) Calculate the resistance R_C of the squib discharge circuit using this formula:

$R_C = RL (V_1/V_2) - RL$.

(a) Make sure that R_C is less than or equal to 5 ohms.

NOTE: If R_C is more than 5 ohms, the circuit can not supply enough current to fire the squib cartridge.

Table 502/26-22-00-993-802 APU Fire Bottle Squib Circuit Test Result

CONNECTOR	RL	V1	V2	Rc
D594				

SUBTASK 26-22-00-860-022

(9) Return the APU fire handle to its usual (stowed) position.

H. APU Fire Extinguishing Bottle Squib Circuit Check with the APU Fire Control Panel

SUBTASK 26-22-00-730-002

(1) Pull the handle, on the P28 APU fire control panel, in the right main gear wheel well.

SUBTASK 26-22-00-860-008

(2) Push and hold the BOTTLE DISCHARGE toggle switch.

(a) Make sure that the BOTTLE DISCHARGE light on the test box comes on.

(b) Write down the voltage shown on the multimeter as (V_1) in Table 503.

(c) Set the switch on the test box to the ON position.

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(d) Write down the voltage shown on the multimeter as (V_2) in Table 503.

SUBTASK 26-22-00-860-009

(3) Release the BOTTLE DISCHARGE toggle switch.

(a) Make sure that the multimeter shows 0 (±2) V.

SUBTASK 26-22-00-760-004

(4) Calculate the resistance R_C of the squib discharge circuit using this formula:

$R_C = RL (V_1/V_2) - RL$.

(a) Make sure that R_C is less than or equal to 5 ohms.

NOTE: If R_C is more than 5 ohms, the circuit can not supply enough current to fire the squib cartridge.

Table 503/26-22-00-993-803 APU Fire Bottle Squib Circuit Test Result

CONNECTOR	RL	V1	V2	Rc
D594				

SUBTASK 26-22-00-720-001

(5) Put the handle, on the P28 APU fire control panel, in the usual position.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 26-22-00-080-003

(1) Disconnect the cable assembly, SPL-10039, from the test box.

SUBTASK 26-22-00-080-001

(2) Disconnect the Fire System Test Equipment, SPL-1663 test box, from the squib connector.

SUBTASK 26-22-00-430-001

(3) Install single strand copper MS20995CY15 lockwire, G50414, on the bottle discharge switch, on the P28 APU fire control panel.

(a) Make sure that the lockwire is positioned at the bottom of the hole in the guard.

SUBTASK 26-22-00-865-001

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

SUBTASK 26-22-00-700-001

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(5) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

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(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-22-00-020-008

(6) Remove the shorting plug, SPL-10040, from the squib.

SUBTASK 26-22-00-430-002

(7) Connect the electrical connector D594 to the squib.

SUBTASK 26-22-00-410-003

(8) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 26-22-00-865-002

(9) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

SUBTASK 26-22-00-710-004

(10) Set the Engines EXT TEST switch to 1.

(a) Make sure that the APU light comes on.

NOTE: The L and R lights should also come on.

SUBTASK 26-22-00-710-005

(11) Set the Engines EXT TEST switch to 2.

(a) Make sure that the APU light comes on.

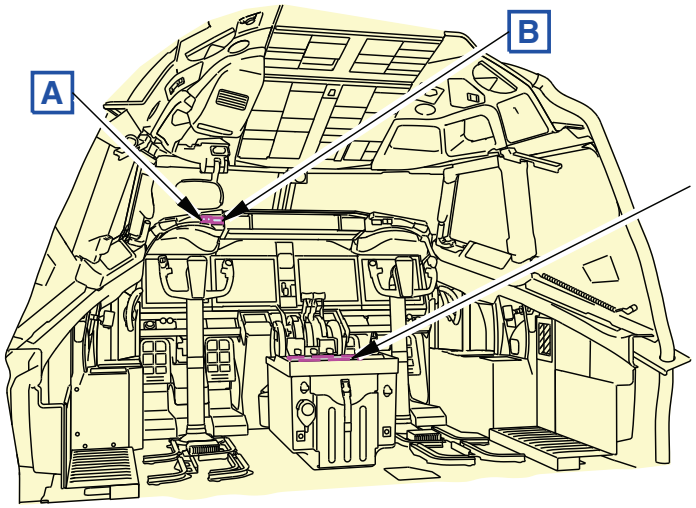
NOTE: The L and R lights should also come on.

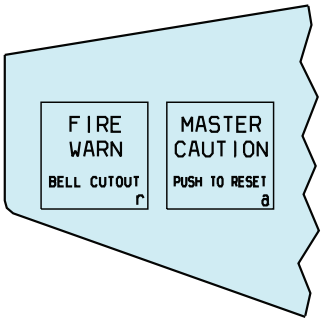
SUBTASK 26-22-00-710-006

(12) Release the Engines EXT TEST switch.

END OF TASK

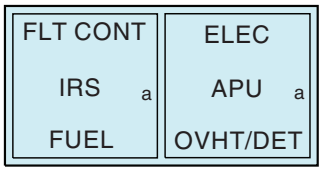
ECCN 9E991 BOEING PROPRIETARY - See title page for details





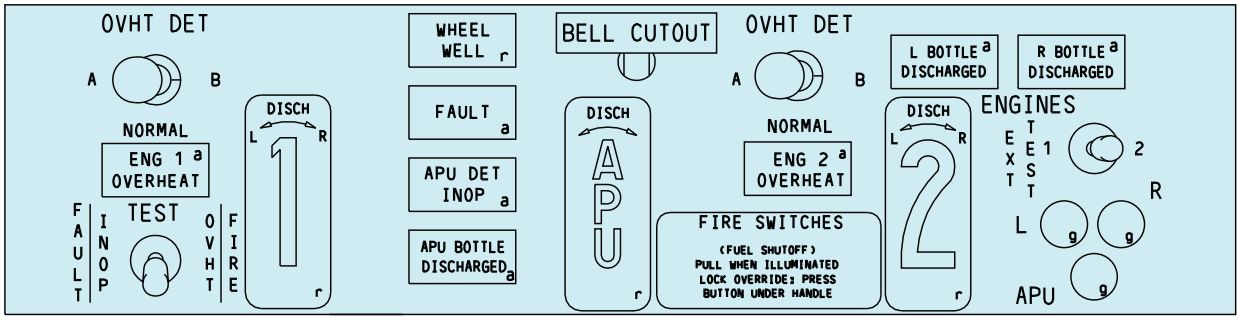
CAPTAIN'S FIRE WARNING SWITCH (P7) (FIRST OFFICER'S IS OPPOSITE)

A



CAPTAIN'S WARNING LIGHTS (P7)

B



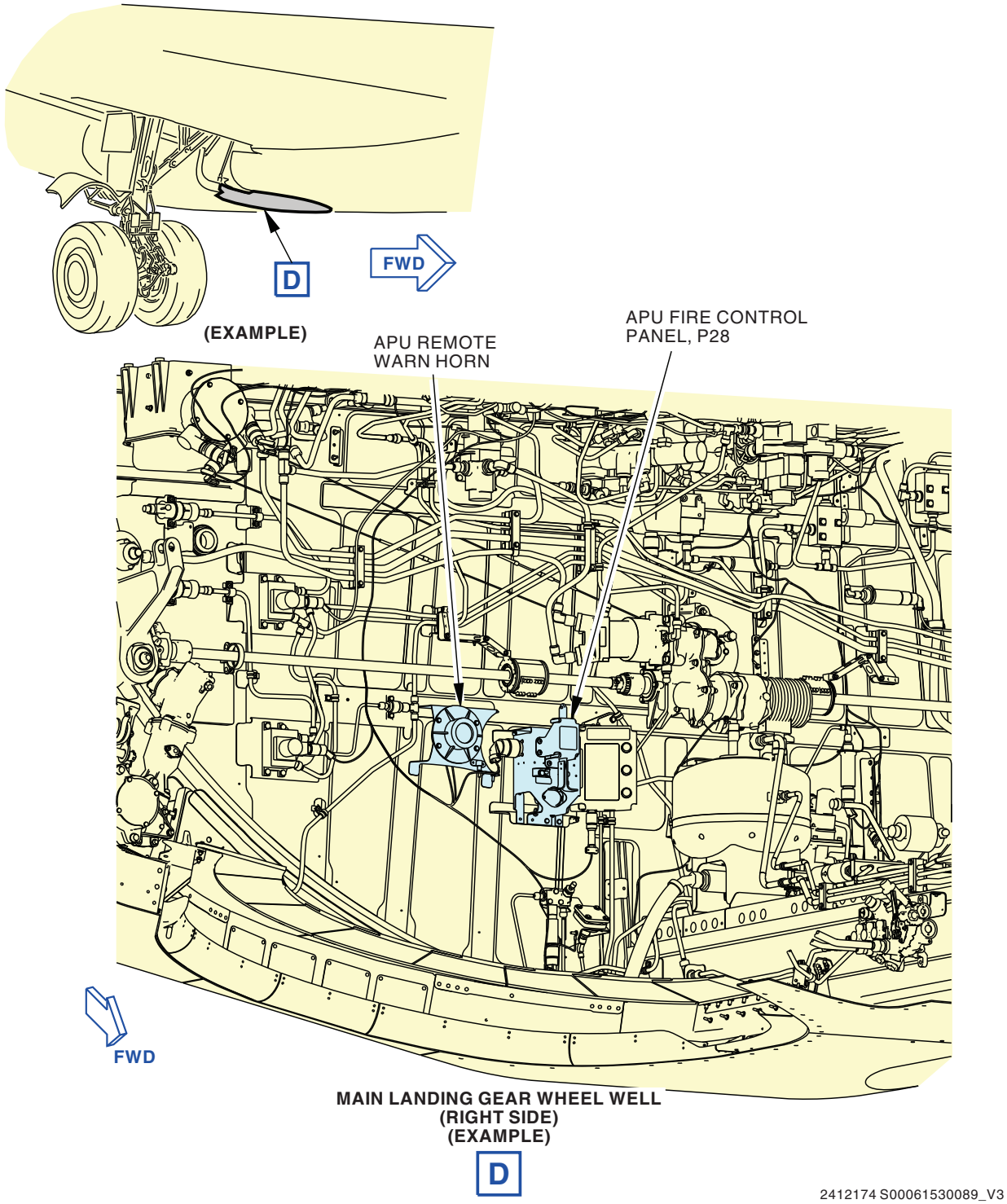
FIRE CONTROL PANEL (P8-1)

C

2412173 S00061530088_V1

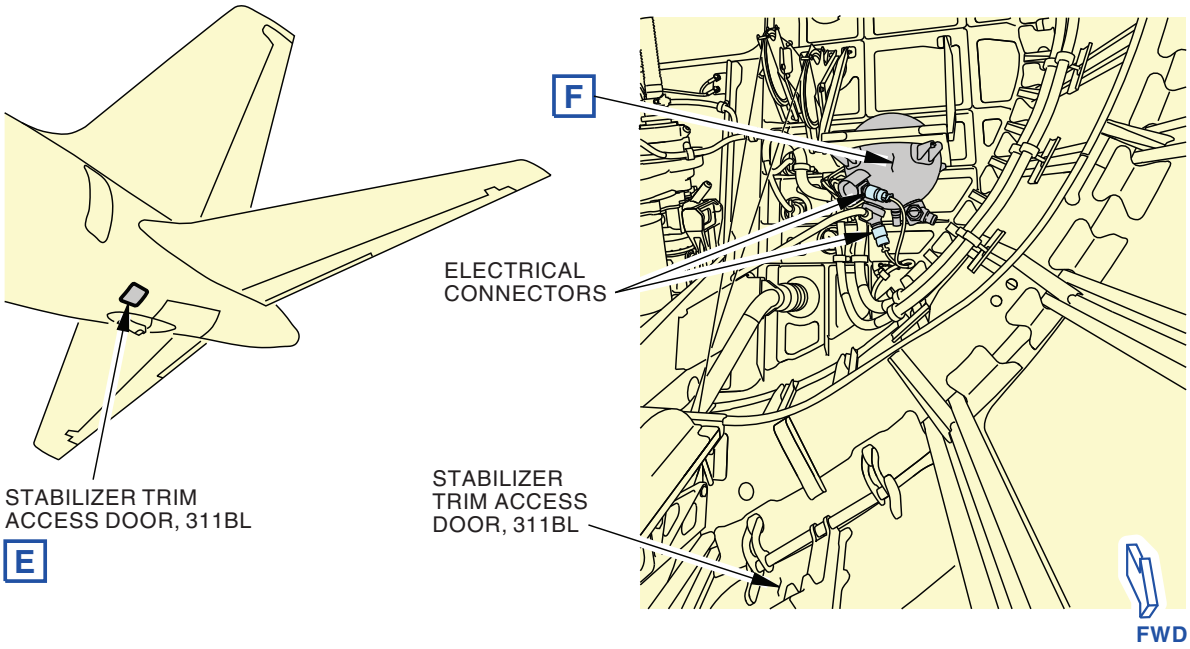
APU Fire Detection Test
Figure 501/26-22-00-990-802 (Sheet 1 of 3)

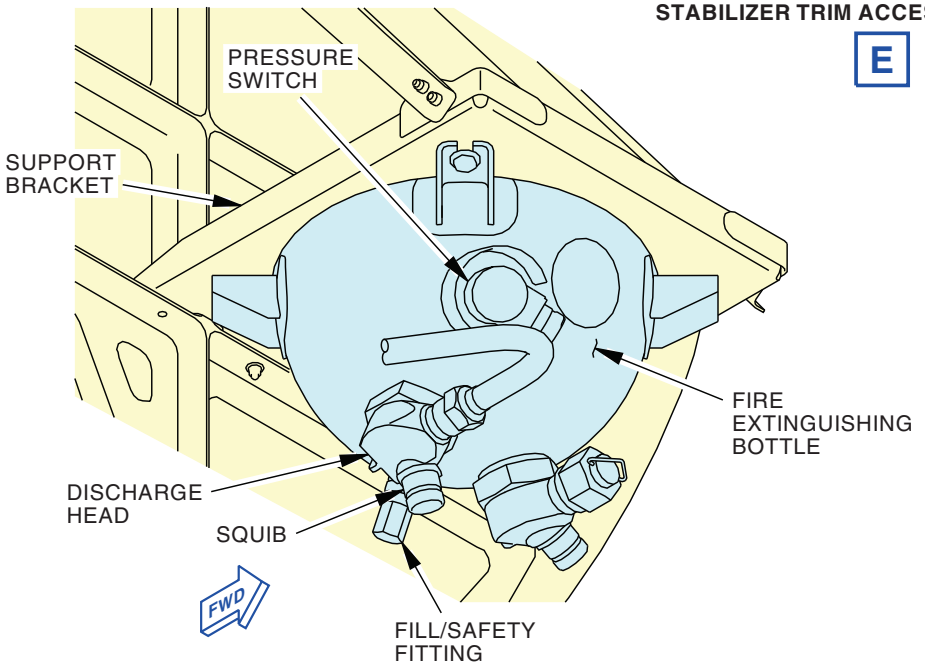
ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fire Detection Test
Figure 501/26-22-00-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2830784 S0000657495_V1

APU Fire Detection Test
Figure 501/26-22-00-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-22-00-730-802

3. APU Fire Extinguishing Bottle Pressure Switch Circuit Test

(Figure 501)

A. General

(1) The pressure switch test should be performed every time the fire control module or pressure switch is disconnected from the airplane wiring.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left

C. APU Fire Extinguishing Bottle Pressure Switch Test

SUBTASK 26-22-00-020-006

(1) Disconnect electrical connector D1176 from the pressure switch on the APU fire bottle.

SUBTASK 26-22-00-730-003

(2) Do a test of the APU fire bottle pressure switch circuit.

(a) Connect a jumper wire between pins 1 and 2 on D1176.

1) Make sure the APU BOTTLE DISCHARGED light on the fire control module comes on.

(b) Remove the jumper from connector D1176.

SUBTASK 26-22-00-420-003

(3) Connect D1176 to the pressure switch on the APU fire bottle.

——— END OF TASK ———

TASK 26-22-00-720-801

4. APU Fire Switch System Shutdown Test

(Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure checks that fuel to the Auxiliary Power Unit (APU) is shut off and the APU generator is disconnected from the airplane power when the fire handle is pulled. It should be performed when the Engine and P8 APU fire control panel or P28 APU fire control panel is removed from the airplane.

B. References

Reference	Title
24-22-00-860-805	Supply APU Generator Power (P/B 201)

C. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
315	APU Compartment - Left
316	APU Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. APU Fire Handle Fuel System Shutdown Test

SUBTASK 26-22-00-860-010

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

SUBTASK 26-22-00-860-011

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	21	C00396	FIRE PROT DETECTION MA WRN & CONT
A	22	C00407	FIRE PROTECTION DETECTION ENG 2
A	23	C00403	FIRE PROTECTION DETECTION APU
A	24	C00405	FIRE PROTECTION DETECTION ENG 1
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	12	C00132	MASTER CAUTION ANNUNCIATOR BUS 1
B	13	C00131	MASTER CAUTION ANNUNCIATOR BAT
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
D	12	C00310	INDICATOR MASTER DIM BAT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
D	15	C01401	LANDING GEAR AIR/GND RELAY
E	11	C00313	INDICATOR MASTER DIM SECT 1
E	12	C00314	INDICATOR MASTER DIM SECT 2
E	13	C00315	INDICATOR MASTER DIM SECT 3
E	14	C00316	INDICATOR MASTER DIM SECT 4
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 26-22-00-860-012

(3) Pull the APU fire handle, on the P8 panel.

(a) Make sure that the fire handle does not extend.

SUBTASK 26-22-00-860-013

(4) Do a test of the APU fire handle fuel shutdown circuit in the flight compartment.

(a) Set the APU master switch, on the P5 panel, to the ON position.

NOTE: The igniter plugs will fire three times. Do not start the APU.

1) Make sure that the APU fuel supply valve, in the left main gear wheel well, is opened (Figure 502).

(b) Push the manual override button and pull the APU fire handle, on the P8 panel.

1) Make sure that the APU fuel supply valve, in the left main gear wheel well, is closed.

(c) Push in the APU fire handle, on the P8 panel, to the usual position.

(d) Set the APU master switch, on the P5 panel, to the OFF position.

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SUBTASK 26-22-00-860-014

(5) Use the P28 APU fire control panel to check the APU fire handle fuel shutoff circuit.

(a) Set the APU master switch, on the P5 panel, to the ON position.

1) Make sure that the APU fuel supply valve, in the left main gear wheel well, is opened.

(b) Pull the handle, on the P28 APU fire control panel, in the right main gear wheel well.

1) Make sure that the APU fuel supply valve, in the left main gear wheel well, is closed.

(c) Push the handle, on the P28 APU fire control panel, to the usual position.

(d) Set the APU master switch, on the P5 panel, to the OFF position.

E. P8 APU Panel Fire Handle Generator Shutdown Test

SUBTASK 26-22-00-860-015

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

SUBTASK 26-22-00-860-016

(2) Do this task: Supply APU Generator Power, TASK 24-22-00-860-805.

SUBTASK 26-22-00-860-017

(3) Set the AC meter select switch, on the P5-13 panel, to APU GEN.

(a) Make sure that the voltage reads 110-120 Alternating Current Volts (VAC).

(b) Make sure that the frequency reads 390-410.

SUBTASK 26-22-00-860-018

CAUTION

DO NOT TURN THE FIRE HANDLE AFTER YOU PULLED IT UP. YOU WILL CAUSE THE FIRE BOTTLES TO RELEASE THEIR CONTENTS IF YOU TURN THE HANDLE.

(4) Push the manual override button and pull the APU fire handle, on the P8 panel.

NOTE: The APU will do an emergency shutdown. When the APU does an emergency shutdown, it will not do the cool-down cycle.

(a) Make sure that the voltage and frequency read 0 after 7 seconds.

SUBTASK 26-22-00-860-019

(5) Set the APU and fire protection circuits to the usual positions.

(a) Set the APU master switch to the OFF position.

(b) Push the APU fire handle to its usual position.

(c) Set the APU master switch to the ON position.

(d) After 30 seconds, make sure that the APU FAULT light goes off.

(e) Make sure that the APU annunciator light goes off.

(f) Push and release the APU annunciator light.

(g) Make sure that the L and R MASTER CAUTION lights go off.

NOTE: The L and R MASTER CAUTION lights stay on if the airplane finds other system faults.

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(h) Set the APU master switch to the OFF position.

F. P28 APU Fire Control Panel Fire Handle Generator Shutdown Test

SUBTASK 26-22-00-860-027

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

SUBTASK 26-22-00-860-031

(2) Do this task: Supply APU Generator Power, TASK 24-22-00-860-805.

SUBTASK 26-22-00-860-029

(3) Set the AC meter select switch, on the P5-13 panel, to APU GEN.

(a) Make sure that the voltage reads 110-120 VAC.

(b) Make sure that the frequency reads 390-410.

SUBTASK 26-22-00-860-025

(4) Pull the handle, on the P28 APU fire control panel, in the right main gear wheel well.

NOTE: The APU will do an emergency shutdown. When the APU does an emergency shutdown, it will not do the cool-down cycle.

(a) Make sure that the voltage and frequency read 0 after 7 seconds.

SUBTASK 26-22-00-860-030

(5) Set the APU and fire protection circuits to the usual positions.

(a) Set the APU master switch to the OFF position.

(b) Push the APU fire handle to its usual position.

(c) Set the APU master switch to the ON position.

(d) After 30 seconds, make sure that the APU FAULT light goes off.

(e) Make sure that the APU annunciator light goes off.

(f) Push and release the APU annunciator light.

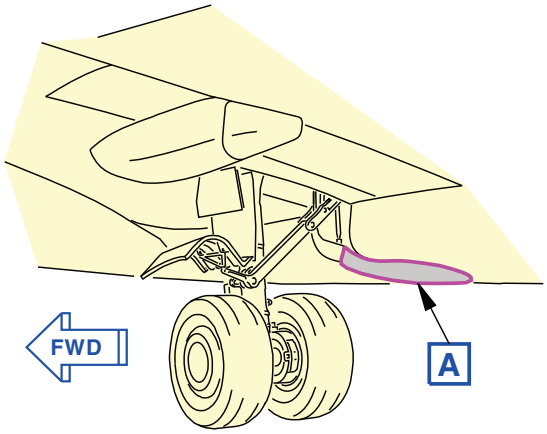
(g) Make sure that the L and R MASTER CAUTION lights go off.

NOTE: The L and R MASTER CAUTION lights stay on if the airplane finds other system faults.

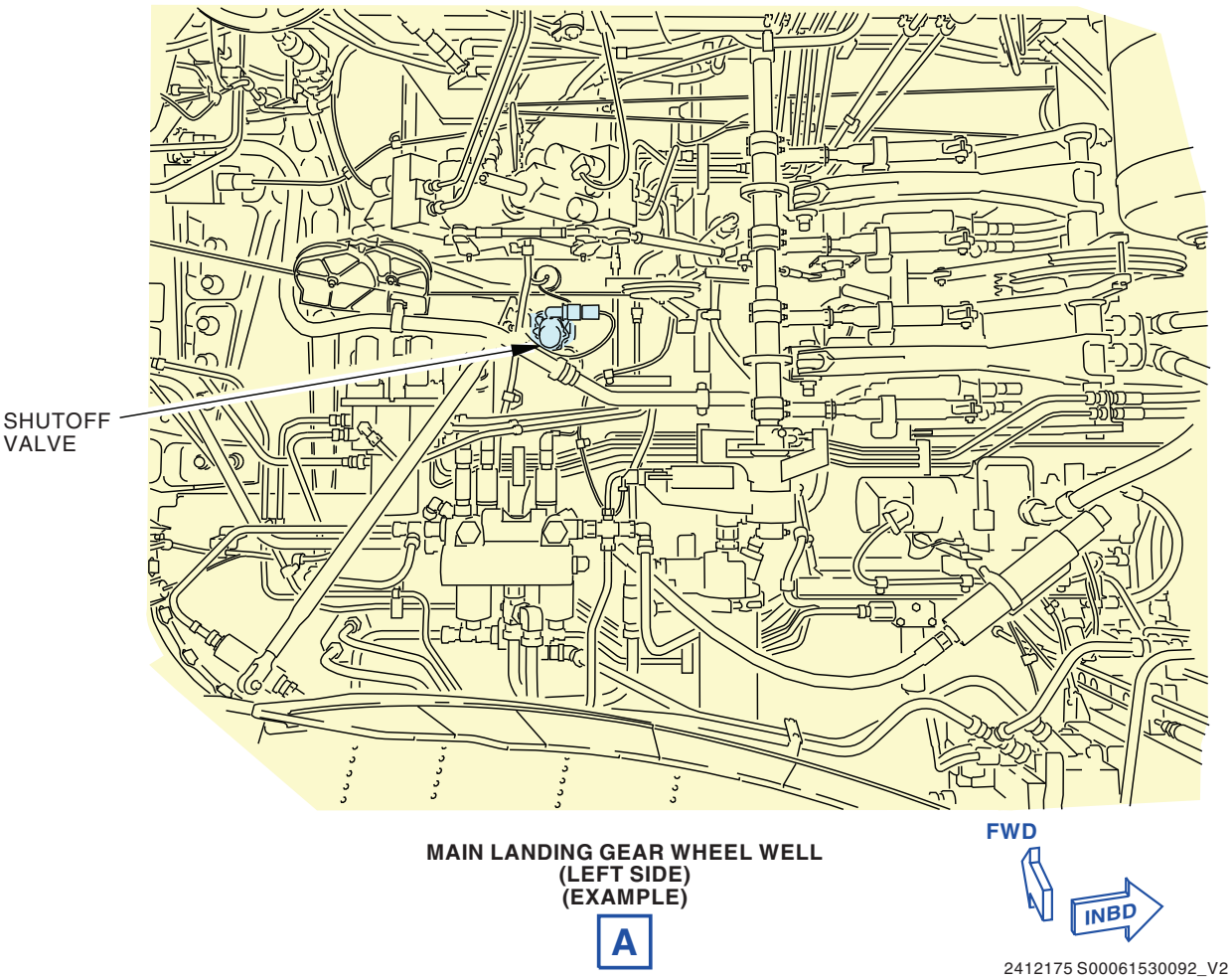
(h) Set the APU master switch to the OFF position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)



MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)
(EXAMPLE)

2412175 S00061530092_V2

APU Fuel Shutoff Valve Actuator Test
Figure 502/26-22-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FIRE EXTINGUISHING BOTTLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) APU fire extinguishing bottle removal

(2) APU fire extinguishing bottle installation.

B. The APU fire extinguishing bottle is installed forward of the APU fire wall. You can get access to the bottle through the Aft Unpressurized Compartment Access Door, 311BL.

TASK 26-22-01-000-801

2. APU Fire Extinguishing Bottle Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to remove the Auxiliary Power Unit (APU) fire extinguishing bottle.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1168	Cap - Shorting or Faraday Cap

D. Location Zones

Zone	Area
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Removal

SUBTASK 26-22-01-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

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SUBTASK 26-22-01-010-001

(2) Open this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. APU Fire Extinguishing Bottle Removal

SUBTASK 26-22-01-020-002

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Before you touch the squib [7], do the procedure for devices that are sensitive to electrostatic discharge.

(a) These are the tasks:

- ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.
- ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-22-01-020-009

(2) Disconnect the electrical connector [1] from the squib [7].

NOTE: The squib is also known as cartridge.

SUBTASK 26-22-01-020-003

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(3) Install the shorting or faraday cap, STD-1168, on the squib [7].

NOTE: The shorting or faraday cap, STD-1168, is attached to the fire extinguishing bottle. The shorting or faraday cap, STD-1168, prevent the squibs from possible activation due to static electricity.

SUBTASK 26-22-01-020-004

(4) Disconnect the electrical connector [2] from the pressure switch.

SUBTASK 26-22-01-020-005

(5) Loosen the nut on the tube assembly [8].

SUBTASK 26-22-01-020-006

(6) Disconnect the tube assembly [8] from the discharge head [11].

SUBTASK 26-22-01-420-005

(7) Install a protective cap [6] on the outlet of the discharge head [11].

SUBTASK 26-22-01-420-006

(8) Install a protective cap on the tube assembly [8].

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SUBTASK 26-22-01-020-007

(9) Remove the bolts [4] and the washers [5] from the mounting lugs.

SUBTASK 26-22-01-020-008

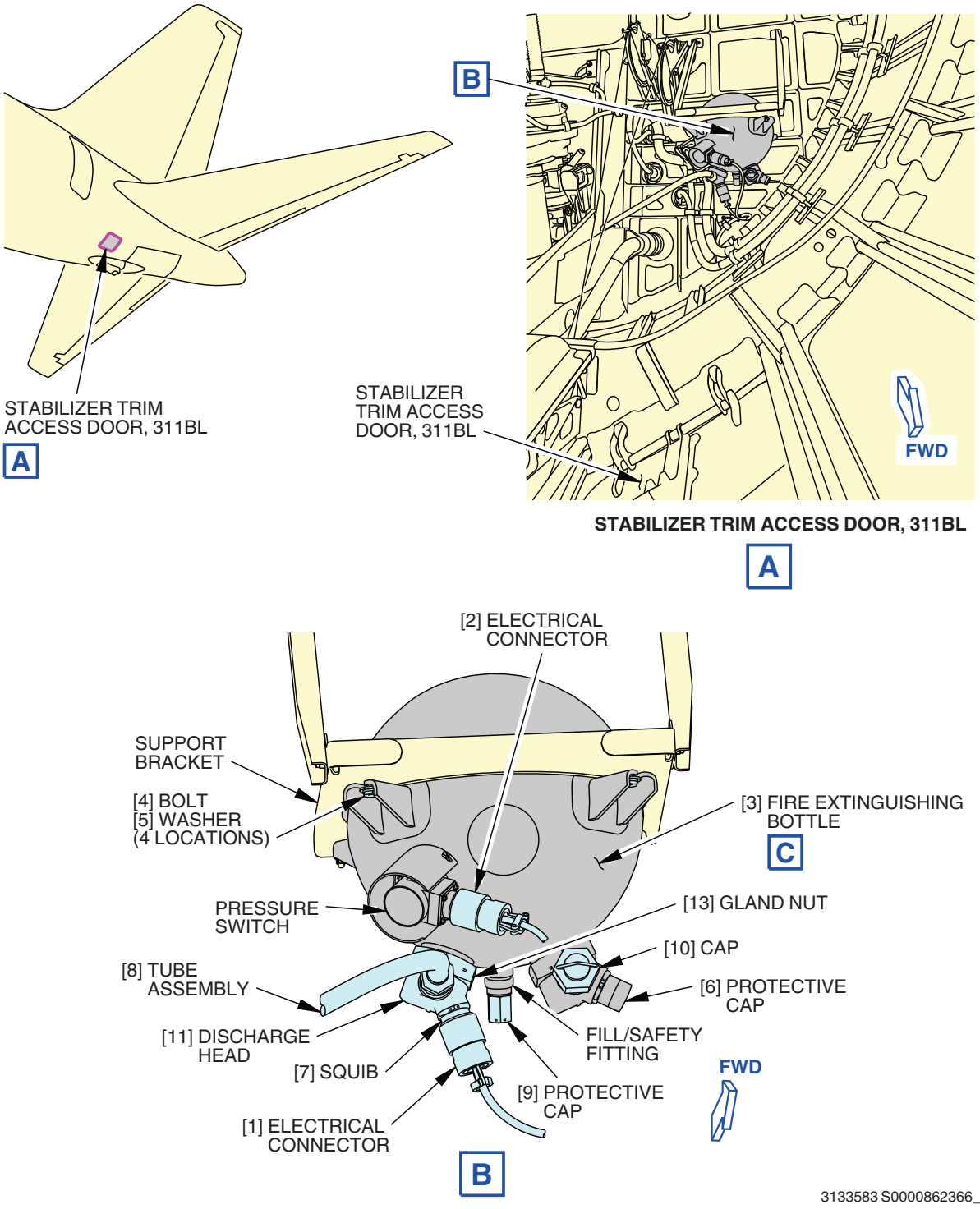
WARNING

BE CAREFUL WHEN YOU MOVE THE FIRE EXTINGUISHING BOTTLE. THE FIRE EXTINGUISHING BOTTLE IS PRESSURIZED AND HAS AN EXPLOSIVE CARTRIDGE AS A COMPONENT. IF THE FIRE EXTINGUISHING BOTTLE RELEASES ITS CONTENTS, IT CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(10) Remove the fire extinguishing bottle [3].

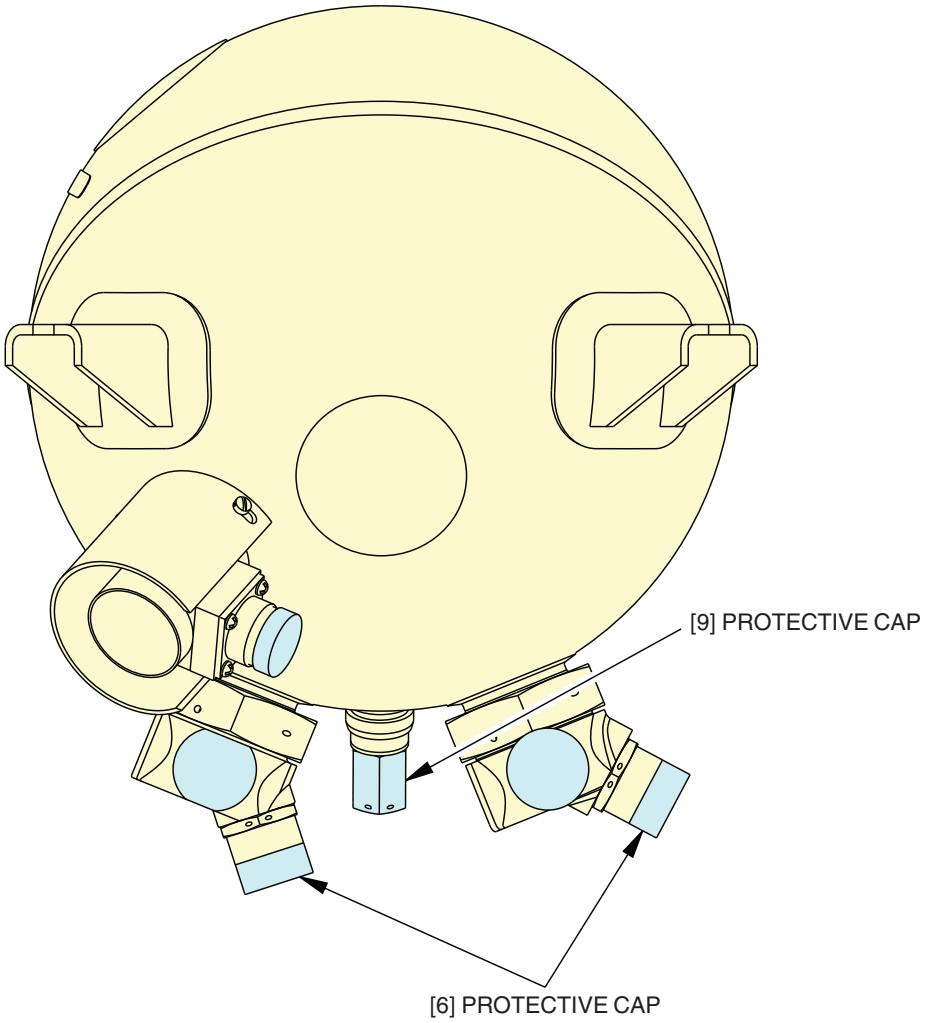
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Auxiliary Power Unit (APU) Fire Extinguishing Bottle Installation
Figure 401/26-22-01-990-807 (Sheet 1 of 2)

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APU ENGINE FIRE EXTINGUISHING BOTTLE

C

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Auxiliary Power Unit (APU) Fire Extinguishing Bottle Installation
Figure 401/26-22-01-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-22-01-400-801

3. APU Fire Extinguishing Bottle Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to install the Auxiliary Power Unit (APU) fire extinguishing bottle.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-20-00-210-802	APU Fire Extinguishing Bottle Check (P/B 601)
26-22-00-730-802	APU Fire Extinguishing Bottle Pressure Switch Circuit Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-4907	Force Gauge, 10 Lbs. with Resolution of 0.1 lb, +/-0.3% or Better Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: LG-5KG Supplier: 92456 Part #: PS-10 Supplier: 1GHM7 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
STD-1079	Resistor - 10K Ohm or Greater
STD-1168	Cap - Shorting or Faraday Cap

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B50073	Alcohol - Isopropyl	ASTM D 770
D50004	Compound - Antiseize	BMS3-28
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50260	Sandpaper - Aluminum Oxide, 220 Grit	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Fire extinguishing bottle	26-22-01-02-080	PTF ALL

F. Location Zones

Zone	Area
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left

G. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Prepare for the Installation

SUBTASK 26-22-01-700-001

(1) Use the force gauge, COM-4907, to weigh the fire extinguishing bottle [3].

NOTE: The bottle assembly includes the charged bottle, discharge heads, squibs and shorting or faraday caps, STD-1168, on the squibs.

(a) Weigh the shorting or faraday caps, STD-1168, including its caps and record as "Value A".

(b) Record the weight marked on the fire extinguishing bottle [3] as "Value B".

(c) Replace the squib protective caps [6] on the squibs [7] with shorting or faraday caps, STD-1168.

(d) Make sure that the shorting or faraday caps, STD-1168, are installed on the squibs [7] during handling, weighing and installation of the fire extinguishing bottle [3]

(e) Remove the remaining protective caps from the fire extinguishing bottle [3].

(f) Weigh the assembly, including the charged bottle, discharge heads [11], squibs [7], shorting or faraday caps, STD-1168 (including shorting plugs' caps), and the loose squib [7] protective caps [6] and record as "Value C'.

(g) Find the measured weight of the fire extinguishing bottle [3] by subtracting "Value A" from "Value C" and record as "Value D" (Value D = Value C - Value A).

SUBTASK 26-22-01-280-001

(2) Replace the fire extinguishing bottle [3] if the measured weight ("Value D") deviates from the weight marked on the fire extinguishing bottle [3] (“Value B”) by more than 0.00 +0.12 / -0.06 lb (0.00 +0.05 / -0.03 kg).

(3) Install the protective caps to the discharge outlets on the fire extinguishing bottle [3].

SUBTASK 26-22-01-860-002

(4) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-22-01-710-002

(5) If the previous bottle was discharged, check the tube assembly [8] for any debris.

SUBTASK 26-22-01-200-001

(6) Do this task: APU Fire Extinguishing Bottle Check, TASK 26-20-00-210-802.

I. APU Fire Extinguishing Bottle Installation

SUBTASK 26-22-01-420-001

(1) Do these steps to install the fire extinguishing bottle [3]:

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WARNING BE CAREFUL WHEN YOU MOVE THE FIRE EXTINGUISHING BOTTLE. THE FIRE EXTINGUISHING BOTTLE IS HIGHLY PRESSURIZED AND HAS AN EXPLOSIVE SQUIB AS A COMPONENT. ACCIDENTAL DISCHARGE OF THE FIRE EXTINGUISHING BOTTLE CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Prepare the fire extinguisher bottle lugs and support bracket mounting surfaces for installation.

1) Remove any old sealant from the bottle lugs and the support bracket.

(b) Use sandpaper, G50260, to sand the surface of the support bracket mounting surface until any surface finish is removed.

1) Avoid removing surface finish outside of the bonding area.

(c) Use cotton wiper, G00034, and alcohol, B50073, to clean the exposed surfaces of the support mounting bracket and the surfaces of the bottle mounting lugs.

(d) Immediately dry the surfaces with a new wiper.

(e) Apply one of these sealants to the surface of the bottle mounting lugs:

- sealant, A50110
- sealant, A50155
- sealant, A50231
- sealant, A50296.

(f) Make sure that sealing compound is applied on the full area of the bottle lug surface.

(g) Install the bolt [4] and washer [5] into the support bracket.

1) Tighten the bolt [4] with your fingers.

(h) Put the fire extinguishing bottle [3] in the mounted position on the support bracket.

1) Slide the bottle lug between previously installed bolt [4], washer [5], and support bracket.

(i) Tighten the bolt [4] to 150 in-lb (16.9 N·m).

(j) Install a second bolt [4] and washer [5] in a cross corner from the first bolt [4].

1) Tighten the bolt [4] with your fingers.

2) Do not install the remaining bolts [4] and washers [5] at this time.

3) Make sure that the bolts [4] are fully engaged in the fire bottle lugs.

NOTE: Movement of the bottle is allowed for installation of the discharge line.

(k) If the outlet of the discharge head [11] does not align with the tube assembly [8], or if the lockwire is not installed between the gland nuts [13], do these steps:

1) Loosen the gland nut [13].

2) Turn the discharge head [11] so that the outlet aligns with the tube assembly [8].

3) Tighten the gland nut [13] to 540 in-lb (61.0 N·m) - 660 in-lb (74.6 N·m)

NOTE: Even if the discharge head is already aligned, it could be necessary to torque the gland nut.

NOTE: It is not necessary to remove the lockwire and torque the gland nut if the gland nut is torqued, and the lockwire installed.

4) Install MS20995C32 lockwire, G01048, from the gland nut [13] to the gland nut [13] on the other discharge head [11].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) Remove the protective cap from the tube assembly [8].

(m) Make sure that MS20995C32 lockwire, G01048, is installed between the squibs [7] and the discharge heads [11].

1) Remove the protective cap from the discharge outlet.

WARNING

YOU MUST WEAR THE APPLICABLE GLOVES WHEN YOU USE HIGH-TEMPERATURE ANTISEIZE COMPOUND. IF YOU GET THE HIGH-TEMPERATURE ANTISEIZE COMPOUND ON YOUR SKIN, REMOVE IT WITH SOAP AND WATER. IF THE HIGH-TEMPERATURE ANTISEIZE COMPOUND GOES THROUGH YOUR SKIN, INJURY TO PERSONNEL CAN OCCUR.

(n) Apply a layer of compound, D50004, to the external threads of the discharge head [11], and to the shoulder of the sleeve that contacts the coupling nut on the tube assembly [8].

NOTE: Do not allow antiseize to enter the discharge head or the tube assembly. Do not apply antiseize to the sealing surface of the sleeve on the tube assembly.

(o) Connect the tube assembly [8] to the discharge head [11].

(p) Remove all unwanted compound.

(q) Install the remaining bolts [4] and washers [5] that attach the fire extinguishing bottle [3].

1) Tighten the bolts [4] snug, then back off 1/2 turn.

2) Make sure that the tube assembly [8] does not bind.

(r) Tighten the tube assembly [8] coupling nut to 360 ±18 in-lb (40.7 ±2.0 N·m).

(s) Tighten all of the bolts [4] to 150 in-lb (16.9 N·m).

1) Make sure that there is sufficient sealant coming out at each edge of the bottle lugs.

2) Make sure that sealant squeeze out is smooth and approximately 0.06 in. (1.52 mm) along the joint.

(t) Install the cap [10] to the outlet of the discharge head that is not used:

1) Remove the protective cap from the discharge outlet.

2) Install the cap [10].

a) Tighten the cap [10] to 450 ±23 in-lb (50.8 ±2.6 N·m).

(u) Install the protective cap [9] to the fill fitting.

1) Secure the protective cap [9] with MS20995C32 lockwire, G01048.

(v) Measure the electrical bonding resistance between the fire extinguishing bottle [3] lugs and support bracket (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the resistance is 0.0025 ohm (2.5 milliohm) or less.

3) If the resistance is more than 0.0025 ohm (2.5 milliohm), clean the bonding surfaces between the fire extinguishing bottle [3] and support bracket with solvent, B00083, and retest.

SUBTASK 26-22-01-710-001

(2) Do these steps to connect the electrical connector [1] to the squib [7] on the fire extinguishing bottle [3]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Before you touch the squib [7], do the procedure for devices that are sensitive to electrostatic discharge, as follows:

- ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.
- ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-22-01-700-002

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(3) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-22-01-420-007

(4) Remove the shorting or faraday cap, STD-1168, from the applicable squib [7].

(a) Make sure that the squib [7] that is not used has a protective cap [6] installed.

SUBTASK 26-22-01-420-002

CAUTION MAKE SURE THAT THE SQUIB PINS CORRECTLY ENGAGE THE ELECTRICAL CONNECTOR. IF NOT, DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(5) Connect the electrical connector [1] to the squib [7].

SUBTASK 26-22-01-420-003

(6) Do this task to test the electrical wiring up to the pressure switch: APU Fire Extinguishing Bottle Pressure Switch Circuit Test, TASK 26-22-00-730-802.

SUBTASK 26-22-01-420-004

(7) Connect the electrical connector [2] to the pressure switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. APU Fire Extinguishing Bottle Installation Test

SUBTASK 26-22-01-860-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 26-22-01-741-001

(2) Set the Engines EXT TEST switch, on the P8-1 panel, to 1 position.

(a) Make sure that the APU light comes on.

NOTE: The L and R engine lights should also come on.

SUBTASK 26-22-01-741-002

(3) Set the Engines EXT TEST switch, on the P8-1 panel, to 2 position.

(a) Make sure that the APU light comes on.

NOTE: The L and R engine lights should also come on.

SUBTASK 26-22-01-741-003

(4) Release the Engines EXT TEST switch.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 26-22-01-010-002

(1) Close this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FIRE EXTINGUISHING BOTTLE - INSPECTION/CHECK

1. General

A. This procedure has this inspection task:

(1) APU Fire Bottle.

TASK 26-22-01-210-801

2. APU Fire Bottle Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
313	Stabilizer Torsion Box Compartment - Left
314	Stabilizer Torsion Box Compartment - Right
315	APU Compartment - Left

B. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
315A	APU Cowl Door

C. Prepare for the Inspection

SUBTASK 26-22-01-010-003

(1) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
315A	APU Cowl Door

D. APU Fire Bottle Inspection

SUBTASK 26-22-01-210-001

(1) Make sure that the fire bottle is not damaged.

SUBTASK 26-22-01-210-002

(2) Make sure that the extinguisher tube is not damaged.

SUBTASK 26-22-01-210-004

(3) Make sure that the drain hole is not blocked and water is not accumulated.

SUBTASK 26-22-01-210-003

(4) Make sure that the tubing outlet is not blocked with unwanted material.

E. Put the Airplane Back to Its Usual Condition

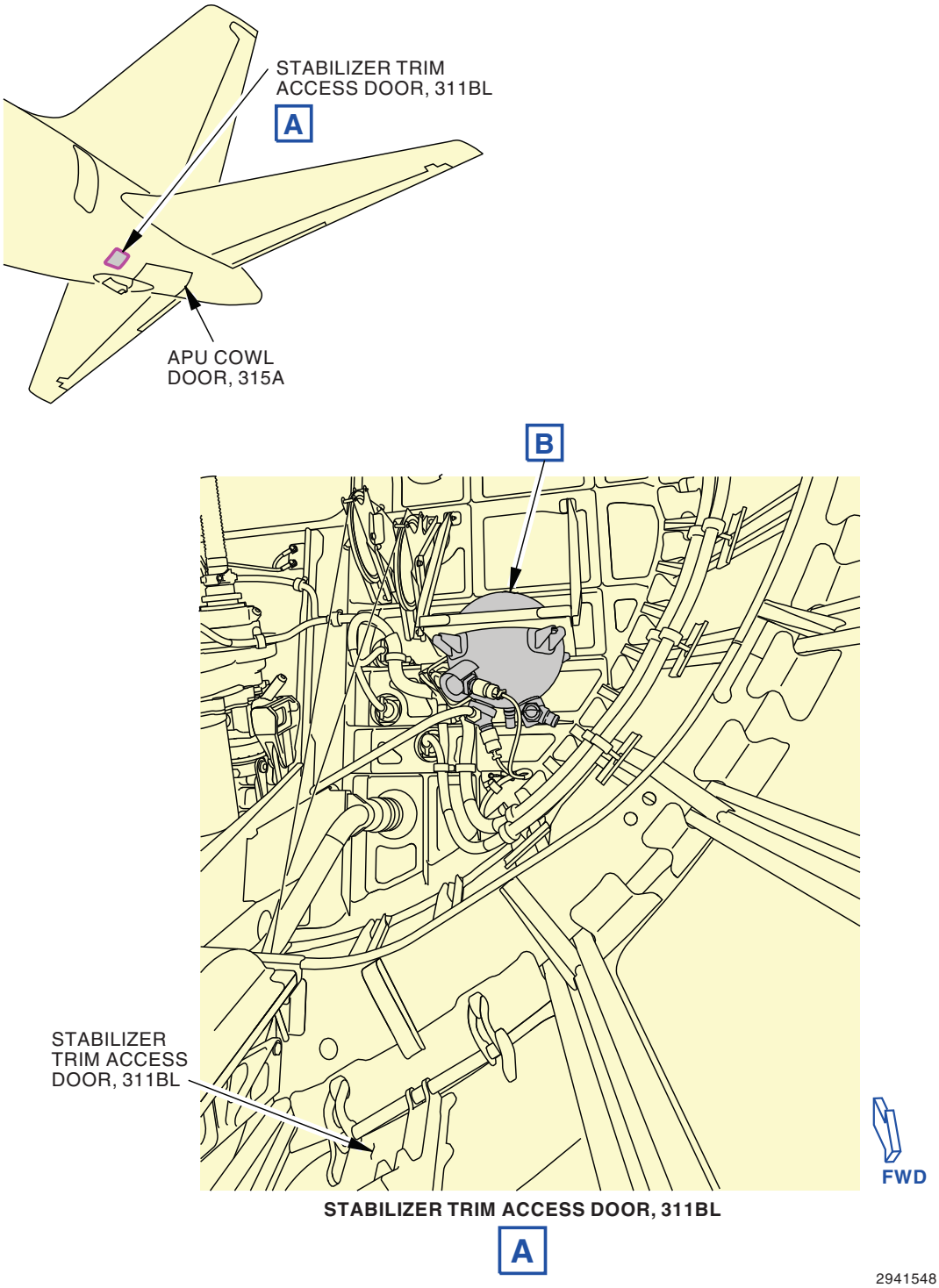
SUBTASK 26-22-01-010-004

(1) Close these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
315A	APU Cowl Door

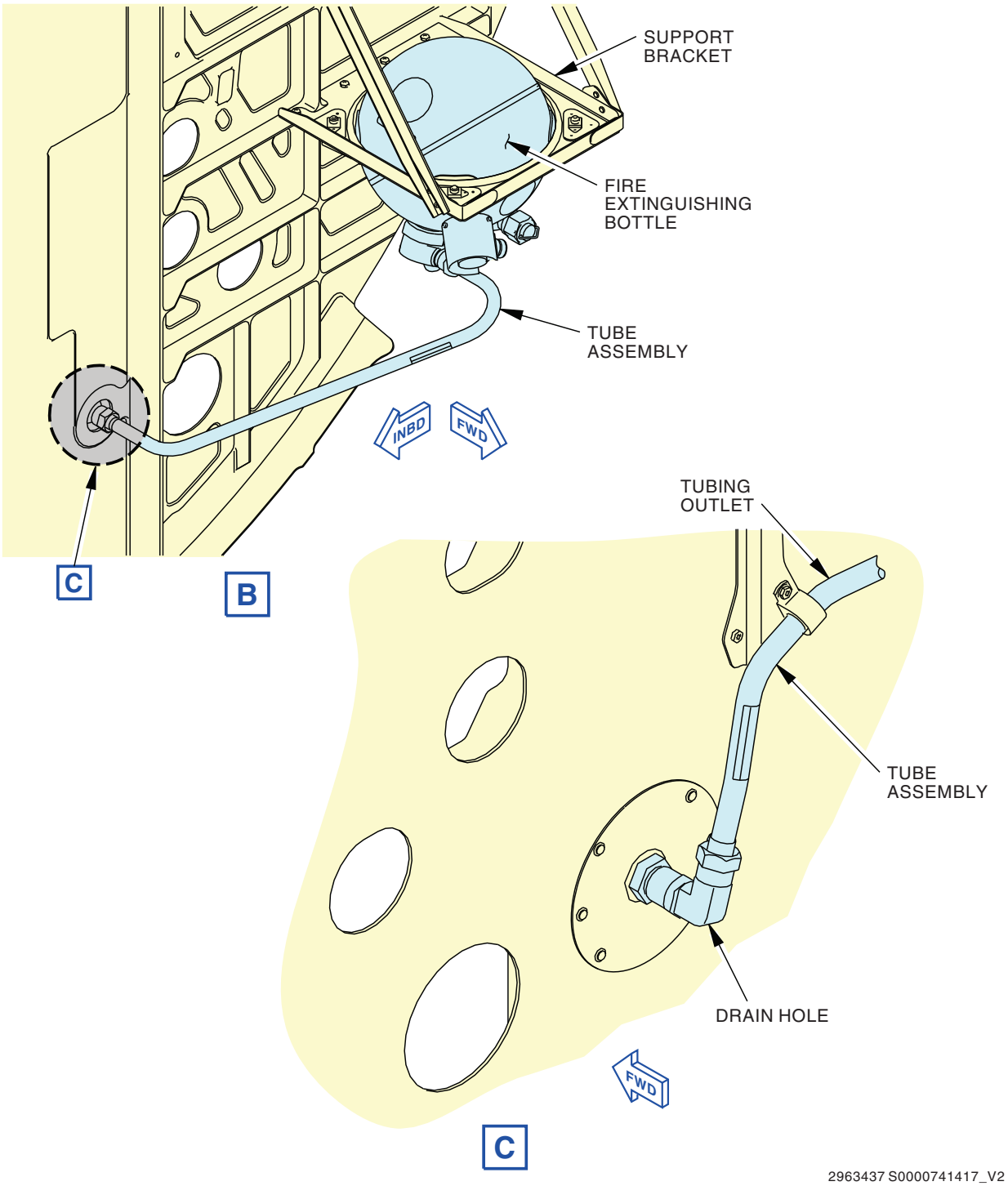
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Auxiliary Power Unit (APU) Fire Extinguishing Bottle Inspection
Figure 601/26-22-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Auxiliary Power Unit (APU) Fire Extinguishing Bottle Inspection
Figure 601/26-22-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FIRE EXTINGUISHING BOTTLE SQUIB - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) APU fire extinguishing bottle squib removal

(2) APU fire extinguishing bottle squib installation.

TASK 26-22-02-000-801

2. APU Fire Extinguishing Bottle Squib Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to remove the Auxiliary Power Unit (APU) fire extinguishing bottle squib.

(2) The APU fire extinguishing squibs are installed on the fire extinguishing bottle. The APU fire extinguishing bottle is installed forward of the APU fire wall.

(a) There are two fire extinguishing squibs on the bottle.

(b) One is not used and has a protective cap installed.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1168	Cap - Shorting or Faraday Cap

D. Location Zones

Zone	Area
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Removal

SUBTASK 26-22-02-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-22-02-010-001

(2) Open this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 26-22-02-860-004

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Before the squib is touched, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

G. APU Fire Extinguishing Bottle Squib Removal

SUBTASK 26-22-02-020-005

(1) Do these steps to remove the squib [3]:

(a) Disconnect the electrical connector [1] from the squib [3].

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(b) Install the shorting or faraday cap, STD-1168, on the squib [3].

NOTE: The shorting caps are attached to the fire extinguishing bottles. These shorting caps prevent the squibs from possible activation due to static electricity.

(c) Remove the lockwire from the squib [3].

(d) Remove the squib [3] from the discharge head.

(e) Discard the O-ring [5] from the removed squib [3].

SUBTASK 26-22-02-020-006

(2) If it is necessary, do these steps to remove the squib [4]:

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION

DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(a) Install the shorting or faraday cap, STD-1168, on the squib [4].

NOTE: The shorting caps are attached to the fire extinguishing bottles. These shorting caps prevent the squibs from possible activation due to static electricity.

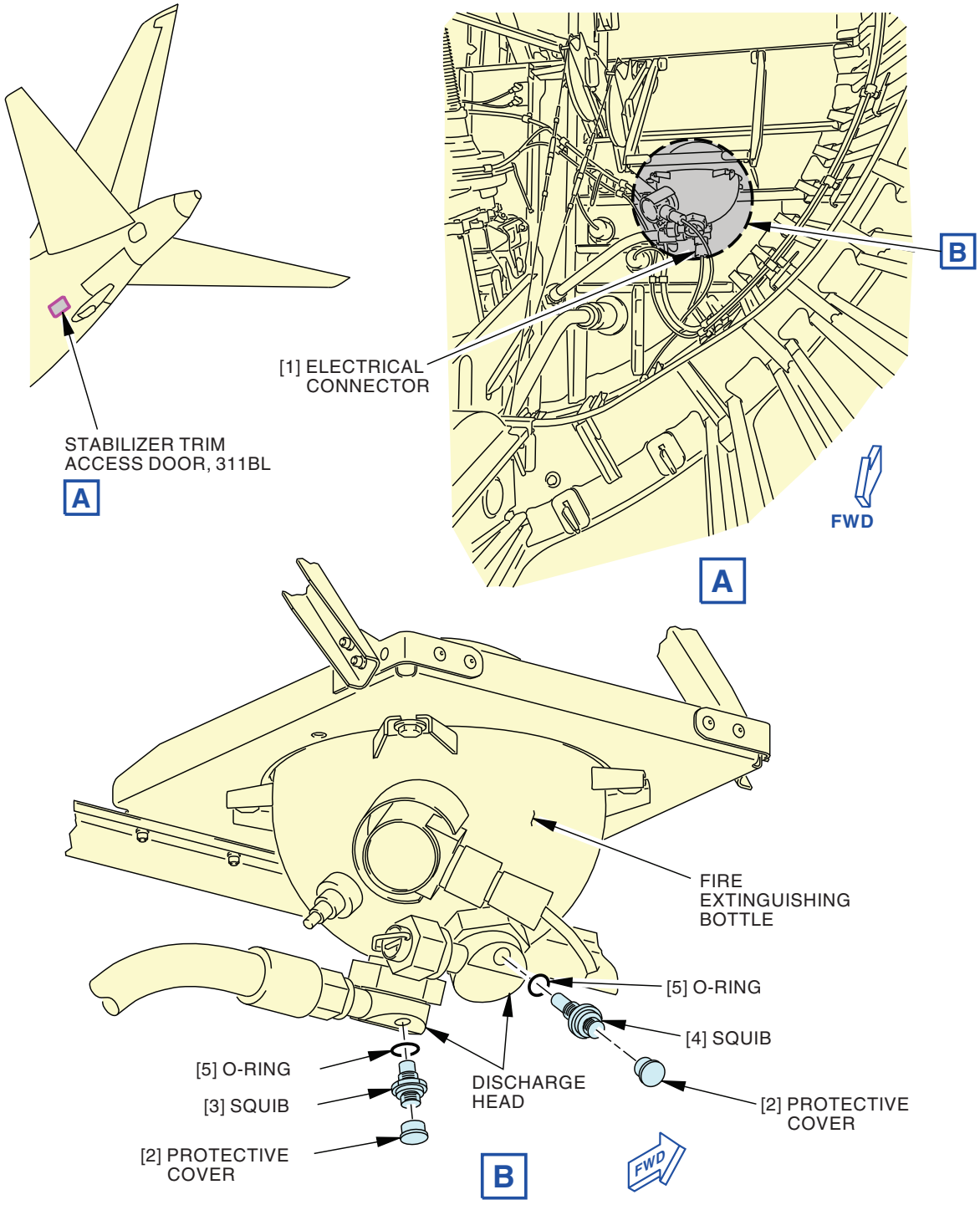
(b) Remove the lockwire from the squib [4].

(c) Remove the squib [4] from the discharge head.

(d) Discard the O-ring [5] from the removed squib [4].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Auxiliary Power Unit (APU) Fire Extinguishing Bottle Squib Installation
Figure 401/26-22-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-22-02-400-801

3. APU Fire Extinguishing Bottle Squib Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
STD-1079	Resistor - 10K Ohm or Greater
STD-1168	Cap - Shorting or Faraday Cap

C. Consumable Materials

Reference	Description	Specification
D00652	Lubricant - O-Ring - Parker Super-O-Lube	
D50287	Lubricant - O-Ring, Drilube 822 (MMS-N306A)	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G02392 [P05-262]	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8121 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Squib	26-22-01-02-105	PTF ALL
4	Squib	26-22-01-02-100	PTF ALL
5	O-ring	26-22-01-02-120	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. APU Fire Extinguishing Bottle Squib Installation

SUBTASK 26-22-02-420-001

(1) Do these steps to install the squib [3] in the fire extinguishing bottle:

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Before the squib [3] is touched, do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

(b) Lubricate a new O-ring [5] with O-ring lubricant, D50287, or Parker Super-O-Lube lubricant, D00652.

(c) Put the O-ring [5] on the squib [3].

(d) Install the squib [3] on the fire bottle.

1) Torque the squib [3] to 90 in-lb (10.2 N·m) - 100 in-lb (11.3 N·m).

2) Install MS20995C32 lockwire, G02392 [P05-262], between the squib [3] and discharge outlet.

SUBTASK 26-22-02-700-001

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(2) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

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SUBTASK 26-22-02-420-004

(3) Remove the shorting or faraday cap, STD-1168, from the squib [3].

SUBTASK 26-22-02-420-002

CAUTION MAKE SURE THAT THE SQUIB PINS CORRECTLY ENGAGE THE ELECTRICAL CONNECTOR. IF NOT, DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(4) Connect the electrical connector [1] to the squib [3].

SUBTASK 26-22-02-420-003

(5) If it is necessary to install the squib [4], do these steps:

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Before the squib [4] is touched, do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

(b) Lubricate the new O-ring [5] with O-ring lubricant, D50287, or Parker Super-O-Lube lubricant, D00652.

(c) Install the O-ring [5] on the squib [4].

(d) Install the squib [4] in the fire bottle.

1) Torque the squib [4] to 90 in-lb (10.2 N·m) - 100 in-lb (11.3 N·m).

2) Install MS20995C32 lockwire, G01048, between the squib [4] and discharge outlet.

3) Remove the shorting or faraday cap, STD-1168, from the squib [4].

SUBTASK 26-22-02-410-001

(6) Make sure that a protective cap is installed on the squib [4].

SUBTASK 26-22-02-860-003

(7) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

H. APU Fire Extinguishing Bottle Squib Installation Test

SUBTASK 26-22-02-740-001

(1) Set the EXT TEST switch, on the P8-1 panel, to the 1 position.

(a) Make sure that the APU light comes on.

NOTE: The L and R lights will also come on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-22-02-740-002

(2) Set the EXT TEST switch, on the P8-1 panel, to the 2 position.

(a) Make sure that the APU light comes on.

NOTE: The L and R lights will also come on.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 26-22-02-010-002

(1) Close this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FIRE CONTROL PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) APU fire control panel removal

(2) APU fire control panel installation.

B. The APU Fire Control Panel is located on the aft bulkhead of the right main wheel well.

TASK 26-22-03-000-801

2. APU Fire Control Panel Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the APU Fire Control Panel.

B. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. Prepare for the Removal

SUBTASK 26-22-03-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

D. APU Fire Control Panel Removal

SUBTASK 26-22-03-020-001

(1) Disconnect the electrical connector from the APU fire control panel [1].

SUBTASK 26-22-03-020-004

(2) Remove the bolt [4] and washers [5] that attach the jumper [6] to the APU fire control panel [1].

SUBTASK 26-22-03-020-005

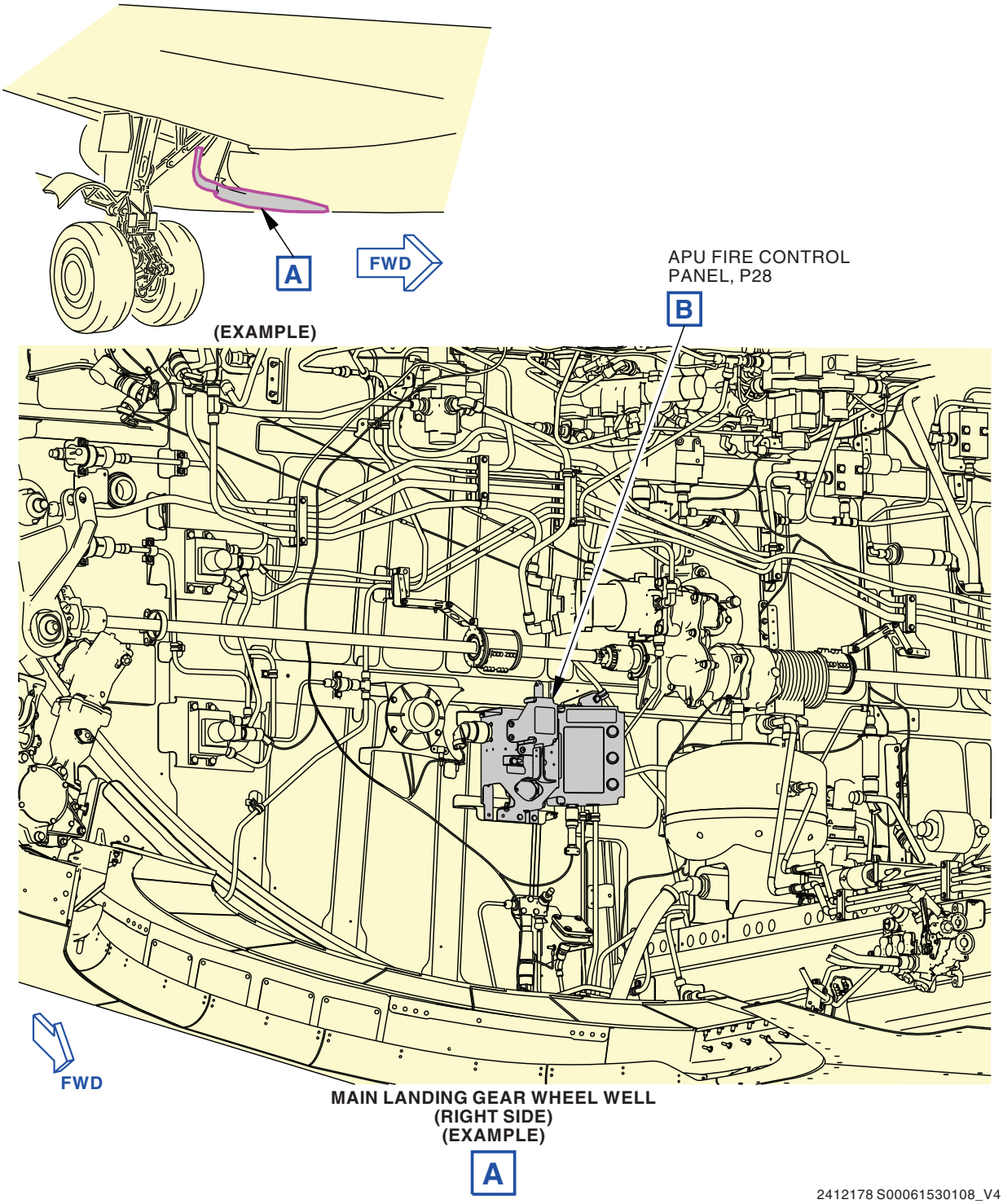
(3) Remove the bolts [7] and washers [8] that attach the APU fire control panel [1] to the nitrogen generation system operability indicator.

SUBTASK 26-22-03-020-002

(4) Remove the bolts [2] and washers [3] that hold the upper and lower flanges of the APU fire control panel [1].

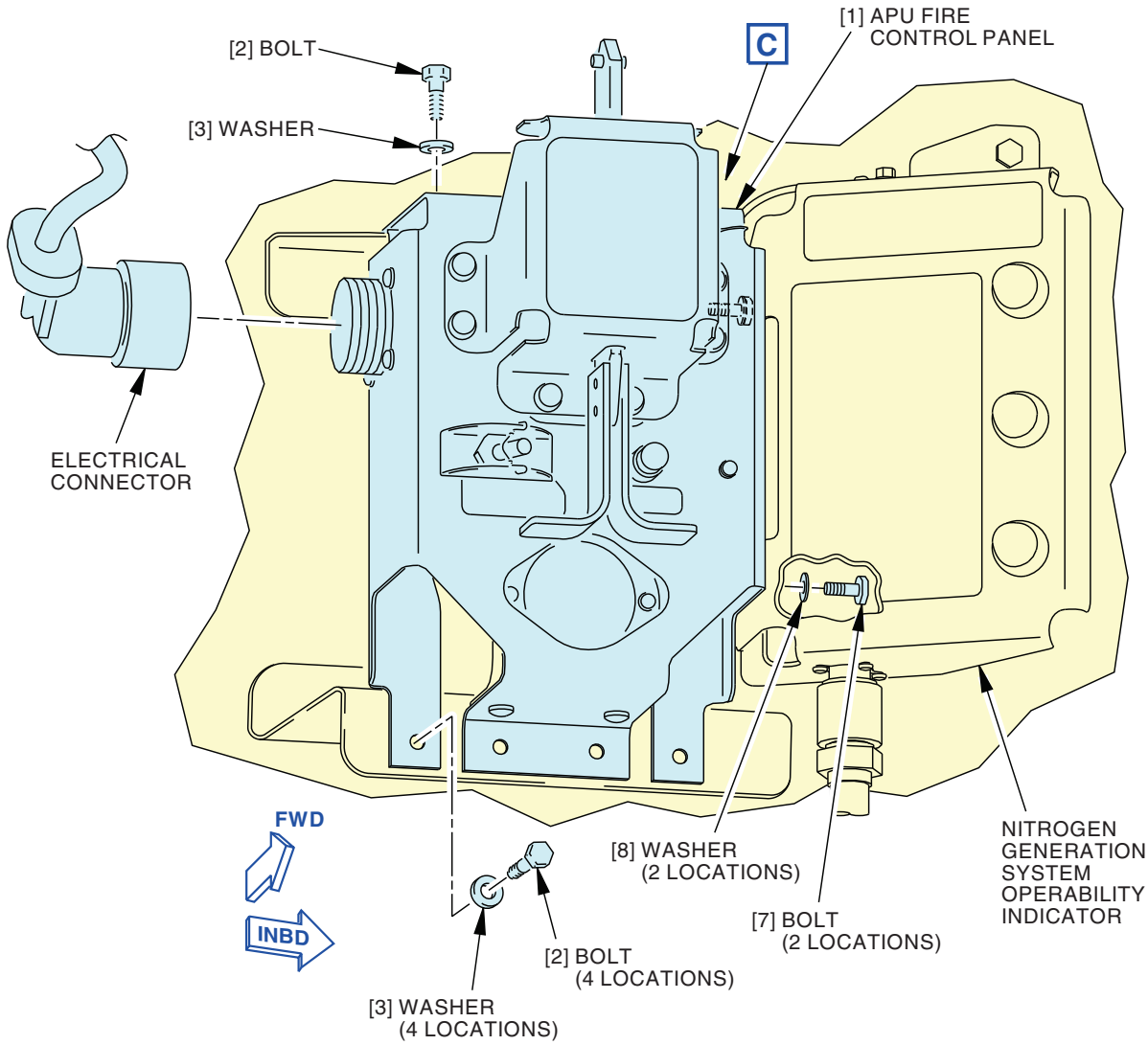
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fire Control Panel Installation
Figure 401/26-22-03-990-801 (Sheet 1 of 3)

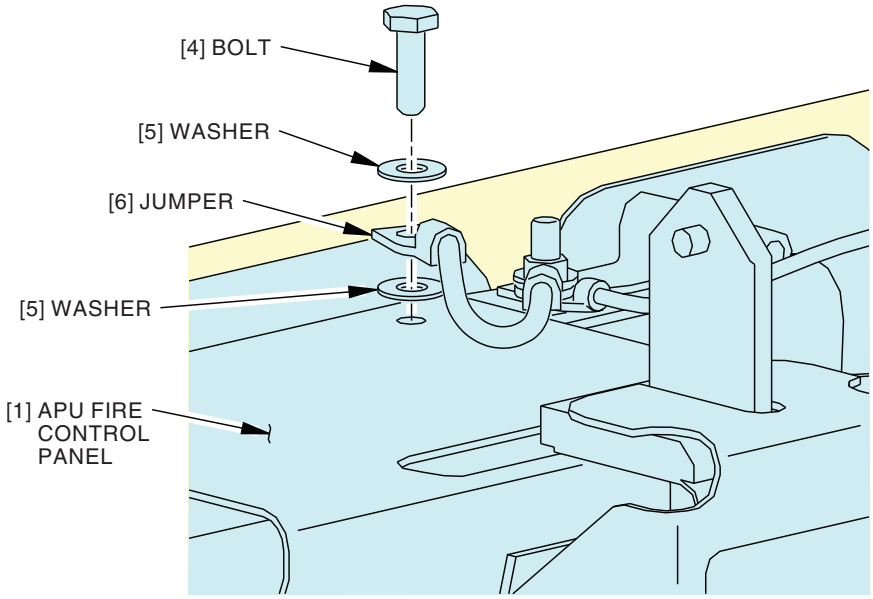
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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APU Fire Control Panel Installation
Figure 401/26-22-03-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



C

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APU Fire Control Panel Installation
Figure 401/26-22-03-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-22-03-400-801

3. APU Fire Control Panel Installation

(Figure 401)

A. General

(1) This task gives instructions to install the APU Fire Control Panel.

B. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
26-22-00-720-801	APU Fire Switch System Shutdown Test (P/B 501)
26-22-00-730-801	APU Fire Extinguishing Bottle Squib Circuit Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-765	Scraper - Plastic
STD-10596	Tool - Sealant Fairing

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	APU fire control panel	26-22-03-03-015	PTF ALL

F. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

G. Prepare for the Installation

SUBTASK 26-22-03-100-001

(1) Prepare the surface of the APU fire control panel [1] where the jumper [6] will be attached.

(a) Use plastic scraper, STD-765, to remove any existing sealant.

(b) Do this task: Hand Clean Metal Surfaces with Abrasives, TASK 20-10-34-120-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-22-03-100-002

(2) Use solvent, B00184, to clean all surfaces of the bolt [4], washers [5], jumper [6] terminals and APU fire control panel [1] surface.

H. APU Fire Control Panel Installation

SUBTASK 26-22-03-420-001

(1) Attach the APU fire control panel [1] to the bulkhead with the bolts [2] and washers [3].

SUBTASK 26-22-03-420-004

(2) Attach the APU fire control panel [1] to the nitrogen generation system operability indicator with the bolts [7] and washers [8].

SUBTASK 26-22-03-420-002

(3) Apply sealant, A50110, to all mating surfaces of the washers [5], jumper [6], APU fire control panel [1] and the shank and threads of the bolt [4].

SUBTASK 26-22-03-420-003

(4) Attach the jumper [6] to the APU fire control panel [1] with the bolt [4] and washers [5].

(a) Smooth out sealant squeeze-out with sealant fairing tool, STD-10596.

SUBTASK 26-22-03-700-001

(5) Measure the bonding resistance between the APU fire control panel [1] and the jumper [6] with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00, Paragraph 21).

(a) Make sure that the measured resistance is 1.0 milliohms (0.0010 ohms) or less.

SUBTASK 26-22-03-600-001

(6) Do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

SUBTASK 26-22-03-020-003

(7) Connect the electrical connector to the APU fire control panel [1].

SUBTASK 26-22-03-860-003

(8) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	21	C00452	FIRE PROTECTION EXTINGUISHERS APU

I. APU Fire Control Panel Installation Test

SUBTASK 26-22-03-710-001

(1) Do this task: APU Fire Extinguishing Bottle Squib Circuit Test, TASK 26-22-00-730-801.

SUBTASK 26-22-03-710-002

(2) Do this task: APU Fire Switch System Shutdown Test, TASK 26-22-00-720-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

APU FIRE CONTROL PANEL - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) APU Fire Control Panel Inspection.

TASK 26-22-03-210-801

2. APU Fire Control Panel Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Consumable Materials

Reference	Description	Specification
G50414	Lockwire - MS20995CY15, Copper - 0.015 Inch (0.381 mm) Diameter	NASM20995

B. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. APU Fire Control Panel Inspection

SUBTASK 26-22-03-210-001

(1) Make sure that the APU fire control panel is not damaged.

SUBTASK 26-22-03-210-002

(2) Make sure that the APU fire control panel is securely attached.

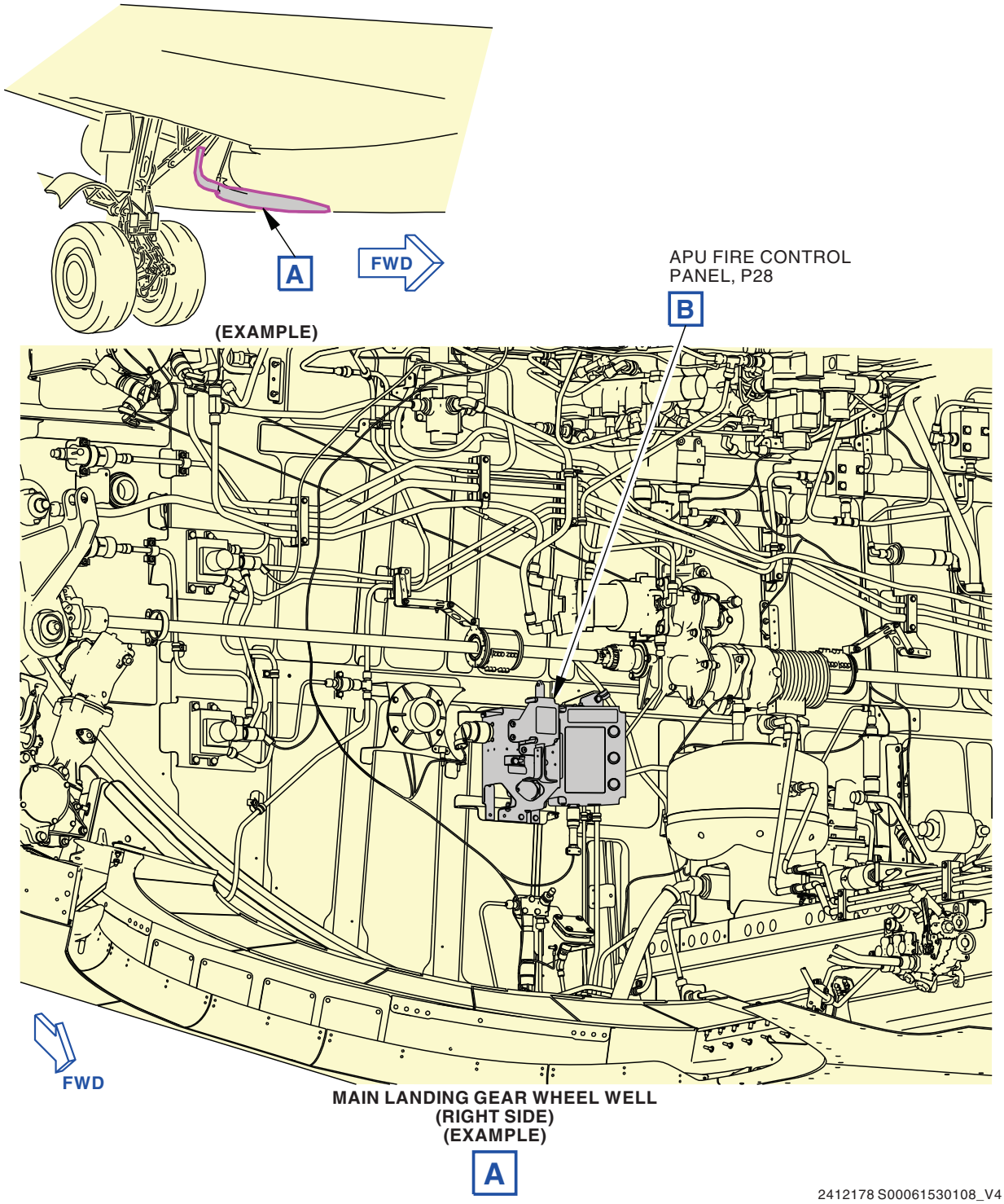
SUBTASK 26-22-03-210-003

(3) Make sure that the copper MS20995CY15 lockwire, G50414, is installed on the bottle discharge switch on the APU fire control panel.

(a) Make sure that the lockwire is positioned at the bottom of the hole in the guard.

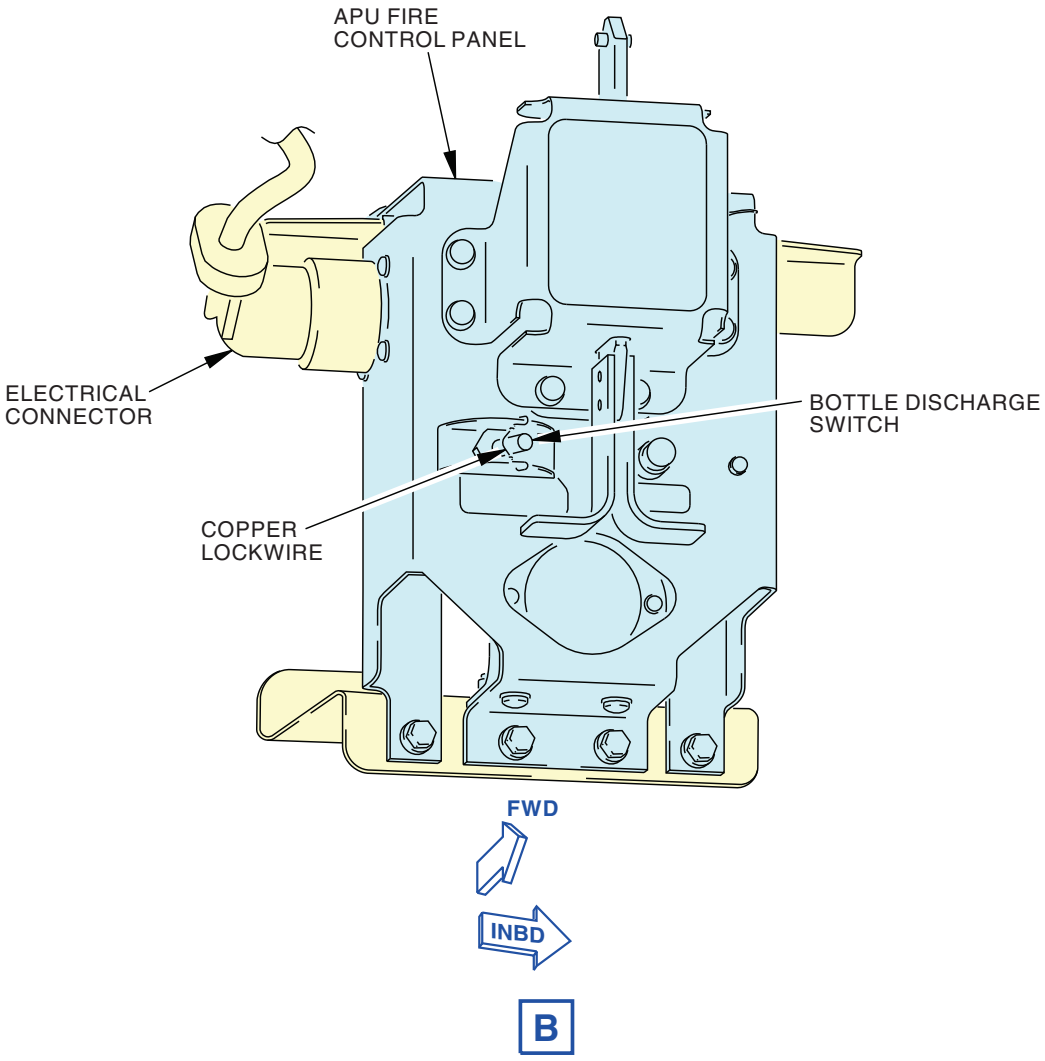
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



APU Fire Control Panel Installation
Figure 601/26-22-03-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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APU Fire Control Panel Installation
Figure 601/26-22-03-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO FIRE EXTINGUISHING - MAINTENANCE PRACTICES

1. General

A. This procedure contains these tasks:

(1) Cargo compartment fire extinguisher bottles and squibs - deactivation.

(2) Cargo compartment fire extinguisher bottles and squibs - activation.

TASK 26-23-00-040-801

2. Cargo Compartment Fire Extinguishers and Squibs - Deactivation

A. General

(1) This task will deactivate the 28 VDC electrical power to the Cargo Compartment Fire Extinguisher Bottle and Squib system.

B. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

C. Procedure

SUBTASK 26-23-00-020-001

(1) Remove the 28 VDC battery electrical power from the Cargo Compartment Fire Extinguisher and Squib system.

SUBTASK 26-23-00-020-002

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

D. Cargo Compartment Fire Extinguishers and Squibs - Tryout

NOTE: This tryout is to make sure that the electrical power to the Cargo Compartment Fire Extinguisher Bottle and Squib system is in zero state.

SUBTASK 26-23-00-860-001

(1) On the cargo fire control panel:

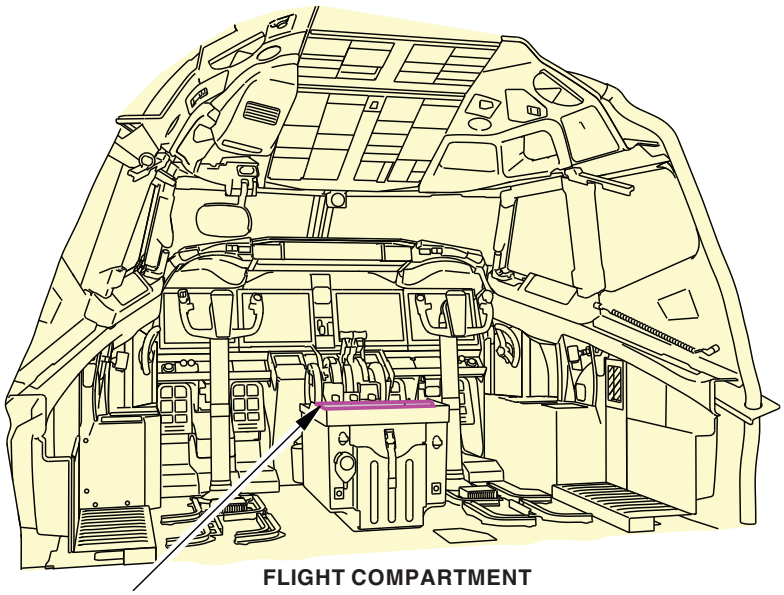
(a) Push and hold the TEST switch.

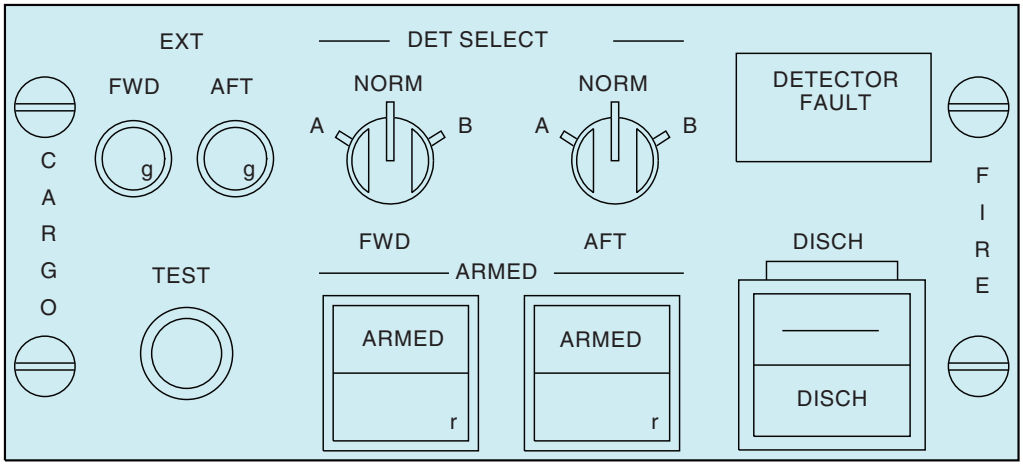
1) Make sure that the green FWD and AFT EXT lights on the cargo fire control panel do not come on.

(b) Release the TEST switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





CARGO BAY FIRE CONTROL PANEL (P8)

A

2412181 S00061530118_V1

Cargo Fire Control Panel
Figure 201/26-23-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-23-00-440-801

3. Cargo Compartment Fire Extinguishers and Squibs - Activation

(Figure 201)

A. General

(1) This task will activate the 28 VDC electrical power to the Cargo Compartment Fire Extinguisher Bottle and Squib system.

B. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

C. Procedure

SUBTASK 26-23-00-860-009

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

SUBTASK 26-23-00-440-001

(2) Do these steps to make sure the squibs are powered.

NOTE: This is to make sure that the electrical power to the Cargo Compartment Fire Extinguisher Bottle and Squib system is in zero state.

(a) On the cargo fire control panel:

1) Push and hold the TEST switch.

a) Make sure that the green FWD and AFT EXT lighton the cargo fire control panel come on.

2) Release the TEST switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO FIRE EXTINGUISHING - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) A test of the cargo fire bottle squib circuit

(2) A test of the cargo fire bottle pressure switch circuit

PTF ALL POST SB 737-26-1173

(3) A test of the cargo fire low rate discharge extinguishing bottle on ground inhibit test.

PTF ALL

(4) An airflow test through the discharge tubes

(5) A pressure test of the cargo fire bottle discharge tubes

26-CMR-01

TASK 26-23-00-730-801

2. Cargo Fire Extinguishing Bottle Squib - Circuit Test

(Figure 501, Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-1664	Test Box - Cargo Fire Extinguishing System Part #: C26006-70 Supplier: 81205 Opt Part #: C26006-1 Supplier: 81205
STD-1079	Resistor - 10K Ohm or Greater
STD-1168	Cap - Shorting or Faraday Cap

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Circuit Test

SUBTASK 26-23-00-860-007

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-00-010-002

(2) Remove the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-00-020-003

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(3) Before you touch the squib, do the procedure for devices that are sensitive to electrostatic discharge (TASK 20-40-12-000-802).

SUBTASK 26-23-00-010-003

(4) Remove the shields to get access to the squib electrical connectors.

SUBTASK 26-23-00-020-011

(5) Disconnect the electrical connectors from these squibs:

(a) Disconnect electrical connector D16578 from the forward squib on bottle 1.

(b) Disconnect electrical connector D16580 from the aft squib on bottle 1.

(c) Disconnect electrical connector D16584 from the forward squib on bottle 2.

(d) Disconnect electrical connector D16586 from the aft squib on bottle 2.

SUBTASK 26-23-00-020-004

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(6) Install the shorting or faraday cap, STD-1168, on the squibs.

NOTE: The shorting caps are attached to the fire extinguishing bottles. These shorting caps prevent the squibs from possible activation due to static electricity.

SUBTASK 26-23-00-730-001

(7) On the test box, SPL-1664, set these switches to the NO DISCHARGE position:

(a) BOTTLE 2 EXT 2 FORWARD S1

(b) BOTTLE 2 EXT 2 AFT S2

(c) BOTTLE 1 EXT 1 FORWARD S3

(d) BOTTLE 1 EXT 1 AFT S4.

SUBTASK 26-23-00-860-008

(8) On the test box, SPL-1664, open these circuit breakers:

(a) CB1, CB2, CB3, CB4, CB5, CB6, CB7, CB8.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-00-480-001

(9) Connect the electrical connectors to the connectors in the test box, SPL-1664:

(a) Connect electrical connector D16578 to connector J3.

(b) Connect electrical connector D16580 to connector J4.

(c) Connect electrical connector D16584 to connector J1.

(d) Connect electrical connector D16586 to connector J2.

SUBTASK 26-23-00-760-001

(10) Set the LIGHTS switch, on the P1-3 captain's forward panel, to the BRT position.

E. Cargo Fire Extinguishing Bottle Squib - Circuit Test

SUBTASK 26-23-00-860-003

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-00-860-006

(2) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 26-23-00-730-002

WARNING MAKE SURE THE AREA OF THE AFT OUTFLOW VALVE IS CLEAR OF PERSONS AND EQUIPMENT. DURING THIS TEST THE AFT OUTFLOW VALVE DOORS CAN START TO OPERATE. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do a test of the forward cargo fire bottle squib circuit:

(a) On the test box, SPL-1664, set the BOTTLE 1 EXT 1 FORWARD S3 switch to the DISCHARGE position.

(b) On the test box, SPL-1664, close these circuit breakers:

1) CB1, CB2, CB3, CB4, CB5, CB6, CB7, CB8.

(c) Push the FWD ARM button, on the P8-75 cargo fire control module, to the ARMED position.

1) Make sure that the ARMED indication on the FWD ARM button comes on.

2) Make sure that these lights, on the P8-75 cargo fire control module, do not come on:

a) FWD light

b) AFT light

c) DISCH light

d) EXT FWD light

e) EXT AFT light.

3) Make sure that none of the circuit breakers on the test box, SPL-1664, opened.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Push and hold the DISCH button, on the P8-75 cargo fire control module, for a minimum of one second.

1) Make sure that these lights do not come on:

NOTE: The DETECTOR FAULT indicator light, on the P8-75 cargo fire control module, can turn on.

a) FWD light

b) AFT light

c) DISCH light

d) EXT FWD light

e) EXT AFT light.

(e) Release the DISCH button.

1) On the test box, SPL-1664, make sure that this circuit breaker opens:

a) CB5.

2) Make sure that these circuit breakers stay closed:

a) CB1, CB2, CB3, CB4, CB6, CB7, CB8.

(f) On the test box, SPL-1664, do these steps:

1) Close this circuit breaker:

a) CB5.

2) Set the BOTTLE 2 EXT 2 FORWARD S1 switch to the DISCHARGE position.

3) Set the BOTTLE 1 EXT 1 FORWARD S3 switch to the NO DISCHARGE position.

a) Make sure that these circuit breakers stay closed:

<1> CB1, CB2, CB3, CB4, CB5, CB6, CB7, CB8.

4) Wait approximately 16.5 minutes for CB1 to open.

a) Make sure that these circuit breakers stay closed:

<1> CB2, CB3, CB4, CB5, CB6, CB7, CB8.

5) Close this circuit breaker:

a) CB1.

SUBTASK 26-23-00-730-003

WARNING MAKE SURE THE AREA OF THE AFT OUTFLOW VALVE IS CLEAR OF PERSONS AND EQUIPMENT. DURING THIS TEST THE AFT OUTFLOW VALVE DOORS CAN START TO OPERATE. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Do a test of the aft cargo fire bottle squib circuit:

(a) On the test box, SPL-1664, set the BOTTLE 2 EXT 2 AFT S2 switch to the DISCHARGE position.

(b) On the test box, SPL-1664, set the BOTTLE 2 EXT 2 FORWARD S1 switch to the NO DISCHARGE position.

1) Make sure that these circuit breakers stay closed:

a) CB1, CB2, CB3, CB4, CB5, CB6, CB7, CB8.

(c) Push the AFT ARM button, on the P8-75 cargo fire control module, to the ARMED position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Wait 5 seconds, and then push the FWD ARM button.
  1) Make sure that the ARMED indication on the FWD ARM button goes off.
  2) Make sure that the ARMED indication on the AFT ARM button comes on.
  3) Make sure that these lights do not come on:
    a) FWD light
    b) AFT light
    c) DISCH light
    d) EXT FWD light
    e) EXT AFT light.
  4) Make sure that none of the circuit breakers on the test box, SPL-1664, opened.

(e) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.
  1) On the test box, SPL-1664, make sure that this circuit breaker opens:
    a) CB3.
  2) Make sure that these circuit breakers stay closed:
    a) CB1, CB2, CB4, CB5, CB6, CB7, CB8.

(f) On the test box, SPL-1664, do these steps:
  1) Close this circuit breaker:
    a) CB3.
  2) Set the BOTTLE 1 EXT 1 AFT S4 switch to the DISCHARGE position.
  3) Set the BOTTLE 2 EXT 2 AFT S2 switch to the NO DISCHARGE position.
    a) Make sure that these circuit breakers stay closed:
      <1> CB1, CB2, CB3, CB4, CB5, CB6, CB7, CB8.
  4) Push and hold the DISCH push-button for a minimum of one second.
  5) Release the DISCH push-button.
    a) Make sure that this circuit breaker opens:
      <1> CB7.
    b) Make sure that these circuit breakers stay closed:
      <1> CB1, CB2, CB3, CB4, CB5, CB6, CB8.
  6) Close this circuit breaker:
    a) CB7.
  7) Set the BOTTLE 1 EXT 1 AFT S4 switch to the NO DISCHARGE position.

(g) Push the AFT ARM button, on the P8-75 cargo fire control module.
  1) Make sure that the ARMED indication on the AFT ARM button goes off.

SUBTASK 26-23-00-710-001

(5) Push and hold the TEST push-button, on the P8-75 cargo fire control module.
  (a) Make sure that these lights come on:
    1) DISCH indicator light
    2) EXT FWD indicator light
    3) EXT AFT indicator light.

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(b) Set the LIGHTS switch, on the P1-3 captain's forward panel, to the DIM position.

(c) Make sure that these lights dim:

1) DISCH indicator light
2) EXT FWD indicator light
3) EXT AFT indicator light.

(d) Set the LIGHTS switch to the BRT position.

(e) Make sure that these lights are brighter:

1) DISCH indicator light
2) EXT FWD indicator light
3) EXT AFT indicator light.

SUBTASK 26-23-00-710-002

(6) Release the TEST push-button.

(a) Make sure that these lights go off:

1) DISCH indicator light
2) EXT FWD indicator light
3) EXT AFT indicator light.

SUBTASK 26-23-00-860-004

(7) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-00-020-016

(8) Disconnect the electrical squib connectors from the test equipment.

SUBTASK 26-23-00-700-001

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(9) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

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(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-23-00-020-005

(10) Remove the protective cap from the squibs.

SUBTASK 26-23-00-420-002

(11) Install the electrical connectors:

(a) Connect electrical connector D16578 to the forward squib on bottle 1.

(b) Connect electrical connector D16584 to the forward squib on bottle 2.

(c) Connect electrical connector D16580 to the aft squib on bottle 1.

(d) Connect electrical connector D16586 to the aft squib on bottle 2.

SUBTASK 26-23-00-860-005

(12) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-00-730-009

(13) Do a test of the cargo fire extinguishing system, as follows:

(a) Make sure that the FWD and AFT DET SELECT switches, on the P8 cargo fire control panel, are set to the NORM position.

(b) Push and hold the TEST switch on the cargo fire control panel.

1) Make sure that the fire warning bell comes on.

2) Make sure that the FIRE WARN lights, on the P7 pilot's glareshield, come on.

(c) Push the FIRE WARN light, on the P7 pilot's glareshield.

1) Make sure that the FIRE WARN lights go off.

2) Make sure that the fire warning bell stops.

3) Make sure that these lights, on the P8 cargo fire control panel, are on:

a) EXT FWD light

b) EXT AFT light

c) FWD light

d) AFT light

e) DISCH light.

4) Make sure that the DETECTOR FAULT light, on the P8 cargo fire control panel, does not come on.

(d) Release the TEST switch on the cargo fire control panel.

1) Make sure that these lights, on the P8 cargo fire control panel, go off:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) EXT FWD light

b) EXT AFT light

c) FWD light

d) AFT light

e) DISCH light.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-00-410-004

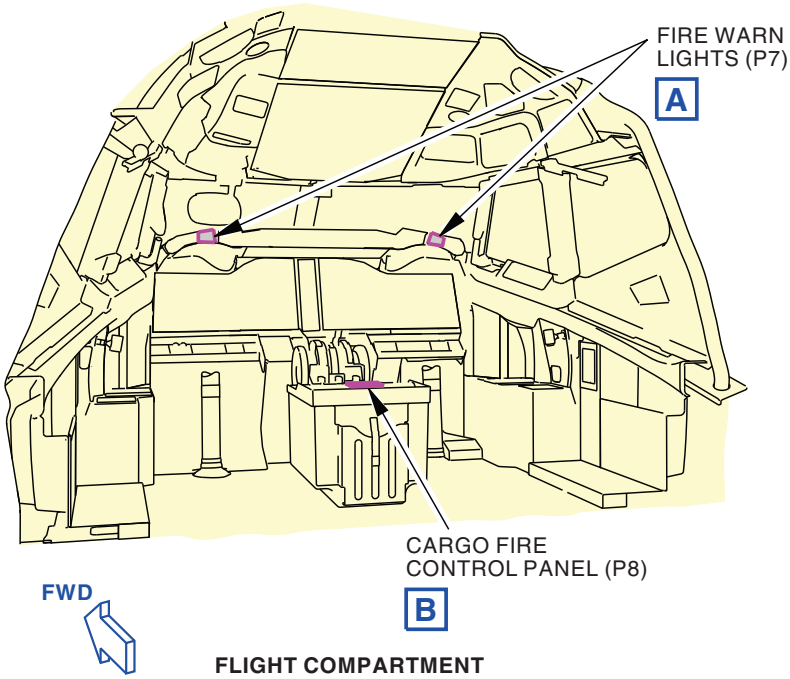
(1) Install the shields to protect the squib electrical connectors.

SUBTASK 26-23-00-410-001

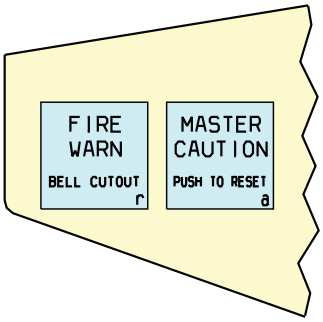
(2) Install the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

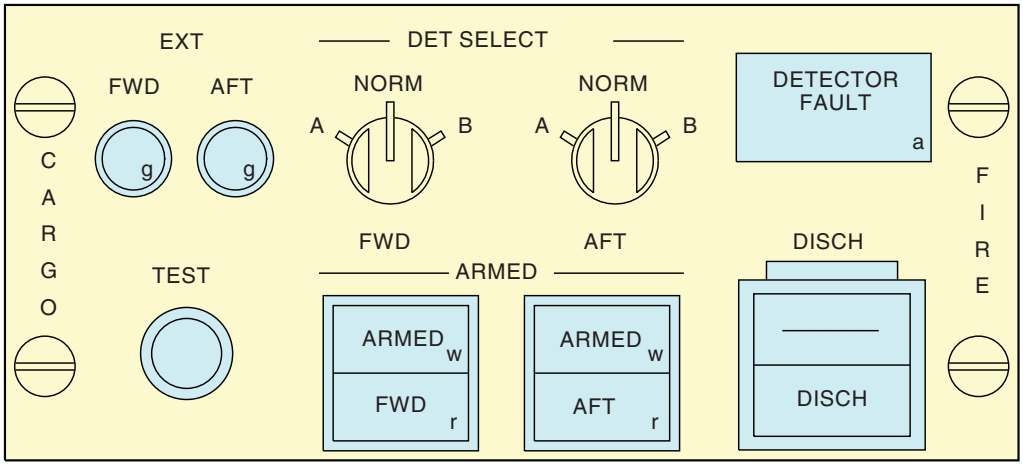


FLIGHT COMPARTMENT



FIRE WARN LIGHTS (P7)
(EXAMPLE)

A



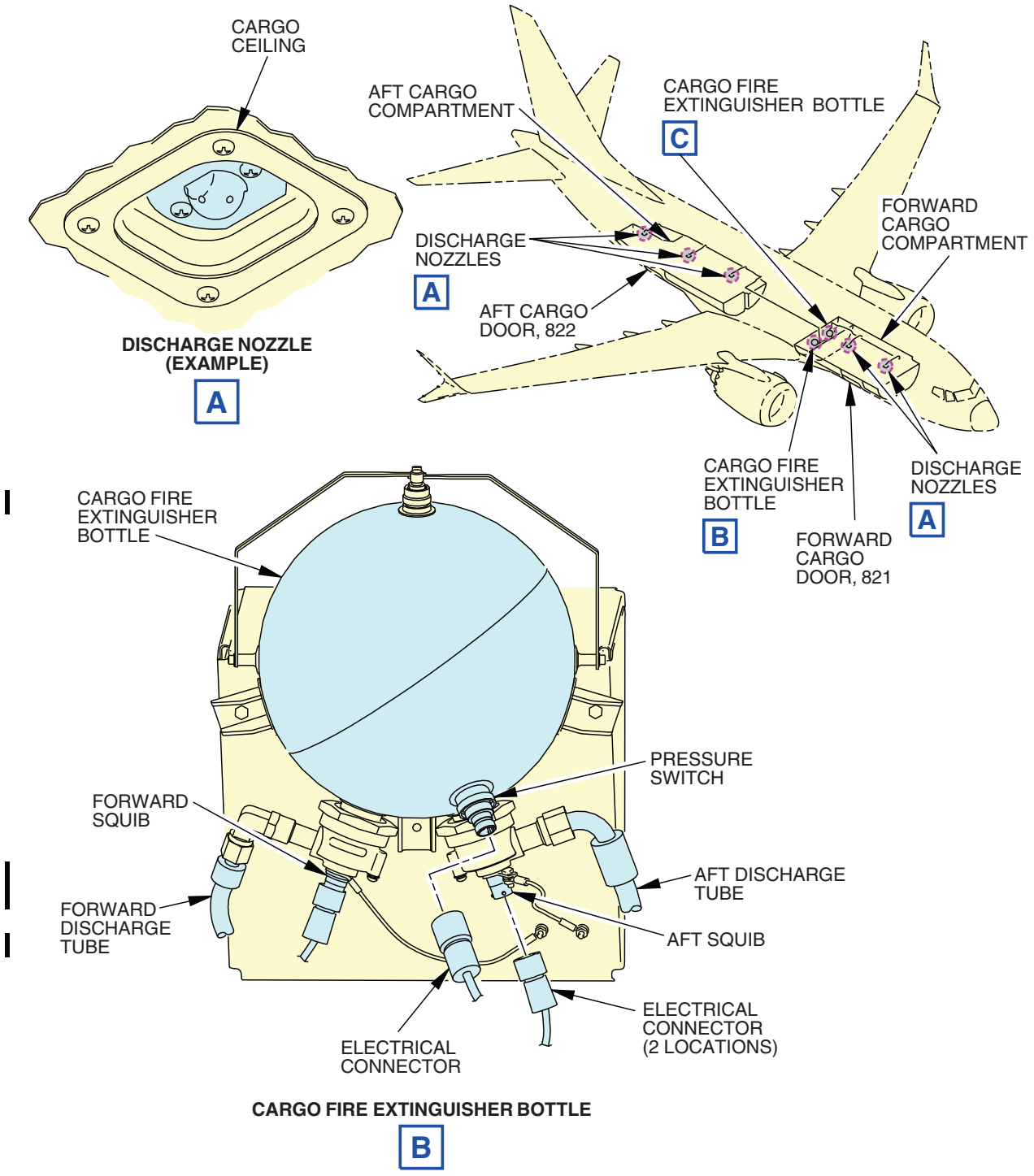
CARGO FIRE CONTROL MODULE (P8-75)

B

2412182 S00061530122_V2

Cargo Fire Control Module (P8-75)
Figure 501/26-23-00-990-803

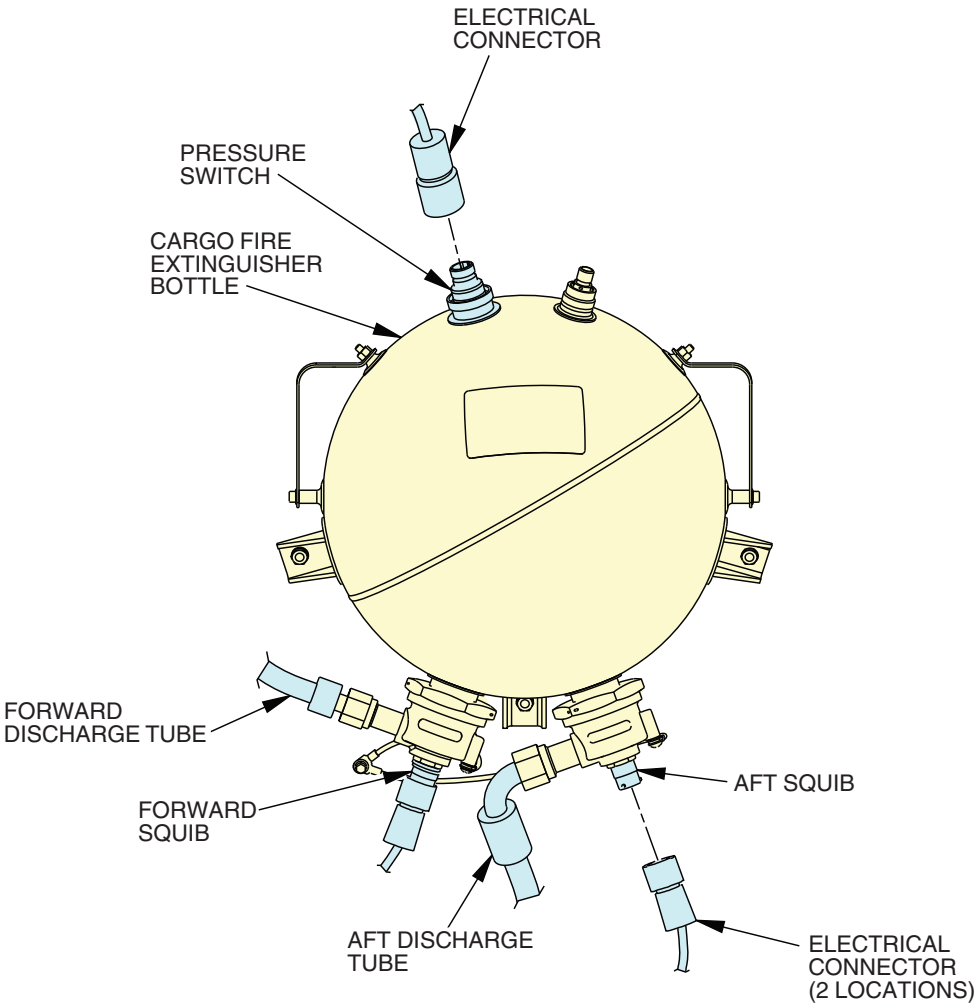
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2565773 S0000613697_V2

Cargo Bay Fire Extinguishing Test
Figure 502/26-23-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CARGO FIRE EXTINGUISHER BOTTLE

C

2546754 S0000604847_V3

Cargo Bay Fire Extinguishing Test
Figure 502/26-23-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-23-00-730-802

3. Cargo Fire Extinguishing Bottle Pressure Switch Circuit Test

(Figure 502)

A. General

(1) The pressure switch circuit test should be done every time the cargo fire control panel, or a pressure switch, is disconnected from the airplane wiring.

B. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-6286	Jumper Wire

D. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Test

SUBTASK 26-23-00-010-006

(1) Remove the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

F. Cargo Fire Extinguishing Bottle Pressure Switch Circuit Test

SUBTASK 26-23-00-020-006

(1) Do a test of the fire extinguisher pressure switch circuit:

(a) Disconnect connector D16576 from the pressure switch on cargo fire bottle 1.

(b) Disconnect connector D16588 from the pressure switch on cargo fire bottle 2.

(c) Do a test of the cargo fire bottle pressure switch circuit:

1) Connect a Jumper Wire, STD-6286, between pins 2 and 3 on connector D16576.

2) Make sure that the DISCH light, on the P8 cargo fire control panel, comes on.

3) Remove the Jumper Wire, STD-6286, from connector D16576.

4) Connect a Jumper Wire, STD-6286, between pins 2 and 3 on connector D16588.

5) Make sure that the DISCH light, on the P8 cargo fire control panel, comes on.

6) Remove the Jumper Wire, STD-6286, from connector D16588.

(d) Connect connector D16576 to the pressure switch on the cargo fire bottle 1.

(e) Connect connector D16588 to the pressure switch on the cargo fire bottle 2.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-00-410-005

(1) Install the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

——— END OF TASK ———

PTF ALL POST SB 737-26-1173

TASK 26-23-00-730-804

4. Cargo Fire Extinguishing Bottle Low Rate Discharge on Ground Inhibit Test

(Figure 501, Figure 502)

A. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
26-23-01-000-801	Cargo Fire Extinguisher Bottle Removal (P/B 401)
26-23-01-400-801	Cargo Fire Extinguisher Bottle Installation (P/B 401)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-1664	Test Box - Cargo Fire Extinguishing System Part #: C26006-70 Supplier: 81205 Opt Part #: C26006-1 Supplier: 81205

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PTF ALL POST SB 737-26-1173 (Continued)

(Continued)

Reference	Description
STD-1079	Resistor - 10K Ohm or Greater
STD-1168	Cap - Shorting or Faraday Cap

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Circuit Test

SUBTASK 26-23-00-860-010

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-00-010-009

(2) Remove the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

SUBTASK 26-23-00-020-012

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(3) Before you touch the squib, do the procedure for devices that are sensitive to electrostatic discharge (TASK 20-40-12-000-802).

SUBTASK 26-23-00-010-010

(4) Remove the lower shields to get access to the squib electrical connectors (TASK 26-23-01-000-801).

SUBTASK 26-23-00-020-013

(5) Disconnect the electrical connectors from these squibs:

(a) Disconnect electrical connector D16578 from bottle 1.

(b) Disconnect electrical connector D16580 from bottle 1.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-26-1173 (Continued)

(c) Disconnect electrical connector D16584 from bottle 2.

(d) Disconnect electrical connector D16586 from bottle 2.

SUBTASK 26-23-00-020-014

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(6) Install the shorting or faraday cap, STD-1168, on the squibs.

NOTE: The shorting caps are attached to the fire extinguishing bottles. These shorting caps prevent the squibs from possible activation due to static electricity.

SUBTASK 26-23-00-730-006

(7) On the test box, SPL-1664, set the following switches to the NO DISCHARGE position:

(a) BOTTLE 2 EXT 2 FORWARD S1

(b) BOTTLE 2 EXT 2 AFT S2

(c) BOTTLE 1 EXT 1 FORWARD S3

(d) BOTTLE 1 EXT 1 AFT S4.

SUBTASK 26-23-00-860-011

(8) On the test box, SPL-1664, open these circuit breakers:

(a) CB1, CB2, CB3, CB4, CB5, CB6, CB7, CB8.

SUBTASK 26-23-00-480-008

(9) Connect the electrical connectors to the connectors in the test box, SPL-1664:

(a) Connect electrical connector D16578 to connector J3.

(b) Connect electrical connector D16580 to connector J4.

(c) Connect electrical connector D16584 to connector J1.

(d) Connect electrical connector D16586 to connector J2.

SUBTASK 26-23-00-760-002

(10) Set the LIGHTS switch, on the P1-3 captain's forward panel, to the BRT position.

E. Cargo Fire Extinguishing Bottle Low Rate Discharge on Ground Inhibit Test

SUBTASK 26-23-00-860-012

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-26-1173 (Continued)

(Continued)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-00-860-013

(2) Make sure the airplane is in ground mode (TASK 32-09-00-860-802).

SUBTASK 26-23-00-730-007

WARNING MAKE SURE THE AREA OF THE AFT OUTFLOW VALVE IS CLEAR OF PERSONS AND EQUIPMENT. DURING THIS TEST THE AFT OUTFLOW VALVE DOORS CAN START TO OPERATE. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do a test of the forward cargo fire bottle squib circuit:

(a) On the test box, SPL-1664, set the BOTTLE 1 EXT 1 FORWARD S3 switch to the DISCHARGE position.

(b) On the test box, SPL-1664, close these circuit breakers:

1) CB1, CB2, CB3, CB4, CB5, CB6, CB7, CB8.

(c) Push the FWD ARM push-button, on the P8-75 cargo fire control module, to the ARMED position.

1) Make sure that the ARMED light for the FWD ARM push-button, on the P8-75 cargo fire control module, comes on.

2) Make sure that these lights, on the P8-75 cargo fire control module, do not come on:

a) FWD light

b) AFT light

c) DISCH light

d) EXT FWD light

e) EXT AFT light.

3) Make sure that none of the circuit breakers on the test box, SPL-1664, opened.

(d) Push and hold the DISCH push-button, on the P8-75 cargo fire control module, for a minimum of one second.

1) Make sure that these lights do not come on:

NOTE: The DETECTOR FAULT indicator light, on the P8-75 cargo fire control module, can turn on.

a) FWD light

b) AFT light

c) DISCH light

d) EXT FWD light

e) EXT AFT light.

(e) Release the DISCH push-button.

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PTF ALL POST SB 737-26-1173 (Continued)

1) Make sure that this circuit breaker opens:
   a) CB5.
2) Make sure that these circuit breakers stay closed:
   a) CB1, CB2, CB3, CB4, CB6, CB7, CB8.
3) Close this circuit breaker:
   a) CB5.

(f) On the test box, SPL-1664, set the BOTTLE 2 EXT 2 FORWARD S1 switch to the DISCHARGE position.

(g) Set the BOTTLE 1 EXT 1 FORWARD S3 switch to the NO DISCHARGE position.

(h) Make sure that these circuit breakers stay closed:
1) CB1, CB2, CB3, CB4, CB5, CB6, CB7, CB8.

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(i) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

(j) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	16	C01356	LANDING GEAR AIR/GND SYS 1

(k) Wait approximately 16.5 minutes for CB1 to remain closed.

NOTE: The BOTTLE 2 EXT 2 FORWARD CB1 circuit breaker will remain closed approximately 16.5 minutes after you push the DISCH push-button.

(l) Make sure that these circuit breakers stay closed:
1) CB1, CB2, CB3, CB4, CB5, CB6, CB7, CB8.

(m) Set the BOTTLE 2 EXT 2 FORWARD S1 switch to the NO DISCHARGE position.

(n) Push the FWD ARM push-button.
1) Make sure that the ARMED light for the FWD ARM push-button goes off.

SUBTASK 26-23-00-730-008

(4) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-00-020-017

(5) Disconnect the electrical squib connectors from the test equipment.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-26-1173 (Continued)

SUBTASK 26-23-00-700-002

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(6) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-23-00-020-015

(7) Remove the shorting or faraday cap, STD-1168, from the squibs.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-00-420-007

(1) Install the electrical connectors:

(a) Connect D16578 to forward squib on bottle 1.

(b) Connect D16584 to forward squib on bottle 2.

(c) Connect D16580 to aft squib on bottle 1.

(d) Connect D16586 to aft squib on bottle 2.

SUBTASK 26-23-00-410-008

(2) Install the lower shields to protect the squib electrical connectors (TASK 26-23-01-400-801).

SUBTASK 26-23-00-410-009

(3) Install the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 26-23-00-860-014

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

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PTF ALL POST SB 737-26-1173 (Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	16	C01356	LANDING GEAR AIR/GND SYS 1

SUBTASK 26-23-00-840-001

(5) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

PTF ALL

——— END OF TASK ———

TASK 26-23-00-720-801

5. Cargo Fire Extinguishing Discharge Line Flow Test

(Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The cargo fire extinguishing discharge line flow test makes sure that there is nothing blocking the discharge lines.

(2) There are two bottles on the airplane. The contents of bottle 1 are released when the discharge switch is pushed. Bottle 2 is discharged 15 minutes later through a flow filter/dryer. In this test, the discharge lines will be removed from bottle 1, and from the output of the filter/dryers. The flow will be checked from the output of bottle 1 through the discharge nozzles and the filter/dryer output line.

B. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1647	Test Equipment - Engine Fire Extinguishing System Part #: C26004-83 Supplier: 81205 Part #: C26004-84 Supplier: 81205
STD-1115	Source - Nitrogen, 0-100 PSIG

D. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

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E. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

F. Prepare for the Test

SUBTASK 26-23-00-010-007

(1) Remove the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

G. Cargo Fire Extinguishing Discharge Line Flow Test

SUBTASK 26-23-00-010-004

(1) If necessary, remove lower shield to get access to the bottle connections.

SUBTASK 26-23-00-030-001

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE FITTING. IF YOU TWIST THE TUBE OR FITTING, YOU CAN CAUSE DAMAGE TO THEM.

(2) Remove the discharge tubes from the discharge outlets on cargo fire bottle 1.

SUBTASK 26-23-00-020-008

(3) Remove the discharge tubes from the outputs of the two pressure filter/dryers connected to bottle 2.

SUBTASK 26-23-00-480-002

CAUTION

DO NOT APPLY AIR PRESSURE TO THE OUTPUT OF THE PRESSURE FILTER/DRIER. AIR PRESSURE CAN CAUSE DAMAGE TO THE FILTER/DRIER.

(4) Connect a 0-100 PSIG nitrogen source, STD-1115, to the forward (yellow) cargo bottle discharge line at bottle 1.

NOTE: Alternate pressure sources can be used. The pressure source must have an accuracy of 1 PSIG and be capable of applying a pressure of 35 to 50 PSIG.

(a) If it is necessary, use equipment, SPL-1647, to connect a source to the discharge line.

(b) Pressurize the discharge line to 15 psig (103 kPa).

(c) Make sure that air is flowing through all open nozzles in the forward cargo compartment.

(d) Make sure that air is flowing through the output line that was removed from the forward filter/dryer of bottle 2.

SUBTASK 26-23-00-080-001

(5) Remove the pressure source from the discharge line.

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SUBTASK 26-23-00-480-003

CAUTION

DO NOT APPLY AIR PRESSURE TO THE OUTPUT OF THE PRESSURE FILTER/DRIER. AIR PRESSURE CAN CAUSE DAMAGE TO THE FILTER/DRIER.

(6) Connect a 0-100 PSIG nitrogen source, STD-1115, to the aft (blue) cargo bottle discharge line at bottle 1.

NOTE: Alternate pressure sources can be used. The pressure source must have an accuracy of 1 PSIG and be capable of applying a pressure of 35 to 50 PSIG.

(a) If it is necessary, use equipment, SPL-1647, to connect a source to the discharge line.

(b) Pressurize the discharge line to 15 psig (103 kPa).

(c) Make sure that air is flowing through all open nozzles in the aft cargo compartment.

(d) Make sure that air is flowing through the output line that was removed from the aft filter/dryer connected to fire bottle 2.

SUBTASK 26-23-00-080-002

(7) Remove the pressure source from the discharge line.

SUBTASK 26-23-00-420-003

(8) Connect the discharge tubes to the discharge outlets on bottle 1.

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Apply a layer of compound, D50004, to the outlet threads, outlet end, and outlet inner diameter where the tube assembly attaches to the discharge head.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Permitted alternative anti-corrosion compounds to use in this location are Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136.

(b) Connect the tube assemblies to the discharge heads.

NOTE: The tube assemblies that are marked in yellow supply the extinguishing agent to the forward cargo compartment and the ones marked in blue supply the aft cargo compartment.

(c) Remove any unwanted compound.

(d) Tighten the nuts on the tube assemblies to these torque values:

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Discharge Tube Torque Values

Nut Size	Tube Outer Diameter	Torque (MIN)	Torque (MAX)
7/8 in	1/2 in	266 in-lb (30 N·m)	294 in-lb (33 N·m)
1 in	5/8 in	342 in-lb (39 N·m)	378 in-lb (43 N·m)

SUBTASK 26-23-00-420-006

(9) Connect the discharge tubes to the discharge outlets on the filter/dryers connected to fire bottle 2.

WARNING DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Apply a layer of compound, D50004, to the outlet threads, outlet end, and outlet inner diameter where the tube assemblies attach to the filter/dryers.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Permitted alternative anti-corrosion compounds to use in this location are Cor-Ban 27L Compound, G50237.

(b) Connect the tube assemblies to the filter/dryers.

NOTE: The tube assemblies that are marked in yellow supply the extinguishing agent to the forward cargo compartment and the ones marked in blue supply the aft cargo compartment.

(c) Remove any unwanted compound.

(d) Tighten the nuts on the tube assemblies to these torque values:

Discharge Tube Torque Values

Nut Size	Tube Outer Diameter	Torque (MIN)	Torque (MAX)
7/8 in	1/2 in	266 in-lb (30 N·m)	294 in-lb (33 N·m)
1 in	5/8 in	342 in-lb (39 N·m)	378 in-lb (43 N·m)

SUBTASK 26-23-00-410-002

(10) If removed, install lower shield in front of the bottle connections.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-00-410-006

(1) Install the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

END OF TASK

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TASK 26-23-00-730-803

6. Cargo Fire Extinguishing Discharge Line Pressure Test

(Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The pressure test is the same for all cargo compartments.

B. References

Reference	Title
26-23-01-200-801	Discharge Head Leak Check (Method 1) (P/B 601)
26-23-01-200-802	Discharge Head Leak Check (Method 2) (P/B 601)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-25014	Adapter - Pressure Discharge Head (7/8-inch fitting at one end and air shop fitting at the other end) Part #: 4-10XHU6-SS MS33514 Supplier: 30780
COM-25015	Adapter - Pressure Female End (3/4-inch fitting at one end and air shop fitting at the other end) Part #: 4-8XHU6-SS MS33514 Supplier: 30780
SPL-1647	Test Equipment - Engine Fire Extinguishing System Part #: C26004-83 Supplier: 81205 Part #: C26004-84 Supplier: 81205
SPL-1648	Discharge Lines - Cargo Fire Extinguisher Nozzle Caps Part #: B26002-1 Supplier: 81205 Part #: B26002-13 Supplier: 81205
STD-1115	Source - Nitrogen, 0-100 PSIG

D. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

E. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left

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(Continued)

Zone	Area
142	Aft Cargo Compartment - Right

F. Prepare for the Test

SUBTASK 26-23-00-200-001

(1) Do a leak check of the FWD and AFT discharge heads of each replaced cargo fire extinguisher bottle as follows:

(a) Do a leak check for the discharge head (TASK 26-23-01-200-801).

(b) If the discharge head pressure adapter, COM-25014, and female end pressure adapter, COM-25015, are not available, do these steps for an alternative procedure of the discharge head leak check:

NOTE: The alternative procedure increases the number of positions where the cargo fire extinguish tubing system is disconnected.

1) Connect the discharge tubes to the discharge ports.

NOTE: Use the discharge tube connection steps below.

2) Do the alternate leak check for the discharge head (TASK 26-23-01-200-802).

SUBTASK 26-23-00-860-015

(2) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

SUBTASK 26-23-00-010-008

(3) If necessary, remove the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

G. Cargo Fire Extinguishing Discharge Line Pressure Test

SUBTASK 26-23-00-010-005

(1) If it is necessary, remove the lower shield to get access to the cargo fire bottle discharge lines.

SUBTASK 26-23-00-020-007

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE FITTING. IF YOU TWIST THE TUBE OR FITTING, YOU CAN CAUSE DAMAGE TO THEM.

(2) Disconnect the forward discharge tube from the discharge outlet on cargo fire bottle 1.

(a) Put an Electrostatic Discharge Sensitive (ESDS) protective cap on the discharge outlet.

SUBTASK 26-23-00-020-009

(3) Disconnect the forward discharge tube from the filter/dryer output on cargo fire bottle 2.

(a) Use equipment, SPL-1647, to plug the forward discharge tube.

(b) Put an ESDS protective cap on the filter/dryer output.

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SUBTASK 26-23-00-480-004

CAUTION

DO NOT APPLY AIR PRESSURE TO THE OUTPUT OF THE PRESSURE FILTER/DRIER. AIR PRESSURE CAN CAUSE DAMAGE TO THE FILTER/DRIER.

(4) Connect a 0-100 PSIG nitrogen source, STD-1115, to the forward (yellow) cargo bottle 1 discharge line.

NOTE: Alternate pressure sources can be used. The pressure source must have an accuracy of 1 PSIG and be capable of applying a pressure of up to 55 PSIG.

(a) If it is necessary, use equipment, SPL-1647, to connect a source to the discharge line.

SUBTASK 26-23-00-480-005

(5) Install the nozzle caps, SPL-1648, on the fire extinguishing nozzles in the forward cargo compartment.

SUBTASK 26-23-00-780-001

(6) Pressurize the fire extinguishing plumbing to 50 psig (344.7 kPa) - 55 psig (379.2 kPa).

(a) Wait at least 3 minutes for the air pressure to stabilize.

SUBTASK 26-23-00-780-002

(7) Turn off the pressure source.

(a) Record the stabilized air pressure.

(b) Wait 10 minutes, then check the pressure.

1) Make sure that the pressure loss during the 10 minute waiting period is not more than 5 psi (34.5 kPa).

SUBTASK 26-23-00-780-003

(8) Bleed the fire extinguishing plumbing to 0 psi.

SUBTASK 26-23-00-020-018

(9) Disconnect the 0-100 PSIG nitrogen source, STD-1115, from the forward (yellow) cargo bottle 1 discharge line.

SUBTASK 26-23-00-020-019

(10) Remove the equipment, SPL-1647, from the forward discharge tube for cargo fire bottle 2.

SUBTASK 26-23-00-020-020

(11) Remove the ESDS protective caps from the discharge outlet on cargo fire bottle 1 and the filter/dryer output on cargo fire bottle 2.

SUBTASK 26-23-00-420-004

(12) Connect the forward discharge tubes to the cargo fire bottle 1 and 2.

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Apply a layer of compound, D50004, (preferred) to these locations:

- Threads of the discharge outlet and filter/dryer output

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- End of the discharge outlet and filter/dryer output
- Shoulder sleeve of the discharge tube.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Permitted alternative anti-corrosion compounds for use in this location are Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136.

(b) Connect the discharge tubes to the discharge outlet on cargo fire bottle 1 and the filter/dryer output on cargo fire bottle 2.

NOTE: The discharge lines that are marked in yellow supply the extinguishing agent to the forward cargo compartment.

(c) Remove any unwanted compound.

(d) Tighten the nuts on the discharge tubes to these torque values:

Discharge Tube Torque Values

Nut Size	Tube Outer Diameter	Torque (MIN)	Torque (MAX)
7/8 in	1/2 in	266 in-lb (30 N·m)	294 in-lb (33 N·m)
1 in	5/8 in	342 in-lb (39 N·m)	378 in-lb (43 N·m)

SUBTASK 26-23-00-080-003

(13) Remove the nozzle caps, SPL-1648, from the discharge nozzles in the forward cargo compartment.

SUBTASK 26-23-00-020-021

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE FITTING. IF YOU TWIST THE TUBE OR FITTING, YOU CAN CAUSE DAMAGE TO THEM.

(14) Disconnect the aft discharge tube from the discharge outlet on cargo fire bottle 1.

(a) Put an ESDS protective cap on the discharge outlet.

SUBTASK 26-23-00-020-022

(15) Disconnect the aft discharge tube from the filter/dryer output on cargo fire bottle 2.

(a) Use equipment, SPL-1647, to plug the aft discharge tube.

(b) Put an ESDS protective cap on the filter/dryer output.

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SUBTASK 26-23-00-480-006

CAUTION

DO NOT APPLY AIR PRESSURE TO THE OUTPUT OF THE PRESSURE FILTER/DRIER. AIR PRESSURE CAN CAUSE DAMAGE TO THE FILTER/DRIER.

(16) Connect a 0-100 PSIG nitrogen source, STD-1115, to the aft (blue) cargo bottle 1 discharge line.

NOTE: Alternate pressure sources can be used. The pressure source must have an accuracy of 1 PSIG and be capable of applying a pressure of up to 55 PSIG.

(a) If it is necessary, use equipment, SPL-1647, to connect a source to the discharge line.

SUBTASK 26-23-00-480-007

(17) Install the nozzle caps, SPL-1648, on the fire extinguishing nozzles in the aft cargo compartment.

SUBTASK 26-23-00-780-004

(18) Pressurize the fire extinguishing plumbing to 50 psig (344.7 kPa) - 55 psig (379.2 kPa).

(a) Wait at least 3 minutes for the air pressure to stabilize.

SUBTASK 26-23-00-780-005

(19) Turn off the pressure source.

(a) Record the stabilized air pressure.

(b) Wait 10 minutes, then check the pressure.

1) Make sure that the pressure loss during the 10 minute waiting period is not more than 5 psi (34.5 kPa).

SUBTASK 26-23-00-780-006

(20) Bleed the fire extinguishing plumbing to 0 psi.

SUBTASK 26-23-00-020-023

(21) Disconnect the 0-100 PSIG nitrogen source, STD-1115, from the aft (blue) cargo bottle 1 discharge line.

SUBTASK 26-23-00-020-024

(22) Remove the equipment, SPL-1647, from the aft discharge tube for cargo fire bottle 2.

SUBTASK 26-23-00-020-025

(23) Remove the ESDS protective caps from the discharge outlet on cargo fire bottle 1 and the filter/dryer output on cargo fire bottle 2.

SUBTASK 26-23-00-420-005

(24) Connect the aft discharge tubes to the cargo fire bottle 1 and 2.

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Apply a layer of compound, D50004, (preferred) to these locations:

- Threads of the discharge outlet and filter/dryer output

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- End of the discharge outlet and filter/dryer output
- Shoulder sleeve of the discharge tube.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Permitted alternative anti-corrosion compounds for use in this location are Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136.

(b) Connect the discharge tubes to the discharge outlet on cargo fire bottle 1 and the filter/dryer output on cargo fire bottle 2.

NOTE: The discharge lines that are marked in blue supply the extinguishing agent to the aft cargo compartment.

(c) Remove any unwanted compound.

(d) Tighten the nuts on the discharge tubes to these torque values:

Discharge Tube Torque Values

Nut Size	Tube Outer Diameter	Torque (MIN)	Torque (MAX)
7/8 in	1/2 in	266 in-lb (30 N·m)	294 in-lb (33 N·m)
1 in	5/8 in	342 in-lb (39 N·m)	378 in-lb (43 N·m)

SUBTASK 26-23-00-020-026

(25) Remove the nozzle caps, SPL-1648, from the discharge nozzles in the aft cargo compartment.

SUBTASK 26-23-00-410-003

(26) If removed, install the lower shield in front of the cargo fire bottle discharge lines.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-00-410-007

(1) Install the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 26-23-00-860-016

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

END OF TASK

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CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the cargo fire extinguisher bottle

(2) An installation of the cargo fire extinguisher bottle

(3) A removal of the cargo fire extinguisher bottle with tool

(4) An installation of the cargo fire extinguisher bottle with tool.

B. Two cargo fire extinguisher bottles are installed in the mix manifold bay, in the aft end of the lower fwd cargo compartment.

TASK 26-23-01-000-801

2. Cargo Fire Extinguisher Bottle Removal

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to remove the cargo fire extinguisher bottles.

(2) When either extinguisher bottle discharges, replace the filter/dryers.

B. References

Reference	Title
20-10-33-000-801	Power Device Cartridge (Squib) Removal (P/B 401)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
WDM 26-23-11	Wiring Diagram Manual

C. Tools/Equipment

Reference	Description
STD-1168	Cap - Shorting or Faraday Cap

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
821	Forward Cargo Door

F. Prepare for the Removal

SUBTASK 26-23-01-860-017

(1) Make sure that the ARM FWD and ARM AFT switches, on the P8 cargo bay fire control panel, are in the up position.

NOTE: The ARMED lights will be off to indicate that the system is disarmed.

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SUBTASK 26-23-01-860-001

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-01-010-001

(3) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 26-23-01-860-002

(4) Remove the lining at the aft end of the forward cargo compartment, to get access to the air conditioning distribution bay (TASK 50-11-02-000-801).

SUBTASK 26-23-01-860-003

(5) If it is necessary, remove the air conditioning duct that is in front of bottle 2 [2].

SUBTASK 26-23-01-010-005

(6) Remove the lower shield [3] that covers the bottle connections.

(a) Remove the screws [4] and washers [5] that attach the lower shield [3].

G. Cargo Fire Extinguisher Bottle Removal

SUBTASK 26-23-01-020-001

(1) Disconnect the electrical connectors from bottle 1 [1] or bottle 2 [2] (WDM 26-23-11).

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(a) Before touching the squibs, do the task for devices that are sensitive to electrostatic discharge (TASK 20-10-33-000-801).

(b) Disconnect the pressure switch connector, D16576 or D17810 from bottle 1 [1].

(c) Disconnect the FWD squib connector, D16578 or D17811 from bottle 1 [1].

(d) Disconnect the AFT squib connector, D16580 or D17812 from bottle 1 [1].

(e) Disconnect the pressure switch connector, D16588 from bottle 2 [2].

(f) Disconnect the FWD squib connector, D16584 or D17807 from bottle 2 [2].

(g) Disconnect the AFT squib connector, D16586 or D17808 from bottle 2 [2].

SUBTASK 26-23-01-480-001

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

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(WARNING PRECEDES)

CAUTION

DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(2) Install the shorting or faraday cap, STD-1168, on the squibs.

NOTE: The shorting caps are attached to the fire extinguishing bottles. These shorting caps prevent the squibs from possibly activation due to static electricity.

SUBTASK 26-23-01-020-002

(3) Remove the nut [6] and washer [7] that attach the ground straps to the ground lug.

SUBTASK 26-23-01-860-004

(4) Disconnect the discharge tubes from the bottle discharge ports.

SUBTASK 26-23-01-020-019

(5) Remove and discard the lockwires that attach the anti-thrust caps to the discharge tubes.

SUBTASK 26-23-01-420-001

(6) Install the anti-thrust caps on each of the bottle discharge ports.

SUBTASK 26-23-01-420-002

(7) Install the protective caps on each of the discharge tubes.

SUBTASK 26-23-01-020-003

(8) Remove the bolts [8] and washers [9] that attach the bottle 1 [1] and/or bottle 2 [2] to the support bracket.

SUBTASK 26-23-01-020-010

WARNING

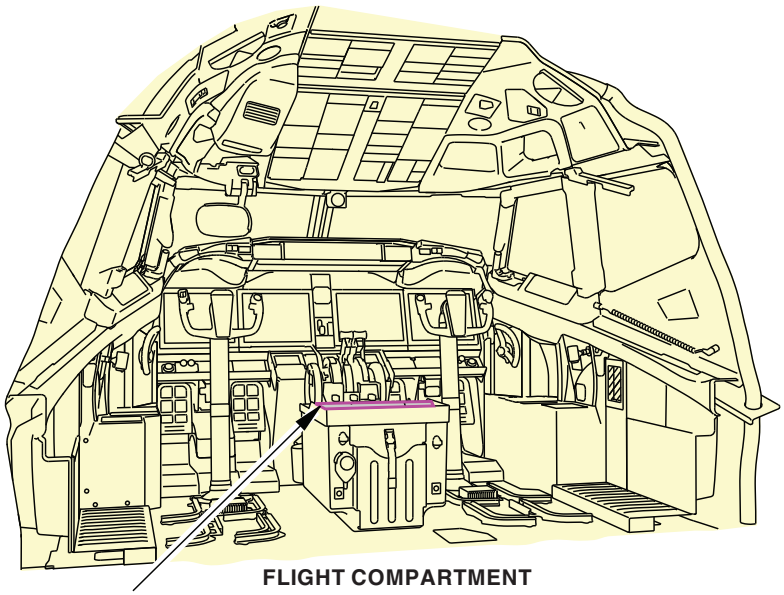
USE SAFE PROCEDURES TO LIFT, REMOVE AND INSTALL THE FIRE BOTTLES. FIRE BOTTLES ARE HEAVY AND CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

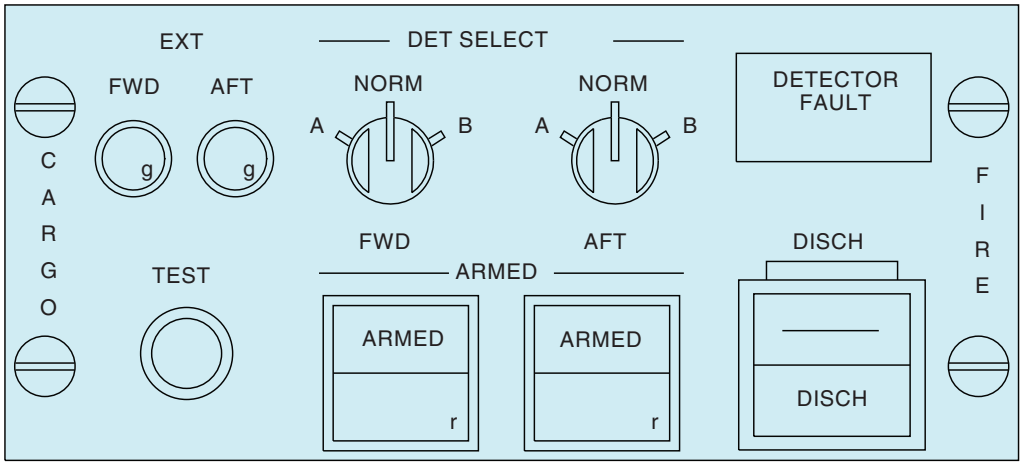
(9) Remove the bottle 1 [1] or bottle 2 [2].

NOTE: Fire bottles can weigh up to 44 lb (20 kg).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





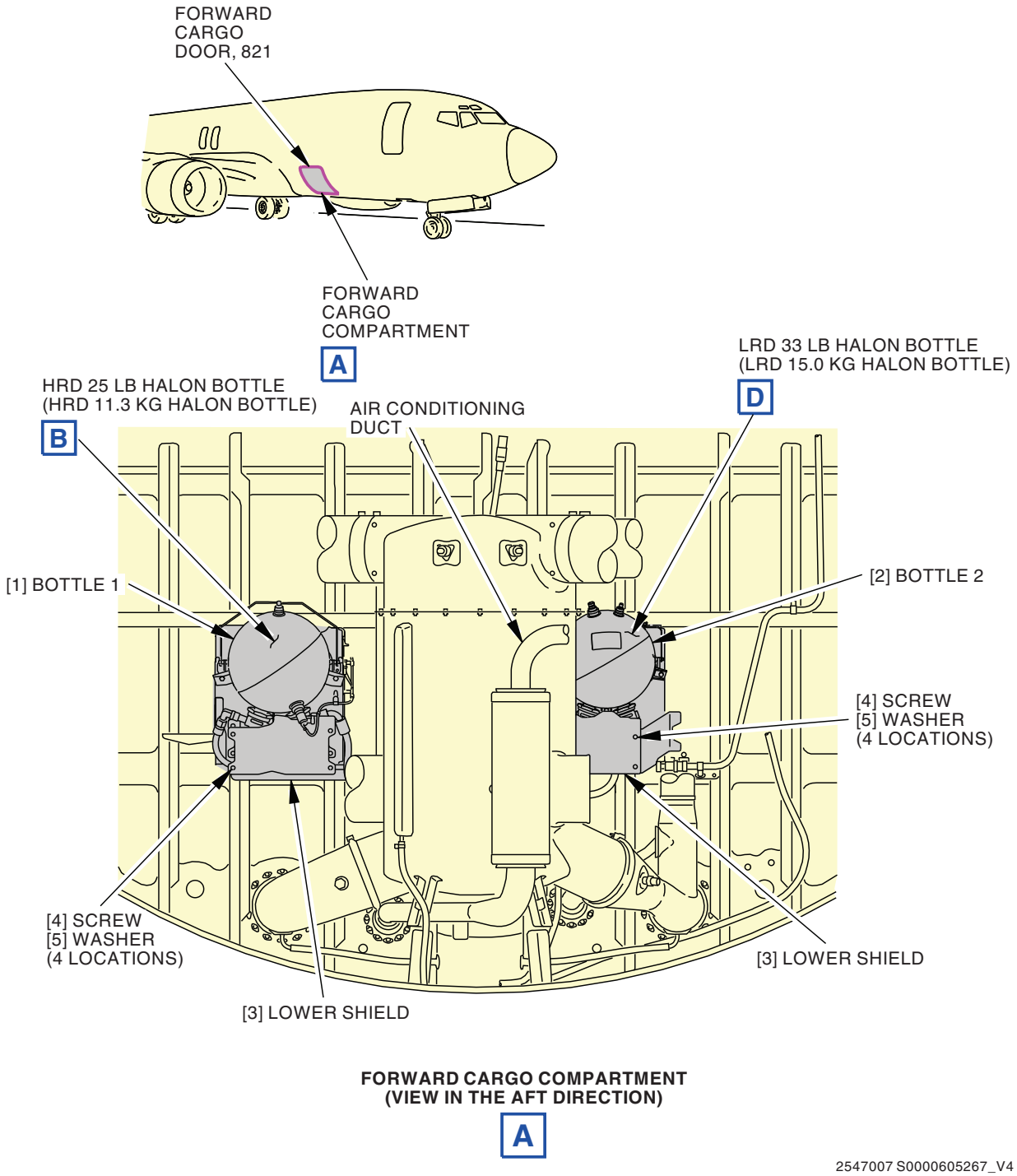
CARGO BAY FIRE CONTROL PANEL (P8)

A

2534661 S0000600502_V1

Cargo Bay Fire Control Panel
Figure 401/26-23-01-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details



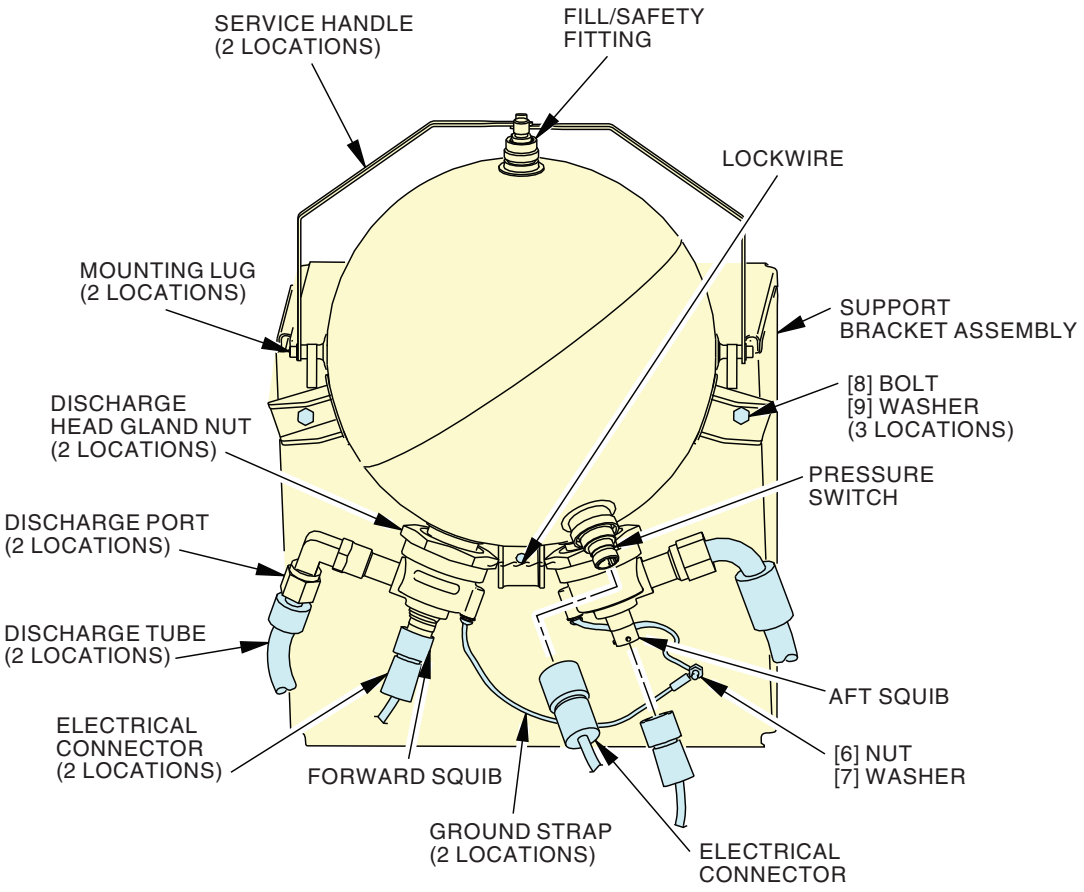
FORWARD CARGO COMPARTMENT
(VIEW IN THE AFT DIRECTION)

A

2547007 S0000605267_V4

Cargo Compartment Fire Extinguisher Bottle Installation
Figure 402/26-23-01-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



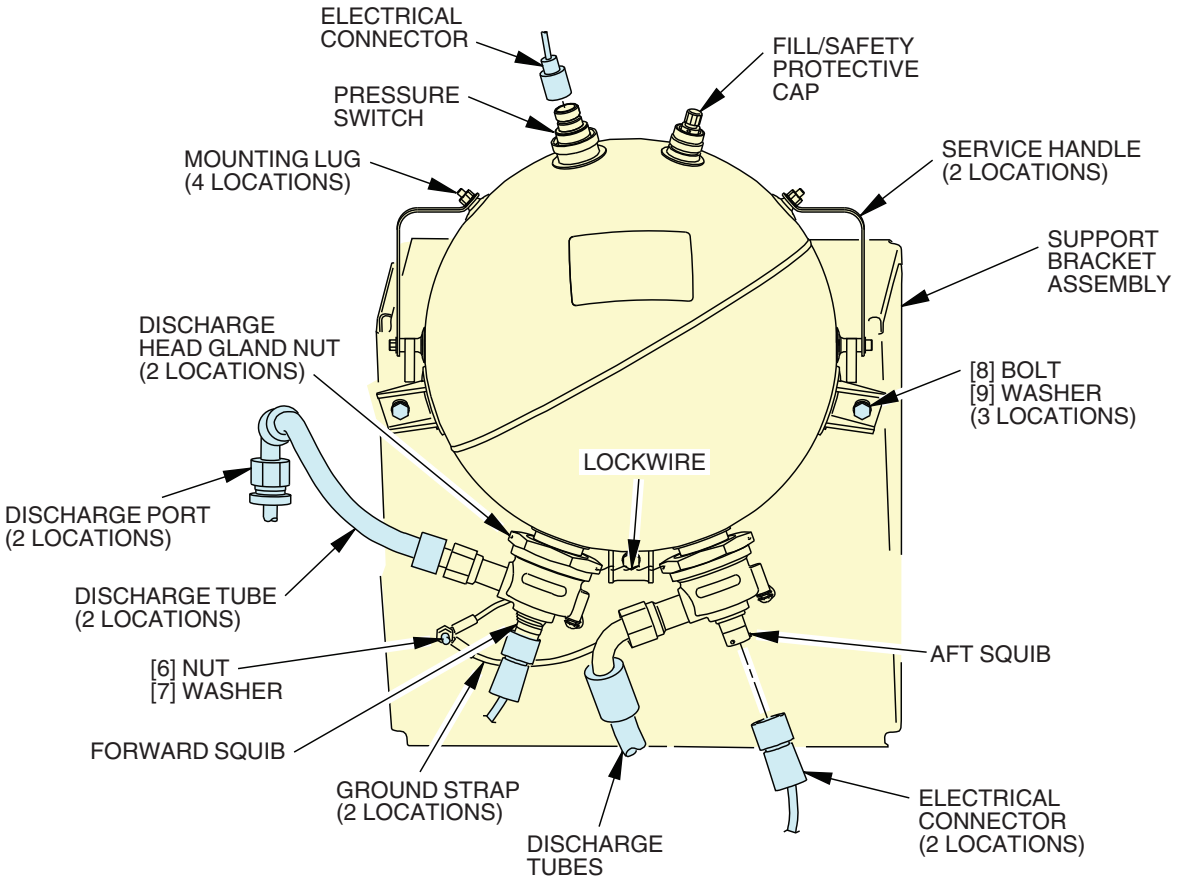
HRD 25 LB HALON BOTTLE
(HRD 11.3 KG HALON BOTTLE)

B

2546795 S0000605273_V4

Cargo Compartment Fire Extinguisher Bottle Installation
Figure 402/26-23-01-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LRD 33 LB HALON BOTTLE
(LRD 15.0 KG HALON BOTTLE)

D

2546975 S0000605268_V4

Cargo Compartment Fire Extinguisher Bottle Installation
Figure 402/26-23-01-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-23-01-400-801

3. Cargo Fire Extinguisher Bottle Installation

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the installation of the cargo fire extinguisher bottles.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-23-01-100-801	Cargo Fire Extinguishing System Cleaning (P/B 701)
26-23-01-200-801	Discharge Head Leak Check (Method 1) (P/B 601)
26-23-01-200-802	Discharge Head Leak Check (Method 2) (P/B 601)
26-23-04-000-801	Cargo Fire Extinguishing Filter/Dryer Removal (P/B 401)
26-23-04-400-801	Cargo Fire Extinguishing Filter/Dryer Installation (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
WDM 26-23-11	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
COM-2529	Gauge - Force, Push/Pull/Spring Scale, 50 Pound Capacity Part #: FDIX 50 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
COM-25014	Adapter - Pressure Discharge Head (7/8-inch fitting at one end and air shop fitting at the other end) Part #: 4-10XHU6-SS MS33514 Supplier: 30780
COM-25015	Adapter - Pressure Female End (3/4-inch fitting at one end and air shop fitting at the other end) Part #: 4-8XHU6-SS MS33514 Supplier: 30780
STD-1079	Resistor - 10K Ohm or Greater
STD-1168	Cap - Shorting or Faraday Cap

D. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
D50004	Compound - Antiseize	BMS3-28
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Bottle 1	26-23-01-05-190	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Bottle 2	26-23-01-05-175	PTF ALL

F. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

G. Access Panels

Number	Name/Location
821	Forward Cargo Door

H. Cargo Fire Extinguisher Bottle Installation

SUBTASK 26-23-01-200-001

(1) If the bottle 1 [1] or bottle 2 [2] discharged, do these steps:

(a) Replace the cargo fire extinguishing filter/dryer (TASK 26-23-04-000-801 and TASK 26-23-04-400-801).

(b) Do this task: Cargo Fire Extinguishing System Cleaning, TASK 26-23-01-100-801.

SUBTASK 26-23-01-200-003

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

WARNING DO NOT MOVE THE BOTTLE WITHOUT A CAP ON THE PORTS. DO NOT LET THE BOTTLE HIT THE AIRPLANE. BE CAREFUL NOT TO DAMAGE THE BOTTLE. IF THE BOTTLE IS ACCIDENTLY DISCHARGED, IT CAN CAUSE INJURY TO PERSONS.

(2) Before you install the bottle 1 [1] or bottle 2 [2], do these steps:

(a) Use gauge, COM-2529, to weigh the bottle 1 [1] or bottle 2 [2].

1) Make sure that the squibs are installed with shorting or faraday cap, STD-1168, when handling the bottle assembly.

(b) Make sure that its weight did not decrease more than the weight shown in Table 401.

NOTE: Depending on the bottle manufacturer or bottle part number, the measured weight marked on the bottle may or may not include some of the protective caps.

Table 401/26-23-01-993-802

Bottle Part Number	Weight Tolerance
475976-X or 473957-X	0.25 lb (0.11 kg)
475975-X	0.13 lb (0.06 kg)

SUBTASK 26-23-01-420-003

(3) Install the bottle 1 [1] or bottle 2 [2] in its position in the airplane, as follows:

(a) Leave the fill/safety protective cap on the bottle 1 [1] or bottle 2 [2].

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WARNING USE SAFE PROCEDURES TO LIFT, REMOVE AND INSTALL THE FIRE BOTTLES. FIRE BOTTLES ARE HEAVY AND CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Lift the bottle 1 [1] or bottle 2 [2] by the service handles and install the mounting lugs on the support bracket assembly.

SUBTASK 26-23-01-420-004

(4) Install the bolts [8] and washers [9] that attach the bottle 1 [1] or bottle 2 [2] to the support bracket assembly.

SUBTASK 26-23-01-420-005

(5) Install the nut [6] and washer [7] that attach the ground straps to the support bracket assembly.

SUBTASK 26-23-01-420-006

(6) Measure the electrical bonding resistance between the bottle 1 [1] or bottle 2 [2] and support bracket assembly (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is 0.001 ohm (1 milliohm) or less.

(c) If the resistance is more than 0.001 ohm (1 milliohm), do these steps:

1) Clean the bonding surfaces between the bottle 1 [1] or bottle 2 [2] and ground straps with cotton wiper, G00034, moisten with alcohol, B00130.

2) Immediately dry the surfaces with a new cotton wiper.

3) Measure resistance again between the bottle 1 [1] or bottle 2 [2] and support bracket assembly.

a) Use an intrinsically safe approved bonding meter, COM-1550.

<1> Make sure that the resistance is 0.001 ohm (1 milliohm) or less.

SUBTASK 26-23-01-020-004

(7) If it is installed, remove and discard the lockwire from the applicable bottle 1 [1] or bottle 2 [2] discharge head gland nuts.

SUBTASK 26-23-01-020-005

(8) If necessary, align the applicable discharge heads of the bottle 1 [1] or bottle 2 [2] to the discharge tubes, do these steps:

(a) Loosen the discharge head gland nuts.

(b) Adjust the discharge heads to get access to the tube connections.

SUBTASK 26-23-01-420-013

(9) Tighten the applicable discharge head gland nuts of the bottle 1 [1] or bottle 2 [2] to 720 ±60 in-lb (81 ±7 N·m).

SUBTASK 26-23-01-080-001

(10) Remove the anti-thrust caps from each of the discharge ports.

SUBTASK 26-23-01-420-007

(11) Use a lockwire to hang the anti-thrust caps on the discharge tubes.

SUBTASK 26-23-01-200-005

(12) Do a leak check of the FWD and AFT discharge heads of each replaced cargo fire extinguisher bottle as follows:

(a) Do a leak check for the discharge head (TASK 26-23-01-200-801).

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(b) If the discharge head pressure adapter, COM-25014, and female end pressure adapter, COM-25015, are not available, do these steps for an alternative procedure of the discharge head leak check:

NOTE: The alternative procedure increases the number of positions where the cargo fire extinguish tubing system is disconnected.

1) Connect the discharge tubes to the discharge ports.

NOTE: Use the discharge tube connection steps below.

2) Do the alternate leak check for the discharge head (TASK 26-23-01-200-802).

SUBTASK 26-23-01-420-014

(13) Install MS20995C32 lockwire, G01048, between the applicable bottle 1 [1] or bottle 2 [2] discharge head gland nuts with the double-twist method.

SUBTASK 26-23-01-420-008

(14) Connect the FWD (Yellow) and AFT (Blue) discharge tubes to the discharge ports, as follows:

(a) Remove the protective caps from each of the discharge tubes.

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

DO NOT APPLY COMPOUND TO THE INNER DIAMETER OF THE DISCHARGE TUBES. COMPOUND CAN CAUSE A BLOCKAGE DAMAGE IN THE FIRE DISCHARGE SYSTEM.

(b) Apply a layer of compound, D50004 (preferred), to the following locations:

- Outlet threads of the tube assembly
- Outlet end of the tube assembly
- Shoulder sleeve of the discharge tube

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Use Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136, as permitted alternative anti-corrosion compounds in this location.

(c) Refer to the outlet identification above each discharge outlet, to make sure that the tubes are connected to the correct ports.

(d) Tighten the nuts on the tube assemblies to these torque values:

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Discharge Tube Torque Values

Nut Size	Tube Outer Diameter	Torque (MIN)	Torque (MAX)
7/8 in	1/2 in	266 in-lb (30 N·m)	294 in-lb (33 N·m)
1 in	5/8 in	342 in-lb (39 N·m)	378 in-lb (43 N·m)

I. Electrical Connector Installation

SUBTASK 26-23-01-420-011

(1) Install the electrical connector to the pressure switch on the bottle 1 [1] (WDM 26-23-11).

(a) Connect a jumper wire between pins 2 and 3 on D16576 or D17810.

1) Make sure that the DISCH light, on the P8 cargo fire control panel, comes on.

2) Remove the jumper wire.

(b) Install the electrical connector to the pressure switch, D16576 or D17810 on the bottle 1 [1].

SUBTASK 26-23-01-420-021

(2) Install the electrical connector to the pressure switch on bottle 2 [2].

(a) Connect a jumper wire between pins 2 and 3 on D16588.

1) Make sure that the DISCH light, on the P8 cargo fire control panel, comes on.

2) Remove the jumper wire.

(b) Install the electrical connector to the pressure switch, D16588 on the bottle 2 [2].

SUBTASK 26-23-01-500-001

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(3) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-23-01-700-001

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(4) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

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SUBTASK 26-23-01-420-015

(5) Make sure that the electrical connector is not damaged.

NOTE: The squib pins can cause damage to the connector, if the pins do not enter the electrical connector receptacles.

SUBTASK 26-23-01-420-016

(6) Remove the shorting or faraday cap, STD-1168, from the bottle 1 [1] squibs.

SUBTASK 26-23-01-420-041

(7) Make sure that the squib pins are not bent or damaged.

SUBTASK 26-23-01-420-045

(8) Connect electrical connector D16578 or D17811 to the forward fire extinguisher bottle squib on the bottle 1 [1].

SUBTASK 26-23-01-420-047

(9) Connect electrical connector D16580 or D17812 to the aft fire extinguisher bottle squib on the bottle 1 [1].

SUBTASK 26-23-01-700-003

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(10) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-23-01-420-042

(11) Make sure that the electrical connector is not damaged.

NOTE: The squib pins can cause damage to the connector, if the pins do not engage the electrical connector receptacles.

SUBTASK 26-23-01-420-055

(12) Remove the shorting or faraday cap, STD-1168, from the bottle 2 [2] squibs.

SUBTASK 26-23-01-420-059

(13) Make sure that the squib pins are not bent or damaged.

SUBTASK 26-23-01-420-046

(14) Connect electrical connector D16584 or D17807 to the forward fire extinguisher bottle squib on the bottle 2 [2].

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SUBTASK 26-23-01-420-048

(15) Connect electrical connector D16586 or D17808 to the aft fire extinguisher bottle squib on the bottle 2 [2].

J. Cargo Fire Extinguisher Bottle Installation Test

SUBTASK 26-23-01-860-033

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-01-700-010

(2) Do a test of the cargo fire extinguishing system, as follows:

(a) Push and hold the TEST switch, on the cargo fire control panel.

(b) Make sure that the green FWD and AFT EXT lights, on the cargo fire control panel, come on.

(c) Release the TEST switch.

(d) Make sure that the green FWD and AFT EXT lights go off.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-01-410-004

(1) Install the lower shield [3] that covers the bottle connections.

(a) Install the screws [4] and washers [5] that attach the lower shield [3].

SUBTASK 26-23-01-410-001

(2) If it is necessary, install the air conditioning duct in front of the bottle 2 [2].

SUBTASK 26-23-01-410-002

(3) Install the liner at the aft end of the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 26-23-01-410-003

(4) Close this access door:

Number	Name/Location
821	Forward Cargo Door

——— END OF TASK ———

TASK 26-23-01-000-802

4. Cargo Fire Extinguisher Bottle Removal - With Tool

(Figure 401, Figure 402)

A. General

(1) This task gives instructions to remove the cargo fire extinguisher bottles using fire extinguisher installation/removal equipment, SPL-5038.

(2) When either extinguisher bottle discharges, replace the filter/dryers.

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B. References

Reference	Title
20-10-33-000-801	Power Device Cartridge (Squib) Removal (P/B 401)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
WDM 26-23-11	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-5038	Instl/Removal Equipment - Fire Extinguisher Bottle (Lwr. Cargo Compt) Part #: C26008-148 Supplier: 81205 Opt Part #: C26008-1 Supplier: 81205
STD-1168	Cap - Shorting or Faraday Cap

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
821	Forward Cargo Door

F. Prepare for the Removal

SUBTASK 26-23-01-860-018

(1) Make sure that the ARM FWD and ARM AFT switches, on the P8 cargo bay fire control panel, are in the up position.

NOTE: The ARMED lights will be off to indicate that the system is disarmed.

SUBTASK 26-23-01-860-019

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-01-010-007

(3) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-01-860-020

(4) Remove the lining at the aft end of the forward cargo compartment to get access to the air conditioning distribution bay (TASK 50-11-02-000-801).

SUBTASK 26-23-01-860-021

(5) If it is necessary, remove the air conditioning duct that is in front of bottle 2 [2].

G. Cargo Fire Extinguisher Bottle Removal

SUBTASK 26-23-01-020-011

(1) Disconnect the electrical connectors from bottle 1 [1] or bottle 2 [2] (WDM 26-23-11).

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(a) Before touching the squibs, do the procedure for devices that are sensitive to electrostatic discharge (TASK 20-10-33-000-801).

(b) Disconnect the pressure switch connector, D16576 or D17810 from bottle 1 [1].

(c) Disconnect the FWD squib connector, D16578 or D17811 from bottle 1 [1].

(d) Disconnect the AFT squib connector, D16580 or D17812 from bottle 1 [1].

(e) Disconnect the pressure switch connector, D16588 from bottle 2 [2].

(f) Disconnect the FWD squib connector, D16584 or D17807 from bottle 2 [2].

(g) Disconnect the AFT squib connector, D16586 or D17808 from bottle 2 [2].

SUBTASK 26-23-01-480-003

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(2) Install the shorting or faraday cap, STD-1168 on the squibs.

NOTE: The shorting caps are attached to the fire extinguishing bottles. These shorting caps prevent the squibs from possible activation due to static electricity.

SUBTASK 26-23-01-020-012

(3) Remove the nut [6] and washer [7] that attach the ground straps to the ground lug.

SUBTASK 26-23-01-860-022

(4) Disconnect the discharge tubes from the bottle discharge ports.

SUBTASK 26-23-01-020-017

(5) Remove and discard the lockwires that attach the anti-thrust caps to the discharge tubes.

SUBTASK 26-23-01-420-023

(6) Install the anti-thrust caps on each of the bottle discharge ports.

SUBTASK 26-23-01-420-024

(7) Install the protective caps on each of the discharge tubes.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-01-020-013

(8) Remove the bolts [8] and washers [9] that attach the bottle 1 [1] and/or bottle 2 [2] to the support bracket.

SUBTASK 26-23-01-020-014

(9) Use the fire extinguisher installation/removal equipment, SPL-5038, to remove bottle 1 [1] or bottle 2 [2].

(a) Use the service handles on the bottle to help transport it to and from the fire extinguisher installation/removal equipment, SPL-5038.

——— END OF TASK ———

TASK 26-23-01-400-802

5. Cargo Fire Extinguisher Bottle Installation - With Tool

(Figure 401, Figure 402)

A. General

(1) This task is for the installation of the cargo fire extinguisher bottles with a tool.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-23-01-100-801	Cargo Fire Extinguishing System Cleaning (P/B 701)
26-23-01-200-801	Discharge Head Leak Check (Method 1) (P/B 601)
26-23-01-200-802	Discharge Head Leak Check (Method 2) (P/B 601)
26-23-04-000-801	Cargo Fire Extinguishing Filter/Dryer Removal (P/B 401)
26-23-04-400-801	Cargo Fire Extinguishing Filter/Dryer Installation (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
WDM 26-23-11	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
COM-2529	Gauge - Force, Push/Pull/Spring Scale, 50 Pound Capacity Part #: FDIX 50 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
COM-25014	Adapter - Pressure Discharge Head (7/8-inch fitting at one end and air shop fitting at the other end) Part #: 4-10XHU6-SS MS33514 Supplier: 30780
COM-25015	Adapter - Pressure Female End (3/4-inch fitting at one end and air shop fitting at the other end) Part #: 4-8XHU6-SS MS33514 Supplier: 30780
SPL-5038	Instl/Removal Equipment - Fire Extinguisher Bottle (Lwr. Cargo Compt) Part #: C26008-148 Supplier: 81205 Opt Part #: C26008-1 Supplier: 81205
STD-1079	Resistor - 10K Ohm or Greater
STD-1168	Cap - Shorting or Faraday Cap

D. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
D50004	Compound - Antiseize	BMS3-28
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Bottle 1	26-23-01-05-190	PTF ALL
2	Bottle 2	26-23-01-05-175	PTF ALL

F. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

G. Access Panels

Number	Name/Location
821	Forward Cargo Door

H. Cargo Fire Extinguisher Bottle Installation

SUBTASK 26-23-01-200-002

(1) If the bottle 1 [1] or bottle 2 [2] discharged, do these steps:

(a) Replace the cargo fire extinguishing filter/dryer (TASK 26-23-04-000-801 and TASK 26-23-04-400-801).

(b) Do this task: Cargo Fire Extinguishing System Cleaning, TASK 26-23-01-100-801.

SUBTASK 26-23-01-200-004

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

WARNING DO NOT MOVE THE BOTTLE WITHOUT A CAP ON THE PORTS. DO NOT LET THE BOTTLE HIT THE AIRPLANE. BE CAREFUL NOT TO DAMAGE THE BOTTLE. IF THE BOTTLE IS ACCIDENTLY DISCHARGED, IT CAN CAUSE INJURY TO PERSONS.

(2) Before you install the bottle 1 [1] or bottle 2 [2], do these steps:

(a) Use gauge, COM-2529, to weigh the bottle 1 [1] or bottle 2 [2].

1) Removal all protective caps before weighing.

2) Make sure that the squibs are installed with shorting or faraday cap, STD-1168 when handling the bottle assembly, except at weighing.

(b) Make sure that its weight did not decrease more than the weight shown in Table 402.

NOTE: Depending on the bottle manufacturer or bottle part number, the measured weight marked on the bottle may or may not include some of the protective caps.

Table 402/26-23-01-993-803

Bottle Part Number	Weight Tolerance
475976-X or 473957-X	0.25 lb (0.11 kg)
475975-X	0.13 lb (0.06 kg)

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SUBTASK 26-23-01-420-025

(3) Install the bottle 1 [1] or bottle 2 [2] with the tool, as follows:

(a) Use the fire extinguisher installation/removal equipment, SPL-5038, to put the bottle 1 [1] or bottle 2 [2] in its position in the airplane.

(b) Leave the fill/safety protective cap on the bottle 1 [1] or bottle 2 [2].

WARNING USE SAFE PROCEDURES TO LIFT, REMOVE AND INSTALL THE FIRE BOTTLES. FIRE BOTTLES ARE HEAVY AND CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(c) Lift the bottle 1 [1] or bottle 2 [2] by the service handles and install the mounting lugs on the support bracket assembly.

SUBTASK 26-23-01-420-026

(4) Install the bolts [8] and washers [9] that attach the bottle 1 [1] or bottle 2 [2] to the support bracket assembly.

SUBTASK 26-23-01-420-027

(5) Install the nut [6] and washer [7] that attach the ground straps to the support bracket assembly.

SUBTASK 26-23-01-420-028

(6) Measure the resistance between the bottle 1 [1] or bottle 2 [2] and support bracket assembly (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is 0.001 ohm (1 milliohm) or less.

(c) If the resistance is more than 0.001 ohm, do these steps:

1) Clean the bonding surfaces between the bottle 1 [1] or bottle 2 [2] and ground straps with cotton wiper, G00034, moisten with alcohol, B00130.

2) Immediately dry the surfaces with a new cotton wiper.

3) Measure resistance again between the bottle 1 [1] or bottle 2 [2] and support bracket assembly (SWPM 20-20-00).

a) Use an intrinsically safe approved bonding meter, COM-1550.

<1> Make sure that the resistance is 0.001 ohm (1 milliohm) or less.

SUBTASK 26-23-01-020-016

(7) If it is installed, remove and discard the lockwire from the applicable bottle 1 [1] or bottle 2 [2] discharge head gland nuts.

SUBTASK 26-23-01-020-015

(8) If necessary, align the applicable discharge heads of the bottle 1 [1] or bottle 2 [2] to the discharge tubes, do these steps:

(a) Loosen the discharge head gland nuts.

(b) Adjust the discharge heads to get access to the tube connections.

SUBTASK 26-23-01-420-061

(9) Tighten the applicable discharge head gland nuts of the bottle 1 [1] or bottle 2 [2] to 720 ±60 in-lb (81 ±7 N·m).

SUBTASK 26-23-01-080-003

(10) Remove the anti-thrust caps from each of the discharge ports.

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SUBTASK 26-23-01-420-029

(11) Use a lockwire to hang the anti-thrust caps on the discharge tubes.

SUBTASK 26-23-01-200-006

(12) Do a leak check of the FWD and AFT discharge heads of each replaced cargo fire extinguisher bottle as follows:

(a) Do this task (TASK 26-23-01-200-801).

(b) If the discharge head pressure adapter, COM-25014, and female end pressure adapter, COM-25015, are not available, do these steps for an alternative procedure of the discharge head leak check:

NOTE: The alternative procedure increases the number of positions where the cargo fire extinguish tubing system is disconnected.

1) Connect the discharge tubes to the discharge ports.

NOTE: Use the discharge tube connection steps below.

2) Do the alternate leak check for the discharge head (TASK 26-23-01-200-802).

SUBTASK 26-23-01-420-063

(13) Install MS20995C32 lockwire, G01048, between the applicable bottle 1 [1] or bottle 2 [2] discharge head gland nuts with the double-twist method.

SUBTASK 26-23-01-020-020

(14) Connect the FWD (Yellow) and AFT (Blue) discharge tubes to the discharge ports.

(a) Remove the protective caps from each of the discharge tubes.

WARNING DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT APPLY COMPOUND TO THE INNER DIAMETER OF THE DISCHARGE TUBES. COMPOUND CAN CAUSE A BLOCKAGE DAMAGE IN THE FIRE DISCHARGE SYSTEM.

(b) Apply a layer of compound, D50004 (preferred), to the following locations:

- Outlet threads of the tube assembly
- Outlet end of the tube assembly
- Shoulder sleeve of the discharge tube

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Use Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136, as permitted alternative anti-corrosion compounds in this location.

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(c) Refer to the outlet identification above each discharge outlet to make sure that the tubes are connected to the correct ports.

(d) Tighten the nuts on the tube assemblies to these torque values:

Discharge Tube Torque Values

Nut Size	Tube Outer Diameter	Torque (MIN)	Torque (MAX)
7/8 in	1/2 in	266 in-lb (30 N·m)	294 in-lb (33 N·m)
1 in	5/8 in	342 in-lb (39 N·m)	378 in-lb (43 N·m)

I. Electrical Connector Installation

SUBTASK 26-23-01-420-034

(1) Install the electrical connector to the pressure switch on the bottle 1 [1] (WDM 26-23-11).

(a) Connect a jumper wire between pins 2 and 3 on D16576 or D17810.

1) Make sure that the DISCH light, on the P8 cargo fire control panel, comes on.

2) Remove the jumper wire.

(b) Install the electrical connector to the pressure switch, D16576 or D17810 on the bottle 1 [1].

SUBTASK 26-23-01-420-032

(2) Install the electrical connector to the pressure switch on the bottle 2 [2].

(a) Connect a jumper wire between pins 2 and 3 on D16588.

1) Make sure that the DISCH light, on the P8 cargo fire control panel, comes on.

2) Remove the jumper wire.

(b) Install the electrical connector to the pressure switch, D16588 on the bottle 2 [2].

SUBTASK 26-23-01-910-001

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(3) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 26-23-01-700-005

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(4) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

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(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-23-01-420-049

(5) Make sure that the electrical connector is not damaged.

NOTE: The squib pins can cause damage to the connector if the pins do not enter the electrical connector receptacles.

SUBTASK 26-23-01-420-056

(6) Remove the shorting or faraday cap, STD-1168 from the bottle 1 [1] squibs.

SUBTASK 26-23-01-420-038

(7) Make sure that the squib pins are not bent or damaged.

SUBTASK 26-23-01-420-050

(8) Connect electrical connector D16578 or D17811 to the forward fire extinguisher bottle squib on the bottle 1 [1].

SUBTASK 26-23-01-420-052

(9) Connect electrical connector D16580 or D17812 to the aft fire extinguisher bottle squib on the bottle 1 [1].

SUBTASK 26-23-01-700-006

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(10) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-23-01-420-043

(11) Make sure that the electrical connector is not damaged.

NOTE: The squib pins can cause damage to the connector if the pins do not engage the electrical connector receptacles.

SUBTASK 26-23-01-420-057

(12) Remove the shorting or faraday cap, STD-1168 from the bottle 2 [2] squibs.

SUBTASK 26-23-01-420-040

(13) Make sure that the squib pins are not bent or damaged.

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SUBTASK 26-23-01-420-051

(14) Connect electrical connector D16584 or D17807 to the forward fire extinguisher bottle squib on the bottle 2 [2].

SUBTASK 26-23-01-420-053

(15) Connect electrical connector D16586 or D17808 to the aft fire extinguisher bottle squib on the bottle 2 [2].

J. Cargo Fire Extinguisher Bottle Installation Test

SUBTASK 26-23-01-860-025

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-01-700-007

(2) Do a test of the cargo fire extinguishing system, as follows:

(a) Push and hold the TEST switch on the cargo fire control panel.

(b) Make sure that the green FWD and AFT EXT lights on the cargo fire control panel come on.

(c) Release the TEST switch.

(d) Make sure that the green FWD and AFT EXT lights go off.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-01-410-007

(1) If it is necessary, install the air conditioning duct in front of the bottle 2 [2].

SUBTASK 26-23-01-410-008

(2) Install the liner at the aft end of the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 26-23-01-410-019

(3) Close this access door:

Number	Name/Location
821	Forward Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE - INSPECTION/CHECK

1. General

A. This procedure has these tasks for the Cargo Fire Extinguisher Bottle:

(1) Discharge Head Leak Check (Method 1).

NOTE: This method is used to connect the air source to the discharge head when the pressure adapters, COM-25014 and COM-25015, are available.

(2) Discharge Head Leak Check (Method 2).

NOTE: This method is used to connect the air source to the far opposite end of the discharge tube, not connected to the discharge head. Use test equipment, SPL-1647.

TASK 26-23-01-200-801

2. Discharge Head Leak Check (Method 1)

(Figure 601, Figure 602)

A. General

(1) This task gives instructions to do the leak check for the discharge heads of the High Rate Discharge and Low Rate Discharge cargo fire extinguisher bottles.

(a) This method is used to connect the air source to the discharge head ports.

(2) The discharge head leak check must be done every time the cargo fire extinguisher bottle is installed.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-23-01 P/B 401	CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE - REMOVAL/INSTALLATION
26-23-01-000-801	Cargo Fire Extinguisher Bottle Removal (P/B 401)
26-23-01-400-801	Cargo Fire Extinguisher Bottle Installation (P/B 401)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
COM-25014	Adapter - Pressure Discharge Head (7/8-inch fitting at one end and air shop fitting at the other end) Part #: 4-10XHU6-SS MS33514 Supplier: 30780
COM-25015	Adapter - Pressure Female End (3/4-inch fitting at one end and air shop fitting at the other end) Part #: 4-8XHU6-SS MS33514 Supplier: 30780
STD-1079	Resistor - 10K Ohm or Greater
STD-1115	Source - Nitrogen, 0-100 PSIG
STD-1168	Cap - Shorting or Faraday Cap
STD-13740	Gauge - Pressure, 0-60 PSIG, +/-0.5% Accuracy
STD-18519	Air Source - Regulated, Dry Filter, 0-60 PSIG

D. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28
G00091	Compound - Oxygen System Leak Detection - Snoop Leak Detector	MIL-PRF-25567
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

E. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

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F. Access Panels

Number	Name/Location
821	Forward Cargo Door

G. Prepare for the Leak Check

SUBTASK 26-23-01-860-027

(1) Make sure that the FWD ARM and AFT ARM switches, on the P8 cargo bay fire control panel, are in the unarmed (up) position.

NOTE: The ARMED lights will be off to indicate that the system is disarmed.

SUBTASK 26-23-01-860-028

(2) Make sure that these circuit breakers are open and install the safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-01-010-009

(3) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 26-23-01-010-010

(4) Remove the lining at the aft end of the forward cargo compartment, to get access to the air conditioning distribution bay (TASK 50-11-02-000-801).

SUBTASK 26-23-01-010-011

(5) If necessary, remove the air conditioning ducts near bottle 2 [2].

SUBTASK 26-23-01-010-012

(6) If necessary, remove the lower shield [3] that covers the applicable bottle connections.

NOTE: It is possible to remove 3 screws from the lower shield to gain access to the bottle connections.

(a) Remove the screws [4] and washers [5] that attach the lower shield [3].

SUBTASK 26-23-01-020-023

(7) Disconnect the electrical connector [10] and electrical connectors [11] from the applicable cargo fire extinguisher bottle (TASK 26-23-01-000-801).

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(a) Before touching the squibs, do the task for handling devices that are sensitive to electrostatic discharge (TASK 20-40-12-000-802).

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SUBTASK 26-23-01-420-064

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(8) Install the shorting or faraday cap, STD-1168, on the applicable squibs.

NOTE: These shorting caps prevent the squibs from a possible activation because of static electricity. The two squibs on the fire extinguishing bottle use different size caps.

H. Discharge Head Leak Check

SUBTASK 26-23-01-020-024

(1) Disconnect the discharge flex tubes [12] from the applicable cargo fire extinguisher bottle discharge ports.

SUBTASK 26-23-01-480-004

(2) Connect the 0-60 psig dry filtered regulated air source, STD-18519, or 0-100 PSIG nitrogen source, STD-1115, to the discharge head.

(a) Use discharge head pressure adapter, COM-25014, or female end pressure adapter, COM-25015, to connect the air supply hose to the FWD or AFT discharge head.

(b) If necessary, connect pressure gauge, STD-13740, between the air source and the discharge head.

NOTE: It is necessary that the pressure gauge accurately reads a 0.5 psig change.

SUBTASK 26-23-01-780-001

(3) Slowly apply pressure 54 psig (372 kPa) - 56 psig (386 kPa) to the discharge head.

SUBTASK 26-23-01-790-001

(4) With a Snoop Leak Detector compound, G00091, make sure that there are no leaks in the air supply hose and the discharge head pressure adapter, COM-25014, or female end pressure adapter, COM-25015.

SUBTASK 26-23-01-780-002

(5) Do a check for changes in air pressure.

(a) Isolate the dry filtered air or nitrogen source from the cargo fire extinguisher bottle and hold 54 psig (372 kPa) - 56 psig (386 kPa).

(b) Record the initial pressure value.

(c) After a minimum of 5 minutes, record the final pressure value.

(d) Make sure that the pressure has not dropped more than 0.5 psig (3.4 kPa).

(e) If the pressure decreases more than 0.5 psig (3.4 kPa), then do the following:

1) Use Snoop Leak Detector compound, G00091, to check for leakage in the locations that follow (View C, Figure 602):

- Around the gland nut
- Around the neck of the discharge head
- Around the squib

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2) Make the necessary adjustments to the discharge head or gland nut to correct the leakage.

3) If necessary, replace the applicable internal O-ring.

4) Repeat steps to check for changes in air pressure.

5) If the pressure does not hold within 0.5 psig (3.4 kPa) of the initial pressure, then replace the applicable cargo fire extinguisher bottle (PAGEBLOCK 26-23-01/401).

SUBTASK 26-23-01-780-003

(6) Reduce the dry filtered air or nitrogen source pressure to zero.

SUBTASK 26-23-01-080-004

(7) Remove the pressure adapter and the dry filtered air or nitrogen source from the fire extinguisher bottle discharge head.

SUBTASK 26-23-01-700-008

(8) Repeat the leak check for the remaining discharge heads.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-01-420-069

(1) Make sure a MS20995C32 lockwire, G01048, is installed between the gland nuts.

SUBTASK 26-23-01-600-001

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Apply a layer of compound, D50004 (preferred), to the following locations:

- Outlet threads of the discharge port
- Outlet end of the discharge port
- Shoulder sleeve of the discharge flex tube

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Use Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136, as permitted alternative compounds in this location.

SUBTASK 26-23-01-420-065

(3) Connect the FWD (Yellow) and AFT (Blue) discharge flex tubes [12] to the discharge ports.

(a) Refer to the outlet identification above each discharge outlet to make sure that the discharge flex tubes [12] are connected to the correct ports.

SUBTASK 26-23-01-420-066

(4) Tighten the nuts on the discharge flex tubes [12] to these torque values:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Discharge Tube Torque Values

Nut Size	Tube Outer Diameter	Torque (MIN)	Torque (MAX)
7/8 in	1/2 in	266 in-lb (30 N·m)	294 in-lb (33 N·m)
1 in	5/8 in	342 in-lb (39 N·m)	378 in-lb (43 N·m)

SUBTASK 26-23-01-420-067

(5) Install the electrical connector [10] to the pressure switch on bottle 1 [1] or bottle 2 [2](TASK 26-23-01-400-801).

SUBTASK 26-23-01-500-002

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(6) Before you touch the squibs, do the task for handling devices that are sensitive to electrostatic discharge (TASK 20-40-12-400-802).

SUBTASK 26-23-01-760-001

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(7) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-23-01-420-068

(8) Install the electrical connectors [11] to the squibs on bottle 1 [1] or bottle 2 [2] (TASK 26-23-01-400-801).

(a) Make sure that the electrical connectors [11] are not damaged.

NOTE: The squib pins can cause damage to the connector if the pins do not enter the electrical connector receptacles.

(b) Remove the shorting or faraday cap, STD-1168, from the squibs.

(c) Make sure that the squib pins are not bent or damaged.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-01-860-029

(9) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-01-710-002

(10) Do a test of the cargo fire extinguishing system, as follows:

(a) Push and hold the TEST switch, on the cargo fire control panel.

(b) Make sure that the green FWD and AFT EXT lights, on the cargo fire control panel, come on.

(c) Release the TEST switch.

(d) Make sure that the green FWD and AFT EXT lights go off.

SUBTASK 26-23-01-410-011

(11) If it was removed, install the lower shield [3] that covers the bottle connections.

(a) Install the screws [4] and washers [5] that attach the lower shield [3].

SUBTASK 26-23-01-410-012

(12) If it was removed, install the air conditioning ducts near bottle 2 [2].

SUBTASK 26-23-01-410-013

(13) Install the liner at the aft end of the forward cargo compartment (TASK 50-11-02-400-801).

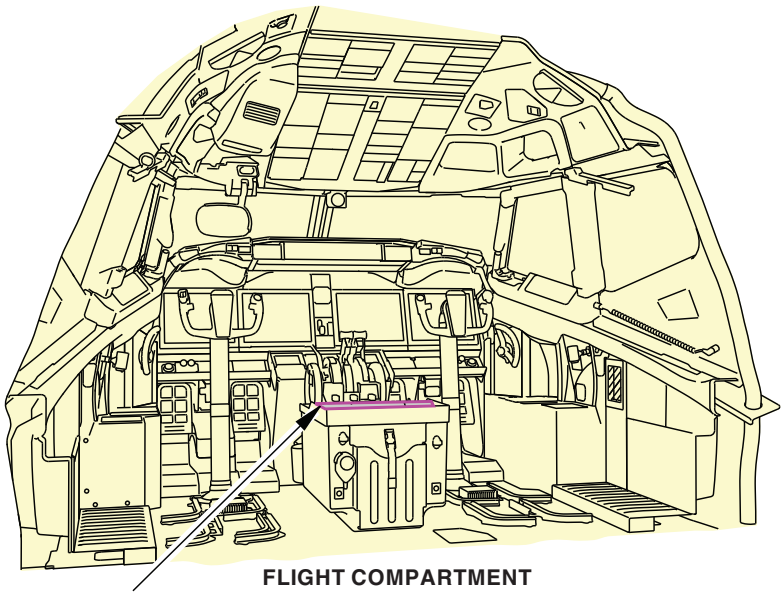
SUBTASK 26-23-01-410-014

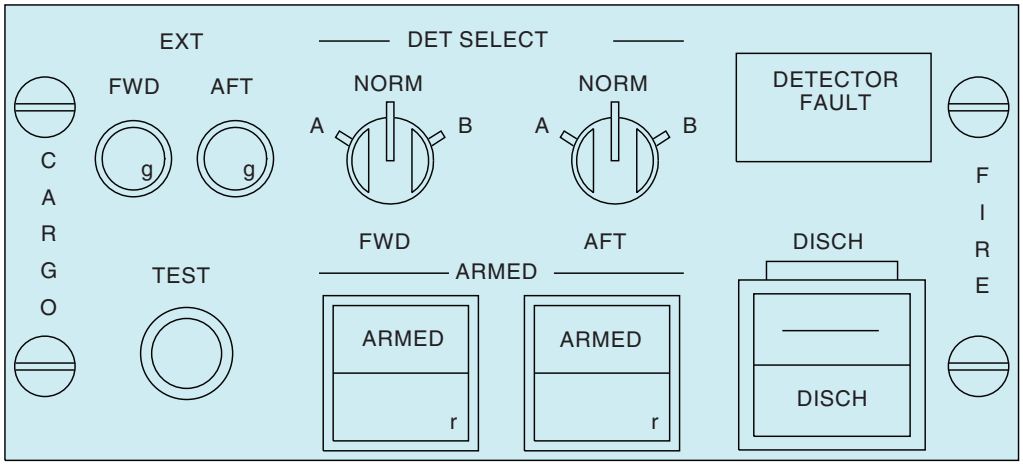
(14) Close this access door:

Number	Name/Location
821	Forward Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





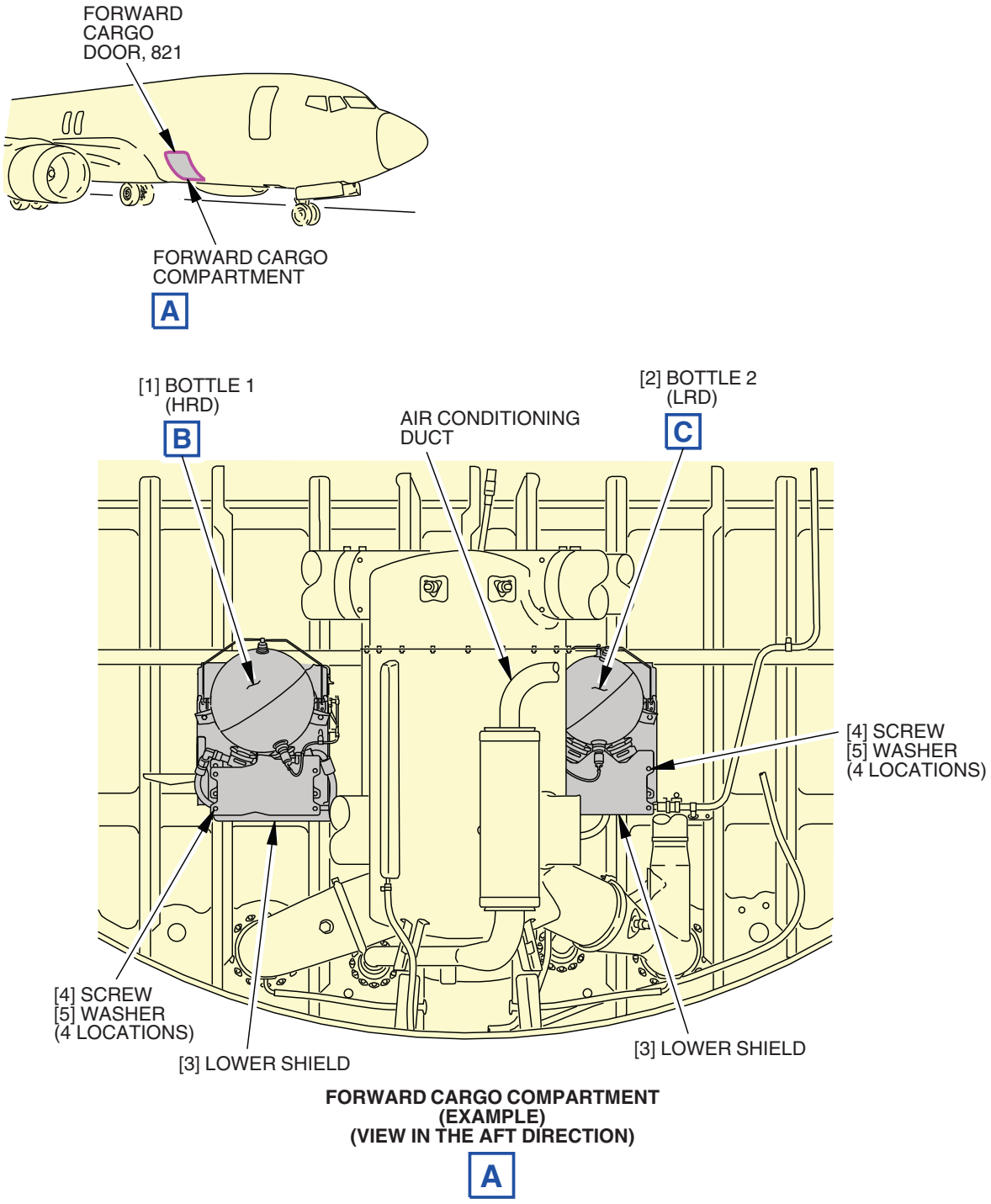
CARGO BAY FIRE CONTROL PANEL (P8)

A

2534661 S0000600502_V1

Cargo Bay Fire Control Panel
Figure 601/26-23-01-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details



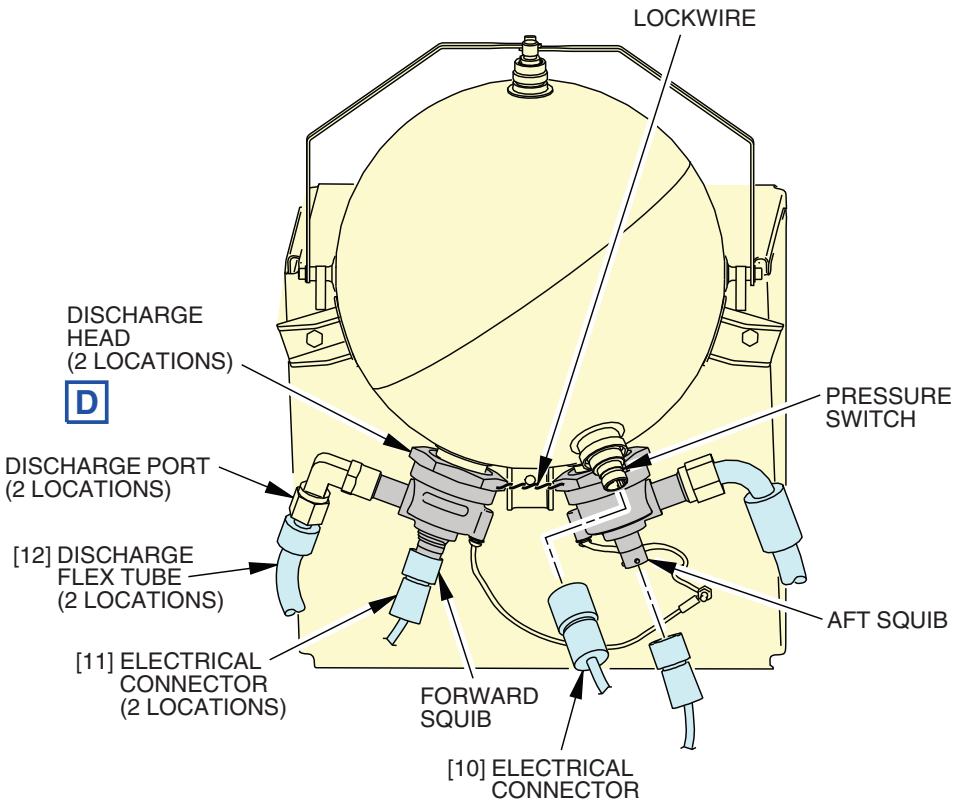
FORWARD CARGO COMPARTMENT
(EXAMPLE)
(VIEW IN THE AFT DIRECTION)

A

3101815 S0000845193_V1

Cargo Fire Extinguisher Bottle - Discharge Head Leak Check
Figure 602/26-23-01-990-809 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



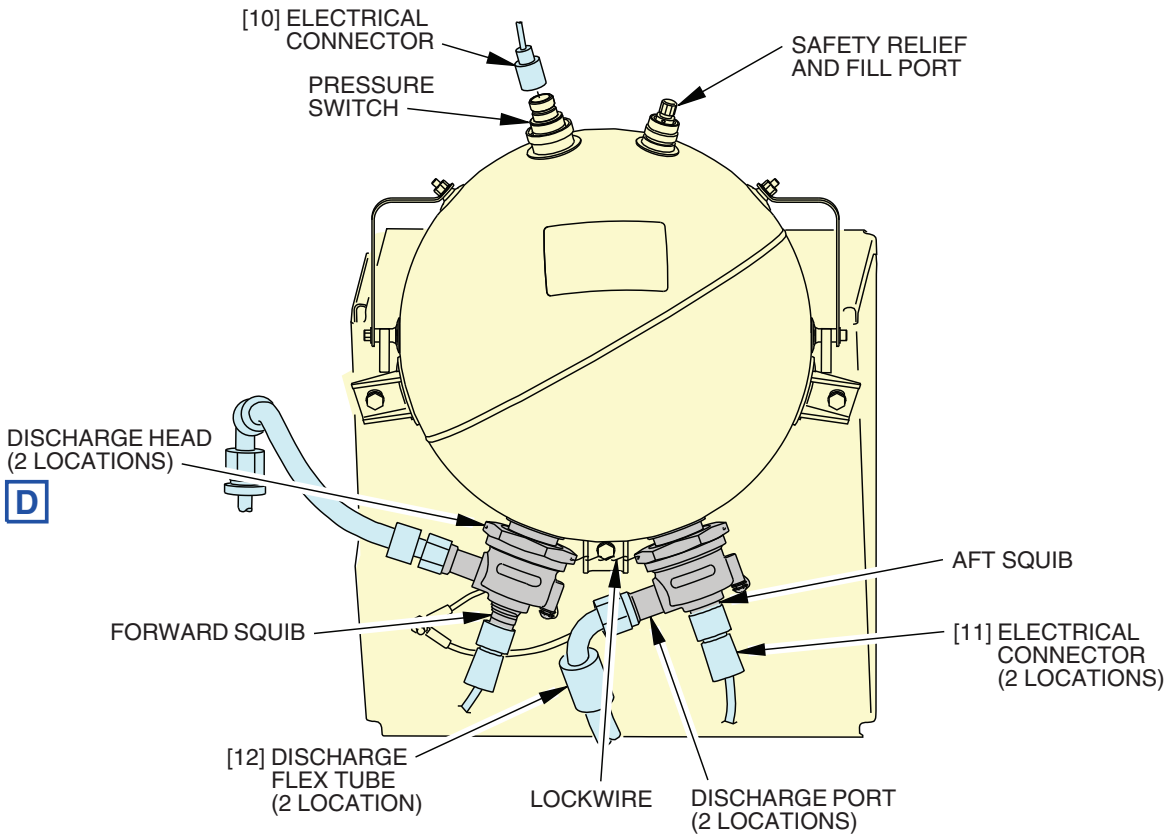
CARGO FIRE EXTINGUISHER BOTTLE (HRD)
(EXAMPLE)

B

3101826 S0000845194_V1

Cargo Fire Extinguisher Bottle - Discharge Head Leak Check
Figure 602/26-23-01-990-809 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



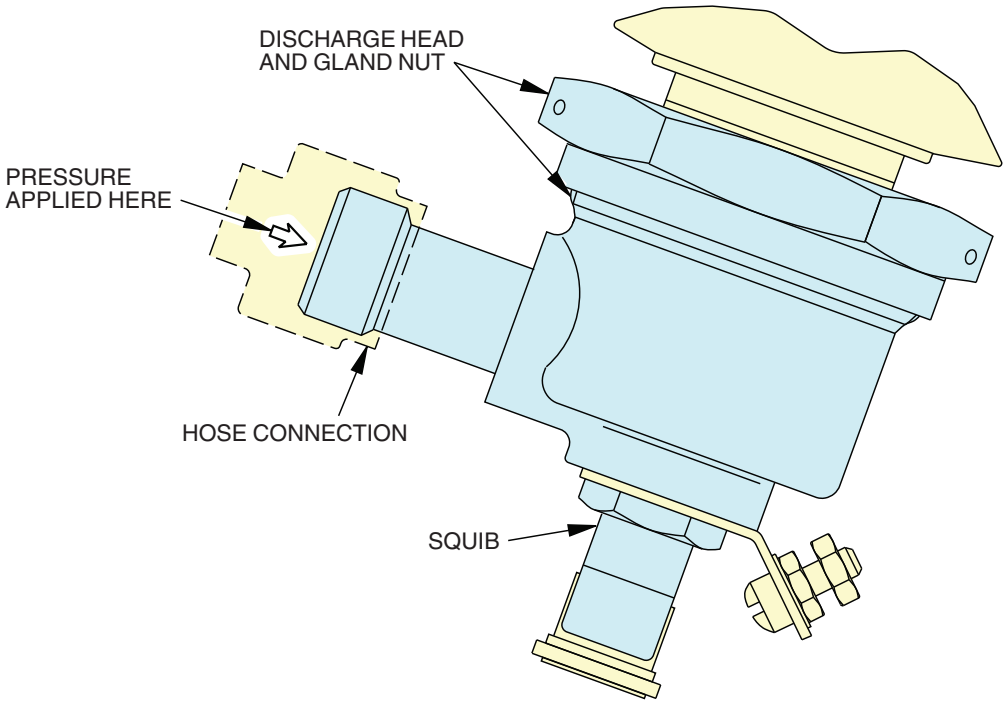
CARGO FIRE EXTINGUISHER BOTTLE (LRD)
(EXAMPLE)

C

3113383 S0000851457_V1

Cargo Fire Extinguisher Bottle - Discharge Head Leak Check
Figure 602/26-23-01-990-809 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



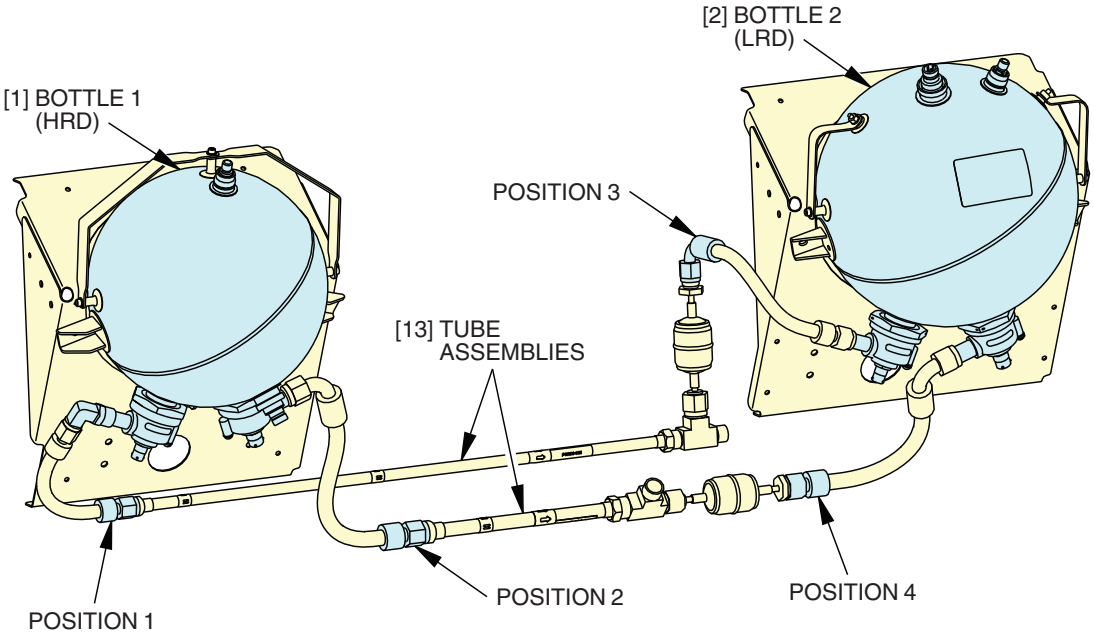
DISCHARGE HEAD LEAK LOCATIONS

D

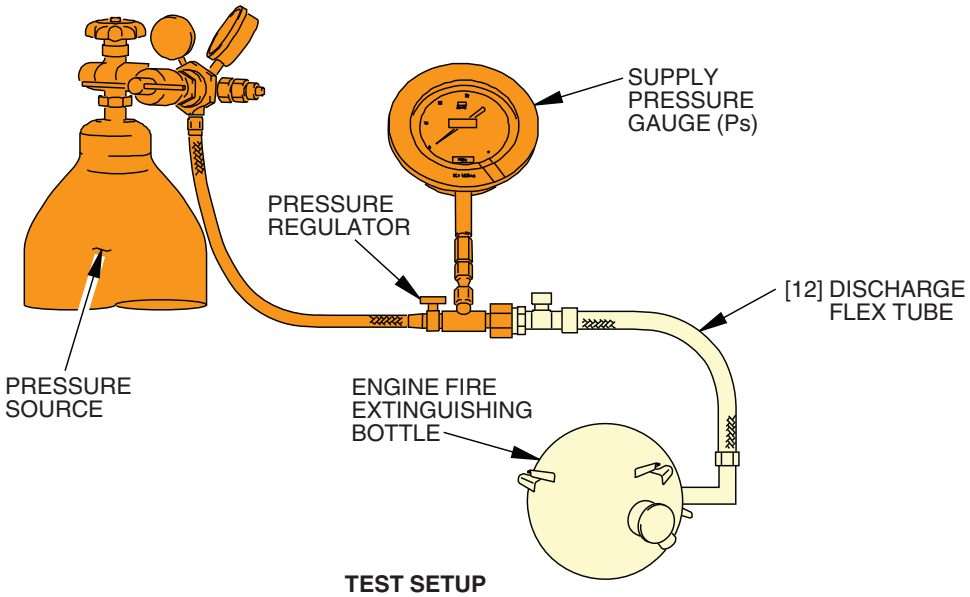
3079215 S0000835679_V2

Cargo Fire Extinguisher Bottle - Discharge Head Leak Check
Figure 602/26-23-01-990-809 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ALTERNATE CONNECTION POSITIONS



TEST SETUP

3101970 S0000845195_V1

Alternate Method - Discharge Head Leak Check
Figure 603/26-23-01-990-810

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-23-01-200-802

3. Discharge Head Leak Check (Method 2)

(Figure 601, Figure 602, Figure 603)

A. General

(1) This task gives alternative instructions to do the leak check for the discharge head of the High Rate Discharge and Low Rate Discharge cargo fire extinguisher bottle.

(a) This method is used to connect the air source to the far opposite end of the discharge flex tube, not connected to the discharge head.

NOTE: Use the alternate method when fittings are not available to connect directly to the discharge heads.

(2) The discharge head leak check must be done every time the cargo fire extinguisher bottle is installed.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
26-23-01 P/B 401	CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE - REMOVAL/INSTALLATION
26-23-01-000-801	Cargo Fire Extinguisher Bottle Removal (P/B 401)
26-23-01-400-801	Cargo Fire Extinguisher Bottle Installation (P/B 401)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-25874	Fitting - Reducer (size 10 to size 8) Part #: AS5230V1008P Supplier: 30974 Part #: MS21916V10-8P Supplier: 81205
SPL-1647	Test Equipment - Engine Fire Extinguishing System Part #: C26004-83 Supplier: 81205 Part #: C26004-84 Supplier: 81205
STD-1079	Resistor - 10K Ohm or Greater
STD-1115	Source - Nitrogen, 0-100 PSIG
STD-1168	Cap - Shorting or Faraday Cap
STD-18519	Air Source - Regulated, Dry Filter, 0-60 PSIG

D. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28
G00091	Compound - Oxygen System Leak Detection - Snoop Leak Detector	MIL-PRF-25567
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

E. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
821	Forward Cargo Door

G. Prepare for the Leak Check

SUBTASK 26-23-01-860-030

(1) Make sure that the FWD ARM and AFT ARM switches, on the P8 cargo bay fire control panel, are in the unarmed (up) position.

NOTE: The ARMED lights will be off to indicate that the system is disarmed.

SUBTASK 26-23-01-860-031

(2) Make sure that these circuit breakers are open and install the safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 26-23-01-010-013

(3) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 26-23-01-010-014

(4) Remove the lining at the aft end of the forward cargo compartment, to get access to the air conditioning distribution bay (TASK 50-11-02-000-801).

SUBTASK 26-23-01-010-015

(5) If necessary, remove the air conditioning ducts near bottle 2 [2].

SUBTASK 26-23-01-010-016

(6) If necessary, remove the lower shield [3] that covers the applicable bottle connections.

NOTE: It is possible to remove 3 screws from the lower shield to gain access to the bottle connections.

(a) Remove the screws [4] and washers [5] that attach the lower shield [3].

SUBTASK 26-23-01-020-025

(7) Disconnect the electrical connector [10] and electrical connectors [11] from the applicable cargo fire extinguisher bottle (TASK 26-23-01-000-801).

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(a) Before touching the squibs, do the task for handling devices that are sensitive to electrostatic discharge (TASK 20-40-12-000-802).

SUBTASK 26-23-01-420-076

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(8) Install the shorting or faraday cap, STD-1168, on the applicable squibs.

NOTE: These shorting caps prevent the squibs from possible activation due to static electricity. The two squibs on the fire extinguishing bottle use different size caps.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Discharge Head Leak Check

SUBTASK 26-23-01-420-070

(1) Make sure the discharge flex tubes [12] are connected to the discharge heads.

SUBTASK 26-23-01-020-026

(2) Disconnect the discharge flex tubes [12], of the applicable cargo fire extinguisher bottle, from the tube assemblies [13] (Figure 603).

SUBTASK 26-23-01-480-005

(3) Connect the 0-60 psig dry filtered regulated air source, STD-18519, or 0-100 PSIG nitrogen source, STD-1115, to the applicable discharge flex tube [12] positions.

(a) Use test equipment, SPL-1647, to connect the air supply hose to the discharge flex tube [12].

(b) If necessary, use reducer, COM-25874, to connect equipment, SPL-1647, to the discharge flex tube [12].

NOTE: The discharge flex tube that connects to the forward squib is smaller. The reducer and SPL-1647 both have male connections. The reducer can be attached to SPL-1647 with a coupler or a hose with female connections on both ends.

SUBTASK 26-23-01-780-004

(4) Slowly apply pressure 54 psig (372 kPa) - 56 psig (386 kPa) to the discharge flex tube.

SUBTASK 26-23-01-790-002

(5) With a Snoop Leak Detector compound, G00091, make sure that there are no leaks in the air supply hose and tubing connections.

SUBTASK 26-23-01-780-005

(6) Do a check for changes in air pressure.

(a) Isolate the dry filtered air or nitrogen source from the cargo fire extinguisher bottle and hold 54 psig (372 kPa) - 56 psig (386 kPa).

(b) Record the initial pressure value.

(c) After a minimum of 5 minutes, record the final pressure value.

(d) Make sure that the pressure has not dropped more than 0.5 psig (3.4 kPa).

(e) If the pressure decreases more than 0.5 psig (3.4 kPa), then do the following:

1) Use Snoop Leak Detector compound, G00091, to check for leakage in the locations that follow (View C, Figure 602):

- Around the gland nut
- Around the neck of the discharge head
- Around the squib

2) Make the necessary adjustments to the discharge head or gland nut to correct the leakage.

3) If necessary, replace the applicable internal O-ring.

4) Repeat steps to check for changes in air pressure.

5) If the pressure does not hold within 0.5 psig (3.4 kPa) of the initial pressure, then replace the applicable cargo fire extinguisher bottle (PAGEBLOCK 26-23-01/401).

SUBTASK 26-23-01-780-006

(7) Reduce the dry filtered air or nitrogen source pressure to zero.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-01-080-005

(8) Remove the test equipment, SPL-1647, and the dry filtered air or nitrogen source from the discharge flex tube [12].

SUBTASK 26-23-01-700-009

(9) Repeat the leak check for the remaining discharge heads.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-01-420-071

(1) Make sure a MS20995C32 lockwire, G01048, is installed between the gland nuts.

SUBTASK 26-23-01-600-002

WARNING DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Apply a layer of compound, D50004 (preferred), to the following locations:

- Outlet threads of the tube assembly
- Outlet end of the tube assembly
- Shoulder sleeve of the discharge flex tube

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Use Cor-Ban 27L Compound, G50237, or corrosion inhibiting material, G50136, as permitted alternative compounds in this location.

SUBTASK 26-23-01-420-072

(3) Connect the FWD (Yellow) and AFT (Blue) discharge flex tubes [12] to the tube assemblies [13].

(a) Refer to the outlet identification above each discharge outlet to make sure that the discharge flex tubes [12] are connected to the correct tube assemblies [13].

SUBTASK 26-23-01-420-073

(4) Tighten the nuts on the discharge flex tubes [12] to these torque values:

Discharge Tube Torque Values

Nut Size	Tube Outer Diameter	Torque (MIN)	Torque (MAX)
7/8 in	1/2 in	266 in-lb (30 N·m)	294 in-lb (33 N·m)
1 in	5/8 in	342 in-lb (39 N·m)	378 in-lb (43 N·m)

SUBTASK 26-23-01-420-074

(5) Install the electrical connector [10] to the pressure switch on bottle 1 [1] or bottle 2 [2] (TASK 26-23-01-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-01-500-003

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(6) Before you touch the squibs, do the task for handling devices that are sensitive to electrostatic discharge (TASK 20-40-12-400-802).

SUBTASK 26-23-01-760-002

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(7) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-23-01-420-075

(8) Install the electrical connectors [11] to the squibs on bottle 1 [1] or bottle 2 [2] (TASK 26-23-01-400-801).

(a) Make sure that the electrical connectors [11] are not damaged.

NOTE: The squib pins can cause damage to the connector if the pins do not enter the electrical connector receptacles.

(b) Remove the shorting or faraday cap, STD-1168, from the squibs.

(c) Make sure that the squib pins are not bent or damaged.

SUBTASK 26-23-01-860-034

(9) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-01-710-003

(10) Do a test of the cargo fire extinguishing system, as follows:

(a) Push and hold the TEST switch, on the cargo fire control panel.

(b) Make sure that the green FWD and AFT EXT lights, on the cargo fire control panel, come on.

(c) Release the TEST switch.

(d) Make sure that the green FWD and AFT EXT lights go off.

SUBTASK 26-23-01-410-015

(11) If it was removed, install the lower shield [3] that covers the bottle connections.

(a) Install the screws [4] and washers [5] that attach the lower shield [3].

SUBTASK 26-23-01-410-016

(12) If it was removed, install the air conditioning ducts near bottle 2 [2].

SUBTASK 26-23-01-410-017

(13) Install the liner at the aft end of the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 26-23-01-410-018

(14) Close this access door:

Number	Name/Location
821	Forward Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE SYSTEM - CLEANING/PAINTING

1. General

A. The cargo fire extinguishing system should be cleaned whenever the cargo compartment fire bottle is discharged.

B. Only the compartment in which the bottle was discharged requires cleaning.

C. This procedure has a task to clean the cargo fire extinguishing system.

TASK 26-23-01-100-801

2. Cargo Fire Extinguishing System Cleaning

(Figure 701)

A. General

(1) There are two bottles on the airplane. In this task, the discharge tubes will be removed from bottle 1 and from the output of the filter/dryers of bottle 2. The flow will be from the output of bottle 1 through the discharge nozzles and the filter/dryer output line.

B. References

Reference	Title
26-23-01-000-801	Cargo Fire Extinguisher Bottle Removal (P/B 401)
26-23-01-400-801	Cargo Fire Extinguisher Bottle Installation (P/B 401)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1325	Source - Pressure, Capable of Supplying 35 +/- 5 PSIG

D. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28

E. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

F. Access Panels

Number	Name/Location
821	Forward Cargo Door
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Cleaning

SUBTASK 26-23-01-860-011

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

SUBTASK 26-23-01-010-003

(2) To access the discharge nozzles in the fwd cargo bay, open this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 26-23-01-010-004

(3) To access the discharge nozzles in the aft cargo bay, open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 26-23-01-860-012

(4) Remove the lining at the aft end of the forward cargo compartment to get access to the air conditioning distribution bay (TASK 50-11-02-000-801).

SUBTASK 26-23-01-860-013

(5) If it is necessary, remove the air conditioning duct that is in front of the fire extinguishing bottle (TASK 26-23-01-000-801).

SUBTASK 26-23-01-010-006

(6) Remove the lower shields that cover the bottle connections (TASK 26-23-01-000-801).

SUBTASK 26-23-01-020-007

(7) Disconnect the applicable discharge tube from the fire extinguisher bottle 1 [1] discharge head.

SUBTASK 26-23-01-020-009

(8) Disconnect the applicable discharge tube from the output side of the filter/dryer of fire extinguisher bottle 2 [2].

SUBTASK 26-23-01-020-008

(9) Remove the discharge nozzles [6] from the discharge tubes in the applicable cargo bay.

(a) Remove the four screws [3] which attach the retainer [4] to the bracket.

(b) Remove the cargo ceiling as required to gain access to the nut.

(c) Remove the spacer [9].

(d) Remove the two screws [5] that attach the discharge nozzle [6] to the pan cover assembly [7].

(e) Remove the discharge nozzle [6].

H. Cargo Fire Extinguishing System Cleaning

SUBTASK 26-23-01-100-001

(1) Make sure that the nozzle [6] is clear of unwanted materials.

SUBTASK 26-23-01-480-002

(2) Connect a pressure source, STD-1325, to the applicable cargo bottle 1 [1] discharge tube.

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(a) Make sure that air is flowing through all the discharge tubes in the cargo compartment and through the filter/dryer output line.

SUBTASK 26-23-01-080-002

(3) Remove the pressure source, STD-1325, from the discharge tube.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-01-420-017

(1) Connect the discharge tube to the discharge outlet on bottle 1 [1].

(a) Apply a layer of compound, D50004, to the outlet threads, outlet end, and outlet inner diameter where the tube assembly attaches to the discharge head.

(b) Connect the tube assembly to the discharge head.

NOTE: Yellow is used to show the plumbing which supplies extinguishant to the forward cargo compartment. Blue is used to show the plumbing which supplies extinguishant to aft cargo compartment.

(c) Remove any unwanted compound.

(d) Tighten the nut on the applicable tube assembly to 266 in-lb (30 N·m) - 294 in-lb (33 N·m) for the FWD (1/2 inch) nut, or to 342 in-lb (39 N·m) - 378 in-lb (43 N·m) for the AFT (5/8 inch) nut.

SUBTASK 26-23-01-420-020

(2) Connect the discharge tube to the filter/dryer outlet on bottle 2 [2].

(a) Connect the tube assembly to the filter/dryer outlet.

NOTE: Yellow is used to show the plumbing which supplies extinguishant to the forward cargo compartment. Blue is used to show the plumbing which supplies extinguishant to aft cargo compartment.

(b) Tighten the nut on the applicable tube assembly to 266 in-lb (30 N·m) - 294 in-lb (33 N·m) for the FWD (1/2 inch) nut, or to 342 in-lb (39 N·m) - 378 in-lb (43 N·m) for the AFT (5/8 inch) nut.

SUBTASK 26-23-01-860-014

(3) Install the air conditioning duct that is in front of the fire extinguishing bottle, if it was previously removed (TASK 26-23-01-400-801).

SUBTASK 26-23-01-410-005

(4) Install the lower shields that cover the bottle connections (TASK 26-23-01-400-801).

SUBTASK 26-23-01-860-015

(5) Install the lining at the aft end of the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 26-23-01-420-018

(6) Install the discharge nozzles [6] on the discharge tubes.

(a) Install the discharge nozzle [6], spacer [8], and pan cover assembly [7] on the discharge tube and hand tighten.

NOTE: Make sure that the pan cover line up with those in the bracket. Make sure that the holes in the discharge nozzle and spacer line up with those in the pan cover assembly.

(b) Tighten the nut to 360 ±18 in-lb (41 ±2 N·m).

(c) Secure the discharge nozzle [6] with screws [5].

(d) Position the spacer [9] on the pan cover assembly [7].

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(e) Install the cargo ceiling.

(f) Position the retainer [4] and secure with screw [3].

SUBTASK 26-23-01-860-016

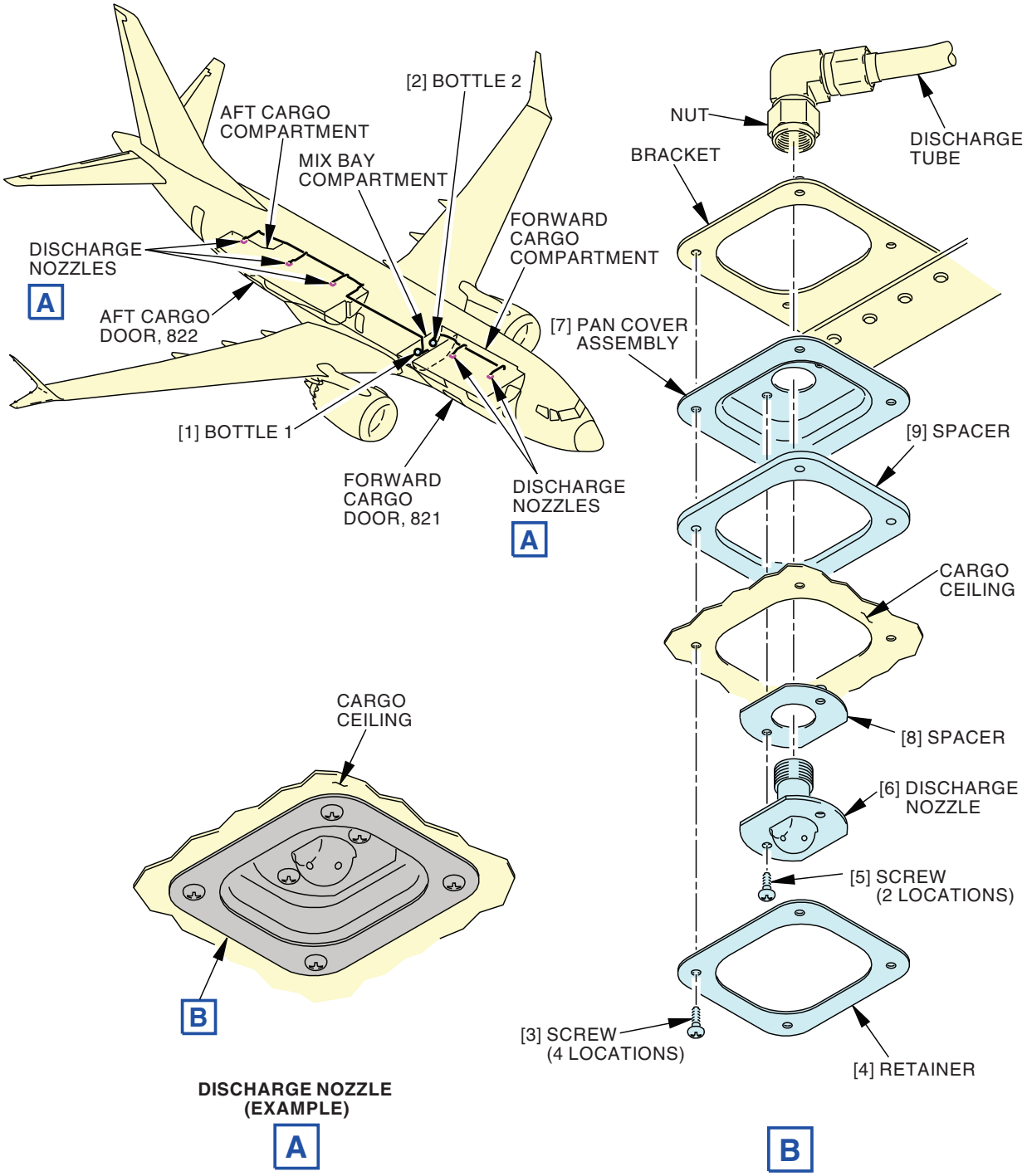
(7) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cargo Fire Extinguishing System Cleaning
Figure 701/26-23-01-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO COMPARTMENT FIRE EXTINGUISHER BOTTLE SQUIB - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the cargo fire extinguisher bottle squibs

(2) An installation of the cargo fire extinguisher bottle squibs.

B. Two cargo fire extinguisher bottles are installed in the mix manifold bay, in the aft end of the lower fwd cargo compartment. Each fire extinguisher bottle has two discharge cartridges (squibs).

TASK 26-23-02-000-801

2. Cargo Fire Extinguisher Bottle Squib Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions for the removal of the cargo fire extinguisher bottle squib.

B. References

Reference	Title
20-10-33-000-801	Power Device Cartridge (Squib) Removal (P/B 401)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
WDM 26-23-11	Wiring Diagram Manual

C. Tools/Equipment

Reference	Description
STD-1168	Cap - Shorting or Faraday Cap

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right

E. Access Panels

Number	Name/Location
821	Forward Cargo Door

F. Prepare for the Removal

SUBTASK 26-23-02-860-002

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

SUBTASK 26-23-02-010-001

(2) Open this access door:

Number	Name/Location
821	Forward Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-02-010-002

(3) Remove the lining at the aft end of the forward cargo compartment to get access to the air conditioning distribution bay (TASK 50-11-02-000-801).

SUBTASK 26-23-02-860-003

(4) If it is necessary, remove the air conditioning duct that is in front of the fire extinguishing bottle.

SUBTASK 26-23-02-010-005

(5) Remove the lower shield that covers the bottle connections

G. Cargo Fire Extinguisher Bottle Squib Removal

WARNING

DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

SUBTASK 26-23-02-860-004

(1) Before you touch the squibs [1], do this task: Power Device Cartridge (Squib) Removal, TASK 20-10-33-000-801.

SUBTASK 26-23-02-020-001

WARNING

DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(2) Disconnect the electrical connectors [5] from the applicable squib as follows (WDM 26-23-11):

(a) For bottle 1 FWD Squib, disconnect connector D16578 or D17811.

(b) For bottle 1 AFT Squib, disconnect connector D16580 or D17812.

(c) For bottle 2 FWD Squib, disconnect connector D16584 or D17807.

(d) For bottle 2 AFT Squib, disconnect connector D16586 or D17808.

SUBTASK 26-23-02-480-001

WARNING

DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

CAUTION

DO NOT PUT SHUNT PLUGS ON THE FIRE BOTTLE SQUIBS. THE SHUNT PLUGS CAN CAUSE DAMAGE TO THE SQUIB PINS.

(3) Install the shorting or faraday cap, STD-1168 on the squib [1].

NOTE: The shorting caps are attached to the fire extinguishing bottles. These shorting caps prevent the squibs from possible activation due to static electricity.

SUBTASK 26-23-02-020-002

(4) Remove the lockwire from the squib if present.

SUBTASK 26-23-02-020-003

(5) Remove the squib [1] from the discharge head [3].

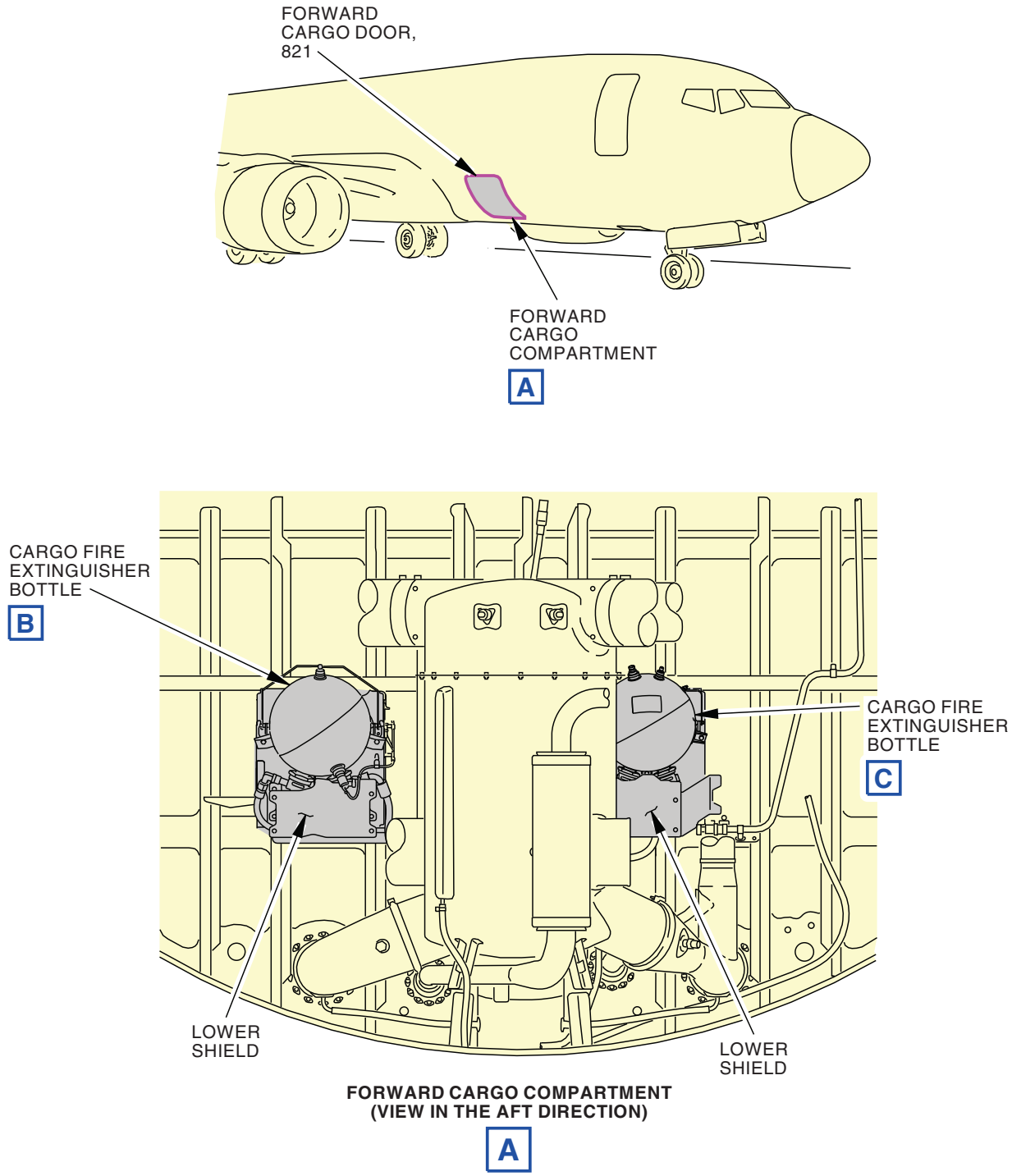
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-02-910-001

(6) Discard the O-ring [2].

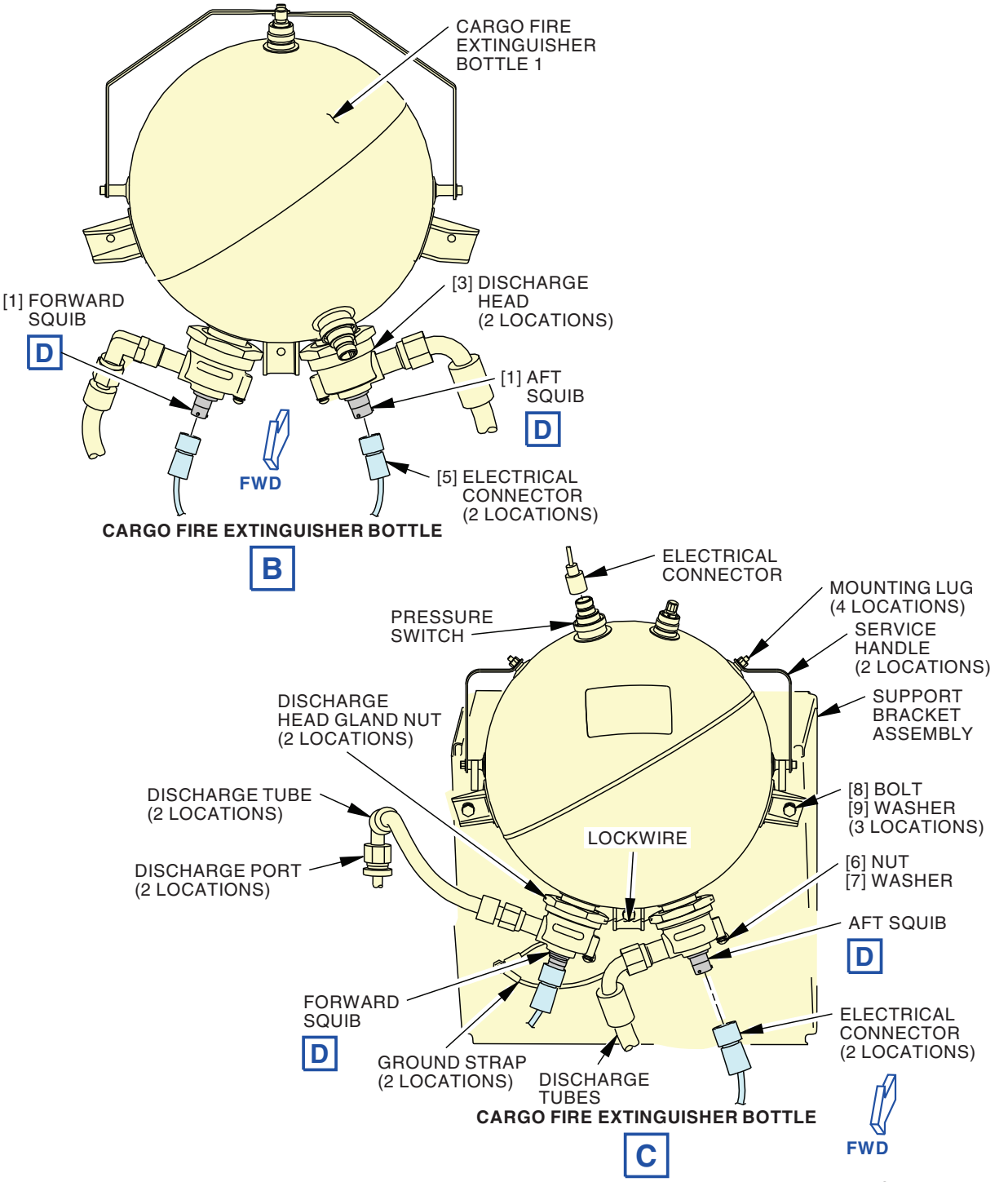
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



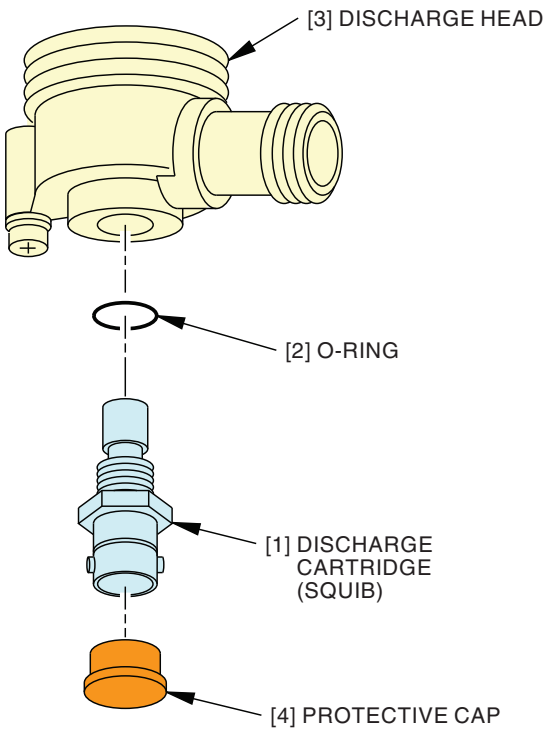
Cargo Compartment Fire Extinguisher Bottle Squib Installation
Figure 401/26-23-02-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cargo Compartment Fire Extinguisher Bottle Squib Installation
Figure 401/26-23-02-990-801 (Sheet 2 of 3)

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Cargo Compartment Fire Extinguisher Bottle Squib Installation
Figure 401/26-23-02-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-23-02-400-801

3. Cargo Fire Extinguisher Bottle Squib Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to install the cargo fire extinguisher bottle squib.

B. References

Reference	Title
20-10-33-400-801	Power Device Cartridge (Squib) Installation (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
WDM 26-23-11	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
STD-1079	Resistor - 10K Ohm or Greater
STD-1168	Cap - Shorting or Faraday Cap

D. Consumable Materials

Reference	Description	Specification
D50284	Lubricant - O-Ring - Silicone	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	O-ring	26-23-01-06-020	PTF ALL
		26-23-01-06-025	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
2 (cont.)		26-23-01-07-030	PTF ALL
		26-23-01-07-045	PTF ALL

F. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right

G. Access Panels

Number	Name/Location
821	Forward Cargo Door

H. Prepare for the Installation

SUBTASK 26-23-02-860-006

(1) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

SUBTASK 26-23-02-010-003

(2) Open this access door:

Number	Name/Location
821	Forward Cargo Door

I. Cargo Fire Extinguisher Bottle Squib Installation

WARNING DO NOT TOUCH THE SQUIB BEFORE YOU DO THE PROCEDURES FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

WARNING DO NOT MOVE OR TOUCH THE CARTRIDGE WITHOUT A SHORTING CAP OR FARADAY CAP INSTALLED ON THE ENDS OF THE SQUIB. THE SQUIB IS AN EXPLOSIVE DEVICE. ACCIDENTAL OPERATION OF THE SQUIB CAN CAUSE INJURY TO PERSONS.

SUBTASK 26-23-02-420-001

(1) Before you touch the squibs [1], do this task: Power Device Cartridge (Squib) Installation, TASK 20-10-33-400-801.

SUBTASK 26-23-02-640-001

(2) Lubricate new O-ring [2] with SLUBE 884-2, D50284.

(a) Install a new O-ring [2] on the squib [1].

(b) Install the squib [1] in the discharge head [3].

1) Tighten the squib [1] to 80 in-lb (9 N·m) - 100 in-lb (11 N·m).

2) Install MS20995C32 lockwire, G01048, between the squib [1] and the discharge outlet.

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SUBTASK 26-23-02-700-001

WARNING MAKE SURE THERE IS NO VOLTAGE AT THE ELECTRICAL CONNECTOR. IF THERE IS A VOLTAGE AT THE ELECTRICAL CONNECTOR, THE SQUIB CAN FIRE SUDDENLY AND CAUSE INJURY TO PERSONNEL.

(3) Do these steps to make sure that there is no stray voltage at the electrical squib connector:

Fire Bottle Squib Electrical Connector

	Engine	APU	Cargo
Connector Pins	3 and 4, and 6 and 7	3 and 4	1 and 2

(a) Connect the digital/analog multimeter, COM-1793, across the two pins.

(b) Make sure that the current is less than 5 milliamps AC, and less than 5 milliamps DC.

(c) Connect a resistor, STD-1079, across the multimeter probes.

(d) Make sure that the voltage is less than 0.250 volts AC, and less than 0.250 volts DC.

(e) If it is necessary, repeat the steps above for the other squib electrical connectors.

(f) Disconnect the multimeter.

SUBTASK 26-23-02-200-001

(4) Make sure that the electrical connector [5] is not damaged.

NOTE: The squib pins can cause damage to the connector if the pins do not enter the electrical connector receptacles.

SUBTASK 26-23-02-420-005

WARNING DO NOT TOUCH THE PINS ON THE SQUIBS. ELECTROSTATIC DISCHARGE CAN CAUSE THE FIRE BOTTLE TO RELEASE ITS CONTENTS SUDDENLY AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT

(5) Remove the shorting or faraday cap, STD-1168 from the squib [1].

SUBTASK 26-23-02-420-003

(6) Make sure that the squib pins are not bent or damaged.

SUBTASK 26-23-02-420-004

(7) Connect the electrical connector [5] to the applicable fire bottle squib [1] (WDM 26-23-11):

(a) Bottle 1 FWD Squib connector, D16578 or D17811.

(b) Bottle 1 AFT Squib connector, D16580 or D17812.

(c) Bottle 2 FWD Squib connector, D16584 or D17807.

(d) Bottle 2 AFT Squib connector, D16586 or D17808.

J. Cargo Fire Extinguisher Bottle Squib Installation Test

SUBTASK 26-23-02-860-007

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	16	C01528	CARGO FIRE EXT 2
B	17	C01526	CARGO FIRE EXT 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-02-710-001

(2) On the cargo fire control panel, push and hold the TEST switch.

(a) Make sure that the green FWD and AFT squib test lights, on the cargo fire control panel, come on.

SUBTASK 26-23-02-710-002

(3) Release the TEST switch.

(a) Make sure that the green FWD and AFT squib test lights go off.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-02-410-004

(1) Install the lower shield that covers the bottle connections.

SUBTASK 26-23-02-410-001

(2) If it was removed, install the air conditioning duct in front of the fire extinguishing bottle.

SUBTASK 26-23-02-410-002

(3) Install the lining at the aft end of the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 26-23-02-410-003

(4) Close this access door:

Number	Name/Location
821	Forward Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO FIRE EXTINGUISHING FILTER/DRYER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the filter/dryer for the cargo fire extinguishing system

(2) Installation of the filter/dryer for the cargo fire extinguishing system

B. The filter/dryers are used to meter and dry the extinguishant. They are installed in the mix bay. The filter/dryers control the rate of flow of extinguishant to the selected cargo compartment and keeps the concentration of extinguishant constant.

C. After the cargo fire extinguishing bottles are discharged, you must replace the fire extinguishing bottles, the squibs, the applicable filter/dryers.

TASK 26-23-04-000-801

2. Cargo Fire Extinguishing Filter/Dryer Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Cargo Fire Extinguishing Filter/Dryer.

B. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right

D. Access Panels

Number	Name/Location
821	Forward Cargo Door

E. Prepare for the Removal

SUBTASK 26-23-04-010-001

(1) Open this access door:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 26-23-04-010-002

(2) Do this task: Forward Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-11-02-000-801.

F. Cargo Fire Extinguishing Filter/Dryer Removal

SUBTASK 26-23-04-020-001

(1) Do these steps to remove the filter/dryer [1]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

BE CAREFUL WHEN YOU DISCONNECT THE TUBES FROM THE FILTER/DRIER. PRESSURE CAN STAY IN THE TUBING AFTER THE FIRE EXTINGUISHING BOTTLE RELEASES ITS CONTENTS. WHEN YOU RELEASE THE PRESSURE IN THE TUBING, YOU CAN CAUSE INJURIES TO PERSONNEL.

(a) Disconnect the outlet tube from the filter/dryer [1].

1) Disconnect the outlet tube before you disconnect the inlet tube.

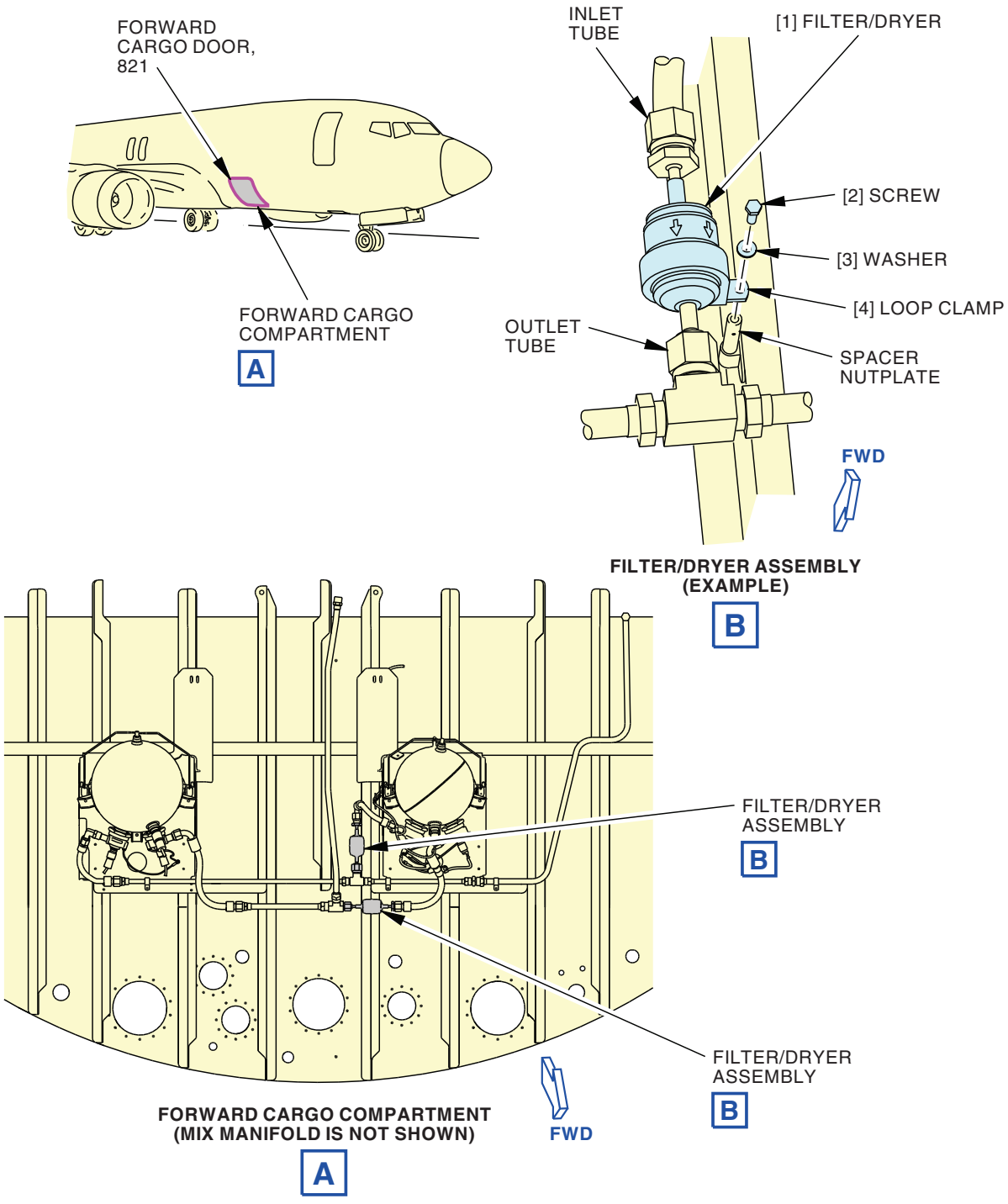
NOTE: If the tubing is pressurized, it is safer to disconnect the outlet tube first.

(b) Disconnect the inlet tube from the filter/dryer [1].

(c) Remove the screw [2], washer [3] and loop clamp [4] that attach the filter/dryer [1] to the spacer nutplate.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Cargo Fire Extinguishing Filter/Dryer Installation
Figure 401/26-23-04-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-23-04-400-801

3. Cargo Fire Extinguishing Filter/Dryer Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Cargo Fire Extinguishing Filter/Dryer.

B. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right

D. Access Panels

Number	Name/Location
821	Forward Cargo Door

E. Cargo Fire Extinguishing Filter/Dryer Installation

SUBTASK 26-23-04-010-003

(1) Open this access door:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 26-23-04-010-004

(2) Make sure aft bulkhead liner is removed (TASK 50-11-02-000-801).

SUBTASK 26-23-04-420-001

(3) Do these steps to install the filter/dryer [1]:

NOTE: Yellow is used to show the plumbing which supplies extinguishant to the forward cargo compartment. Blue is used to show the plumbing which supplies extinguishant to the aft cargo compartment.

(a) Put the filter/dryer [1] in its position and attach it with the screw [2], washer [3] and loop clamp [4].

(b) Connect the inlet tube to the filter/dryer [1].

1) Tighten the inlet tube to 265 in-lb (30 N·m) to 290 in-lb (33 N·m).

(c) Connect the outlet tube to the filter/dryer [1].

1) Tighten the outlet tube to 345 in-lb (39 N·m) to 375 in-lb (42 N·m).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 26-23-04-410-001

(1) Do this task: Forward Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-11-02-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 26-23-04-410-002

(2) Close this access door:

Number	Name/Location
821	Forward Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LAVATORY WASTE COMPARTMENT FIRE EXTINGUISHING - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal/installation of the fire extinguishing bottle in the lavatory waste compartment.

(2) An inspection/check of the fire extinguishing system in the lavatory waste compartment.

B. The fire extinguishing system for the lavatory waste compartment operates automatically when heat is detected in the lavatory waste compartment. When the temperature is hot enough, heat fusible tips on the fire extinguisher bottle melt. The extinguisher fills the waste compartment with an inert gas. A separate temperature indicator inside the waste compartment shows when the temperature has gotten hot.

TASK 26-24-01-900-801

2. Lavatory Waste Compartment Fire Extinguishing Bottle Removal/Installation

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-6250	Gauge - Force Part #: LG-1KG Supplier: 92456

B. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Fire Extinguishing Bottle Removal

SUBTASK 26-24-01-010-001

(1) Open the waste compartment door.

SUBTASK 26-24-01-020-002

(2) Remove the mounting screws [3] and washers [2] that hold the fire extinguishing bottle [1] in its position.

SUBTASK 26-24-01-020-003

(3) Hold the fire extinguishing bottle [1] and carefully remove the nozzles through the holes.

SUBTASK 26-24-01-020-004

(4) Remove the fire extinguishing bottle [1].

D. Fire Extinguishing Bottle Installation

SUBTASK 26-24-01-280-001

(1) Use the force gauge, COM-6250, and weigh the fire extinguishing bottle.

NOTE: The total weight of the halon and the bottle is shown on the label of the fire extinguishing bottle. Review table below for allowable weight deviations for fire extinguisher bottle.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Replace the bottle if the difference between the weight marked on the bottle and the measured weight is more than what is specified in Table 201:

Table 201/26-24-01-993-801

Part Number	Max Permitted Weight Loss (grams)
30100022–X	10
A800100–X	10
A825100–X	10
A826300–X	10
A826700–X	10
474276	10
474453–X	10
800100–X	10
G800100–X	5
G825100–X	5
G826300–X	5
G826700–X	5

SUBTASK 26-24-01-420-001

(2) Put the fire extinguishing bottle [1] in its position and install the nozzles through the holes.

(a) Make sure that the nozzles point into the waste container area.

SUBTASK 26-24-01-420-002

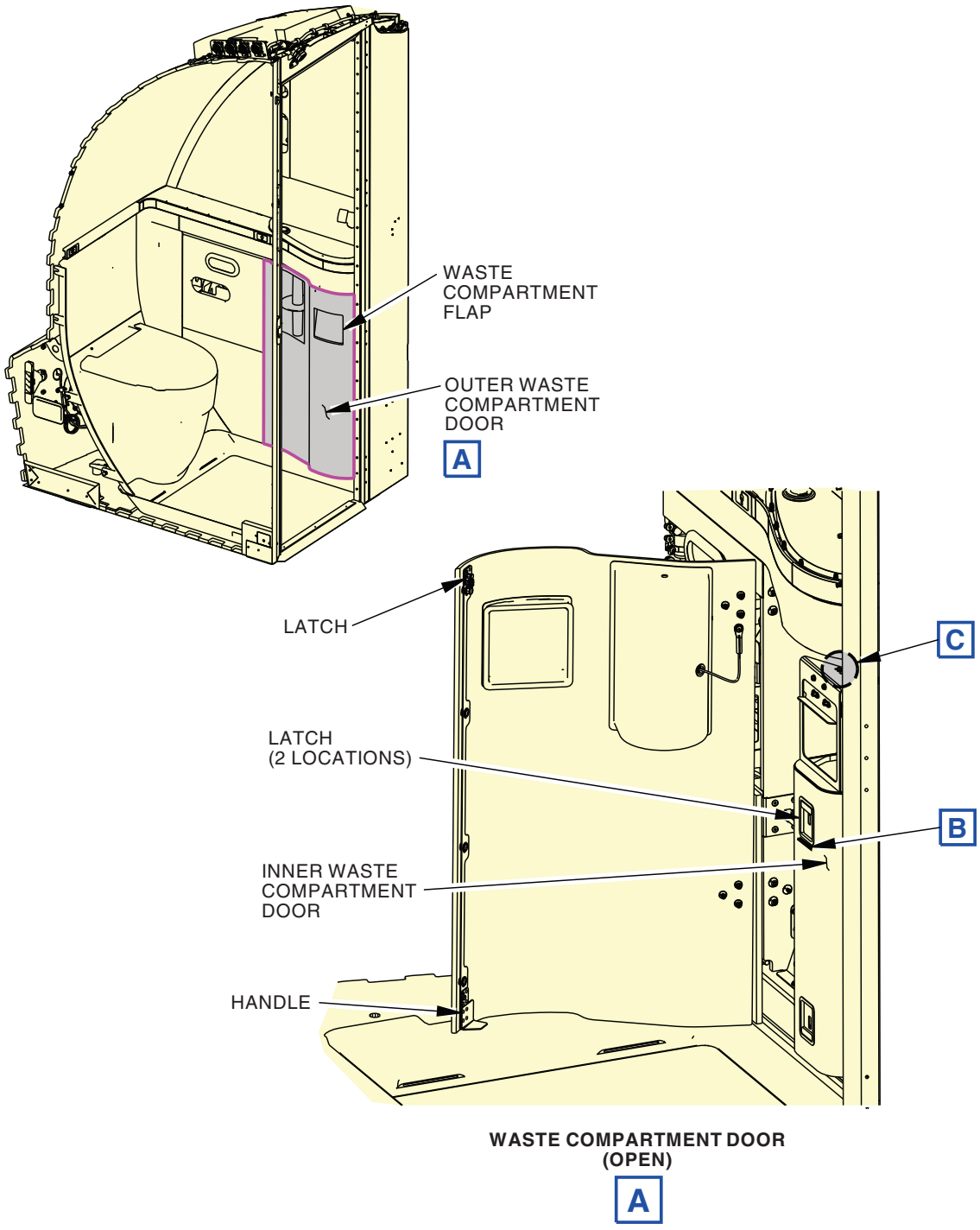
(3) Install the washers [2] and screws [3] that hold the fire extinguishing bottle [1] in its position.

SUBTASK 26-24-01-410-002

(4) Close the waste compartment door.

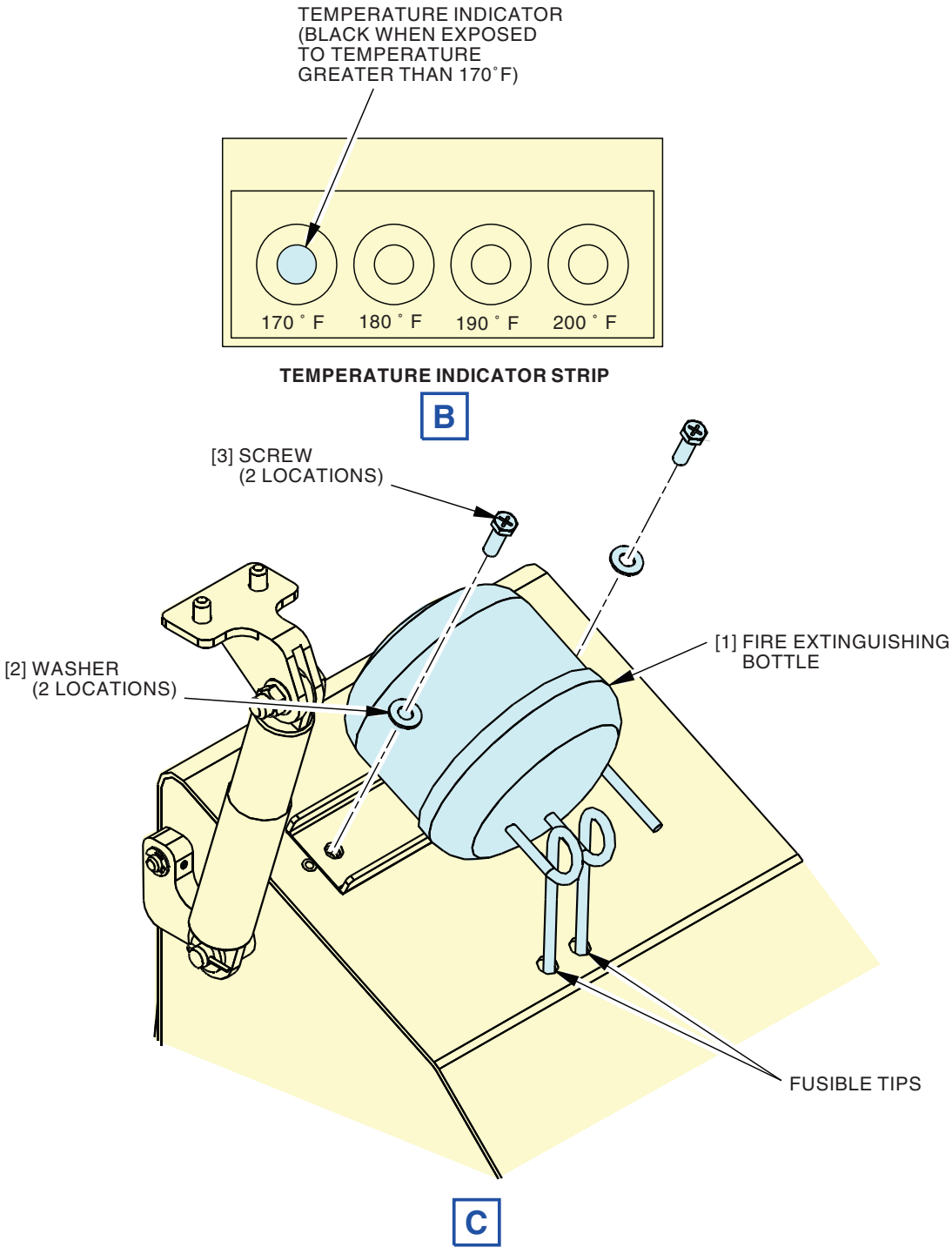
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Waste Compartment Fire Extinguishing - Maintenance Practices
Figure 201/26-24-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Waste Compartment Fire Extinguishing - Maintenance Practices
Figure 201/26-24-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-24-01-200-801

3. Lavatory Waste Compartment Fire Extinguishing Bottle Inspection/Check

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

B. Lavatory Waste Compartment Fire Extinguishing Bottle Inspection/Check

SUBTASK 26-24-01-280-002

(1) Open the waste compartment door.

SUBTASK 26-24-01-280-003

(2) Open the waste container door to get access to the fire extinguishing bottle on the inside of the waste compartment.

SUBTASK 26-24-01-020-005

(3) Remove the waste container.

SUBTASK 26-24-01-960-001

(4) Examine the fusible tips on the discharge tubes.

(a) Replace the fire extinguishing bottle and the temperature indicator strip if the fusible tips are melted.

NOTE: The temperature indicator strip is under the upper handle of the waste container door.

SUBTASK 26-24-01-960-002

(5) Examine the temperature indicator strip to see if one or more of the temperature indicators changed from white to black.

(a) If one or more of the temperature indicator slips changed from white to black, replace the temperature indicator strip.

SUBTASK 26-24-01-280-004

(6) Examine the fire extinguishing bottle for corrosion, scratches, or dents.

(a) Replace the fire extinguishing bottle if dents with a depth equal to or greater than 0.125 in. (3.175 mm) or scratches greater than 0.005 in. (0.127 mm) in depth are found.

SUBTASK 26-24-01-020-006

(7) Install the waste container in the waste compartment.

SUBTASK 26-24-01-280-005

(8) Close the waste container door.

SUBTASK 26-24-01-280-006

(9) Close the waste compartment door.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LAVATORY WASTE COMPARTMENT - INSPECTION/CHECK

1. General

A. This procedure has this inspection task:

(1) Lavatory waste compartment.

TASK 26-24-01-210-801

2. Lavatory Waste Compartment Temperature Indicator Strip Inspection

(Figure 26-24-01-990-801)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
26-24-01-200-801	Lavatory Waste Compartment Fire Extinguishing Bottle Inspection/Check (P/B 201)
26-24-01-990-801	Figure: Lavatory Waste Compartment Fire Extinguishing - Maintenance Practices (P/B 201)

B. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Lavatory Waste Compartment Temperature Indicator Strip Inspection

SUBTASK 26-24-01-010-003

(1) Open the waste compartment door to get access to the temperature indicator strip.

SUBTASK 26-24-01-210-001

(2) Examine the temperature indicator strip to see if one or more of the temperature indicators changed from white to black.

(a) If one or more of the temperature indicators changed from white to black, do this task: Lavatory Waste Compartment Fire Extinguishing Bottle Inspection/Check, TASK 26-24-01-200-801.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 26-24-01-010-004

(1) Close the waste compartment door.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HALON FIRE EXTINGUISHERS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the halon fire extinguisher

(2) An installation of the halon fire extinguisher.

B. The extinguisher is mounted on a bracket that has a quick-release strap.

TASK 26-26-01-020-801

2. Halon Fire Extinguisher Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the halon fire extinguisher.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Halon Fire Extinguisher Removal

SUBTASK 26-26-01-030-001

(1) Release the quick-release strap [2].

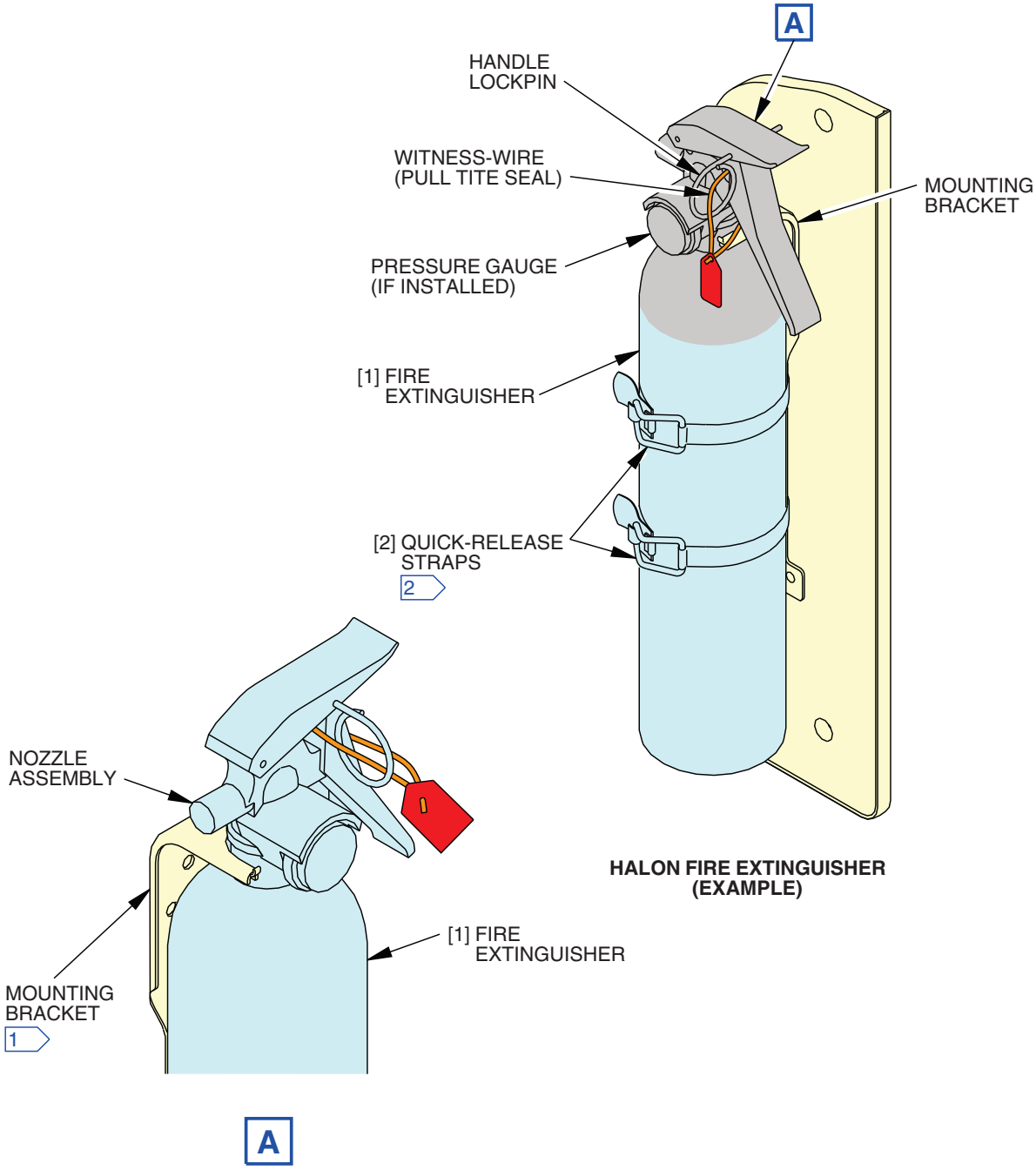
NOTE: There can be one quick-release strap or two quick-release straps.

SUBTASK 26-26-01-030-002

(2) Remove the fire extinguisher [1] from the mounting brackets.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 PLACE METAL NECK OF HALON FIRE EXTINGUISHER INTO MOUNTING BRACKET.

2 THERE CAN BE ONE QUICK-RELEASE STRAP OR TWO QUICK-RELEASE STRAPS.

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Halon Fire Extinguisher Installation
Figure 401/26-26-01-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-26-01-430-801

3. Halon Fire Extinguisher - Installation

(Figure 401)

A. General

(1) This task gives instructions to install the halon fire extinguisher.

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fire extinguisher	26-26-01-01-010	PTF ALL
		26-26-01-04-005	PTF ALL

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Halon Fire Extinguisher Installation

SUBTASK 26-26-01-200-003

(1) Do a general visual inspection of the fire extinguisher [1].

SUBTASK 26-26-01-430-001

(2) Make sure that the fire extinguisher [1] is in the correct position on the mounting brackets.

SUBTASK 26-26-01-430-002

(3) Connect the quick-release strap [2].

NOTE: There can be one quick-release strap or two quick-release straps.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HALON FIRE EXTINGUISHER - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) An inspection of the halon fire extinguisher.

B. The portable fire extinguishers are in some general locations in the passenger compartment and flight compartment. If the extinguisher is not easy to see, a placard will identify the location. The extinguishers are in one or more of these areas: the galley, lavatory stowage areas, closets, or pet containers near attendant seats. There is one halon extinguisher in the flight compartment.

C. Refer to the applicable vendor manuals for servicing or to charge the fire extinguisher.

TASK 26-26-01-200-801

2. Halon Fire Extinguisher - Inspection/Check

(Figure 601)

A. References

Reference	Title
26-26-01-020-801	Halon Fire Extinguisher Removal (P/B 401)
26-26-01-430-801	Halon Fire Extinguisher - Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-373	Gauge - Force, Push-Pull (equivalent or better tool with range 0 to 5 lbs, 0.025 graduations, accuracy 0.5% full scale) Part #: DST-4A Supplier: 1GHM7 Part #: FDIX 5 Supplier: 0BFD9

C. Location Zones

Zone	Area
200	Upper Half of Fuselage
211	Flight Compartment - Left

D. Halon Fire Extinguisher Inspection

SUBTASK 26-26-01-210-001

(1) Make sure that the applicable decals, nameplates, and labels are in good condition.

SUBTASK 26-26-01-210-002

(2) Do this task: Halon Fire Extinguisher Removal, TASK 26-26-01-020-801.

SUBTASK 26-26-01-200-002

(3) Use force gauge, COM-373, to weigh the fire extinguisher and compare the value against the weight shown on the applicable decal, nameplate, or label located on the fire extinguisher.

(a) If the weight of the fire extinguisher has decreased by more than 0.125 lb (0.057 kg) from the weight recorded on the applicable decal, nameplate, or label, the fire extinguisher has a leak, replace the fire extinguisher.

(b) Make sure that the weight of the fire extinguisher is 3.5 lb (1.59 kg) - 3.81 lb (1.73 kg).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If the weight of the fire extinguisher is not correct, replace the fire extinguisher.

NOTE: A complete fire extinguisher includes the pull-out pin, pull-tite seal, hose assembly with hose strap, or nozzle, as applicable.

(c) If it is necessary, write the weight of the fire extinguisher on the applicable decal, nameplate, or label.

SUBTASK 26-26-01-210-003

(4) Make sure that the mounting bracket is in its correct position on the airplane.

SUBTASK 26-26-01-210-006

(5) Make sure that the handle lockpin and witness-wire (pull tite seal) are in their correct positions on the handle.

(a) If the witness-wire is missing or broken, replace the witness-wire or fire extinguisher bottle.

SUBTASK 26-26-01-210-007

(6) If it is necessary, examine the pressure gage.

(a) Make sure that the fire extinguisher [1] has the correct pressure.

SUBTASK 26-26-01-210-008

(7) Make sure that there is no damage to the fire extinguisher [1].

SUBTASK 26-26-01-210-009

(8) If there are other manufacturer inspections or maintenance procedures that show on the fire extinguisher [1], do those procedures.

SUBTASK 26-26-01-210-004

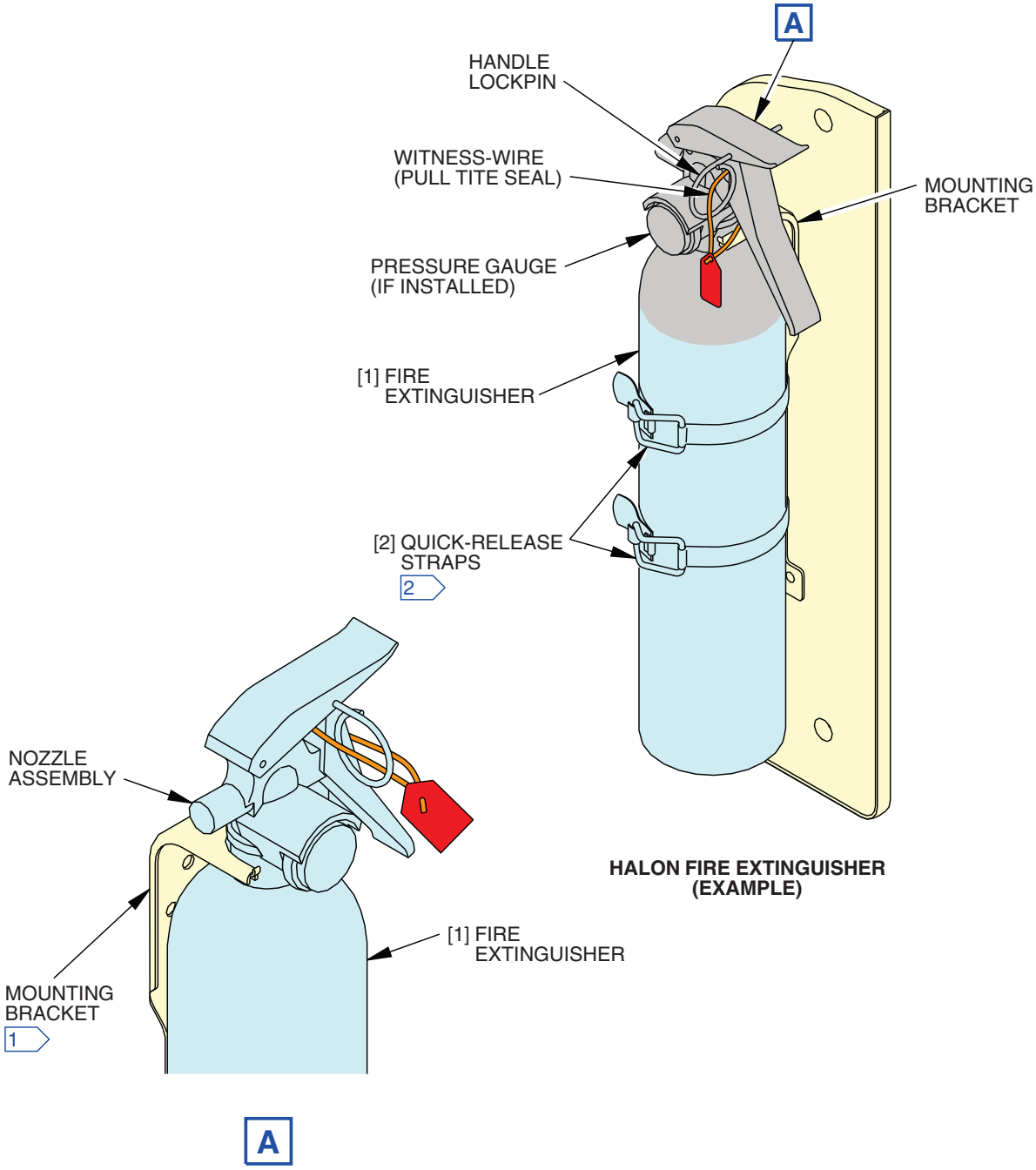
(9) Do this task: Halon Fire Extinguisher - Installation, TASK 26-26-01-430-801.

SUBTASK 26-26-01-210-005

(10) Make sure that the fire extinguisher [1] is in its correct position and tight on the mounting bracket.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HALON FIRE EXTINGUISHER (EXAMPLE)

1 PLACE METAL NECK OF HALON FIRE EXTINGUISHER INTO MOUNTING BRACKET.

2 THERE CAN BE ONE QUICK-RELEASE STRAP OR TWO QUICK-RELEASE STRAPS.

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Halon Fire Extinguisher Inspection
Figure 601/26-26-01-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-26-01-200-802

3. Halon Fire Extinguisher - Check

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task includes steps to check the Halon fire extinguisher

B. References

Reference	Title
26-26-01-020-801	Halon Fire Extinguisher Removal (P/B 401)
26-26-01-430-801	Halon Fire Extinguisher - Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-373	Gauge - Force, Push-Pull (equivalent or better tool with range 0 to 5 lbs, 0.025 graduations, accuracy 0.5% full scale) Part #: DST-4A Supplier: 1GHM7 Part #: FDIX 5 Supplier: 0BFD9

D. Location Zones

Zone	Area
200	Upper Half of Fuselage
211	Flight Compartment - Left

E. Prepare for the Check

SUBTASK 26-26-01-200-004

(1) Do this task: Halon Fire Extinguisher Removal, TASK 26-26-01-020-801

F. Halon Fire Extinguisher Check

SUBTASK 26-26-01-200-005

(1) Use force gauge, COM-373, to weigh the fire extinguisher and compare the value against the weight shown on the applicable decal, nameplate, or label located on the fire extinguisher.

(a) If the weight of the fire extinguisher has decreased by more than 0.125 lb (0.057 kg) from the weight recorded on the applicable decal, nameplate, or label, the fire extinguisher has a leak, replace the fire extinguisher

(b) Make sure that the weight of the fire extinguisher is 3.5 lb (1.59 kg) - 3.81 lb (1.73 kg).

1) If the weight of the fire extinguisher is not correct, replace the fire extinguisher.

NOTE: A complete fire extinguisher includes the pull-out pin, pull-tite seal, hose assembly with hose strap, or nozzle, as applicable.

(c) If it is necessary, write the weight of the fire extinguisher on the applicable decal, nameplate, or label.

SUBTASK 26-26-01-200-006

(2) Make sure that the handle lockpin and witness-wire (pull tite seal) are in their correct positions on the handle.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If the witness-wire is missing or broken, replace the witness-wire or fire extinguisher bottle.

SUBTASK 26-26-01-200-007

(3) Examine the pressure gage.

(a) Make sure that the fire extinguisher [1] has the correct pressure.

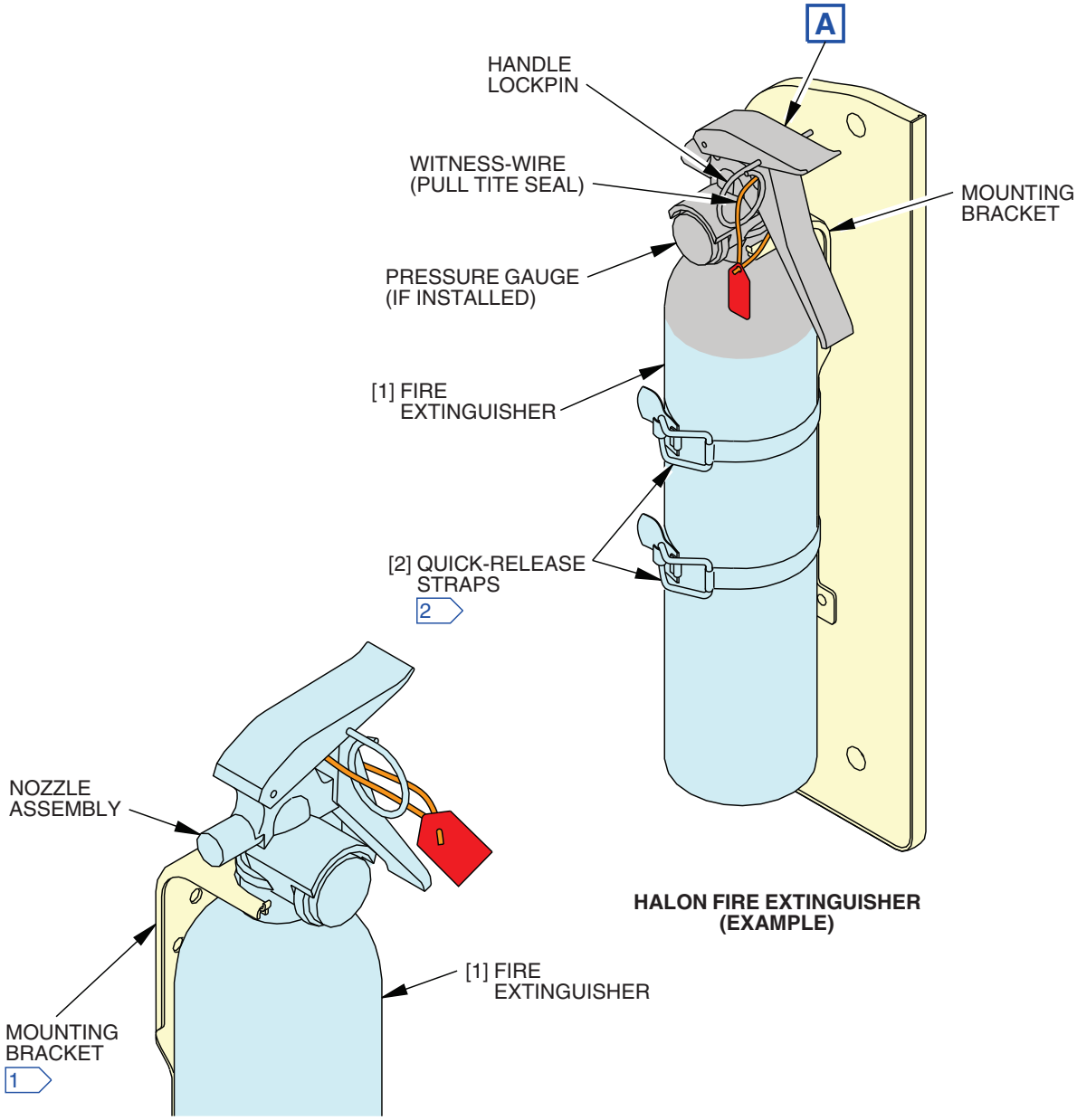
G. Put the Airplane Back to Its Usual Condition

SUBTASK 26-26-01-200-008

(1) Do this task: Halon Fire Extinguisher - Installation, TASK 26-26-01-430-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 PLACE METAL NECK OF HALON FIRE EXTINGUISHER INTO MOUNTING BRACKET.

2 THERE CAN BE ONE QUICK-RELEASE STRAP OR TWO QUICK-RELEASE STRAPS.

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Halon Fire Extinguisher Check
Figure 602/26-26-01-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER FIRE EXTINGUISHERS- REMOVAL/INSTALLATION

1. General

A. The procedure contains the following tasks:

(1) A removal of the water fire extinguisher

(2) An installation of the water fire extinguisher.

B. The extinguisher is mounted on a bracket which has a quick-release strap.

TASK 26-26-02-000-801

2. Water Fire Extinguisher Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the water fire extinguisher.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Water Fire Extinguisher Removal

SUBTASK 26-26-02-020-003

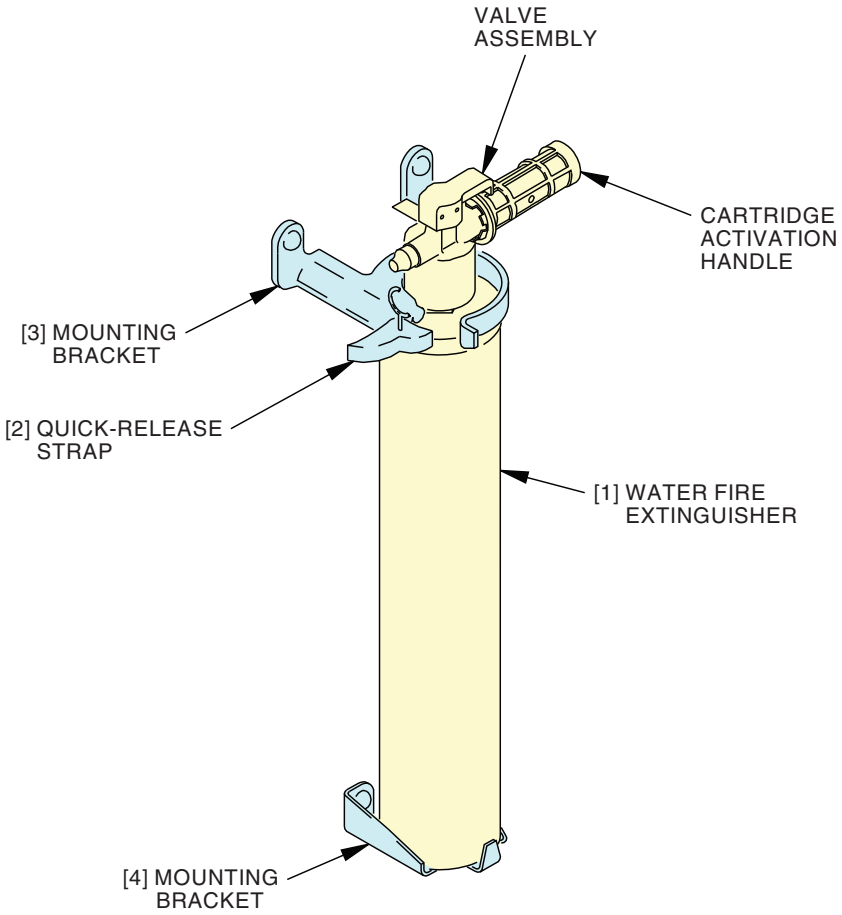
(1) Unfasten the quick-release strap [2].

SUBTASK 26-26-02-020-001

(2) Remove the water fire extinguisher [1] from the mounting bracket [3] and mounting bracket [4].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WATER FIRE EXTINGUISHER (EXAMPLE)

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Water-Type Fire Extinguishers Installation
Figure 401/26-26-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 26-26-02-400-801

3. Water Fire Extinguisher Installation

(Figure 401)

A. General

(1) This task gives instructions to install the water fire extinguisher.

B. References

Reference	Title
26-26-02-200-801	Water Fire Extinguishers - Inspection (P/B 601)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Water fire extinguisher	26-26-02-04-005	PTF ALL

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Water Fire Extinguisher Installation

SUBTASK 26-26-02-780-001

(1) If the water fire extinguisher [1] has a pressure gauge, make sure that the water fire extinguisher [1] has the correct pressure.

SUBTASK 26-26-02-210-020

(2) Do the general visual inspection of the water fire extinguisher [1].

(a) If a used extinguisher is installed, make sure that the used water fire extinguisher [1] is in a serviceable condition (TASK 26-26-02-200-801).

SUBTASK 26-26-02-420-002

(3) Install the water fire extinguisher [1] in its mounting bracket [3] and mounting bracket [4].

SUBTASK 26-26-02-860-001

(4) Make sure that the water fire extinguisher [1] fits snugly.

(a) If it is necessary, adjust the location of the mounting bracket [4] to prevent the quick-release strap [2] latch from snagging with the water fire extinguisher [1] cartridge activation handle.

SUBTASK 26-26-02-860-002

(5) Make sure that the water fire extinguisher [1] is readily accessible and rotated with the activation cartridge handle.

SUBTASK 26-26-02-860-003

(6) Make sure that the quick-release strap [2] latch is on the opposite side of the water fire extinguisher [1] cartridge activation handle.

SUBTASK 26-26-02-420-003

(7) Fasten the quick-release strap [2].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER FIRE EXTINGUISHERS - INSPECTION/CHECK

1. General

A. This procedure has a task to inspect the portable water fire extinguishers.

B. The portable fire extinguishers are installed in several general locations in the passenger compartment. If the extinguisher is not easy to see, the location will be identified with a placard. The extinguishers are usually installed in one or more of these areas: galley or lavatory stowage areas, closets, doghouses, near attendant seats.

C. To service or recharge the fire extinguisher, refer to the applicable vendor manuals.

TASK 26-26-02-200-801

2. Water Fire Extinguishers - Inspection

(Figure 601)

A. General

(1) The portable fire extinguishers are installed in several general locations in the passenger compartment. If the extinguisher is not easy to see, the location will be identified with a placard. The extinguishers are usually installed in one or more of these areas: galley or lavatory stowage areas, closets, doghouses, near attendant seats.

(2) This task has a weight check of the portable water fire extinguisher assembly and the weight check of the CO^2 cartridge.

B. References

Reference	Title
26-26-02-000-801	Water Fire Extinguisher Removal (P/B 401)
26-26-02-400-801	Water Fire Extinguisher Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-11525	Gauge - Force, 10-lb capacity Part #: PS-10 Supplier: 1GHM7
COM-16140	Gauge - Force, 2 lb capacity, +/- 0.3 % full scale accuracy Part #: DST-4A Supplier: 1GHM7
STD-252	Ferrule Crimping Tool
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy

D. Consumable Materials

Reference	Description	Specification
D00013	Grease - Aircraft And Instrument Grease	MIL-PRF-23827 (NATO G-354) (Supersedes MIL-G-23827)
D00062	Lubricant - Pneumatic System	SAE AMS-G-4343 (NATO G-392)
D50354	O-ring, packing P/N 202331	
G51248	Wire - Seal (P/N 204960)	

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E. Location Zones

Zone	Area
200	Upper Half of Fuselage

F. Water Fire Extinguishers Inspection

SUBTASK 26-26-02-210-012

(1) Make sure that the instruction decal and nameplate are in good condition.

SUBTASK 26-26-02-210-002

(2) Do this task: Water Fire Extinguisher Removal, TASK 26-26-02-000-801.

SUBTASK 26-26-02-210-015

(3) Make sure that there is no physical damage to the extinguisher.

SUBTASK 26-26-02-210-016

(4) Make sure that there are no leaks in the extinguisher.

SUBTASK 26-26-02-210-017

(5) If there are other manufacturer inspection or maintenance procedures that show on the extinguisher, do those procedures.

SUBTASK 26-26-02-210-014

(6) Make sure that the lockwire seal [4] (seal wire, G51248) is correctly installed on the handle.

(a) If the lockwire seal [4] (seal wire, G51248) is missing or broken, do one of the following steps:

1) Replace the water fire extinguisher.

2) Weigh the water fire extinguisher and cartridge [6].

a) Do the step to weigh the water fire extinguisher.

b) Do the steps to make sure that the water fire extinguisher is not pressurized and to remove and weigh the CO^2 cartridge [6].

c) If the weight of the water fire extinguisher and the CO^2 cartridge [6] are good, do the step to assemble the cartridge holder [1].

SUBTASK 26-26-02-510-001

(7) Do this step to weigh the water fire extinguisher:

(a) Use the force gauge, COM-11525, push/pull scale 0-25 pound (0-11 kilogram), STD-753, or equivalent to make sure that the weight of the water fire extinguisher is not less than the weight shown on the fire extinguisher nameplate.

SUBTASK 26-26-02-210-018

WARNING BEFORE YOU DISASSEMBLE THE EXTINGUISHER, MAKE SURE THAT YOU REMOVE ALL THE PRESSURE FROM THE FIRE EXTINGUISHER. WHEN YOU DISASSEMBLE A PRESSURIZED FIRE EXTINGUISHER, IT CAN CAUSE INJURY.

(8) Make sure that the water fire extinguisher is not pressurized.

(a) Visually check the water fire extinguisher.

1) Make sure that the lockwire seal [4] (seal wire, G51248) is not broken or missing.

2) Make sure that the red line on the valve assembly and the red line on the cartridge holder [1] are aligned.

(b) If the lockwire seal [4] (seal wire, G51248) is broken or missing, or the cartridge holder [1] alignment is not correct, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Push and hold the valve assembly lever down.
   a) If there is air pressure and/or extinguishing agent discharging at the nozzle, replace the fire extinguisher.

SUBTASK 26-26-02-020-002

(9) Remove the CO^2 cartridge [6].
   (a) Remove and discard the lockwire seal [4].
   (b) Turn the cartridge holder [1] counterclockwise and remove it from the valve assembly.
   (c) Remove the spring [5] and CO^2 cartridge [6] from the cartridge holder [1].
      1) If the CO^2 cartridge [6] is damaged or used, replace the CO^2 cartridge [6].
   (d) Remove the O-ring [7] from the valve assembly.
      1) Discard the O-ring [7].

SUBTASK 26-26-02-210-019

(10) Weight check the CO^2 cartridge [6].
   (a) Put the CO^2 cartridge [6] on a scale (2 lb force gauge, COM-16140, or equivalent).
   (b) Make sure that the CO^2 cartridge [6] weighs a minimum of 0.073 lb (33 g).
      1) If the CO^2 cartridge [6] weighs less than 0.073 lb (33 g), discard the CO^2 cartridge [6] and install new.

SUBTASK 26-26-02-420-001

(11) Assemble the cartridge holder [1].
   (a) Put a thin layer of lubricant, D00062, on the new O-ring [7].
   (b) Install the new O-ring [7] in the valve assembly.
      1) Use O-ring, packing, D50354.
   (c) Put a thin layer of grease, D00013, on the internal threads of the cartridge holder [1].
   (d) Install the spring [5] and CO^2 cartridge [6] in the cartridge holder [1].
   (e) Install the cartridge holder [1] in the valve assembly and turn the cartridge holder [1] clockwise until:
      1) The red line on the cartridge holder [1] aligns with the red line on the valve assembly.
      2) The lockwire seal [4] holes in the lock ring [2] and valve assembly are aligned.
         a) If it is necessary, loosen the setscrews [3] to adjust the position of the lock ring [2].
   (f) Install a new lockwire seal [4] (seal wire, G51248) through the holes in the lock ring [2] and the body of the valve assembly.
   (g) Install the wires through the lead seal.
      1) Pull the wires tight and crimp the lead seal with crimping tool, STD-252.

         NOTE: It is satisfactory to use a standard plier to crimp the lead seal.

SUBTASK 26-26-02-210-004

(12) Do this task: Water Fire Extinguisher Installation, TASK 26-26-02-400-801.

SUBTASK 26-26-02-210-011

(13) Make sure that the mounting bracket is attached correctly to the airplane.

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SUBTASK 26-26-02-210-013

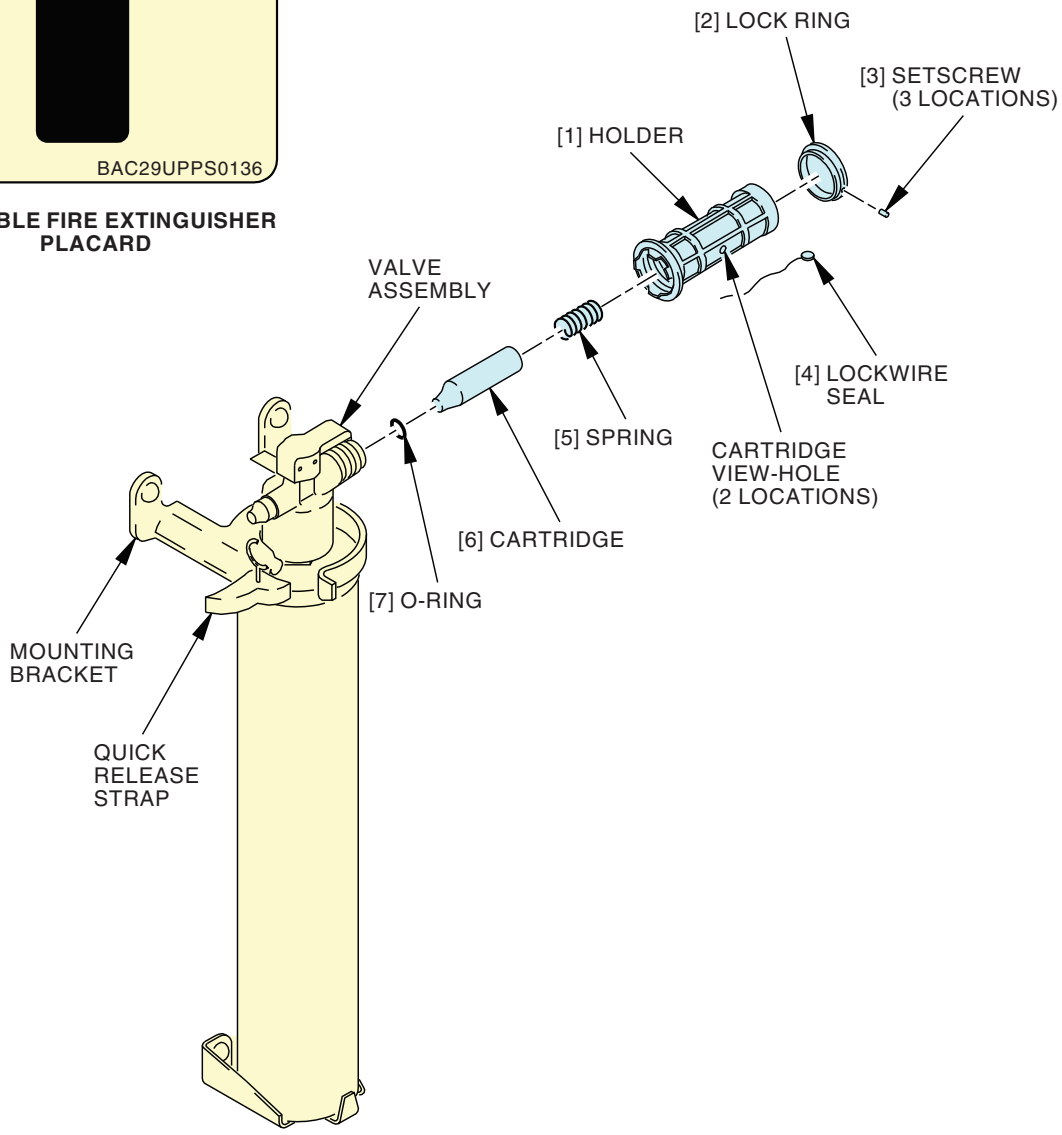
(14) Make sure that the extinguisher is installed tightly to the mounting bracket.

——— END OF TASK ———

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PORTABLE FIRE EXTINGUISHER PLACARD



WATER FIRE EXTINGUISHER (EXAMPLE)

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Water-Type Fire Extinguishers Inspection
Figure 601/26-26-02-990-802

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TASK 26-26-02-200-802

3. Water Fire Extinguishers - Check

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task includes the steps to check the water-type fire extinguisher.

B. References

Reference	Title
26-26-02-000-801	Water Fire Extinguisher Removal (P/B 401)
26-26-02-400-801	Water Fire Extinguisher Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-11525	Gauge - Force, 10-lb capacity Part #: PS-10 Supplier: 1GHM7
COM-16140	Gauge - Force, 2 lb capacity, +/- 0.3 % full scale accuracy Part #: DST-4A Supplier: 1GHM7
STD-252	Ferrule Crimping Tool
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy

D. Consumable Materials

Reference	Description	Specification
D00013	Grease - Aircraft And Instrument Grease	MIL-PRF-23827 (NATO G-354) (Supersedes MIL-G-23827)
D00062	Lubricant - Pneumatic System	SAE AMS-G-4343 (NATO G-392)
D50354	O-ring, packing P/N 202331	
G51248	Wire - Seal (P/N 204960)	

E. Location Zones

Zone	Area
200	Upper Half of Fuselage

F. Prepare for the Check

SUBTASK 26-26-02-200-001

(1) Do this task: Water Fire Extinguisher Removal, TASK 26-26-02-000-801.

G. Water-type Fire Extinguisher Check

SUBTASK 26-26-02-200-002

(1) Make sure that the lockwire seal [4] (seal wire, G51248) is correctly installed on the handle.

(a) If the lockwire seal [4] (seal wire, G51248) is missing or broken, or the alignment of the red line on the cartridge holder is not correct, do one of the following steps:

1) Replace the water fire extinguisher.

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2) Weigh the water fire extinguisher and cartridge [6].
   a) Do the step to weigh the water fire extinguisher.
   b) Make sure that the water fire extinguisher is not pressurized and remove and weigh the CO^2 cartridge [6].
      <1> Push and hold the valve assembly lever down.
         <a> If there is air pressure and/or extinguishing agent discharging at the nozzle, replace the fire extinguisher.
   c) If the weight of the water fire extinguisher and the CO^2cartridge [6] are good, do the step to assemble the cartridge holder [1].

SUBTASK 26-26-02-200-003

(2) Do this step to weigh the water fire extinguisher:
   (a) Use the force gauge, COM-11525, push/pull scale 0-25 pound (0-11 kilogram), STD-753, or equivalent to make sure that the weight of the water fire extinguisher is not less than the weight shown on the fire extinguisher nameplate.

SUBTASK 26-26-02-200-004

(3) Remove the CO^2 cartridge [6].
   (a) Remove and discard the lockwire seal [4].
   (b) Turn the cartridge holder [1] counterclockwise and remove it from the valve assembly.
   (c) Remove the spring [5] and CO^2 cartridge [6] from the cartridge holder [1].
      1) If the CO^2 cartridge [6] is damaged or used, replace the CO^2 cartridge [6].
   (d) Remove the O-ring [7] from the valve assembly.
      1) Discard the O-ring [7].

SUBTASK 26-26-02-200-005

(4) Weight check the CO^2 cartridge [6].
   (a) Put the CO^2 cartridge [6] on a scale (2 lb force gauge, COM-16140, or equivalent).
   (b) Make sure that the CO^2 cartridge [6] weighs a minimum of 0.073 lb (33 g).
      1) If the CO^2 cartridge [6] weighs less than 0.073 lb (33 g), discard the CO^2 cartridge [6] and install new.

SUBTASK 26-26-02-200-006

(5) Assemble the cartridge holder [1].
   (a) Put a thin layer of lubricant, D00062, on the new O-ring [7].
   (b) Install the new O-ring [7] in the valve assembly.
      1) Use O-ring, packing, D50354.
   (c) Put a thin layer of grease, D00013, on the internal threads of the cartridge holder [1].
   (d) Install the spring [5] and CO^2 cartridge [6] in the cartridge holder [1].
   (e) Install the cartridge holder [1] in the valve assembly and turn the cartridge holder [1] clockwise until:
      1) The red line on the cartridge holder [1] aligns with the red line on the valve assembly.
      2) The lockwire seal [4] holes in the lock ring [2] and valve assembly are aligned.
         a) If it is necessary, loosen the setscrews [3] to adjust the position of the lock ring [2].

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(f) Install a new lockwire seal [4] (seal wire, G51248) through the holes in the lock ring [2] and the body of the valve assembly.

(g) Install the wires through the lead seal.

1) Pull the wires tight and crimp the lead seal with crimping tool, STD-252.

NOTE: It is satisfactory to use a standard plier to crimp the lead seal.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 26-26-02-200-007

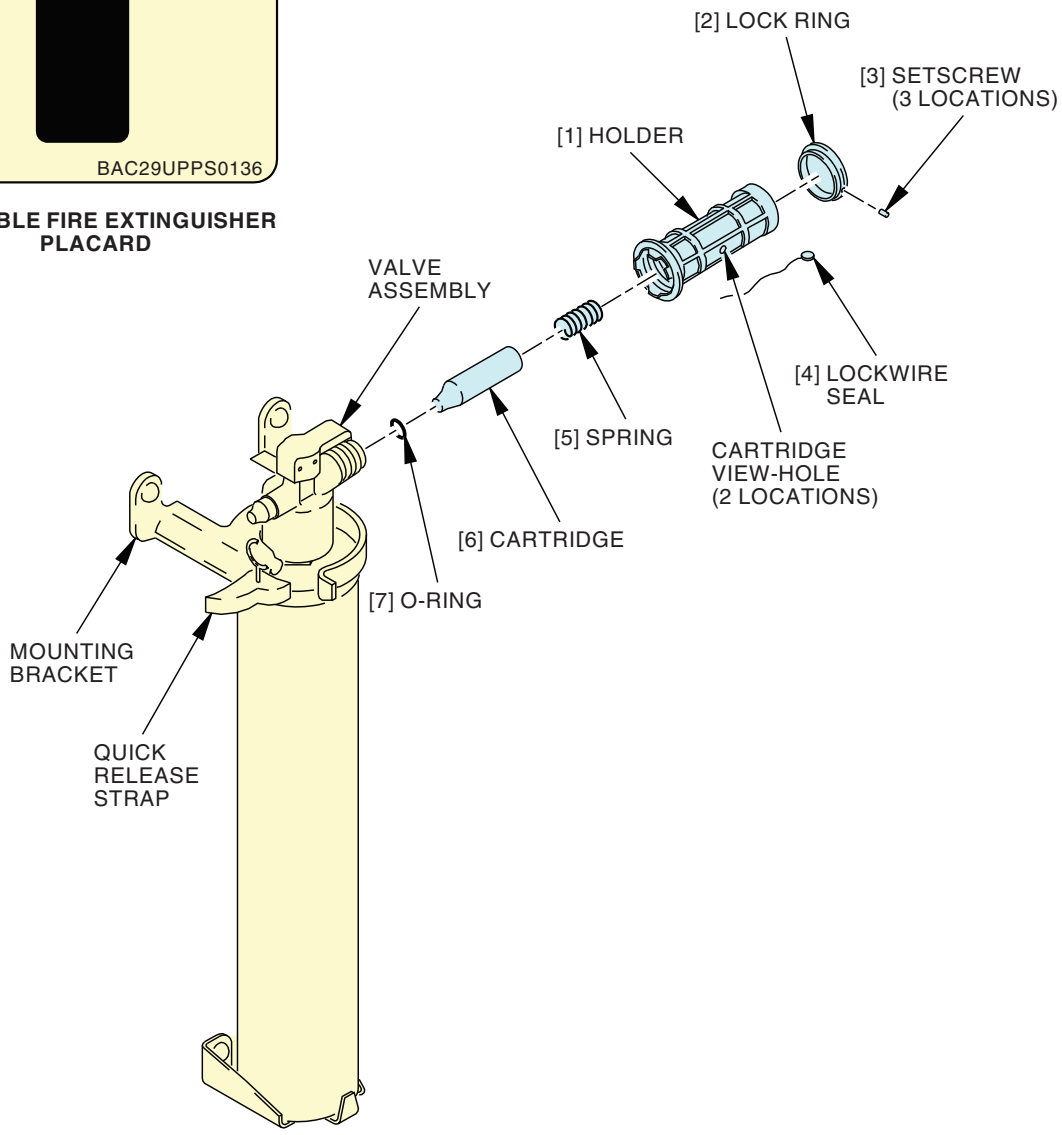
(1) Do this task: Water Fire Extinguisher Installation, TASK 26-26-02-400-801

——— END OF TASK ———

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PORTABLE FIRE EXTINGUISHER PLACARD



WATER FIRE EXTINGUISHER (EXAMPLE)

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Water-type Fire Extinguisher Check
Figure 602/26-26-02-990-803

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